Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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Preface

Note: For any corrections, suggestions, or feedback regarding this solutions manual, readers are encouraged to reach out through the following channels:

- Discord Errata Server: https://discord.gg/qecQk37vfm
- Reddit: https://www.reddit.com/user/ansv9a8fdh3/
- GitHub: https://github.com/blanketism/Axler-Solutions

This solutions manual accompanies the fourth edition of Sheldon Axler's *Linear Algebra Done Right*. While several community-maintained solution sets exist for the third edition, the fourth edition introduces enough new and restructured material to warrant a dedicated resource–one written from scratch with the new text in mind.

This manual is written primarily for undergraduates encountering rigorous linear algebra for the first time. Axler's text assumes little beyond mathematical maturity and a basic familiarity with proof writing. There are some solutions in sections that may have mathematical identities that require additional background or insight, and these are explained as needed to ensure clarity and completeness.

A reader may notice certain notes underneath each exercise statement. A guide is as follows:

- **Axler Hint.** *These are hints* (*explicitly indicated, or otherwise*) *provided by Axler to guide the reader and assist in solving the exercise.*
- **Axler Note.** *Small excerpts written by Axler that either provide a reference to some term used in the problem, or offer additional context or clarification relevant to understanding the exercise. I have abstained from including every note provided.*
- **Manual Remark.** *These are remarks provided by myself to assist the reader in understanding the exercise, offering additional insights, clarifications, or alternative perspectives that may not be immediately apparent from the problem statement alone.*

The goal here is not to replace the work of reading and thinking. Linear algebra, particularly in Axler's determinant-free, operator-theoretic style, rewards patience and independent struggle in ways that are difficult to replicate by reading someone else's solution. This manual is best used after a genuine attempt at each problem, as a means of checking reasoning, identifying gaps, or becoming unstuck–not as a first resort.

Despite best efforts, mistakes are inevitable in any document of this scope: incorrect arguments, incomplete cases, notational slips, and the occasional solution that is simply wrong. It has been some time since I first worked through Axler's text, and while I have made every effort to ensure accuracy, I recognize that errors may still exist. Readers who encounter such mistakes are encouraged to report them through the channels listed above, and all contributions to improving this manual are greatly appreciated.

Finally, I am grateful to Professor Axler for writing a transformative text. I recall my first experience with this text as an undergraduate, and it profoundly shaped my understanding and appreciation of linear algebra. This solutions manual is my way of giving back and supporting future students in their journey through Axler's material. I hope that it serves as a helpful companion, providing clarity and guidance while encouraging readers to engage deeply with the subject on their own terms.

1 Vector Spaces

1A $\mathbb{R}^n$ and $\mathbb{C}^n$

Exercise 1A1

Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$ and $\beta = c + di$ for some $a, b, c, d \in \mathbb{R}$. Then

$$\begin{aligned}\alpha + \beta &= (a + bi) + (c + di) \\ &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + di) + (a + bi) = \beta + \alpha.\end{aligned}$$

■

Exercise 1A2

Show that $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha + \beta) + \gamma &= ((a + bi) + (c + di)) + (e + fi) \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= (a + c + e) + (b + d + f)i \\ &= (a + (c + e)) + (b + (d + f))i \\ &= (a + bi) + ((c + di) + (e + fi)) = \alpha + (\beta + \gamma).\end{aligned}$$

■

Exercise 1A3

Show that $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}(\alpha\beta)\gamma &= ((a + bi)(c + di))(e + fi) \\ &= ((ac - bd) + (ad + bc)i)(e + fi) \\ &= (ace - bde - adf - bcf) + (acf - bdf + ade + bce)i \\ &= (ace - adf - bde - bcf) + (acf + ade + bce - bdf)i \\ &= (a + bi)((ce - df) + (cf + de)i) \\ &= (a + bi)((c + di)(e + fi)) = \alpha(\beta\gamma).\end{aligned}$$

■

Exercise 1A4

Show that $\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$ for all $\alpha, \beta, \gamma \in \mathbb{C}$.

Solution. Let $\alpha = a + bi$, $\beta = c + di$, and $\gamma = e + fi$ for some $a, b, c, d, e, f \in \mathbb{R}$. Then

$$\begin{aligned}\gamma(\alpha + \beta) &= (e + fi)((a + bi) + (c + di)) \\ &= (e + fi)((a + c) + (b + d)i) \\ &= (e(a + c) - f(b + d)) + (e(b + d) + f(a + c))i \\ &= (ea - fb + ec - fd) + (eb + fa + ed + fc)i \\ &= (ea - fb) + (eb + fa)i + (ec - fd) + (ed + fc)i \\ &= (e + fi)(a + bi) + (e + fi)(c + di) = \gamma\alpha + \gamma\beta.\end{aligned}$$

■

Exercise 1A5

Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$. We want to find $\beta = c + di$ such that $\alpha + \beta = 0$. This gives

$$(a + bi) + (c + di) = (a + c) + (b + d)i = 0 + 0i.$$

Solving for c and d, we get $c = -a$ and $d = -b$. Hence, $\beta = c + di = -a - bi = -\alpha$.

To see that β is unique, suppose there is another $\gamma \in \mathbb{C}$ such that $\alpha + \gamma = 0$. Then

$$\beta = \beta + 0 = \beta + (\alpha + \gamma) = (\beta + \alpha) + \gamma = 0 + \gamma = \gamma.$$

Hence, β is unique. ■

Exercise 1A6

Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Solution. Let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$ with $\alpha \neq 0$. Then at least one of a or b is nonzero. We want to find $\beta = c + di$ such that $\alpha\beta = 1$. If $a = 0$, then $b \neq 0$, and $\beta = 1/(bi) = -i/b$. If $b = 0$, then $a \neq 0$, and $\beta = 1/a$. Otherwise, both a and b are nonzero. We then have

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i = 1 + 0i.$$

Solving for c and d, we get the system of equations

$$\begin{cases} ac - bd = 1 \\ ad + bc = 0 \end{cases} \implies \begin{cases} a^2c - abd = a \\ b^2c + abd = 0 \end{cases}$$

Adding the two equations, we get

$$(a^2 + b^2)c = a \implies c = \frac{a}{a^2 + b^2}.$$

Substituting this into the second equation, we get

$$abd + b^2\frac{a}{a^2 + b^2} = 0 \implies d = \frac{-b}{a^2 + b^2}.$$

Hence,

$$\beta = c + di = \frac{a}{a^2 + b^2} + \frac{-b}{a^2 + b^2} i.$$

If $\gamma \in \mathbb{C}$ is another complex number such that $\alpha\gamma = 1$, then

$$\gamma = \gamma 1 = \gamma(\alpha\beta) = (\gamma\alpha)\beta = 1\beta = \beta.$$

Hence, β is unique. ∎

Exercise 1A7

Show that

$$\frac{-1 + \sqrt{3}i}{2}$$

is a cube root of 1.

Solution. Note that the complex number given can be written as

$$\frac{1}{2}(-1 + \sqrt{3}i).$$

Cubing $-1 + \sqrt{3}i$, we get

$$\begin{aligned}
(-1 + \sqrt{3}i)^3 &= (-1)^3 + 3(-1)^2(\sqrt{3}i) + 3(-1)(\sqrt{3}i)^2 + (\sqrt{3}i)^3 \\
&= -1 + 3\sqrt{3}i + 3(-1)(-3) + (\sqrt{3}i)^3 \\
&= -1 + 3\sqrt{3}i + 9 - 3\sqrt{3}i \\
&= 8.
\end{aligned}$$

Therefore,

$$\left(\frac{-1 + \sqrt{3}i}{2}\right)^3 = \frac{(-1 + \sqrt{3}i)^3}{8} = \frac{8}{8} = 1.$$ ∎

Exercise 1A8

Find two distinct square roots of i.

Solution. Let $z = a + bi$ for some $a, b \in \mathbb{R}$. We want to find z such that $z^2 = i$. This gives

$$(a + bi)^2 = a^2 - b^2 + 2abi = 0 + 1i.$$

Solving for a and b, we get the system of equations

$$\begin{cases} a^2 - b^2 = 0 \\ 2ab = 1 \end{cases}$$

The first equation yields $a = b$ or $a = -b$. If $a = b$, then the second equation gives

$$2b^2 = 1 \implies b = \pm\frac{1}{\sqrt{2}} \implies a = \pm\frac{1}{\sqrt{2}}.$$

If $a = -b$, this results in $-2b^2 = 1$ which has no solution when $b \in \mathbb{R}$. Hence, the two distinct square roots of i are

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \quad \text{and} \quad -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i. \qquad \blacksquare$$

Exercise 1A9

Find $x \in \mathbb{R}^4$ such that

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

Solution. Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$. The equation becomes

$$(4, -3, 1, 7) + 2(x_1, x_2, x_3, x_4) = (5, 9, -6, 8),$$

which simplifies to the system of equations

$$\begin{cases} 5 = 4 + 2x_1 \\ 9 = -3 + 2x_2 \\ -6 = 1 + 2x_3 \\ 8 = 7 + 2x_4 \end{cases} \implies \begin{cases} 2x_1 = 1 \\ 2x_2 = 12 \\ 2x_3 = -7 \\ 2x_4 = 1 \end{cases}$$

Therefore,

$$x = \left(\frac{1}{2}, 6, -\frac{7}{2}, \frac{1}{2}\right). \qquad \blacksquare$$

Exercise 1A10

Explain why there does not exist $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

Solution. Suppose there exists $\lambda \in \mathbb{C}$ such that

$$\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i).$$

This gives the system of equations

$$\begin{cases} \lambda(2 - 3i) = 12 - 5i \\ \lambda(5 + 4i) = 7 + 22i \\ \lambda(-6 + 7i) = -32 - 9i \end{cases}$$

Solving the first equation for λ, we get

$$\lambda = \frac{12 - 5i}{2 - 3i} = \frac{(12 - 5i)(2 + 3i)}{(2 - 3i)(2 + 3i)} = \frac{24 + 36i - 10i - 15i^2}{4 + 9} = \frac{24 + 26i + 15}{13} = 3 + 2i.$$

However, using this value for λ in the third equation, we get

$$(3 + 2i)(-6 + 7i) = -18 + 21i - 12i + 14i^2 = -18 + 9i - 14 = -32 + 9i \neq -32 - 9i.$$

Hence, no such λ exists. $\blacksquare$

Exercise 1A11

Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n)$, $y = (y_1, y_2, \ldots, y_n)$, and $z = (z_1, z_2, \ldots, z_n)$. Then

$$\begin{aligned}(x + y) + z &= ((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) + (z_1, z_2, \ldots, z_n)\\ &= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) + (z_1, z_2, \ldots, z_n)\\ &= ((x_1 + y_1) + z_1, (x_2 + y_2) + z_2, \ldots, (x_n + y_n) + z_n)\\ &= (x_1 + (y_1 + z_1), x_2 + (y_2 + z_2), \ldots, x_n + (y_n + z_n))\\ &= (x_1, x_2, \ldots, x_n) + ((y_1, y_2, \ldots, y_n) + (z_1, z_2, \ldots, z_n))\\ &= x + (y + z).\end{aligned}$$

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Exercise 1A12

Show that $(ab)x = a(bx)$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned}(ab)x &= (ab)(x_1, x_2, \ldots, x_n)\\ &= ((ab)x_1, (ab)x_2, \ldots, (ab)x_n)\\ &= (a(bx_1), a(bx_2), \ldots, a(bx_n))\\ &= a((bx_1, bx_2, \ldots, bx_n))\\ &= a(bx).\end{aligned}$$

■

Exercise 1A13

Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$. Then

$$\begin{aligned}1x &= 1(x_1, x_2, \ldots, x_n)\\ &= (1x_1, 1x_2, \ldots, 1x_n)\\ &= (x_1, x_2, \ldots, x_n)\\ &= x.\end{aligned}$$

■

Exercise 1A14

Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and $x, y \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$ be elements of $\mathbb{F}^n$ and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned}
\lambda(x+y) &= \lambda((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) \\
&= \lambda(x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) \\
&= (\lambda(x_1 + y_1), \lambda(x_2 + y_2), \ldots, \lambda(x_n + y_n)) \\
&= (\lambda x_1 + \lambda y_1, \lambda x_2 + \lambda y_2, \ldots, \lambda x_n + \lambda y_n) \\
&= (\lambda x_1, \lambda x_2, \ldots, \lambda x_n) + (\lambda y_1, \lambda y_2, \ldots, \lambda y_n) \\
&= \lambda(x_1, x_2, \ldots, x_n) + \lambda(y_1, y_2, \ldots, y_n) = \lambda x + \lambda y.
\end{aligned}$$

■

Exercise 1A15

Show that $(a+b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and $x \in \mathbb{F}^n$.

Solution. Let $x = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$ and $a, b \in \mathbb{F}$. Then

$$\begin{aligned}
(a+b)x &= (a+b)(x_1, x_2, \ldots, x_n) \\
&= ((a+b)x_1, (a+b)x_2, \ldots, (a+b)x_n) \\
&= (ax_1 + bx_1, ax_2 + bx_2, \ldots, ax_n + bx_n) \\
&= (ax_1, ax_2, \ldots, ax_n) + (bx_1, bx_2, \ldots, bx_n) \\
&= ax + bx.
\end{aligned}$$

■

1B Definition of Vector Space

Exercise 1B1

Prove that $-(-v) = v$ for every $v \in V$.

Solution. Let $v \in V$. By the definition of additive inverse, we have

$$v + (-v) = 0 = -(-v) + (-v).$$

Since additive inverses are unique, we conclude that $-(-v) = v$. ■

Exercise 1B2

Suppose $a \in \mathbb{F}, v \in V$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

Solution. If $a = 0$, then we are done. Suppose $a \neq 0$. Then

$$v = 1v = \frac{1}{a}(av) = \frac{1}{a} \cdot 0 = 0.$$ ■

Exercise 1B3

Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.
Manual Remark. *The solution says "dividing by 3" in the sense that we're dividing by the scalar 3 in the field $\mathbb{F}$, not a vector in V.*

Solution. Solving for x explicitly, we obtain

$$3x = w - v \implies x = \frac{1}{3}(w - v).$$

This shows that there exists $x \in V$ such that $v + 3x = w$. To see that x is unique, suppose there exists another $y \in V$ such that $v + 3y = w$. Then dividing by 3, we get

$$3x = w - v = 3y \implies x = y.$$ ■

Exercise 1B4

The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

Solution. The empty set fails to satisfy the requirement that there exists a zero vector $0 \in V$ such that $v + 0 = v$ for all $v \in V$ since it contains no elements. ■

Exercise 1B5

Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here, the 0 on the left side is the number 0, and the 0 on the right side is the additive identity in V.

Solution. Suppose that $0v = 0$ for all $v \in V$. Let $v \in V$. Then

$$v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0.$$

Hence, $(-1)v$ is the additive inverse of v.

Conversely, suppose that for every $v \in V$, there exists an additive inverse $-v \in V$ such that $v + (-v) = 0$. Then $0v = (0 + 0)v = 0v + 0v$. Adding $-0v$ to both sides, we get $0 = 0v$. Hence, $0v = 0$ for all $v \in V$. ■

Exercise 1B6

Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$, define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \qquad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty = \infty + t = \infty + \infty &= \infty, \\ t + (-\infty) = (-\infty) + t = (-\infty) + (-\infty) &= -\infty, \\ \infty + (-\infty) = (-\infty) + \infty &= 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbb{R}$? Explain.

Solution. No, $\mathbb{R} \cup \{\infty, -\infty\}$ is not a vector space over $\mathbb{R}$ as it does not satisfy the associative property of addition. Let $t \in \mathbb{R} \setminus \{0\}$. Then

$$(t + \infty) + (-\infty) = \infty + (-\infty) = 0 \neq t = t + 0 = t + (\infty + (-\infty)).$$ ■

Exercise 1B7

Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S, and show that V^S is a vector space with these definitions.

Solution. Let $f, g \in V^S$ and $t \in \mathbb{F}$. Define addition and scalar multiplication as follows:

$$(f+g)(s) = f(s) + g(s), \quad (tf)(s) = t(f(s))$$

for all $s \in S$. We will show that V^S is a vector space with these operations.

(1) **Commutativity**: Let $f, g \in V^S$. Then for all $s \in S$,

$$(f+g)(s) = f(s) + g(s) = g(s) + f(s) = (g+f)(s).$$

(2) **Associativity**: Let $f, g, h \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$\begin{aligned} ((f+g)+h)(s) &= (f+g)(s) + h(s) \\ &= (f(s)+g(s)) + h(s) \\ &= f(s) + (g(s)+h(s)) \\ &= f(s) + (g+h)(s) \\ &= (f+(g+h))(s), \end{aligned}$$

and

$$\begin{aligned} ((tu)f)(s) &= (tu)(f(s)) \\ &= t(u(f(s))) \\ &= t(uf)(s) \\ &= (t(uf))(s). \end{aligned}$$

(3) **Additive identity**: Let $0 \in V$ be the additive identity in V. Define the zero function $0_S \in V^S$ by $0_S(s) = 0$ for all $s \in S$. Then for all $f \in V^S$ and $s \in S$,

$$(f+0_S)(s) = f(s) + 0_S(s) = f(s) + 0 = f(s) = 0 + f(s) = (0_S + f)(s).$$

Hence, 0_S is the additive identity in V^S.

(4) **Additive inverse**: Let $f \in V^S$. Define the function $-f \in V^S$ by $(-f)(s) = -f(s)$ for all $s \in S$. Then for all $s \in S$,

$$(f+(-f))(s) = f(s) + (-f)(s) = f(s) + (-f(s)) = 0 = 0_S(s).$$

Hence, $-f$ is the additive inverse of f in V^S.

(5) **Multiplicative identity**: For all $f \in V^S$ and $s \in S$,

$$(1f)(s) = 1(f(s)) = f(s).$$

(6) **Distributive properties**: Let $f, g \in V^S$ and $t, u \in \mathbb{F}$. Then for all $s \in S$,

$$\begin{aligned} ((t+u)f)(s) &= (t+u)(f(s)) \\ &= t(f(s)) + u(f(s)) \\ &= (tf)(s) + (uf)(s) \\ &= (tf+uf)(s), \end{aligned}$$

and

$$\begin{aligned} (t(f+g))(s) &= t((f+g)(s)) \\ &= t(f(s)+g(s)) \\ &= t(f(s)) + t(g(s)) \\ &= (tf)(s) + (tg)(s) \\ &= (tf+tg)(s). \end{aligned}$$

■

Exercise 1B8

Suppose V is a real vector space.

- The **complexification** of V, denoted by $V_{\mathbb{C}}$, equals $V \times V$. An element of $V_{\mathbb{C}}$ is an ordered pair (u, v), where $u, v \in V$, but we write this as $u + iv$.
- Addition on $V_{\mathbb{C}}$ is defined by
$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$
for all $u_1, u_2, v_1, v_2 \in V$.
- Complex scalar multiplication on $V_{\mathbb{C}}$ is defined by
$$(a + ib)(u + iv) = (au - bv) + i(av + bu)$$
for all $a, b \in \mathbb{R}$ and $u, v \in V$.

Prove that with the definitions of addition and scalar multiplication as above, $V_{\mathbb{C}}$ is a complex vector space.

Solution. We prove that $V_{\mathbb{C}}$ is a complex vector space by verifying the vector space axioms.

(1) **Commutativity**: Let $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then
$$(u_1 + v_1 i) + (u_2 + v_2 i) = (u_1 + u_2) + i(v_1 + v_2) = (u_2 + u_1) + i(v_2 + v_1) = (u_2 + v_2 i) + (u_1 + v_1 i).$$

(2) **Associativity**: Let $u_1 + v_1 i, u_2 + v_2 i, u_3 + v_3 i \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((u_1 + v_1 i) + (u_2 + v_2 i)) + (u_3 + v_3 i) &= ((u_1 + u_2) + i(v_1 + v_2)) + (u_3 + v_3 i) \\ &= ((u_1 + u_2) + u_3) + i((v_1 + v_2) + v_3) \\ &= (u_1 + (u_2 + u_3)) + i(v_1 + (v_2 + v_3)) \\ &= (u_1 + v_1 i) + ((u_2 + v_2 i) + (u_3 + v_3 i)).\end{aligned}$$

Similarly, let $a + bi, c + di \in \mathbb{C}$ and $u + vi \in V_{\mathbb{C}}$. Then
$$\begin{aligned}((a + bi)(c + di))(u + vi) &= ((ac - bd) + i(ad + bc))(u + vi) \\ &= ((ac - bd)u - (ad + bc)v) + i((ac - bd)v + (ad + bc)u) \\ &= (a(cu - dv) - b(du + cv)) + i(a(du + cv) + b(cu - dv)) \\ &= (a + bi)((cu - dv) + i(du + cv)) \\ &= (a + bi)((c + di)(u + vi)).\end{aligned}$$

(3) **Additive identity**: Let $0 \in V$ be the additive identity in V. Then for all $u + vi \in V_{\mathbb{C}}$,
$$(u + vi) + (0 + 0i) = (u + 0) + i(v + 0) = u + vi = (0 + 0i) + (u + vi).$$

Hence, $0 + 0i$ is the additive identity in $V_{\mathbb{C}}$.

(4) **Additive inverse**: Let $u + vi \in V_{\mathbb{C}}$. Then
$$(u + vi) + (-u + (-v)i) = (u - u) + i(v - v) = 0 + 0i.$$

Hence, $-u + (-v)i$ is the additive inverse of $u + vi$ in $V_{\mathbb{C}}$.

(5) **Multiplicative identity**: For all $u + vi \in V_{\mathbb{C}}$,
$$(1 + 0i)(u + vi) = (1u - 0v) + i(1v + 0u) = u + vi.$$

(6) **Distributive properties**: Let $a + bi, c + di \in \mathbb{C}$ and $u_1 + v_1 i, u_2 + v_2 i \in V_{\mathbb{C}}$. Then

$$\begin{aligned}
(a+bi)((u_1+v_1 i)+(u_2+v_2 i)) &= (a+bi)((u_1+u_2)+i(v_1+v_2)) \\
&= (a(u_1+u_2)-b(v_1+v_2))+i(a(v_1+v_2)+b(u_1+u_2)) \\
&= (au_1-bv_1)+i(av_1+bu_1)+(au_2-bv_2)+i(av_2+bu_2) \\
&= (a+bi)(u_1+v_1 i)+(a+bi)(u_2+v_2 i),
\end{aligned}$$

and

$$\begin{aligned}
((a+bi)+(c+di))(u_1+v_1 i) &= ((a+c)+(b+d)i)(u_1+v_1 i) \\
&= ((a+c)u_1-(b+d)v_1)+i((a+c)v_1+(b+d)u_1) \\
&= (au_1-bv_1)+i(av_1+bu_1)+(cu_1-dv_1)+i(cv_1+du_1) \\
&= (a+bi)(u_1+v_1 i)+(c+di)(u_1+v_1 i).
\end{aligned}$$

■

1C Subspaces

Exercise 1C1

For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.

(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 0\}$
(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 4\}$
(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 x_2 x_3 = 0\}$
(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 = 5x_3\}$

Solution. Let U be the subset of $\mathbb{F}^3$ in question.

(a) Since the zero vector $(0,0,0)$ satisfies $0 + 2(0) + 3(0) = 0$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = (u_1 + 2u_2 + 3u_3) + (v_1 + 2v_2 + 3v_3) = 0 + 0 = 0.$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 + 2(au_2) + 3(au_3) = a(u_1 + 2u_2 + 3u_3) = a \cdot 0 = 0.$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$.

(b) The zero vector $(0,0,0)$ does not satisfy $0 + 2(0) + 3(0) = 4$, so U does not contain the zero vector. Hence, it is not a subspace of $\mathbb{F}^3$.

(c) Suppose $u = (1,0,0)$ and $v = (0,1,1)$. Then u and v are in U since $1 \cdot 0 \cdot 0 = 0$ and $0 \cdot 1 \cdot 1 = 0$. However, $u + v = (1,1,1)$ is not in U since $1 \cdot 1 \cdot 1 = 1 \neq 0$. Hence, U is not closed under addition and is not a subspace of $\mathbb{F}^3$.

(d) Since the zero vector $(0,0,0)$ satisfies $0 = 5(0)$, U is nonempty. Let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be elements of U. Then

$$u_1 + v_1 = 5u_3 + 5v_3 = 5(u_3 + v_3).$$

Hence, $u + v$ is in U. Let $a \in \mathbb{F}$. Then

$$au_1 = a(5u_3) = 5(au_3).$$

Hence, au is in U. Therefore, U is a subspace of $\mathbb{F}^3$. ■

Exercise 1C2

Verify all assertions about subspaces in Example 1.35.

(a) If $b \in \mathbb{F}$, then
$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4 + b\}$$
is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.
(b) The set of continuous, real-valued functions on the interval $[0,1]$ is a subspace of $\mathbb{R}^{[0,1]}$.
(c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.
(d) The set of differentiable real-valued functions f on the interval $(0,3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.
(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^{\infty}$.

Solution. Let U be the subset of $\mathbb{F}^4$ in question.

(a) Suppose U is a subspace of $\mathbb{F}^4$. Then it must contain the zero vector $(0,0,0,0)$. Then $0 = 5(0) + b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4\}$. The proof is similar to that of Exercise 1C1 (d), hence U is a subspace of $\mathbb{F}^4$.

(b) Note that the 0 function on $[0,1]$ is continuous, so $0 \in U$. Let $f, g \in U$ be continuous functions. Then $f + g$ is continuous, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is continuous, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$.

(c) Since the derivative of the 0 function is the 0 function, $0 \in U$. Let $f, g \in U$ be differentiable functions. Then $f + g$ is differentiable, so $f + g \in U$. Let $a \in \mathbb{R}$. Then af is differentiable, so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Suppose U is a subspace of $\mathbb{R}^{(0,3)}$. Then it must contain the 0 function on $(0,3)$. Then $0' = 0 = b$, which implies $b = 0$.

Conversely, suppose $b = 0$. Then $U = \{f \in \mathbb{R}^{(0,3)} \mid f'(2) = 0\}$. Let $f, g \in U$ be differentiable functions such that $f'(2) = g'(2) = 0$. Then

$$(f+g)'(2) = f'(2) + g'(2) = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(2) = af'(2) = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(0,3)}$.

(e) Let U be the set of all sequences of complex numbers with limit 0. Then the zero sequence is in U. Let $(z_n), (w_n) \in U$ be sequences with limit 0. Then

$$\lim_{n\to\infty}(z_n + w_n) = \lim_{n\to\infty} z_n + \lim_{n\to\infty} w_n = 0 + 0 = 0,$$

so $(z_n) + (w_n) \in U$. Let $a \in \mathbb{C}$. Then

$$\lim_{n\to\infty}(az_n) = a \lim_{n\to\infty} z_n = a \cdot 0 = 0,$$

so $a(z_n) \in U$. Therefore, U is a subspace of $\mathbb{C}^\infty$. ■

Exercise 1C3

Show that the set of differentiable real-valued functions f on the interval $(-4,4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

Solution. Let U be the set in question. Then the zero function is in U since $0'(-1) = 0 = 3 \cdot 0 = 3 \cdot 0(2)$. Let $f, g \in U$ be differentiable functions such that $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$. Then

$$(f+g)'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f+g)(2),$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$(af)'(-1) = af'(-1) = a \cdot 3f(2) = 3(af)(2),$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{(-4,4)}$. ■

Exercise 1C4

Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0,1]$ such that

$$\int_0^1 f = b$$

is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Solution. Let U be the set in question. Suppose U is a subspace of $\mathbb{R}^{[0,1]}$. Then it must contain the zero function on $[0,1]$. Then

$$\int_0^1 0 = 0 = b,$$

which implies $b = 0$.

Conversely, suppose $b = 0$. Then

$$U = \left\{ f \in \mathbb{R}^{[0,1]} \;\middle|\; \int_0^1 f = 0 \right\}.$$

The zero function is in U since $\int_0^1 0 = 0$. Let $f, g \in U$ be continuous functions such that $\int_0^1 f = \int_0^1 g = 0$. Then

$$\int_0^1 (f+g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0,$$

so $f + g \in U$. Let $a \in \mathbb{R}$. Then

$$\int_0^1 (af) = a \int_0^1 f = a \cdot 0 = 0,$$

so $af \in U$. Therefore, U is a subspace of $\mathbb{R}^{[0,1]}$. ■

Exercise 1C5

Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Solution. No, $\mathbb{R}^2$ is not a subspace of $\mathbb{C}^2$. Let $u = (1,0) \in \mathbb{R}^2$ and $a = i \in \mathbb{C}$. Then $au = (i,0) \notin \mathbb{R}^2$, so $\mathbb{R}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{C}^2$. ■

Exercise 1C6

(a) Is $\{(a,b,c) \in \mathbb{R}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
(b) Is $\{(a,b,c) \in \mathbb{C}^3 \mid a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Manual Remark. *Here, I present an elementary argument that does not require the use of De Moivre's Theorem; rather, I arrive at the conclusion directly. However, readers who are familiar with it may find it easier to note that the complex number 1 has three cube roots of unity. Other readers may also use Vieta's formulas to arrive at the same polynomial in ω to obtain $1 + \omega = -\omega^2$.*

Solution. Let U be the subset in question.

(a) Since the zero vector $(0,0,0)$ satisfies $0^3 = 0^3$, U is nonempty. Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be elements of U. Then $x_1^3 = x_2^3$ and $y_1^3 = y_2^3$ both imply that $x_1 = x_2$ and $y_1 = y_2$. Hence,

$$(x_1 + y_1)^3 = (x_2 + y_2)^3,$$

so $x + y \in U$. Let $a \in \mathbb{R}$. Then

$$(ax_1)^3 = a^3 x_1^3 = a^3 x_2^3 = (ax_2)^3,$$

so $ax \in U$. Therefore, U is a subspace of $\mathbb{R}^3$.

(b) No, U is not a subspace of $\mathbb{C}^3$. To see this, consider the cubic polynomial $x^3 - 1 = 0$. This polynomial has three distinct roots in $\mathbb{C}$ as follows:

$$x^3 - 1 = (x-1)(x^2 + x + 1) = 0$$

Applying the quadratic formula to $x^2 + x + 1 = 0$, we get

$$x = \frac{-1 \pm \sqrt{(-1)^2 - 4(1)(1)}}{2(1)} = \frac{-1 \pm i\sqrt{3}}{2}.$$

Let

$$\omega = \frac{-1 + i\sqrt{3}}{2}.$$

Then $\omega^3 = 1$, and

$$\omega^2 = \frac{-1 - i\sqrt{3}}{2}.$$

Moreover, it is not hard to see that $\omega + \omega^2 = -1$. Let $u = (1,1,0)$ and $v = (\omega, 1, 0)$. Then $u, v \in U$ since $1^3 = 1^3$ and $\omega^3 = 1^3$. Then $u + v = (1 + \omega, 2, 0)$. However,

$$(1+\omega)^3 = 1 + 3\omega + 3\omega^2 + \omega^3 = 1 + 3(-1) + 1 = -1 \neq 2^3 = 8.$$

Hence, $u + v \notin U$, so U is not closed under addition and is not a subspace of $\mathbb{C}^3$. ■

Exercise 1C7

Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Solution. Consider the subset $\mathbb{Q}^2$ of $\mathbb{R}^2$. Then $\mathbb{Q}^2$ is nonempty because $(0,0) \in \mathbb{Q}^2$, closed under addition since the sum of any two rational numbers is rational, and closed under taking additive inverses since the additive inverse of a rational number is rational. However, for any irrational number $r \in \mathbb{R}$, the vector $(r, 0) = r(1,0) \in \mathbb{R}^2$ is not in $\mathbb{Q}^2$. Hence, $\mathbb{Q}^2$ is not closed under scalar multiplication and is not a subspace of $\mathbb{R}^2$. Therefore, the statement is false. ■

Exercise 1C8

Give an example of a nonempty subset U of $\mathbb{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbb{R}^2$.

Solution. Consider the subset

$$U = \{(x, 0) \mid x \in \mathbb{R}\} \cup \{(0, y) \mid y \in \mathbb{R}\}.$$

Then U is nonempty since $(1, 0) \in U$, and it is closed under scalar multiplication since any scalar multiple of $(1, 0)$ or $(0, 1)$ is still in U. However, U is not closed under addition since $(1, 0) + (0, 1) = (1, 1) \notin U$. Hence, U is not a subspace of $\mathbb{R}^2$. ■

Exercise 1C9

A function $f : \mathbb{R} \to \mathbb{R}$ is called **periodic** if there exists a positive number p such that $f(x+p) = f(x)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$. Explain.

Solution. It is not a subspace of $\mathbb{R}^{\mathbb{R}}$. Consider the periodic functions $f(x) = \cos x$ and $g(x) = \cos \sqrt{2}x$. Then the sum $f + g$ attains a maximum value of 2 at 0. If $f + g$ were periodic with period p, then $(f + g)(p) = (f + g)(0) = 2$. This implies $\cos p = \cos p\sqrt{2} = 1$. Then $p = 2\pi m$ and $p\sqrt{2} = 2\pi n$ for positive integers m and n. Since $p\sqrt{2} = 2\pi n$ implies $p = \sqrt{2}\pi n$, we have

$$2\pi m = \sqrt{2}\pi n \implies \sqrt{2} = \frac{n}{m},$$

which is a contradiction, since $\sqrt{2}$ is irrational. Then $f + g$ is not periodic. ■

Exercise 1C10

Suppose V_1 and V_2 are subspaces of V. Prove that the intersection $V_1 \cap V_2$ is a subspace of V.

Solution. Let $U = V_1 \cap V_2$. Since both V_1 and V_2 are subspaces of V, they both contain the zero vector of V. Hence, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in V_1$ and $u, v \in V_2$. Since both V_1 and V_2 are subspaces, they are closed under addition. Therefore, $u + v \in V_1$ and $u + v \in V_2$, which implies that $u + v \in U$. Let $a \in \mathbb{F}$. Then since both V_1 and V_2 are closed under scalar multiplication, we have that $au \in V_1$ and $au \in V_2$, which implies that $au \in U$. Therefore, U is a subspace of V. ■

Exercise 1C11

Prove that the intersection of every collection of subspaces of V is a subspace of V.

Solution. Let C be a collection of subspaces of V, and let

$$U = \bigcap_{W \in C} W.$$

Since each subspace in C contains the zero vector, $0 \in U$, so U is nonempty. Let $u, v \in U$. Then $u, v \in W$ for all $W \in C$. Since each W is a subspace, they are closed under addition, so $u + v \in W$ for all $W \in C$, which implies $u + v \in U$. Let $a \in \mathbb{F}$. Then $au \in W$ for all $W \in C$, which implies $au \in U$. Therefore, U is a subspace of V. ■

Exercise 1C12

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Solution. Let U, W be subspaces of V. Suppose $U \subseteq W$. Then $U \cup W = W$, which is a subspace of V. Similarly, if $W \subseteq U$, then $U \cup W = U$, which is a subspace of V.

Conversely, suppose $U \cup W$ is a subspace of V. If neither $U \subseteq W$ nor $W \subseteq U$, then there exist $u \in U \setminus W$ and $w \in W \setminus U$. Since $U \cup W$ is a subspace, it must be closed under addition. Then $u + w \in U \cup W$. However, $u + w \notin U$ because if it were, then $w = (u + w) - u \in U$, which is a contradiction. Similarly, $u + w \notin W$ because if it were, then $u = (u + w) - w \in W$, which is also a contradiction. Hence, $u + w \notin U \cup W$, which contradicts the assumption that $U \cup W$ is a subspace. Therefore, one of the subspaces must be contained in the other. ■

Exercise 1C13

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Axler Note. *This result fails when the field has only two elements.*

Manual Remark. *The proof of this exercise is more involved than the proof of* Exercise 1C12 *in that it requires a more careful analysis of the possible relationships between the three subspaces.*

Appendix. *See* Appendix 1C13 *for a more elegant proof of this result, which implicitly uses the result that* $\operatorname{char}(\mathbb{F}) \neq 2$.

Solution. Let U_1, U_2, U_3 be subspaces of V. Without loss of generality, suppose $U_2 \subseteq U_1$ and $U_3 \subseteq U_1$. Then $U_1 \cup U_2 \cup U_3 = U_1$, which is a subspace of V.

Conversely, suppose $U_1 \cup U_2 \cup U_3$ is a subspace of V. Suppose that some pair satisfies containment, say $U_3 \subseteq U_2$. Let $W = U_2 \cup U_3 = U_2$. Then $U_1 \cup W = U_1 \cup U_2 \cup U_3$. By Exercise 1C12, either $U_1 \subseteq U_2$ or $U_2 \subseteq U_1$. If the former occurs, then $U_1 \cup U_2 = U_2$, which contains U_3. If the latter occurs, then $U_1 \cup U_2 = U_1$. Since $U_3 \subseteq U_2$, then $U_3 \subseteq U_1$. Therefore, one of the subspaces must contain the other two.

Now suppose no subspace is contained in another. Then there exist $u \in U_2 \setminus U_3$ and $v \in U_3 \setminus U_2$. Since $U_1 \cup U_2 \cup U_3$ is a subspace, $u + \alpha v \in U_1 \cup U_2 \cup U_3$ for any nonzero $\alpha \in \mathbb{F}$. If $u + \alpha v \in U_2$, then $\alpha v = (u + \alpha v) - u \in U_2$, which implies $v \in U_2$, a contradiction. If $u + \alpha v \in U_3$, then $u = (u + \alpha v) - \alpha v \in U_3$, a contradiction. Then $u + \alpha v \in U_1$ for all nonzero $\alpha \in \mathbb{F}$. Now choose nonzero $\beta, \gamma \in \mathbb{F}$ with $\beta \neq \gamma$. Then $u + \beta v, u + \gamma v \in U_1$, so

$$(u + \beta v) - (u + \gamma v) = (\beta - \gamma)v \in U_1.$$

Since $\beta - \gamma \neq 0$, we have $v \in U_1$. Similarly, $(u + \beta v) - \beta v = u \in U_1$. Since $u \in U_2 \setminus U_3$ and $v \in U_3 \setminus U_2$ were arbitrary, $U_2 \setminus U_3 \subseteq U_1$ and $U_3 \setminus U_2 \subseteq U_1$.

Now let $w \in U_2 \cap U_3$. Since $U_2 \not\subseteq U_3$, there exists $x \in U_2 \setminus U_3 \subseteq U_1$. Then $w + x \in U_2$. If $w + x \in U_3$, then $x = (w + x) - w \in U_3$, a contradiction. Hence, $w + x \in U_2 \setminus U_3 \subseteq U_1$, so $w = (w + x) - x \in U_1$. Therefore, $U_2 \cap U_3 \subseteq U_1$, which gives $U_2 = (U_2 \setminus U_3) \cup (U_2 \cap U_3) \subseteq U_1$ and similarly $U_3 \subseteq U_1$, contradicting our assumption. Hence, one of the subspaces must contain the other two. ■

Exercise 1C14

Suppose
$$U = \{(x, -x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\} \quad \text{and} \quad W = \{(x, x, 2x) \in \mathbb{F}^3 \mid x \in \mathbb{F}\}.$$
Describe $U + W$ using symbols, and also give a description of $U + W$ that uses no symbols.

Solution. Let $u = (x, -x, 2x) \in U$ and $w = (y, y, 2y) \in W$. Then
$$u + w = (x + y, -x + y, 2x + 2y) = (x + y, y - x, 2(x + y)).$$
Observe that $x + y$ can take any value in $\mathbb{F}$, and $y - x$ can also take any value in $\mathbb{F}$. Hence, we can write
$$U + W = \{(a, b, 2a) \in \mathbb{F}^3 \mid a, b \in \mathbb{F}\}.$$
In words, $U + W$ is the set of all vectors in $\mathbb{F}^3$ whose third component is twice the first component, and whose second component can be any value in $\mathbb{F}$. ■

Exercise 1C15

Suppose U is a subspace of V. What is $U + U$?

Solution. Let $u_1, u_2 \in U$. Then $u_1 + u_2 \in U$ since U is closed under addition. Hence, $U + U \subseteq U$. Conversely, let $u \in U$. Then $u = u + 0$, where 0 is the additive identity in U. Since $0 \in U$, we have that $u \in U + U$. Hence, $U \subseteq U + U$. Therefore, $U + U = U$. ■

Exercise 1C16

Is the operation of addition on subspaces of V commutative? In other words, if U and W are subspaces of V, is $U + W = W + U$?

Solution. Yes, the operation of addition on subspaces of V is commutative. Let $u \in U$ and $w \in W$. Then $u + w \in U + W$. Because addition in V is commutative, we have that $u + w = w + u$, where $w \in W$ and $u \in U$. Hence, $w + u \in W + U$. Therefore, every element of $U + W$ is also an element of $W + U$, which implies that $U + W \subseteq W + U$. By a similar argument, we can show that $W + U \subseteq U + W$. Hence, $U + W = W + U$. ■

Exercise 1C17

Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is
$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Solution. Yes, the operation of addition on subspaces of V is associative. Let $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Then $(u_1 + u_2) + u_3 \in (V_1 + V_2) + V_3$. Because addition in V is associative, we have that $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$, where $u_1 \in V_1$, $u_2 \in V_2$, and $u_3 \in V_3$. Hence, $u_1 + (u_2 + u_3) \in V_1 + (V_2 + V_3)$. Therefore, every element of $(V_1 + V_2) + V_3$ is also an element of $V_1 + (V_2 + V_3)$, which implies that

$(V_1+V_2)+V_3 \subseteq V_1+(V_2+V_3)$. By a similar argument, we can show that $V_1+(V_2+V_3) \subseteq (V_1+V_2)+V_3$. Hence, $(V_1+V_2)+V_3 = V_1+(V_2+V_3)$. ■

Exercise 1C18

Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

Solution. Yes, the operation of addition on the subspaces of V has an additive identity. The additive identity is the zero subspace $\{0\}$, which contains only the zero vector of V. For any subspace U of V, we have that $U + \{0\} = U$.

If a subspace U of V had an additive inverse W, then we would have that $U + W = \{0\}$. However, $U \subseteq U + W$ since $u = u + 0$ for any $u \in U$. Hence, $U + W = \{0\}$ implies that $U = \{0\}$. Therefore, the only subspace of V that has an additive inverse is the zero subspace $\{0\}$. ■

Exercise 1C19

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U,$$

then $V_1 = V_2$.

Solution. The statement is false. Consider the vector space $\mathbb{R}^2$ and let

$$V_1 = \{(x,0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}, \quad V_2 = \{(0,y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\}, \quad U = \{(x,x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}.$$

Then $V_1 + U = \mathbb{R}^2$ and $V_2 + U = \mathbb{R}^2$, but $V_1 \neq V_2$. Therefore, the statement is false. ■

Exercise 1C20

Suppose

$$U = \{(x,x,y,y) \in \mathbb{F}^4 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

Solution. Let

$$W = \{(0,z,0,w) \in \mathbb{F}^4 \mid z, w \in \mathbb{F}\}.$$

If $(x,x,y,y) \in U$ and $(0,z,0,w) \in W$, then $(x,x,y,y) = (0,z,0,w)$ if and only if $x = 0$, $y = 0$, $z = 0$, and $w = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a,b,c,d) \in \mathbb{F}^4$, we can write

$$(a,b,c,d) = (x,x,y,y) + (0,z,0,w),$$

where

$$x = a, \quad y = c, \quad z = b - x = b - a, \quad w = d - y = d - c.$$

Hence, every vector in $\mathbb{F}^4$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^4 = U \oplus W$. ■

Exercise 1C21

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W$.

Solution. Let

$$W = \{(0, 0, z, w, v) \in \mathbb{F}^5 \mid z, w, v \in \mathbb{F}\}.$$

If $(x, y, x + y, x - y, 2x) \in U$ and $(0, 0, z, w, v) \in W$, then $(x, y, x + y, x - y, 2x) = (0, 0, z, w, v)$ if and only if $x = 0$, $y = 0$, $z = 0$, $w = 0$, and $v = 0$. Hence, $U \cap W = \{0\}$. Moreover, for any $(a, b, c, d, e) \in \mathbb{F}^5$, we can write

$$(a, b, c, d, e) = (x, y, x + y, x - y, 2x) + (0, 0, z, w, v),$$

which implies

$$\begin{cases} a = x + 0 \\ b = y + 0 \\ c = (x + y) + z \\ d = (x - y) + w \\ e = 2x + v \end{cases} \implies \begin{cases} x = a \\ y = b \\ z = c - (x + y) = c - (a + b) \\ w = d - (x - y) = d - (a - b) \\ v = e - 2x = e - 2a \end{cases}$$

Hence, every vector in $\mathbb{F}^5$ can be written as a sum of a vector in U and a vector in W. Therefore, $\mathbb{F}^5 = U \oplus W$. ■

Exercise 1C22

Suppose

$$U = \{(x, y, x + y, x - y, 2x) \in \mathbb{F}^5 \mid x, y \in \mathbb{F}\}.$$

Find three subspaces W_1, W_2, W_3 of $\mathbb{F}^5$, none of which equals $\{0\}$, such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.
Manual Remark. *The derivation of z, w, e is from* Exercise 1C21.

Solution. Observe that U can be expressed as

$$U = \{(x, 0, x, x, 2x) \in \mathbb{F}^5 \mid x \in \mathbb{F}\} + \{(0, y, y, -y, 0) \in \mathbb{F}^5 \mid y \in \mathbb{F}\}.$$

Define the subspaces

$$\begin{aligned} W_1 &= \{(0, 0, z, 0, 0) \in \mathbb{F}^5 \mid z \in \mathbb{F}\}, \\ W_2 &= \{(0, 0, 0, w, 0) \in \mathbb{F}^5 \mid w \in \mathbb{F}\}, \\ W_3 &= \{(0, 0, 0, 0, v) \in \mathbb{F}^5 \mid v \in \mathbb{F}\}. \end{aligned}$$

Consider $(a, b, c, d, e) \in \mathbb{F}^5$. We can write

$$(a, b, c, d, e) = (x, 0, x, x, 2x) + (0, y, y, -y, 0) + (0, 0, z, 0, 0) + (0, 0, 0, w, 0) + (0, 0, 0, 0, v),$$

where

$$\begin{aligned} x &= a, \\ y &= b, \\ z &= c - a - b, \\ w &= d - a + b, \\ v &= e - 2a. \end{aligned}$$

Hence, every vector in $\mathbb{F}^5$ can be written as a sum of a vector in U and vectors in W_1, W_2, W_3. Moreover, if $a = b = c = d = e = 0$, then $x = y = z = w = v = 0$ so that the only way to write the zero vector as a sum of vectors in U, W_1, W_2, W_3 is to take all of them to be the zero vector. Therefore, $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$. ■

Exercise 1C23

Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that

$$V = V_1 \oplus U \quad \text{and} \quad V = V_2 \oplus U,$$

then $V_1 = V_2$.

Solution. See Exercise 1C19 for a counterexample. For $V_1 \oplus U$, observe that $V_1 \cap U = \{0\}$. Moreover, for any $(a, b) \in \mathbb{R}^2$, we can write

$$(a, b) = (a - b, 0) + (b, b) \in V_1 \oplus U.$$

The proof for $V_2 \oplus U$ is similar. Hence, $V = V_1 \oplus U = V_2 \oplus U$, but $V_1 \neq V_2$. Therefore, the statement is false. ■

Exercise 1C24

A function $f : \mathbb{R} \to \mathbb{R}$ is called **even** if

$$f(-x) = f(x)$$

for all $x \in \mathbb{R}$. A function $f : \mathbb{R} \to \mathbb{R}$ is called **odd** if

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Let V_e denote the set of real-valued even functions on $\mathbb{R}$, and let V_o denote the set of real-valued odd functions on $\mathbb{R}$. Show that $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.

Solution. Let $f \in \mathbb{R}^{\mathbb{R}}$. Define

$$f_e(x) = \frac{f(x) + f(-x)}{2} \quad \text{and} \quad f_o(x) = \frac{f(x) - f(-x)}{2}.$$

Then f_e is even since

$$f_e(-x) = \frac{f(-x) + f(x)}{2} = f_e(x),$$

and f_o is odd since

$$f_o(-x) = \frac{f(-x) - f(x)}{2} = -\frac{f(x) - f(-x)}{2} = -f_o(x).$$

Moreover, we have that

$$f(x) = f_e(x) + f_o(x).$$

Hence, every function in $\mathbb{R}^{\mathbb{R}}$ can be written as a sum of an even function and an odd function.

Now, suppose $g \in V_e \cap V_o$. Then g is both even and odd. Then for any $x \in \mathbb{R}$, we have

$$g(x) = g(-(-x)) = -g(-x) = -g(x),$$

which implies that $2g(x) = 0$, or $g(x) = 0$. Hence, the only function in the intersection of V_e and V_o is the zero function. Therefore, $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$. ■

2 Finite-Dimensional Vector Spaces

2A Span and Linear Independence

Exercise 2A1

Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$\{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}$$

Solution. Clearly, the span of the list involves two vectors, since we may rewrite z in terms of x and y as $z = -x - y$. Thus, we may choose the following two vectors:

$$v_1 = (1, 0, -1), \quad v_2 = (0, 1, -1).$$

To find two more vectors, we may pick any two distinct linear combinations of v_1 and v_2. For example, we may select

$$v_3 = v_1 + v_2 = (1, 1, -2), \quad v_4 = 2v_1 + 3v_2 = (2, 3, -5).$$

Then

$$\operatorname{span}(v_1, v_2, v_3, v_4) = \operatorname{span}(v_1, v_2) = \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$ ∎

Exercise 2A2

Prove or give a counterexample: If v_1, v_2, v_3, v_4 spans V, then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V.

Solution. Let $U = \{v_1, v_2, v_3, v_4\}$ and $W = \{v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4\}$. Clearly, $v_4 \in \operatorname{span} U$. Then we may write

$$\begin{aligned} v_3 &= (v_3 - v_4) + v_4 \in \operatorname{span} W, \\ v_2 &= (v_2 - v_3) + v_3 \in \operatorname{span} W, \\ v_1 &= (v_1 - v_2) + v_2 \in \operatorname{span} W. \end{aligned}$$

Thus, $v_1, v_2, v_3, v_4 \in \operatorname{span} W$, and so $V = \operatorname{span} U \subseteq \operatorname{span} W$. Since W is a list of vectors in V, we have $\operatorname{span} W \subseteq V$. Therefore, $\operatorname{span} W = V$, and so W spans V. ∎

Exercise 2A3

Suppose $v_1, \dots, v_m$ is a list of vectors in V. For $k \in \{1, \dots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $\operatorname{span}(v_1, \dots, v_m) = \operatorname{span}(w_1, \dots, w_m)$.

Solution. This is a generalization of Exercise 2A2 and a direct consequence of the fact that $v_k = w_k - w_{k-1}$ for $k \in \{2, \ldots, m\}$ and $v_1 = w_1$. Since each v_k is a linear combination of $w_1, \ldots, w_k$, we have $\operatorname{span}(v_1, \ldots, v_m) \subseteq \operatorname{span}(w_1, \ldots, w_m)$. On the other hand, each w_k is a linear combination of $v_1, \ldots, v_k$, so $\operatorname{span}(w_1, \ldots, w_m) \subseteq \operatorname{span}(v_1, \ldots, v_m)$. Therefore, $\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(w_1, \ldots, w_m)$. ■

Exercise 2A4

(a) Show that a list of length one in a vector space is linearly independent if and only if the vector in the list is not 0.
(b) Show that a list of length two in a vector space is linearly independent if and only if neither of the two vectors in the list is a scalar multiple of the other.

Solution.

(a) Let v be the vector in the list. If $v = 0$, then the list is linearly dependent, since $0 \cdot v = 0$. If $v \neq 0$, then the only way to write 0 as a linear combination of v is to take the scalar multiple to be 0. Thus, the list is linearly independent.
(b) Let v_1 and v_2 be the vectors in the list. If v_1 is a scalar multiple of v_2, then there exists a scalar λ such that $v_1 = \lambda v_2$. Then we may write

$$0 = v_1 - \lambda v_2,$$

which shows that the list is linearly dependent. Conversely, if the list is linearly dependent, then there exist scalars α and β, not both zero, such that

$$\alpha v_1 + \beta v_2 = 0.$$

If $\alpha \neq 0$, then

$$v_1 = -\frac{\beta}{\alpha} v_2.$$

Similarly, if $\beta \neq 0$, then

$$v_2 = -\frac{\alpha}{\beta} v_1.$$

Thus, the list is linearly independent if and only if neither vector is a scalar multiple of the other. ■

Exercise 2A5

Find a number t such that

$$(3, 1, 4), (2, -3, 5), (5, 9, t)$$

is not linearly independent in $\mathbb{R}^3$.

Solution. To say that the list is not linearly independent is to say that there exist scalars α, β, γ, not all zero, such that

$$\alpha(3, 1, 4) + \beta(2, -3, 5) + \gamma(5, 9, t) = (0, 0, 0).$$

This is equivalent to the following system of equations:

$$\begin{aligned} 0 &= 3\alpha + 2\beta + 5\gamma, \\ 0 &= \alpha - 3\beta + 9\gamma, \\ 0 &= 4\alpha + 5\beta + t\gamma. \end{aligned}$$

From the second equation, we obtain $\alpha = 3\beta - 9\gamma$. Substituting this into the first equation, we have

$$\begin{aligned} 0 &= 3(3\beta - 9\gamma) + 2\beta + 5\gamma \\ &= 11\beta - 22\gamma. \end{aligned}$$

Thus, $\beta = 2\gamma$. Substituting this into the expression for α, we have $\alpha = 3(2\gamma) - 9\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= 4(-3\gamma) + 5(2\gamma) + t\gamma \\ &= -12\gamma + 10\gamma + t\gamma \\ &= (t-2)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $t - 2 = 0$, or $t = 2$. Then the list is not linearly independent when $t = 2$. ■

Exercise 2A6

Show that the list $(2,3,1),(1,-1,2),(7,3,c)$ is linearly dependent in $\mathbb{F}^3$ if and only if $c = 8$.

Solution. We may proceed similarly to Exercise 2A5 to obtain the following system of equations:

$$\begin{aligned} 0 &= 2\alpha + \beta + 7\gamma, \\ 0 &= 3\alpha - \beta + 3\gamma, \\ 0 &= \alpha + 2\beta + c\gamma. \end{aligned}$$

The second equation gives us $\beta = 3\alpha + 3\gamma$. Substituting this into the first equation, we have

$$\begin{aligned} 0 &= 2\alpha + (3\alpha + 3\gamma) + 7\gamma \\ &= 5\alpha + 10\gamma. \end{aligned}$$

Thus, $\alpha = -2\gamma$. Substituting this into the expression for β, we have $\beta = 3(-2\gamma) + 3\gamma = -3\gamma$. Substituting these values into the third equation, we have

$$\begin{aligned} 0 &= (-2\gamma) + 2(-3\gamma) + c\gamma \\ &= -2\gamma - 6\gamma + c\gamma \\ &= (c-8)\gamma. \end{aligned}$$

Since $\gamma \neq 0$, we must have $c - 8 = 0$, or $c = 8$. Then the list is linearly dependent when $c = 8$.

Conversely, if $c \neq 8$, then $(c-8)\gamma = 0$ implies $\gamma = 0$. Then $\alpha = -2\gamma = 0$ and $\beta = -3\gamma = 0$, so the only solution to the system of equations is $\alpha = \beta = \gamma = 0$. Thus, the list is linearly independent when $c \neq 8$. ■

Exercise 2A7

(a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1+i, 1-i$ is linearly independent.

(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list $1+i, 1-i$ is linearly dependent.

Solution.

(a) Suppose $\alpha, \beta \in \mathbb{R}$ such that
$$\alpha(1+i) + \beta(1-i) = 0.$$
Then we have
$$(\alpha+\beta) + (\alpha-\beta)i = 0.$$
Since $0 = 0 + 0i$, we must have $\alpha + \beta = 0$ and $\alpha - \beta = 0$. Solving this system of equations, we find that $\alpha = \beta = 0$. Thus, the list is linearly independent.

(b) Suppose $a + bi, c + di \in \mathbb{C}$ such that
$$(a+bi)(1+i) + (c+di)(1-i) = 0.$$
Then we have
$$(a-b+c+d) + (a+b-c+d)i = 0.$$
Then we have the system of equations
$$\begin{aligned} 0 &= a - b + c + d, \\ 0 &= a + b - c + d. \end{aligned}$$
Adding the two equations results in $0 = 2a + 2d$, or $a = -d$. Substituting this into the first equation, we obtain $b = c$. Then we have infinitely many nontrivial solutions to the system of equations, such as $a = 1, b = 0, c = 0, d = -1$. Thus, the list is linearly dependent. ■

Exercise 2A8

Suppose v_1, v_2, v_3, v_4 is linearly independent in V. Prove that the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ is also linearly independent.

Solution. Let $\alpha, \beta, \gamma, \delta \in \mathbb{F}$ such that
$$\alpha(v_1 - v_2) + \beta(v_2 - v_3) + \gamma(v_3 - v_4) + \delta v_4 = 0.$$
Then we have
$$\alpha v_1 + (-\alpha+\beta)v_2 + (-\beta+\gamma)v_3 + (-\gamma+\delta)v_4 = 0.$$
Since v_1, v_2, v_3, v_4 is linearly independent, we must have
$$\begin{aligned} 0 &= \alpha, \\ 0 &= -\alpha + \beta, \\ 0 &= -\beta + \gamma, \\ 0 &= -\gamma + \delta. \end{aligned}$$
From the first equation, we have $\alpha = 0$. Substituting this into the second equation gives us $\beta = 0$. Substituting this into the third equation gives us $\gamma = 0$. Finally, substituting this into the fourth equation gives us $\delta = 0$. Thus, the only solution to the original equation is $\alpha = \beta = \gamma = \delta = 0$, and so the list is linearly independent. ■

Exercise 2A9

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V, then $5v_1 - 4v_2, v_2, v_3, \ldots, v_m$ is linearly independent.

Solution. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(5v_1 - 4v_2) + \alpha_2 v_2 + \cdots + \alpha_m v_m = 0.$$

Then we have

$$5\alpha_1 v_1 + (-4\alpha_1 + \alpha_2)v_2 + \alpha_3 v_3 + \cdots + \alpha_m v_m = 0.$$

Since $v_1, v_2, \ldots, v_m$ is linearly independent, we must have

$$\begin{aligned} 0 &= 5\alpha_1, \\ 0 &= -4\alpha_1 + \alpha_2, \\ 0 &= \alpha_3, \\ &\vdots \\ 0 &= \alpha_m. \end{aligned}$$

From the first equation, we have $\alpha_1 = 0$. Substituting this into the second equation gives us $\alpha_2 = 0$. The remaining equations give us $\alpha_k = 0$ for $k = 3, \ldots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \ldots, m$, and so the list is linearly independent. ■

Exercise 2A10

Prove or give a counterexample: If $v_1, v_2, \ldots, v_m$ is a linearly independent list of vectors in V and $\lambda \in \mathbb{F}$ with $\lambda \neq 0$, then $\lambda v_1, \lambda v_2, \ldots, \lambda v_m$ is linearly independent.

Solution. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1(\lambda v_1) + \cdots + \alpha_m(\lambda v_m) = 0.$$

Then we have

$$\lambda(\alpha_1 v_1 + \cdots + \alpha_m v_m) = 0.$$

Since $\lambda \neq 0$, we must have

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0.$$

Since $v_1, v_2, \ldots, v_m$ is linearly independent, we must have $\alpha_k = 0$ for $k = 1, \ldots, m$. Thus, the only solution to the original equation is $\alpha_k = 0$ for $k = 1, \ldots, m$, and so the list is linearly independent. ■

Exercise 2A11

Prove or give a counterexample: If $v_1, \ldots, v_m$ and $w_1, \ldots, w_m$ are linearly independent lists of vectors in V, then the list $v_1 + w_1, \ldots, v_m + w_m$ is linearly independent.

Solution. Consider the following counterexample in $\mathbb{R}^2$:

$$v_1 = (1,0), \quad v_2 = (0,1), \quad w_1 = (0,1), \quad w_2 = (1,0).$$

Then v_1, v_2 and w_1, w_2 are linearly independent lists of vectors in $\mathbb{R}^2$. However,

$$v_1 + w_1 = (1,1), \quad v_2 + w_2 = (1,1),$$

and so the list $v_1 + w_1, v_2 + w_2$ is linearly dependent. Thus, the statement is false. ■

Exercise 2A12

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Prove that if $v_1 + w, \dots, v_m + w$ is linearly dependent, then $w \in \operatorname{span}(v_1, \dots, v_m)$.

Solution. Let $\alpha_1, \dots, \alpha_m \in \mathbb{F}$, not all zero, such that

$$\alpha_1(v_1 + w) + \cdots + \alpha_m(v_m + w) = 0.$$

Then we have

$$(\alpha_1 v_1 + \cdots + \alpha_m v_m) + (\alpha_1 + \cdots + \alpha_m)w = 0.$$

If $\alpha_1 + \cdots + \alpha_m = 0$, then we have

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0,$$

which contradicts the linear independence of $v_1, \dots, v_m$. Thus, we must have $\alpha_1 + \cdots + \alpha_m \neq 0$. Let $\beta = \alpha_1 + \cdots + \alpha_m$. Then we may write

$$w = -\frac{\alpha_1}{\beta} v_1 - \cdots - \frac{\alpha_m}{\beta} v_m.$$

Thus, w is a linear combination of $v_1, \dots, v_m$, and so $w \in \operatorname{span}(v_1, \dots, v_m)$. ■

Exercise 2A13

Suppose $v_1, \dots, v_m$ is linearly independent in V and $w \in V$. Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \operatorname{span}(v_1, \dots, v_m).$$

Solution. Suppose $v_1, \dots, v_m, w$ is linearly independent. If there were scalars $\alpha_1, \dots, \alpha_m, \beta$ such that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m + \beta w = 0,$$

then linear independence would imply that $\alpha_1 = \cdots = \alpha_m = \beta = 0$. Suppose $w \in \operatorname{span}(v_1, \dots, v_m)$. Then there exist scalars $\gamma_1, \dots, \gamma_m$ such that

$$\gamma_1 v_1 + \cdots + \gamma_m v_m = w.$$

Then we may write

$$-\gamma_1 v_1 - \cdots - \gamma_m v_m + 1 \cdot w = 0,$$

which contradicts the linear independence of $v_1, \ldots, v_m, w$. Thus, $w \notin \operatorname{span}(v_1, \ldots, v_m)$.

Conversely, suppose $w \notin \operatorname{span}(v_1, \ldots, v_m)$. Then there do not exist scalars $\alpha_1, \ldots, \alpha_m$ such that

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = w.$$

If $v_1, \ldots, v_m, w$ is linearly dependent, then there exist scalars $\beta_1, \ldots, \beta_m, \gamma$, not all zero, such that

$$\beta_1 v_1 + \cdots + \beta_m v_m + \gamma w = 0.$$

If $\gamma = 0$, then we have

$$\beta_1 v_1 + \cdots + \beta_m v_m = 0,$$

which contradicts the linear independence of $v_1, \ldots, v_m$. Thus, we must have $\gamma \neq 0$. Then we may write

$$w = -\frac{\beta_1}{\gamma} v_1 - \cdots - \frac{\beta_m}{\gamma} v_m,$$

which contradicts the assumption that w is not in the span of $v_1, \ldots, v_m$. Thus, $v_1, \ldots, v_m, w$ is linearly independent. ■

Exercise 2A14

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, \ldots, v_m$ is linearly independent if and only if the list $w_1, \ldots, w_m$ is linearly independent.

Solution. Suppose $v_1, \ldots, v_m$ is linearly independent. Let $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_1 w_1 + \cdots + \alpha_m w_m = 0.$$

Substituting $w_k = v_1 + \cdots + v_k$ for each k, we get

$$\alpha_1 v_1 + \alpha_2(v_1 + v_2) + \alpha_3(v_1 + v_2 + v_3) + \cdots + \alpha_m(v_1 + \cdots + v_m) = 0.$$

Then we may rewrite this as

$$(\alpha_1 + \cdots + \alpha_m)v_1 + (\alpha_2 + \cdots + \alpha_m)v_2 + \cdots + (\alpha_{m-1} + \alpha_m)v_{m-1} + \alpha_m v_m = 0.$$

Since $v_1, \ldots, v_m$ is linearly independent, we must have $\alpha_1 = \cdots = \alpha_m = 0$. Hence, $w_1, \ldots, w_m$ is linearly independent.

Conversely, suppose $w_1, \ldots, w_m$ is linearly independent. Let $\beta_1, \ldots, \beta_m \in \mathbb{F}$ such that

$$\beta_1 v_1 + \cdots + \beta_m v_m = 0.$$

Noting that $v_1 = w_1$ and $v_k = w_k - w_{k-1}$ for $k \in \{2, \ldots, m\}$, we may rewrite the equation as

$$\beta_1 w_1 + \beta_2(w_2 - w_1) + \cdots + \beta_m(w_m - w_{m-1}) = 0,$$
$$(\beta_1 - \beta_2)w_1 + (\beta_2 - \beta_3)w_2 + \cdots + (\beta_{m-1} - \beta_m)w_{m-1} + \beta_m w_m = 0.$$

Since $w_1, \dots, w_m$ is linearly independent, we find that the system of equations

$$\begin{aligned} 0 &= \beta_1 - \beta_2, \\ 0 &= \beta_2 - \beta_3, \\ &\vdots \\ 0 &= \beta_{m-1} - \beta_m \\ 0 &= \beta_m, \end{aligned}$$

has the unique solution $\beta_1 = \cdots = \beta_m = 0$. Hence, $v_1, \dots, v_m$ is linearly independent. ■

Exercise 2A15

Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$.

Solution. Since $\mathcal{P}_4(\mathbb{F})$ can be spanned by the list $1, x, x^2, x^3, x^4$, and the length of a linearly independent list cannot exceed the length of a spanning list, there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A16

Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

Solution. Since the list $1, x, x^2, x^3, x^4$ is a linearly independent list in $\mathcal{P}_4(\mathbb{F})$ and has length 5, and the length of a spanning list cannot be less than the length of a linearly independent list, no list of four polynomials can span $\mathcal{P}_4(\mathbb{F})$. ■

Exercise 2A17

Prove that V is infinite-dimensional if and only if there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m.

Solution. Suppose V is infinite-dimensional. We proceed by induction to construct a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m. Since V is infinite-dimensional, there exists a nonzero vector $v_1 \in V$. Suppose we have constructed a linearly independent list of vectors $v_1, \dots, v_m$ in V. If every $v \in V$ is a linear combination of $v_1, \dots, v_m$, then $V = \operatorname{span}(v_1, \dots, v_m)$, which contradicts the assumption that V is infinite-dimensional. Thus, there exists a vector $v_{m+1} \in V$ that is not in $\operatorname{span}(v_1, \dots, v_m)$. Then by Exercise 2A13, the list $v_1, \dots, v_m, v_{m+1}$ is linearly independent. By induction, we can construct a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m.

Conversely, suppose there exists a sequence of vectors $v_1, v_2, \dots$ in V such that $v_1, \dots, v_m$ is linearly independent for every positive integer m. Then for every positive integer m, there exists a linearly independent list of vectors in V of length m. Since there is no maximum length for a linearly independent list of vectors in V, this implies that V is infinite-dimensional. ■

Exercise 2A18

Prove that $\mathbb{F}^\infty$ is infinite-dimensional.

Solution. By Exercise 2A17, it suffices to construct a sequence of vectors $v_1, v_2, \ldots$ in $\mathbb{F}^\infty$ such that $v_1, \ldots, v_m$ is linearly independent for every positive integer m. Let v_k be the vector in $\mathbb{F}^\infty$ whose k-th coordinate is 1 and all other coordinates are 0. Explicitly, we have

$$v_1 = (1, 0, 0, \ldots), \quad v_2 = (0, 1, 0, \ldots), \quad v_3 = (0, 0, 1, \ldots), \quad \ldots$$

Then for any positive integer m, the list $v_1, \ldots, v_m$ is linearly independent, since the only way to write the zero vector as a linear combination of $v_1, \ldots, v_m$ is to take all coefficients to be zero. Thus, by Exercise 2A17, $\mathbb{F}^\infty$ is infinite-dimensional. ■

Exercise 2A19

Prove that the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional.

Solution. By Exercise 2A17, it suffices to construct a sequence of continuous real-valued functions $f_1, f_2, \ldots$ on the interval $[0, 1]$ such that $f_1, \ldots, f_m$ is linearly independent for every positive integer m. Let $f_k(x) = x^{k-1}$ for each positive integer k. Then for any positive integer m, let $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ such that

$$\alpha_1 f_1(x) + \cdots + \alpha_m f_m(x) = 0$$

for all $x \in [0, 1]$. This is equivalent to the polynomial equation

$$\alpha_1 + \alpha_2 x + \cdots + \alpha_m x^{m-1} = 0$$

for all $x \in [0, 1]$. Since a polynomial of degree at most $m - 1$ can have at most $m - 1$ roots unless it is the zero polynomial, we must have $\alpha_1 = \cdots = \alpha_m = 0$. Thus, the list $f_1, \ldots, f_m$ is linearly independent. Therefore, by Exercise 2A17, the real vector space of all continuous real-valued functions on the interval $[0, 1]$ is infinite-dimensional. ■

Exercise 2A20

Suppose $p_0, p_1, \ldots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for each $k \in \{0, \ldots, m\}$. Prove that $p_0, p_1, \ldots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

Solution. Since $p_k(2) = 0$ for each $k \in \{0, \ldots, m\}$, we have

$$p_k(x) = (x - 2)q_k(x)$$

for some polynomial $q_k(x) \in \mathcal{P}_{m-1}(\mathbb{F})$. Then we may write

$$p_0(x), p_1(x), \ldots, p_m(x) = (x - 2)q_0(x), (x - 2)q_1(x), \ldots, (x - 2)q_m(x).$$

Since $\mathcal{P}_{m-1}(\mathbb{F})$ can be spanned by the m polynomials $1, x, \ldots, x^{m-1}$, the list of $m + 1$ polynomials $q_0, q_1, \ldots, q_m$ must be linearly dependent. Thus, there exist scalars $\alpha_0, \ldots, \alpha_m$, not all zero, such that

$$\alpha_0 q_0(x) + \cdots + \alpha_m q_m(x) = 0.$$

Then we may write

$$\alpha_0 p_0(x) + \cdots + \alpha_m p_m(x) = (x-2)(\alpha_0 q_0(x) + \cdots + \alpha_m q_m(x)) = 0.$$

Since not all the scalars $\alpha_0, \ldots, \alpha_m$ are zero, this shows that $p_0, p_1, \ldots, p_m$ is linearly dependent in $\mathcal{P}_m(\mathbb{F})$. ■

2B Bases

Exercise 2B1

Find all vector spaces that have exactly one basis.
Manual Remark. *The proof of this exercise fails if $\mathbb{F}$ contains only two elements. In this case, an additional vector space with exactly one basis is the vector space of dimension one over $\mathbb{F}$, since the only nonzero scalar is 1, and any nonzero vector v in the vector space is a basis.*

Solution. Since an infinite-dimensional vector space has infinitely many bases, we only need to consider finite-dimensional vector spaces. Let V be a finite-dimensional vector space, and let $v_1, \ldots, v_n$ be a basis of V. Then any other basis of V must have the same length as $v_1, \ldots, v_n$, which is n. If $n \geq 1$, consider $\alpha \in \mathbb{F}$ such that $\alpha \neq 0$ and $\alpha \neq 1$. Then the list $\alpha v_1, v_2, \ldots, v_n$ still spans V, since we may recover $v_1 = \alpha^{-1}(\alpha v_1)$. Moreover, the list $\alpha v_1, v_2, \ldots, v_n$ is linearly independent, since if

$$\beta_1(\alpha v_1) + \beta_2 v_2 + \cdots + \beta_n v_n = 0,$$

then we may write

$$(\alpha\beta_1)v_1 + \beta_2 v_2 + \cdots + \beta_n v_n = 0.$$

Since $v_1, \ldots, v_n$ is linearly independent, we must have $\alpha\beta_1 = 0$, which implies that $\beta_1 = 0$. Then we also have $\beta_2 = \cdots = \beta_n = 0$. Thus, the list $\alpha v_1, v_2, \ldots, v_n$ is linearly independent. Therefore, if $n \geq 1$, then V has more than one basis. If $n = 0$, then $V = \{0\}$. The empty list spans V, and the empty list must be linearly independent, since there are no nonzero scalars to consider. Thus, the empty list is a basis of V. Moreover, this is the only basis of V, since 0 is linearly dependent. Therefore, the only vector space that has exactly one basis is the zero vector space. ■

Exercise 2B2

Verify all assertions in Example 2.27.

(a) The list $(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$ is a basis of $\mathbb{F}^n$, called the *standard basis* of $\mathbb{F}^n$.
(b) The list $(1, 2), (3, 5)$ is a basis of $\mathbb{F}^2$. Note that this list has length two, which is the same as the length of the standard basis of $\mathbb{F}^2$. In the next section, we will see that this is not a coincidence.
(c) The list $(1, 2, -4), (7, -5, 6)$ is linearly independent in $\mathbb{F}^3$ but is not a basis of $\mathbb{F}^3$ because it does not span $\mathbb{F}^3$.
(d) The list $(1, 2), (3, 5), (4, 13)$ spans $\mathbb{F}^2$ but is not a basis of $\mathbb{F}^2$ because it is not linearly independent.
(e) The list $(1, 1, 0), (0, 0, 1)$ is a basis of $\{(x, x, y) \in \mathbb{F}^3 \mid x, y \in \mathbb{F}\}$.
(f) The list $(1, -1, 0), (1, 0, -1)$ is a basis of

$$\{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$

(g) The list $1, z, \ldots, z^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$, called the *standard basis* of $\mathcal{P}_m(\mathbb{F})$.

Solution.

(a) Suppose $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$\alpha_1(1,0,\ldots,0) + \alpha_2(0,1,0,\ldots,0) + \cdots + \alpha_n(0,\ldots,0,1) = 0.$$

Then each $\alpha_i = 0$, so the list is linearly independent. Moreover, any vector $(x_1, \ldots, x_n) \in \mathbb{F}^n$ can be written as

$$(x_1,\ldots,x_n) = x_1(1,0,\ldots,0) + x_2(0,1,0,\ldots,0) + \cdots + x_n(0,\ldots,0,1),$$

so the list spans $\mathbb{F}^n$. Thus, the list is a basis of $\mathbb{F}^n$.

(b) If $(3,5) = a(1,2)$ for some $a \in \mathbb{F}$, then we must have $3 = a$ and $5 = 2a$, which is impossible. Thus, the list is linearly independent. Moreover, for any element $(x,y) \in \mathbb{F}^2$, we can solve the system of equations

$$\begin{aligned} x &= 3\alpha + \beta, \\ y &= 5\alpha + 2\beta \end{aligned}$$

for $\alpha, \beta \in \mathbb{F}$. Solving this system of equations, we find that

$$\alpha = 2x - y, \quad \beta = -5x + 3y.$$

Since we can find $\alpha, \beta \in \mathbb{F}$ for any $(x,y) \in \mathbb{F}^2$, the list spans $\mathbb{F}^2$. Thus, the list is a basis of $\mathbb{F}^2$.

(c) Note that $\mathbb{F}^3$ is spanned by the linearly independent list $(1,0,0),(0,1,0),(0,0,1)$. Since the length of a spanning list must be at least the length of a linearly independent list, the list $(1,2,-4),(7,-5,6)$ cannot span $\mathbb{F}^3$.

(d) From part (b), we saw that $(1,2),(3,5)$ is a basis of $\mathbb{F}^2$. Since $(4,13) = -5(3,5) + 19(1,2)$, the list $(1,2),(3,5),(4,13)$ is linearly dependent. Moreover, since $(1,2),(3,5)$ is a basis of $\mathbb{F}^2$, the list $(1,2),(3,5),(4,13)$ spans $\mathbb{F}^2$. Thus, the list is not a basis of $\mathbb{F}^2$.

(e) Let U be the subspace in question. If $\alpha, \beta \in \mathbb{F}$ such that

$$\alpha(1,1,0) + \beta(0,0,1) = (0,0,0),$$

then we must have $\alpha = \beta = 0$, so the list is linearly independent. Moreover, for any $(x,x,y) \in U$, we can write

$$(x,x,y) = x(1,1,0) + y(0,0,1),$$

so the list spans U. Thus, the list is a basis of U.

(f) Let U be the subspace in question. If $\alpha, \beta \in \mathbb{F}$ such that

$$\alpha(1,-1,0) + \beta(1,0,-1) = (0,0,0),$$

then we must have $\alpha = \beta = 0$, so the list is linearly independent. Moreover, for any $(x,y,z) \in U$, note that $x + y + z = 0$, implying that $x = -y - z$. Then we can write

$$(x,y,z) = (-y-z,y,z) = -y(1,-1,0) - z(1,0,-1),$$

so the list spans U. Thus, the list is a basis of U.

(g) If $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$ such that

$$\alpha_0 + \alpha_1 z + \cdots + \alpha_m z^m = 0,$$

then we must have $\alpha_0 = \cdots = \alpha_m = 0$, so the list is linearly independent. Moreover, any polynomial $p(z) \in \mathcal{P}_m(\mathbb{F})$ can be written as

$$p(z) = \alpha_0 + \alpha_1 z + \cdots + \alpha_m z^m,$$

so the list spans $\mathcal{P}_m(\mathbb{F})$. Thus, the list is a basis of $\mathcal{P}_m(\mathbb{F})$. ■

Exercise 2B3

(a) Let U be the subspace of $\mathbb{R}^5$ defined by

$$U = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 \mid x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathbb{R}^5$.

(c) Find a subspace W of $\mathbb{R}^5$ such that $\mathbb{R}^5 = U \oplus W$.

Solution.

(a) Note that we can rewrite the subspace U as

$$U = \{(3x_2, x_2, 7x_4, x_4, x_5) \in \mathbb{R}^5 \mid x_2, x_4, x_5 \in \mathbb{R}\}.$$

Then we can write any vector in U as a linear combination of the following three vectors:

$$u_1 = (3, 1, 0, 0, 0), \quad u_2 = (0, 0, 7, 1, 0), \quad u_3 = (0, 0, 0, 0, 1).$$

Moreover, these three vectors are linearly independent, since the only way to write the zero vector as a linear combination of u_1, u_2, u_3 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathbb{R}^5$ by adding two more vectors that are linearly independent of u_1, u_2, u_3. For example, we can add the vectors

$$u_4 = (1, 0, 0, 0, 0), \quad u_5 = (0, 0, 1, 0, 0),$$

which are linearly independent of u_1, u_2, u_3: since u_1 is the only vector in the list with a nonzero first component, u_4 cannot be a linear combination of u_1, u_2, u_3. Similarly, since u_2 is the only vector in the list with a nonzero third component, u_5 cannot be a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathbb{R}^5$.

(c) Let W be the subspace of $\mathbb{R}^5$ spanned by the vectors u_4 and u_5. Then we have

$$\mathbb{R}^5 = U \oplus W,$$

since $U \cap W = \{0\}$ and every vector in $\mathbb{R}^5$ can be written as a sum of a vector in U and a vector in W. ■

Exercise 2B4

(a) Let U be the subspace of $\mathbb{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 \mid 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathbb{C}^5$.

(c) Find a subspace W of $\mathbb{C}^5$ such that $\mathbb{C}^5 = U \oplus W$.

Solution.

(a) Note that we can rewrite the subspace U as

$$U = \{(z_1, 6z_1, -2z_4 - 3z_5, z_4, z_5) \in \mathbb{C}^5 \mid z_1, z_4, z_5 \in \mathbb{C}\}.$$

Then we can write any vector in U as a linear combination of the following three vectors:

$$u_1 = (1, 6, 0, 0, 0), \quad u_2 = (0, 0, -2, 1, 0), \quad u_3 = (0, 0, -3, 0, 1).$$

Moreover, these three vectors are linearly independent, since the only way to write the zero vector as a linear combination of u_1, u_2, u_3 is to take all coefficients to be zero. Thus, $\{u_1, u_2, u_3\}$ is a basis of U.

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathbb{C}^5$ by adding two more vectors that are linearly independent of u_1, u_2, u_3. For example, we can add the vectors

$$u_4 = (1, 0, 0, 0, 0), \quad u_5 = (0, 0, 1, 0, 0),$$

which are linearly independent of u_1, u_2, u_3: since u_1 is the only vector in the list with a nonzero first component, u_4 cannot be a linear combination of u_1, u_2, u_3. Similarly, u_2 and u_3 have nonzero third components, but any linear combination of u_2 and u_3 will have nonzero fourth and fifth components, so u_5 cannot be a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathbb{C}^5$.

(c) Let W be the subspace of $\mathbb{C}^5$ spanned by the vectors u_4 and u_5. Then we have

$$\mathbb{C}^5 = U \oplus W,$$

since $U \cap W = \{0\}$ and every vector in $\mathbb{C}^5$ can be written as a sum of a vector in U and a vector in W. ■

Exercise 2B5

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Solution. Let $u_1, \ldots, u_m$ be a basis of U and $w_1, \ldots, w_n$ a basis of W. Then the list $u_1, \ldots, u_m, w_1, \ldots, w_n$ spans V, since any vector in V can be written as a sum of a vector in U and a vector in W. We can extract a basis of V from this list by removing any linearly dependent vectors. Thus, there exists a basis of V consisting of vectors in $U \cup W$. ■

Exercise 2B6

Prove or give a counterexample: If p_0, p_1, p_2, p_3 is a list in $\mathcal{P}_3(\mathbb{F})$ such that none of the polynomials p_0, p_1, p_2, p_3 has degree 2, then the list is not a basis of $\mathcal{P}_3(\mathbb{F})$.

Solution. This is not true. Let $p_0 = 1, p_1 = x, p_2 = x^3, p_3 = x^2 + x^3$. Then none of the polynomials p_0, p_1, p_2, p_3 has degree 2. Moreover, the list p_0, p_1, p_2, p_3 is linearly independent, since the only way to write the zero polynomial as a linear combination of p_0, p_1, p_2, p_3 is to take all coefficients to be zero. Finally, noting that $x^2 = p_3 - p_2$, then for any polynomial $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \in \mathcal{P}_3(\mathbb{F})$, we can write

$$p(x) = a_0p_0 + a_1p_1 + a_2(p_3 - p_2) + a_3p_2 = a_0p_0 + a_1p_1 + (a_3 - a_2)p_2 + a_2p_3,$$

which shows that the list spans $\mathcal{P}_3(\mathbb{F})$. Thus, the list p_0, p_1, p_2, p_3 is a basis of $\mathcal{P}_3(\mathbb{F})$. ■

Exercise 2B7

Suppose v_1, v_2, v_3, v_4 is a basis of V. Prove that

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V.

Manual Remark. *The technique of transforming the scalars $\beta_1, \beta_2, \beta_3, \beta_4$ into the scalars $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ is good to remember. You will later learn that this is called a* change of basis.

Solution. Let $\alpha_1, \ldots, \alpha_4 \in \mathbb{F}$ such that

$$\alpha_1(v_1 + v_2) + \alpha_2(v_2 + v_3) + \alpha_3(v_3 + v_4) + \alpha_4 v_4 = 0.$$

Then we may write

$$\alpha_1 v_1 + (\alpha_1 + \alpha_2)v_2 + (\alpha_2 + \alpha_3)v_3 + (\alpha_3 + \alpha_4)v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is a basis of V, we must have

$$\alpha_1 = 0, \quad \alpha_1 + \alpha_2 = 0, \quad \alpha_2 + \alpha_3 = 0, \quad \alpha_3 + \alpha_4 = 0.$$

From the first equation, we have $\alpha_1 = 0$. Substituting this into the second equation gives us $\alpha_2 = 0$. Substituting this into the third equation gives us $\alpha_3 = 0$. Finally, substituting this into the fourth equation gives us $\alpha_4 = 0$. Thus, the list is linearly independent. Moreover, since v_1, v_2, v_3, v_4 is a basis of V, then any vector $v \in V$ can be written as

$$v = \beta_1 v_1 + \beta_2 v_2 + \beta_3 v_3 + \beta_4 v_4$$

for some $\beta_1, \ldots, \beta_4 \in \mathbb{F}$. Using the expressions from proving the linear independence of the list, we may show that there exist $\alpha_1, \ldots, \alpha_4 \in \mathbb{F}$ such that

$$\begin{cases} \beta_1 = \alpha_1, \\ \beta_2 = \alpha_1 + \alpha_2, \\ \beta_3 = \alpha_2 + \alpha_3, \\ \beta_4 = \alpha_3 + \alpha_4. \end{cases}$$

Then $v \in V$ may be written as

$$v = \alpha_1(v_1 + v_2) + \alpha_2(v_2 + v_3) + \alpha_3(v_3 + v_4) + \alpha_4 v_4,$$

where

$$\alpha_1 = \beta_1, \quad \alpha_2 = \beta_2 - \beta_1, \quad \alpha_3 = \beta_3 - \beta_2 + \beta_1, \quad \alpha_4 = \beta_4 - \beta_3 + \beta_2 - \beta_1.$$

Thus, the list spans V. Therefore, the list is a basis of V. ∎

Exercise 2B8

Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U.

Solution. This is not true. Let $V = \mathbb{R}^4$ and v_1, v_2, v_3, v_4 be the standard basis of $\mathbb{R}^4$. Let U be the subspace of $\mathbb{R}^4$ defined by

$$U = \operatorname{span}(v_1, v_2, v_3 + v_4) = \{(x, y, z, z) \in \mathbb{R}^4 \mid x, y, z \in \mathbb{R}\}.$$

Then $v_1, v_2 \in U$ and $v_3, v_4 \notin U$. However, v_1, v_2 does not span U, since $(0, 0, 1, 1) \in U$ but cannot be written as a linear combination of v_1 and v_2. Thus, v_1, v_2 is not a basis of U. ■

Exercise 2B9

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that $v_1, \ldots, v_m$ is a basis of V if and only if $w_1, \ldots, w_m$ is a basis of V.

Solution. By Exercise 2A14, we know that $v_1, \ldots, v_m$ is linearly independent if and only if $w_1, \ldots, w_m$ is linearly independent. By Exercise 2A3, we know that $v_1, \ldots, v_m$ spans V if and only if $w_1, \ldots, w_m$ spans V. Thus, $v_1, \ldots, v_m$ is a basis of V if and only if $w_1, \ldots, w_m$ is a basis of V. ■

Exercise 2B10

Suppose U and W are subspaces of V such that $V = U \oplus W$. Suppose also that $u_1, \ldots, u_m$ is a basis of U and $w_1, \ldots, w_n$ is a basis of W. Prove that

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

is a basis of V.

Solution. Let $u_1, \ldots, u_m$ be a basis of U and $w_1, \ldots, w_n$ be a basis of W. We claim that the list

$$u_1, \ldots, u_m, w_1, \ldots, w_n$$

is a basis of V. Let $\alpha_1, \ldots, \alpha_m, \beta_1, \ldots, \beta_n \in \mathbb{F}$ such that

$$\alpha_1 u_1 + \cdots + \alpha_m u_m + \beta_1 w_1 + \cdots + \beta_n w_n = 0.$$

Then we may write

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = -\beta_1 w_1 - \cdots - \beta_n w_n.$$

Since the left-hand side is in U and the right-hand side is in W, and $U \cap W = \{0\}$, we must have

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = 0 \quad \text{and} \quad \beta_1 w_1 + \cdots + \beta_n w_n = 0.$$

Since $u_1, \ldots, u_m$ is a basis of U, we have $\alpha_1 = \cdots = \alpha_m = 0$. Similarly, since $w_1, \ldots, w_n$ is a basis of W, we have $\beta_1 = \cdots = \beta_n = 0$. Thus, the list is linearly independent. Moreover, since $V = U \oplus W$, then $v = u + w$ for some $u \in U$ and $w \in W$. Then we may write

$$u = \gamma_1 u_1 + \cdots + \gamma_m u_m \quad \text{and} \quad w = \delta_1 w_1 + \cdots + \delta_n w_n$$

for some $\gamma_1, \dots, \gamma_m, \delta_1, \dots, \delta_n \in \mathbb{F}$. Hence,

$$v = u + w = \gamma_1 u_1 + \cdots + \gamma_m u_m + \delta_1 w_1 + \cdots + \delta_n w_n,$$

which shows that the list spans V. Therefore, the list is a basis of V. ■

Exercise 2B11

Suppose V is a real vector space. Show that if $v_1, \dots, v_n$ is a basis of V (as a real vector space), then $v_1, \dots, v_n$ is also a basis of the complexification $V_{\mathbb{C}}$ (as a complex vector space).
Axler Note. *Please see* Exercise 1B8 *for the definition of the complexification* $V_{\mathbb{C}}$.

Solution. Let $v_1, \dots, v_n$ be a basis of V. We claim that $v_1, \dots, v_n$ is also a basis of $V_{\mathbb{C}}$. Let $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ such that

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0.$$

Then we may write each $\alpha_k = a_k + b_k i$ for some $a_k, b_k \in \mathbb{R}$. Then we have

$$(a_1 + b_1 i)v_1 + \cdots + (a_n + b_n i)v_n = 0.$$

This implies that

$$(a_1 v_1 + \cdots + a_n v_n) + i(b_1 v_1 + \cdots + b_n v_n) = 0.$$

Since the left-hand side is a sum of two vectors in V, and V is a real vector space, we must have

$$a_1 v_1 + \cdots + a_n v_n = 0 \quad \text{and} \quad b_1 v_1 + \cdots + b_n v_n = 0.$$

Since $v_1, \dots, v_n$ is a basis of V, we have $a_1 = \cdots = a_n = 0$ and $b_1 = \cdots = b_n = 0$. Thus, $\alpha_1 = \cdots = \alpha_n = 0$, which shows that the list is linearly independent. Moreover, let $v \in V_{\mathbb{C}}$. Then we may write $v = u + wi$ for some $u, w \in V$. Since $v_1, \dots, v_n$ is a basis of V, we may write

$$u = \gamma_1 v_1 + \cdots + \gamma_n v_n \quad \text{and} \quad w = \delta_1 v_1 + \cdots + \delta_n v_n$$

for some $\gamma_1, \dots, \gamma_n, \delta_1, \dots, \delta_n \in \mathbb{R}$. Hence,

$$v = u + wi = (\gamma_1 + \delta_1 i)v_1 + \cdots + (\gamma_n + \delta_n i)v_n,$$

which shows that the list spans $V_{\mathbb{C}}$. Therefore, the list is a basis of $V_{\mathbb{C}}$. ■

2C Dimension

Manual Remark. *In Exercise 2C3 through Exercise 2C7, each problem gives a set U that satisfies some linear condition on the polynomials in $\mathcal{P}_4(\mathbb{F})$. In each case, the polynomials we exhibit span U, since expressing a general element of U in terms of the free parameters allowed by the defining condition writes it directly as a linear combination of them. They are also linearly independent: since they have different degrees, a linear combination of them can equal the zero polynomial only if every coefficient is zero, as otherwise the highest-degree term present could not cancel. We will therefore omit this reasoning going forward and simply exhibit a basis for each such U.*

Exercise 2C1

Show that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$.

Solution. Since $\dim(\mathbb{R}^2) = 2$, any subspace of $\mathbb{R}^2$ must have dimension 0, 1, or 2. The only subspace of dimension 0 is $\{0\}$, and the only subspace of dimension 2 is $\mathbb{R}^2$. Any subspace with dimension 1 consists of all scalar multiples of some nonzero $(x, y) \in \mathbb{R}^2$, which is precisely the line with direction (x, y) containing the origin. Thus, the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$. ■

Exercise 2C2

Show that the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, all lines in $\mathbb{R}^3$ containing the origin, all planes in $\mathbb{R}^3$ containing the origin, and $\mathbb{R}^3$.

Solution. Since $\dim(\mathbb{R}^3) = 3$, any subspace of $\mathbb{R}^3$ must have dimension 0, 1, 2, or 3. The only subspace of dimension 0 is $\{0\}$, and the only subspace of dimension 3 is $\mathbb{R}^3$. Any subspace with dimension 1 consists of all scalar multiples of some nonzero $(x, y, z) \in \mathbb{R}^3$, which is precisely the line containing the origin in the direction of (x, y, z). Any subspace with dimension 2 consists of all linear combinations of two linearly independent vectors in $\mathbb{R}^3$, which is precisely a plane containing the origin. Thus, the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, all lines in $\mathbb{R}^3$ containing the origin, all planes in $\mathbb{R}^3$ containing the origin, and $\mathbb{R}^3$. ■

Exercise 2C3

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(6) = 0\}$. Find a basis of U.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Manual Remark. *Please see remark at the beginning of this section regarding the basis of U.*

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(6) = 0$, which gives us the equation

$$a_0 + 6a_1 + 36a_2 + 216a_3 + 1296a_4 = 0.$$

We can solve this equation for a_0 in terms of a_1, a_2, a_3, a_4:

$$a_0 = -6a_1 - 36a_2 - 216a_3 - 1296a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = -6 + x, \quad u_2 = -36 + x^2, \quad u_3 = -216 + x^3, \quad u_4 = -1296 + x^4.$$

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = 1,$$

which is linearly independent of u_1, u_2, u_3, u_4: since u_5 has no higher-degree terms, it cannot be expressed as a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} 1$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha$ for some $\alpha \in \mathbb{F}$. Since $p(6) = 0$, we have $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W. \qquad \blacksquare$$

Exercise 2C4

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) \mid p''(6) = 0\}$. Find a basis of U.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

Manual Remark. *Please see remark at the beginning of this section regarding the basis of U.*

Solution.

(a) Let $p = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 \in U$. Then we have

$$p''(x) = 2a_2 + 6a_3 x + 12a_4 x^2,$$

and so $p''(6) = 0$ gives us the equation

$$2a_2 + 36a_3 + 432a_4 = 0.$$

We can solve this equation for a_2 in terms of a_3, a_4:

$$a_2 = -18a_3 - 216a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = 1, \quad u_2 = x, \quad u_3 = -18x^2 + x^3, \quad u_4 = -216x^2 + x^4.$$

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{R})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = x^2,$$

which is linearly independent of u_1, u_2, u_3, u_4: since only u_3 and u_4 have x^2 terms but contain other higher-degree terms, u_5 cannot be expressed as a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} x^2$. Then we have

$$\mathcal{P}_4(\mathbb{R}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{R})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x^2$ for some $\alpha \in \mathbb{R}$. Since $p''(6) = 0$, we have $2\alpha = 0$, which implies that $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{R}) = U \oplus W.$$

■

Exercise 2C5

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(2) = p(5)\}$. Find a basis of U.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Manual Remark. *Please see remark at the beginning of this section regarding the basis of U.*

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(2) = p(5)$, which gives us the equation

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = a_0 + 5a_1 + 25a_2 + 125a_3 + 625a_4.$$

We can simplify this equation to get

$$-3a_1 - 21a_2 - 117a_3 - 609a_4 = 0.$$

We can solve this equation for a_1 in terms of a_2, a_3, a_4:

$$a_1 = -7a_2 - 39a_3 - 203a_4.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = 1, \quad u_2 = -7x + x^2, \quad u_3 = -39x + x^3, \quad u_4 = -203x + x^4.$$

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = x,$$

which is linearly independent of u_1, u_2, u_3, u_4: since only u_2, u_3, u_4 have x terms but contain other higher-degree terms, u_5 cannot be expressed as a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} x$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x$ for some $\alpha \in \mathbb{F}$. Since $p(2) = p(5)$, we have $2\alpha = 5\alpha$, which implies that $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W.$$

■

Exercise 2C6

(a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) \mid p(2) = p(5) = p(6)\}$. Find a basis of U.
(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
(c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Manual Remark. *Please see remark at the beginning of this section regarding the basis of U.*

Solution.

(a) Let $p = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \in U$. Then we have $p(2) = p(5) = p(6)$. We know from Exercise 2C5 that the condition $p(2) = p(5)$ gives us the equation

$$a_1 = -7a_2 - 39a_3 - 203a_4.$$

The condition $p(5) = p(6)$ gives us the equation

$$5a_1 + 25a_2 + 125a_3 + 625a_4 = 6a_1 + 36a_2 + 216a_3 + 1296a_4,$$

which simplifies to

$$a_1 + 11a_2 + 91a_3 + 671a_4 = 0.$$

Since we know that $a_1 = -7a_2 - 39a_3 - 203a_4$, we can substitute this into the second equation to get

$$4a_2 + 52a_3 + 468a_4 = 0.$$

We can solve this equation for a_2 in terms of a_3, a_4:

$$a_2 = -13a_3 - 117a_4.$$

Then $a_1 = 52a_3 + 616a_4$. Thus, we can write any polynomial in U as a linear combination of the following three polynomials:

$$u_1 = 1, \quad u_2 = 52x - 13x^2 + x^3, \quad u_3 = 616x - 117x^2 + x^4.$$

(b) We can extend the basis $\{u_1, u_2, u_3\}$ of U to a basis of $\mathcal{P}_4(\mathbb{F})$ by adding two more polynomials that are linearly independent of u_1, u_2, u_3. For example, we can add the polynomials

$$u_4 = x^2, \quad u_5 = x^3,$$

which are linearly independent of u_1, u_2, u_3: since only u_2 and u_3 have x^2 terms but contain other higher-degree terms, u_4 cannot be expressed as a linear combination of u_1, u_2, u_3. Similarly, u_5 has an x^3 term but no higher-degree terms, so it cannot be expressed as a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$.

(c) Let $W = \operatorname{span}(u_4, u_5) = \operatorname{span}(x^2, x^3)$. Then we have

$$\mathcal{P}_4(\mathbb{F}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{F})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha x^2 + \beta x^3$ for some $\alpha, \beta \in \mathbb{F}$. Since $p(2) = p(5) = p(6)$, we have

$$4\alpha + 8\beta = 25\alpha + 125\beta = 36\alpha + 216\beta.$$

Solving this system of equations, we find that $\alpha = \beta = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{F}) = U \oplus W. \qquad \blacksquare$$

Exercise 2C7

(a) Let

$$U = \left\{ p \in \mathcal{P}_4(\mathbb{R}) \,\middle|\, \int_{-1}^{1} p = 0 \right\}.$$

Find a basis of U.

(b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Find a subspace W of $\mathcal{P}_4(\mathbb{R})$ such that $\mathcal{P}_4(\mathbb{R}) = U \oplus W$.

Manual Remark. *Please see remark at the beginning of this section regarding the basis of U.*

Solution.

(a) Let $p = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 \in U$. Then we have

$$\begin{aligned}
\int_{-1}^{1} p &= \int_{-1}^{1} (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4)\, \mathrm{d}x \\
&= \int_{-1}^{1} (a_0 + a_2 x^2 + a_4 x^4)\, \mathrm{d}x \\
&= 2 \int_{0}^{1} (a_0 + a_2 x^2 + a_4 x^4)\, \mathrm{d}x \\
&= 2\left(a_0 + \frac{a_2}{3} + \frac{a_4}{5}\right) = 0.
\end{aligned}$$

We can solve this equation for a_0 in terms of a_2, a_4:

$$a_0 = -\frac{a_2}{3} - \frac{a_4}{5}.$$

Then we can write any polynomial in U as a linear combination of the following four polynomials:

$$u_1 = x, \quad u_2 = x^2 - \frac{1}{3}, \quad u_3 = x^3, \quad u_4 = x^4 - \frac{1}{5}.$$

(b) We can extend the basis $\{u_1, u_2, u_3, u_4\}$ of U to a basis of $\mathcal{P}_4(\mathbb{R})$ by adding one more polynomial that is linearly independent of u_1, u_2, u_3, u_4. For example, we can add the polynomial

$$u_5 = 1,$$

which is linearly independent of u_1, u_2, u_3, u_4: since u_5 has no higher-degree terms, it cannot be expressed as a linear combination of u_1, u_2, u_3, u_4. Thus, $\{u_1, u_2, u_3, u_4, u_5\}$ is a basis of $\mathcal{P}_4(\mathbb{R})$.

(c) Let $W = \operatorname{span} u_5 = \operatorname{span} 1$. Then we have

$$\mathcal{P}_4(\mathbb{R}) = U + W,$$

since every polynomial in $\mathcal{P}_4(\mathbb{R})$ can be written as a sum of a polynomial in U and a polynomial in W. Moreover, if $p \in U \cap W$, then $p = \alpha$ for some $\alpha \in \mathbb{R}$. Since the integral of a constant over a symmetric interval is nonzero unless the function is identically zero, we have $\alpha = 0$. Thus, $U \cap W = \{0\}$, and so we have

$$\mathcal{P}_4(\mathbb{R}) = U \oplus W.$$

■

Exercise 2C8

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1.$$

Solution. Observe that $(v_i + w) - (v_{i+1} + w) = v_i - v_{i+1}$ for $1 \leq i \leq m - 1$. Then the list of vectors

$$v_1 - v_2, v_2 - v_3, \ldots, v_{m-1} - v_m$$

is contained in $\operatorname{span}(v_1 + w, \ldots, v_m + w)$. We claim that this list is linearly independent. Suppose that

$$\alpha_1(v_1 - v_2) + \alpha_2(v_2 - v_3) + \cdots + \alpha_{m-1}(v_{m-1} - v_m) = 0$$

for some $\alpha_1, \ldots, \alpha_{m-1} \in \mathbb{F}$. Then we can rewrite this equation as

$$\alpha_1 v_1 + (\alpha_2 - \alpha_1)v_2 + \cdots + (\alpha_{m-1} - \alpha_{m-2})v_{m-1} - \alpha_{m-1}v_m = 0.$$

Since $v_1, \ldots, v_m$ is linearly independent, we must have $\alpha_1 = 0$, $\alpha_2 - \alpha_1 = 0$, ..., $\alpha_{m-1} - \alpha_{m-2} = 0$, and $-\alpha_{m-1} = 0$. This implies that $\alpha_1 = \cdots = \alpha_{m-1} = 0$. Thus, the list is linearly independent, and so we have

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1.$$

■

Exercise 2C9

Suppose m is a positive integer and $p_0, p_1, \ldots, p_m \in \mathcal{P}(\mathbb{F})$ are such that each p_k has degree k. Prove that $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Solution. Suppose that

$$\alpha_0 p_0 + \alpha_1 p_1 + \cdots + \alpha_m p_m = 0$$

for some $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$. Since p_m is the only polynomial with nonzero coefficient of x^m, we must have $\alpha_m = 0$. Next, since p_{m-1} is the only polynomial with nonzero coefficient of x^{m-1}, we must have $\alpha_{m-1} = 0$. Continuing in this way, we find that $\alpha_0 = \cdots = \alpha_m = 0$. Thus, the list is linearly independent. Moreover, since $\dim \mathcal{P}_m(\mathbb{F}) = m + 1$, any linearly independent list of $m + 1$ vectors in $\mathcal{P}_m(\mathbb{F})$ is a basis. Therefore, $p_0, p_1, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$. ■

Exercise 2C10

Suppose m is a positive integer. For $0 \leq k \leq m$, let

$$p_k(x) = x^k(1 - x)^{m-k}.$$

Show that $p_0, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$.

Solution. Suppose that

$$\alpha_0 p_0 + \alpha_1 p_1 + \cdots + \alpha_m p_m = 0$$

for some $\alpha_0, \ldots, \alpha_m \in \mathbb{F}$. Since p_0 is the only polynomial with nonzero constant term, we must have $\alpha_0 = 0$, hence the sum above becomes

$$\alpha_1 p_1 + \cdots + \alpha_m p_m = 0.$$

Next, since p_1 is the only polynomial with nonzero coefficient of x in the sum above, we must have $\alpha_1 = 0$. Continuing in this way, we find that $\alpha_0 = \cdots = \alpha_m = 0$. Thus, the list is linearly independent. Moreover, since $\dim \mathcal{P}_m(\mathbb{F}) = m + 1$, any linearly independent list of $m + 1$ vectors in $\mathcal{P}_m(\mathbb{F})$ is a basis. Therefore, $p_0, \ldots, p_m$ is a basis of $\mathcal{P}_m(\mathbb{F})$. ■

Exercise 2C11

Suppose U and W are both four-dimensional subspaces of $\mathbb{C}^6$. Prove that there exist two vectors in $U \cap W$ such that neither of these vectors is a scalar multiple of the other.

Solution. Let U and W be four-dimensional subspaces of $\mathbb{C}^6$. By the dimension formula for the sum of two subspaces, we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 4 + 4 - \dim(U \cap W) = 8 - \dim(U \cap W).$$

Since $U + W$ is a subspace of $\mathbb{C}^6$, we have $\dim(U + W) \leq 6$. Thus, we have

$$8 - \dim(U \cap W) \leq 6,$$

which implies that $\dim(U \cap W) \geq 2$. Therefore, there exist at least two linearly independent vectors in $U \cap W$, and neither of these vectors is a scalar multiple of the other. ■

Exercise 2C12

Suppose that U and W are subspaces of $\mathbb{R}^8$ such that $\dim U = 3$, $\dim W = 5$, and $U + W = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = U \oplus W$.

Solution. By the dimension formula for the sum of two subspaces, we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 3 + 5 - \dim(U \cap W) = 8 - \dim(U \cap W).$$

Since $U + W = \mathbb{R}^8$, we have $\dim(U + W) = 8$. Thus, we have

$$8 - \dim(U \cap W) = 8,$$

which implies that $\dim(U \cap W) = 0$. Therefore, $U \cap W = \{0\}$, and so we have

$$\mathbb{R}^8 = U \oplus W.$$ ■

Exercise 2C13

Suppose U and W are both five-dimensional subspaces of $\mathbb{R}^8$. Prove that $U \cap W \neq \{0\}$.

Solution. By the dimension formula for the sum of two subspaces, we have

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W) = 5 + 5 - \dim(U \cap W) = 10 - \dim(U \cap W).$$

Since $U + W$ is a subspace of $\mathbb{R}^8$, we have $\dim(U+W) \leq 8$. Thus, we have

$$10 - \dim(U \cap W) \leq 8,$$

which implies that $\dim(U \cap W) \geq 2$. Therefore, $U \cap W$ is nontrivial, and so we have $U \cap W \neq \{0\}$. ■

Exercise 2C14

Suppose V is a ten-dimensional vector space and V_1, V_2, V_3 are subspaces of V with $\dim V_1 = \dim V_2 = \dim V_3 = 7$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Solution. Let $W = V_1 \cap V_2$. By the dimension formula for the sum of two subspaces, we have

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim W = 7 + 7 - \dim W = 14 - \dim W.$$

Since $V_1 + V_2$ is a subspace of V, we have $\dim(V_1 + V_2) \leq 10$. Thus, we have $14 - \dim W \leq 10$, which implies that $\dim W \geq 4$. Therefore, $V_1 \cap V_2$ is at least four-dimensional. Now, consider the subspace $W \cap V_3 = V_1 \cap V_2 \cap V_3$. By the dimension formula for the sum of two subspaces, we have

$$\dim(W + V_3) = \dim W + \dim V_3 - \dim(W \cap V_3) = \dim W + 7 - \dim(W \cap V_3).$$

Since $W + V_3$ is a subspace of V, we have $\dim(W + V_3) \leq 10$. Thus, we have

$$\dim W + 7 - \dim(W \cap V_3) \leq 10,$$

which implies that $\dim(W \cap V_3) \geq \dim W + 7 - 10 = \dim W - 3$. Since $\dim W \geq 4$, we have $\dim(W \cap V_3) \geq 1$. Therefore, $V_1 \cap V_2 \cap V_3$ is nontrivial, and so we have $V_1 \cap V_2 \cap V_3 \neq \{0\}$. ■

Exercise 2C15

Suppose V is finite-dimensional and V_1, V_2, V_3 are subspaces of V with $\dim V_1 + \dim V_2 + \dim V_3 > 2 \dim V$. Prove that $V_1 \cap V_2 \cap V_3 \neq \{0\}$.

Solution. Let $W = V_1 \cap V_2$. By the dimension formula for the sum of two subspaces, we have

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim W.$$

Since $V_1 + V_2$ is a subspace of V, we have $\dim(V_1 + V_2) \leq \dim V$. Thus, we have

$$\dim V_1 + \dim V_2 - \dim W \leq \dim V,$$

which implies that $\dim W \geq \dim V_1 + \dim V_2 - \dim V$. Now, consider the subspace $W \cap V_3 = V_1 \cap V_2 \cap V_3$. By the dimension formula for the sum of two subspaces, we have

$$\dim(W + V_3) = \dim W + \dim V_3 - \dim(W \cap V_3).$$

Since $W + V_3$ is a subspace of V, we have $\dim(W + V_3) \leq \dim V$. Thus, we have

$$\dim W + \dim V_3 - \dim(W \cap V_3) \leq \dim V,$$

which implies that $\dim(W \cap V_3) \geq \dim W + \dim V_3 - \dim V$. Since $\dim W \geq \dim V_1 + \dim V_2 - \dim V$, we have

$$\dim(W \cap V_3) \geq (\dim V_1 + \dim V_2 - \dim V) + \dim V_3 - \dim V = \dim V_1 + \dim V_2 + \dim V_3 - 2\dim V.$$

Since $\dim V_1 + \dim V_2 + \dim V_3 > 2\dim V$, we have $\dim(W \cap V_3) > 0$. Therefore, $V_1 \cap V_2 \cap V_3$ is nontrivial, and so we have $V_1 \cap V_2 \cap V_3 \neq \{0\}$. ■

Exercise 2C16

Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V, each of dimension $n - 1$, whose intersection equals U.

Solution. Let U be a subspace of V with $\dim U = m < n = \dim V$. We can extend a basis of U to a basis of V. Let $\{u_1, \dots, u_m\}$ be a basis of U, and extend it to a basis of V by adding vectors $v_1, \dots, v_{n-m}$. Then $\{u_1, \dots, u_m, v_1, \dots, v_{n-m}\}$ is a basis of V. For each $i = 1, \dots, n - m$, let

$$W_i = \operatorname{span}(u_1, \dots, u_m, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}).$$

By construction, W_i does not contain the extended basis vector v_i. Moreover, each W_i is a subspace of V with dimension $n - 1$, since it is spanned by $n - 1$ linearly independent vectors. Additionally, we have

$$U = W_1 \cap W_2 \cap \cdots \cap W_{n-m},$$

since any vector in the intersection containing nonzero v_i would not be in W_i, leaving behind only the basis vectors in U. Therefore, there exist $n - m$ subspaces of V, each of dimension $n - 1$, whose intersection equals U. ■

Exercise 2C17

Suppose that $V_1, \dots, V_m$ are finite-dimensional subspaces of V. Prove that $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

Solution. We will prove the statement by induction on m. For the base case, let $m = 2$. Then the sum $V_1 + V_2$ is finite-dimensional. By the dimension formula for the sum of two subspaces, we have

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2) \leq \dim V_1 + \dim V_2.$$

Hence, the statement holds for $m = 2$. Now, suppose that the statement holds for some positive integer m. That is, suppose that if $V_1, \dots, V_m$ are finite-dimensional subspaces of V, then $V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_m) \leq \dim V_1 + \cdots + \dim V_m.$$

We want to show that the statement holds for $m+1$. Let $V_1, \dots, V_{m+1}$ be finite-dimensional subspaces of V. By the induction hypothesis, we know that $W = V_1 + \cdots + V_m$ is finite-dimensional and

$$\dim W \leq \dim V_1 + \cdots + \dim V_m.$$

Then we have

$$\dim(W + V_{m+1}) = \dim W + \dim V_{m+1} - \dim(W \cap V_{m+1}).$$

Since $W \cap V_{m+1}$ is a subspace of V, we have $\dim(W \cap V_{m+1}) \geq 0$. Thus, we have

$$\dim(W + V_{m+1}) \leq \dim W + \dim V_{m+1} \leq \dim V_1 + \cdots + \dim V_m + \dim V_{m+1}.$$

Therefore, $V_1 + \cdots + V_{m+1}$ is finite-dimensional and

$$\dim(V_1 + \cdots + V_{m+1}) \leq \dim V_1 + \cdots + \dim V_m + \dim V_{m+1}. \qquad \blacksquare$$

Exercise 2C18

Suppose V is finite-dimensional, with $\dim V = n \geq 1$. Prove that there exist one-dimensional subspaces $V_1, \dots, V_n$ of V such that

$$V = V_1 \oplus \cdots \oplus V_n.$$

Solution. Let $v_1, \dots, v_n$ be a basis of V. For each $i = 1, \dots, n$, let $V_i = \operatorname{span} v_i$. Then each V_i is a one-dimensional subspace of V, since it is spanned by a single nonzero vector. Moreover, we have

$$V = V_1 + \cdots + V_n,$$

and the sum is direct because the basis vectors are linearly independent. Therefore, we have

$$V = V_1 \oplus \cdots \oplus V_n. \qquad \blacksquare$$

Exercise 2C19

Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if V_1, V_2, V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ & + \dim(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

Solution. One may guess that the given formula above is true by analogy with the formula for the number of elements in the union of three finite sets. However, this erroneously conflates the idea of subspace sums with the idea of set unions. In fact, the formula is not true in general. For example, let $V = \mathbb{R}^2$, and let

$$V_1 = \operatorname{span}((1,0)), \quad V_2 = \operatorname{span}((0,1)), \quad V_3 = \operatorname{span}((1,1)).$$

It is easy to see that $\dim V_1 = \dim V_2 = \dim V_3 = 1$, but $\dim(V_1 \cap V_2) = \dim(V_1 \cap V_3) = \dim(V_2 \cap V_3) = 0$ and $\dim(V_1 \cap V_2 \cap V_3) = 0$. Thus, the right-hand side of the formula is

$$1 + 1 + 1 - 0 - 0 - 0 + 0 = 3,$$

but the left-hand side is $\dim(V_1 + V_2 + V_3) = \dim \mathbb{R}^2 = 2$. Therefore, the formula is not true in general. ■

Exercise 2C20

Prove that if V_1, V_2, and V_3 are subspaces of a finite-dimensional vector space, then

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ & - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}.\end{aligned}$$

Solution. Note that we need to account for all pairs of subspaces and all pairs of sums of subspaces, since the sum of subspaces is not equivalent to the union of sets. We can use the dimension formula repeatedly to obtain the answer. To that end, define the following subspaces:

$$W_1 = V_1 + V_2, \quad W_2 = V_1 + V_3, \quad W_3 = V_2 + V_3.$$

Applying the dimension formula to each of W_1, W_2, W_3, we have

$$\begin{aligned}\dim W_1 &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2), \\ \dim W_2 &= \dim V_1 + \dim V_3 - \dim(V_1 \cap V_3), \\ \dim W_3 &= \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3).\end{aligned}$$

Next, we can apply the dimension formula to each of the sums $W_1 + V_3, W_2 + V_2, W_3 + V_1$:

$$\begin{aligned}\dim(W_1 + V_3) &= \dim W_1 + \dim V_3 - \dim(W_1 \cap V_3), \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2) + \dim V_3 - \dim(W_1 \cap V_3) \\ \dim(W_2 + V_2) &= \dim W_2 + \dim V_2 - \dim(W_2 \cap V_2), \\ &= \dim V_1 + \dim V_3 - \dim(V_1 \cap V_3) + \dim V_2 - \dim(W_2 \cap V_2) \\ \dim(W_3 + V_1) &= \dim W_3 + \dim V_1 - \dim(W_3 \cap V_1), \\ &= \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) + \dim V_1 - \dim(W_3 \cap V_1).\end{aligned}$$

Note that $W_1 + V_3 = W_2 + V_2 = W_3 + V_1 = V_1 + V_2 + V_3$. Thus, we can add the three equations above and divide by 3 to obtain the desired formula:

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 \\ & - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ & - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}.\end{aligned}$$

■

3 Linear Maps

3A Vector Space of Linear Maps

Exercise 3A1

Suppose $b, c \in \mathbb{R}$. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (2x - 4y + 3z + b, 6x + cxyz).$$

Show that T is linear if and only if $b = c = 0$.

Solution. Suppose T is linear. Then

$$T(0, 0, 0) = (b, 0),$$

which implies that $b = 0$. Now consider the vector $(1, 1, 1) \in \mathbb{R}^3$. Then

$$T(2, 2, 2) = 2T(1, 1, 1) \implies (2, 12 + 8c) = (2, 12 + 2c).$$

Then $12 + 8c = 12 + 2c$, or $c = 0$.

Conversely, if $b = c = 0$, then $T(x, y, z) = (2x - 4y + 3z, 6x)$. Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned} T(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= (2(x_1 + x_2) - 4(y_1 + y_2) + 3(z_1 + z_2), 6(x_1 + x_2)) \\ &= (2x_1 + 2x_2 - 4y_1 - 4y_2 + 3z_1 + 3z_2, 6x_1 + 6x_2) \\ &= (2x_1 - 4y_1 + 3z_1, 6x_1) + (2x_2 - 4y_2 + 3z_2, 6x_2) \\ &= T(x_1, y_1, z_1) + T(x_2, y_2, z_2), \end{aligned}$$

and

$$\begin{aligned} T(\alpha x_1, \alpha y_1, \alpha z_1) &= (2(\alpha x_1) - 4(\alpha y_1) + 3(\alpha z_1), 6(\alpha x_1)) \\ &= (\alpha(2x_1 - 4y_1 + 3z_1), \alpha(6x_1)) \\ &= \alpha(2x_1 - 4y_1 + 3z_1, 6x_1) \\ &= \alpha T(x_1, y_1, z_1). \end{aligned}$$

Therefore, T is linear if and only if $b = c = 0$. ■

Exercise 3A2

Suppose $b, c \in \mathbb{R}$. Define $T : \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x + c\sin p(0)\right).$$

Show that T is linear if and only if $b = c = 0$.

Solution. Suppose T is linear. Then for any $p \in \mathcal{P}(\mathbb{R})$, we have $T(2p) = 2Tp$. Looking at the first component of $T(2p)$, we have

$$3(2p)(4) + 5(2p)'(6) + b(2p)(1)(2p)(2) = 6p(4) + 10p'(6) + 4bp(1)p(2).$$

On the other hand, the first component of $2Tp$ is

$$2(3p(4) + 5p'(6) + bp(1)p(2)) = 6p(4) + 10p'(6) + 2bp(1)p(2).$$

Comparing the two expressions, we get $4bp(1)p(2) = 2bp(1)p(2)$ for all $p \in \mathcal{P}(\mathbb{R})$. Since there exist polynomials p such that $p(1)p(2) \neq 0$, we must have $b = 0$.

The second component of $T(2p)$ must be equal to the second component of $2Tp$. We have

$$\int_{-1}^{2} x^3(2p)(x)\,\mathrm{d}x + c\sin(2p(0)) = 2\int_{-1}^{2} x^3 p(x)\,\mathrm{d}x + 2c\sin(p(0)).$$

The integral terms may be subtracted, so we get $c\sin(2p(0)) = 2c\sin(p(0))$ for all $p \in \mathcal{P}(\mathbb{R})$. Consider $p(x) = \pi/2$. Then $\sin(2p(0)) = \sin(\pi) = 0$ and $2\sin(p(0)) = 2\sin(\pi/2) = 2$. Therefore, $c = 0$.

Conversely, if $b = c = 0$, let $p, q \in \mathcal{P}(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$\begin{aligned}
T(p+q) &= \left(3(p+q)(4) + 5(p+q)'(6), \int_{-1}^{2} x^3(p+q)(x)\,\mathrm{d}x\right) \\
&= \left(3p(4) + 3q(4) + 5p'(6) + 5q'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x + \int_{-1}^{2} x^3 q(x)\,\mathrm{d}x\right) \\
&= \left(3p(4) + 5p'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) + \left(3q(4) + 5q'(6), \int_{-1}^{2} x^3 q(x)\,\mathrm{d}x\right) \\
&= Tp + Tq,
\end{aligned}$$

and

$$\begin{aligned}
T(\alpha p) &= \left(3(\alpha p)(4) + 5(\alpha p)'(6), \int_{-1}^{2} x^3(\alpha p)(x)\,\mathrm{d}x\right) \\
&= \left(\alpha(3p(4) + 5p'(6)), \alpha\int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) \\
&= \alpha\left(3p(4) + 5p'(6), \int_{-1}^{2} x^3 p(x)\,\mathrm{d}x\right) \\
&= \alpha Tp.
\end{aligned}$$

Therefore, T is linear if and only if $b = c = 0$. ■

Exercise 3A3

Suppose that $T \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$. Show that there exist scalars $A_{j,k} \in \mathbb{F}$ for $j = 1, \ldots, m$ and $k = 1, \ldots, n$ such that

$$T(x_1, \ldots, x_n) = (A_{1,1}x_1 + \cdots + A_{1,n}x_n, \ldots, A_{m,1}x_1 + \cdots + A_{m,n}x_n)$$

for every $(x_1, \ldots, x_n) \in \mathbb{F}^n$.

Solution. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{F}^n$. Then for each $k = 1, \ldots, n$, we have

$$T(e_k) = (A_{1,k}, A_{2,k}, \ldots, A_{m,k})$$

for some scalars $A_{j,k} \in \mathbb{F}$. Now let $(x_1, \ldots, x_n) \in \mathbb{F}^n$. Then

$$\begin{aligned} T(x_1, \ldots, x_n) &= T(x_1 e_1 + \cdots + x_n e_n) \\ &= x_1 T(e_1) + \cdots + x_n T(e_n) \\ &= x_1 (A_{1,1}, A_{2,1}, \ldots, A_{m,1}) + \cdots + x_n (A_{1,n}, A_{2,n}, \ldots, A_{m,n}) \\ &= (A_{1,1} x_1 + \cdots + A_{1,n} x_n, A_{2,1} x_1 + \cdots + A_{2,n} x_n, \ldots, A_{m,1} x_1 + \cdots + A_{m,n} x_n). \end{aligned}$$

Therefore,

$$T(x_1, \ldots, x_n) = (A_{1,1} x_1 + \cdots + A_{1,n} x_n, A_{2,1} x_1 + \cdots + A_{2,n} x_n, \ldots, A_{m,1} x_1 + \cdots + A_{m,n} x_n).$$ ■

Exercise 3A4

Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_m$ is a list of vectors in V such that $Tv_1, \ldots, Tv_m$ is a linearly independent list in W. Prove that $v_1, \ldots, v_m$ is linearly independent.

Solution. Suppose that $a_1 v_1 + \cdots + a_m v_m = 0$ for some scalars $a_1, \ldots, a_m \in \mathbb{F}$. Applying T to both sides, we get

$$a_1 T v_1 + \cdots + a_m T v_m = T(0) = 0.$$

Since $Tv_1, \ldots, Tv_m$ is linearly independent, it follows that $a_1 = \cdots = a_m = 0$. Therefore, $v_1, \ldots, v_m$ is linearly independent. ■

Exercise 3A5

Prove that $\mathcal{L}(V, W)$ is a vector space, as was asserted in 3.6.

Solution. We must first ensure that the operations of addition and scalar multiplication are well-defined, i.e., if $S, T \in \mathcal{L}(V, W)$ and $\alpha \in \mathbb{F}$, then $S + T \in \mathcal{L}(V, W)$ and $\alpha T \in \mathcal{L}(V, W)$. We first show that $S + T$ is linear. Let $v, u \in V$. Then

$$\begin{aligned} (S + T)(v + u) &= S(v + u) + T(v + u) \\ &= (Sv + Su) + (Tv + Tu) \\ &= (Sv + Tv) + (Su + Tu) \\ &= (S + T)v + (S + T)u. \end{aligned}$$

Moreover, for any $v \in V$ and $\alpha \in \mathbb{F}$,

$$\begin{aligned} (S + T)(\alpha v) &= S(\alpha v) + T(\alpha v) \\ &= \alpha S v + \alpha T v \\ &= \alpha (S v + T v) \\ &= \alpha (S + T) v. \end{aligned}$$

This shows that $S + T$ is linear. Now we show that αT is linear. Let $v, u \in V$. Then

$$\begin{aligned} (\alpha T)(v + u) &= \alpha T(v + u) \\ &= \alpha (T v + T u) \\ &= \alpha T v + \alpha T u \\ &= (\alpha T) v + (\alpha T) u. \end{aligned}$$

Also, for any $v \in V$ and $\beta \in \mathbb{F}$,

$$\begin{aligned}(\alpha T)(\beta v) &= \alpha T(\beta v)\\ &= \alpha(\beta T v)\\ &= (\alpha\beta)Tv\\ &= (\beta\alpha)Tv\\ &= \beta(\alpha Tv)\\ &= \beta(\alpha T)v.\end{aligned}$$

Therefore, $S + T \in \mathcal{L}(V, W)$ and $\alpha T \in \mathcal{L}(V, W)$. We now check the vector space axioms:

(1) **Commutativity**: Let $S, T \in \mathcal{L}(V, W)$. Then for any $v \in V$,

$$\begin{aligned}(S + T)v &= Sv + Tv\\ &= Tv + Sv\\ &= (T + S)v.\end{aligned}$$

(2) **Associativity**: Let $R, S, T \in \mathcal{L}(V, W)$. Then for any $v \in V$,

$$\begin{aligned}((R + S) + T)v &= (R + S)v + Tv\\ &= (Rv + Sv) + Tv\\ &= Rv + (Sv + Tv)\\ &= Rv + (S + T)v\\ &= (R + (S + T))v.\end{aligned}$$

Similarly, for any $\alpha, \beta \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$,

$$\begin{aligned}((\alpha\beta)T)v &= (\alpha\beta)(Tv)\\ &= \alpha(\beta T v)\\ &= \alpha((\beta T)v)\\ &= (\alpha(\beta T))v.\end{aligned}$$

(3) **Additive identity**: Let $0 \in \mathcal{L}(V, W)$ be the zero map defined by $0v = 0$ for all $v \in V$. Then for any $T \in \mathcal{L}(V, W)$ and $v \in V$,

$$(T + 0)v = Tv + 0v = Tv + 0 = Tv.$$

(4) **Inverse**: Let $T \in \mathcal{L}(V, W)$. Define $-T \in \mathcal{L}(V, W)$ by $(-T)v = -Tv$ for all $v \in V$. Then for any $v \in V$,

$$(T + (-T))v = Tv + (-T)v = Tv - Tv = 0 = 0v.$$

(5) **Multiplicative identity**: Let $1 \in \mathbb{F}$ be the multiplicative identity. Then for any $T \in \mathcal{L}(V, W)$ and $v \in V$,

$$(1T)v = 1(Tv) = Tv.$$

(6) **Distributive properties**: Let $T, S \in \mathcal{L}(V, W)$ and $\alpha, \beta \in \mathbb{F}$. Then for any $v \in V$,

$$\begin{aligned}(\alpha(T + S))v &= \alpha((T + S)v)\\ &= \alpha(Tv + Sv)\\ &= \alpha Tv + \alpha Sv\\ &= (\alpha T)v + (\alpha S)v\\ &= ((\alpha T) + (\alpha S))v,\end{aligned}$$

and

$$\begin{aligned}((\alpha+\beta)T)v &= (\alpha+\beta)(Tv)\\ &= \alpha Tv + \beta Tv\\ &= (\alpha T)v + (\beta T)v\\ &= ((\alpha T) + (\beta T))v.\end{aligned}$$ ■

Exercise 3A6

Prove that multiplication of linear maps has the associative, identity, and distributive properties asserted in 3.8.

(a) **Associativity**: $(T_1T_2)T_3 = T_1(T_2T_3)$ whenever T_1, T_2, and T_3 are linear maps such that the products make sense (meaning T_3 maps into the domain of T_2, and T_2 maps into the domain of T_1).

(b) **Identity**: $TI = IT = T$ whenever $T \in \mathcal{L}(V, W)$; here the first I is the identity operator on V and the second I is the identity operator on W.

(c) **Distributive properties**: $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$ whenever $T, T_1, T_2 \in \mathcal{L}(U, V)$ and $S, S_1, S_2 \in \mathcal{L}(V, W)$.

Solution.

(a) Let $T_1 \in \mathcal{L}(V, W)$, $T_2 \in \mathcal{L}(U, V)$, and $T_3 \in \mathcal{L}(X, U)$. Then for any $x \in X$,

$$\begin{aligned}((T_1T_2)T_3)x &= (T_1T_2)(T_3x)\\ &= T_1(T_2(T_3x))\\ &= T_1((T_2T_3)x)\\ &= (T_1(T_2T_3))x.\end{aligned}$$

(b) Let $T \in \mathcal{L}(V, W)$. Then for any $v \in V$,

$$(TI)v = T(Iv) = Tv \quad \text{and} \quad (IT)v = I(Tv) = Tv.$$

(c) Let $T, T_1, T_2 \in \mathcal{L}(U, V)$ and $S, S_1, S_2 \in \mathcal{L}(V, W)$. Then for any $u \in U$,

$$\begin{aligned}((S_1+S_2)T)u &= (S_1+S_2)(Tu)\\ &= S_1(Tu) + S_2(Tu)\\ &= (S_1T)u + (S_2T)u\\ &= (S_1T + S_2T)u,\end{aligned}$$

and

$$\begin{aligned}(S(T_1+T_2))u &= S((T_1+T_2)u)\\ &= S(T_1u + T_2u)\\ &= S(T_1u) + S(T_2u)\\ &= (ST_1)u + (ST_2)u\\ &= (ST_1 + ST_2)u.\end{aligned}$$ ■

Exercise 3A7

Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

Solution. Suppose $\dim V = 1$ and let $v \in V$ be a nonzero vector. Then $\{v\}$ is a basis of V. Let $T \in \mathcal{L}(V)$. Since $Tv \in V$, there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$. Now let $u \in V$. Then there exists $\alpha \in \mathbb{F}$ such that $u = \alpha v$. Therefore,

$$Tu = T(\alpha v) = \alpha Tv = \alpha(\lambda v) = \lambda(\alpha v) = \lambda u.$$ ∎

Exercise 3A8

Give an example of a function $\varphi : \mathbb{R}^2 \to \mathbb{R}$ such that

$$\varphi(av) = a\varphi(v)$$

for all $a \in \mathbb{R}$ and all $v \in \mathbb{R}^2$ but φ is not linear.
Manual Remark. *This was a particularly difficult exercise because all obvious examples of functions that are homogeneous are also additive. The key idea is to utilize composition of functions.*

Solution. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$\varphi(x, y) = \sqrt[3]{x^3 + y^3}.$$

Then for any $a \in \mathbb{R}$ and $(x, y) \in \mathbb{R}^2$,

$$\varphi(a(x, y)) = \varphi(ax, ay) = \sqrt[3]{(ax)^3 + (ay)^3} = \sqrt[3]{a^3(x^3 + y^3)} = a\sqrt[3]{x^3 + y^3} = a\varphi(x, y).$$

However, φ is not additive. For example, let $v = (1, 0)$ and $w = (0, 1)$. Then

$$\varphi(v + w) = \varphi(1, 1) = \sqrt[3]{1^3 + 1^3} = \sqrt[3]{2},$$

but

$$\varphi(v) + \varphi(w) = \varphi(1, 0) + \varphi(0, 1) = \sqrt[3]{1^3 + 0^3} + \sqrt[3]{0^3 + 1^3} = 1 + 1 = 2.$$

Therefore, φ is not linear. ∎

Exercise 3A9

Give an example of a function $\varphi : \mathbb{C} \to \mathbb{C}$ such that

$$\varphi(w + z) = \varphi(w) + \varphi(z)$$

for all $w, z \in \mathbb{C}$ but φ is not linear. (Here $\mathbb{C}$ is thought of as a complex vector space.)

Solution. Let $\varphi : \mathbb{C} \to \mathbb{C}$ be defined by

$$\varphi(x + iy) = x.$$

Then for any $w = x_1 + iy_1$ and $z = x_2 + iy_2$ in $\mathbb{C}$,

$$\varphi(w + z) = \varphi((x_1 + x_2) + i(y_1 + y_2)) = x_1 + x_2 = \varphi(w) + \varphi(z).$$

However, φ is not homogeneous. For example, let $a = i$ and $z = 1$. Then

$$\varphi(iz) = \varphi(i) = 0,$$

but

$$i\varphi(z) = i\varphi(1) = i(1) = i.$$

Therefore, φ is not linear. ∎

Exercise 3A10

Prove or give a counterexample: If $q \in \mathcal{P}(\mathbb{R})$ and $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is defined by $Tp = q \circ p$, then T is a linear map.

Solution. This is false. Let $q(x) = x^2$ and define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $Tp = q \circ p$. Then for any $p_1, p_2 \in \mathcal{P}(\mathbb{R})$,

$$T(p_1 + p_2) = q \circ (p_1 + p_2) = (p_1 + p_2)^2 = p_1^2 + 2p_1p_2 + p_2^2 \neq p_1^2 + p_2^2 = Tp_1 + Tp_2.$$ ∎

Exercise 3A11

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

Solution. Suppose T is a scalar multiple of the identity, i.e., $T = \lambda I$ for some $\lambda \in \mathbb{F}$. Then for any $S \in \mathcal{L}(V)$,

$$ST = S(\lambda I) = \lambda S = (\lambda I)S = TS.$$

Conversely, suppose $ST = TS$ for every $S \in \mathcal{L}(V)$. We claim that v and Tv are linearly dependent for every $v \in V$. Suppose not. Then there exists $v \in V$ such that v and Tv are linearly independent. Extend $\{v, Tv\}$ to a basis of V. Define $S \in \mathcal{L}(V)$ by sending v to itself and sending $Tv, v_3, \ldots, v_n$ to 0. Then

$$STv = S(Tv) = 0, \quad TSv = Tv \neq 0,$$

which contradicts the assumption that $ST = TS$. Therefore, v and Tv are linearly dependent for every $v \in V$.

Now let $v \in V$ be nonzero. Then there exists $\lambda_v \in \mathbb{F}$ such that $Tv = \lambda_v v$. We claim that λ_v is independent of the choice of v. Let $v_1, \ldots, v_n$ be a basis of V.

- If $n = 1$, then the claim is immediate, since $Tv_1 = \lambda_{v_1} v_1$.
- If $n \geq 2$, let v_i, v_j be two distinct basis vectors. Then their linear independence implies

  $$\lambda_{v_i+v_j} v_i + \lambda_{v_i+v_j} v_j = T(v_i + v_j) = Tv_i + Tv_j = \lambda_{v_i} v_i + \lambda_{v_j} v_j.$$

  Since v_i and v_j are linearly independent, it follows that $\lambda_{v_i+v_j} = \lambda_{v_i} = \lambda_{v_j}$. Therefore, λ_v is independent of the choice of v, and we may denote it by λ.

Let $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ be any vector in V. Then

$$Tv = T(\alpha_1 v_1 + \cdots + \alpha_n v_n) = \alpha_1 T v_1 + \cdots + \alpha_n T v_n = \alpha_1 \lambda v_1 + \cdots + \alpha_n \lambda v_n = \lambda(\alpha_1 v_1 + \cdots + \alpha_n v_n) = \lambda v.$$

Therefore, T is a scalar multiple of the identity. ■

Exercise 3A12

Suppose U is a subspace of V with $U \neq V$. Suppose $S \in \mathcal{L}(U, W)$ and $S \neq 0$ (which means that $Su \neq 0$ for some $u \in U$). Define $T : V \to W$ by

$$Tv = \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U. \end{cases}$$

Prove that T is not a linear map on V.

Solution. Since $U \neq V$, there exists $v \in V$ such that $v \notin U$. Let $u \in U$ be such that $Su \neq 0$. Then

$$T(u + v) = 0,$$

since $u + v \in U$ would imply $v = (v + u) - u \in U$, which is a contradiction. However,

$$Tu + Tv = Su + 0 = Su \neq 0.$$

Therefore, T is not linear. ■

Exercise 3A13

Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V. In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

Solution. Let $u_1, \ldots, u_m$ be a basis of U. Since U is a subspace of V, we can extend this basis to a basis of V, say $u_1, \ldots, u_m, v_1, \ldots, v_n$. Define $T : V \to W$ by

$$Tu_i = Su_i \quad \text{for all } i = 1, \ldots, m,$$

and

$$Tv_k = 0 \quad \text{for all } k = 1, \ldots, n.$$

Then T is well-defined on the basis of V, and we can extend it linearly to all of V. For any $u \in U$, we may write

$$u = \alpha_1 u_1 + \cdots + \alpha_m u_m$$

for some scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$, and 0 as the coefficients for $v_1, \ldots, v_n$. Then

$$Tu = \alpha_1 T u_1 + \cdots + \alpha_m T u_m + 0 \cdot T v_1 + \cdots + 0 \cdot T v_n = \alpha_1 S u_1 + \cdots + \alpha_m S u_m = Su.$$

Therefore, T extends S to all of V. ■

Exercise 3A14

Suppose V is finite-dimensional with $\dim V > 0$, and suppose W is infinite-dimensional. Prove that $\mathcal{L}(V, W)$ is infinite-dimensional.

Solution. Let $v_1, \ldots, v_n$ be a basis of V. Since W is infinite-dimensional, there exists an infinite linearly independent list of vectors $w_1, w_2, \ldots$ in W. For each $k \in \mathbb{N}$, define $T_k \in \mathcal{L}(V, W)$ by

$$T_k v_j = \begin{cases} w_k & \text{if } j = 1, \\ 0 & \text{if } j \neq 1. \end{cases}$$

In other words, T_k sends the first basis vector v_1 to w_k and all other basis vectors to 0. We claim that the list $T_1, T_2, \ldots$ is linearly independent in $\mathcal{L}(V, W)$. Suppose that

$$a_1 T_1 + a_2 T_2 + \cdots + a_m T_m = 0$$

for some scalars $a_1, \ldots, a_m \in \mathbb{F}$. Evaluating both sides at v_1, we get

$$a_1 T_1 v_1 + a_2 T_2 v_1 + \cdots + a_m T_m v_1 = a_1 w_1 + a_2 w_2 + \cdots + a_m w_m = 0.$$

Since $w_1, \ldots, w_m$ is linearly independent in W, it follows that $a_1 = \cdots = a_m = 0$. Therefore, $T_1, T_2, \ldots$ is linearly independent in $\mathcal{L}(V, W)$, and hence $\mathcal{L}(V, W)$ is infinite-dimensional. ■

Exercise 3A15

Suppose $v_1, \ldots, v_m$ is a linearly dependent list of vectors in V. Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \ldots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies $T v_k = w_k$ for each $k = 1, \ldots, m$.

Solution. Since $v_1, \ldots, v_m$ is linearly dependent, there exist scalars $a_1, \ldots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \cdots + a_m v_m = 0.$$

Fix some nonzero a_i among $a_1, \ldots, a_m$. Let $w_1, \ldots, w_m \in W$ be defined by

$$w_k = \begin{cases} 0 & \text{if } k \neq i, \\ w & \text{if } k = i, \end{cases}$$

where w is any nonzero vector in W. Suppose for contradiction that there exists $T \in \mathcal{L}(V, W)$ such that $T v_k = w_k$ for each $k = 1, \ldots, m$. Then

$$T(a_1 v_1 + \cdots + a_m v_m) = a_1 T v_1 + \cdots + a_m T v_m = a_1 w_1 + \cdots + a_m w_m.$$

Since $a_1 v_1 + \cdots + a_m v_m = 0$, we have $T(0) = 0$, so

$$0 = a_1 w_1 + \cdots + a_m w_m.$$

However, by definition of w_k, we have $a_k w_k = a_i w$ if $k = i$ and $a_k w_k = 0$ if $k \neq i$. Therefore, the sum $a_1 w_1 + \cdots + a_m w_m$ is a nonzero scalar multiple of w, which is nonzero since $w \neq 0$. This is a contradiction. Therefore, no such $T \in \mathcal{L}(V, W)$ exists. ■

Exercise 3A16

Suppose V is finite-dimensional with $\dim V > 1$. Prove that there exist $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$.

Solution. Let v_1, v_2 be linearly independent vectors in V. We may extend this list to a basis of V, say $v_1, v_2, \ldots, v_n$. Define $S, T \in \mathcal{L}(V)$ by

$$Sv_1 = v_2, \quad Sv_2 = v_1, \quad Sv_k = 0 \text{ for } k > 2,$$

and

$$Tv_1 = v_1, \quad Tv_2 = 0, \quad Tv_k = 0 \text{ for } k > 2.$$

Then

$$STv_1 = S(Tv_1) = Sv_1 = v_2,$$

but

$$TSv_1 = T(Sv_1) = Tv_2 = 0.$$

Therefore, $ST \neq TS$. ■

Exercise 3A17

Suppose V is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$.
Axler Note. *A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a two-sided ideal of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.*

Solution. Let $\mathcal{E}$ be a two-sided ideal of $\mathcal{L}(V)$. If $\mathcal{E} = \{0\}$, then we are done. Otherwise, there exists $E \in \mathcal{E}$ such that $E \neq 0$. Then there exists $v_1 \in V$ such that $Ev_1 \neq 0$. Since V is finite-dimensional, we may extend the list $\{v\}_1$ to a basis of V, say $v_1, v_2, \ldots, v_n$. Let $w_1 = Ev_1 \neq 0$. We may also extend the list $\{w\}_1$ to a basis of V, say $w_1, w_2, \ldots, w_n$. Define $T_1 \in \mathcal{L}(V)$ by

$$T_1 w_1 = v_1, \quad T_1 w_k = 0 \text{ for } k > 1.$$

Then $T_1 E \in \mathcal{E}$ since $\mathcal{E}$ is a two-sided ideal. Observe that $T_1 E v_1 = T_1(Ev_1) = T_1 w_1 = v_1 \neq 0$. Therefore, $T_1 E \neq 0$. Consider $S_1 \in \mathcal{L}(V)$ defined by

$$S_1 v_1 = v_1, \quad S_1 v_k = 0 \text{ for } k > 1.$$

Then $T_1 E S_1 \in \mathcal{E}$ since $\mathcal{E}$ is a two-sided ideal. Observe that $T_1 E S_1 v_1 = T_1 E v_1 = T_1 w_1 = v_1 \neq 0$. Therefore, $T_1 E S_1 \neq 0$. We now discuss the construction of the identity map $I \in \mathcal{L}(V)$.

Construct $T_i, S_i \in \mathcal{L}(V)$ for each $i = 1, \ldots, n$ as follows:

$$T_i w_1 = v_i, \quad T_i w_k = 0 \text{ for } k \neq 1,$$

and

$$S_i v_i = v_1, \quad S_i v_k = 0 \text{ for } k \neq i.$$

Then $T_i E S_i \in \mathcal{E}$ and $T_i E S_i v_i = v_i \neq 0$. Since $\mathcal{E}$ is a subspace, it contains all linear combinations of these maps. In particular, for any $v \in V$, we can write $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$ for some scalars $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{F}$. Observe that

$$T_i E S_i v = T_i E S_i(\alpha_1 v_1) + \cdots + T_i E S_i(\alpha_i v_i) + \cdots + T_i E S_i(\alpha_n v_n) = \alpha_i v_i.$$

Consider the map

$$E^* = T_1ES_1 + T_2ES_2 + \cdots + T_nES_n \in \mathcal{E},$$

Then for any $v = \alpha_1 v_1 + \cdots + \alpha_n v_n \in V$,

$$E^*v = T_1ES_1v + T_2ES_2v + \cdots + T_nES_nv = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = v.$$

Hence, $E^* = I \in \mathcal{E}$. Since $\mathcal{E}$ is a two-sided ideal, $I = E^* \in \mathcal{E}$ implies that for any $T \in \mathcal{L}(V)$, we have $TI = T \in \mathcal{E}$. Therefore, $\mathcal{E} = \mathcal{L}(V)$. ■

3B Null Spaces and Ranges

Exercise 3B1

Give an example of a linear map T with $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

Solution. Let $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^2)$ be defined by

$$T(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2).$$

Then $\operatorname{null} T = \{(0, 0, x_3, x_4, x_5) \mid x_3, x_4, x_5 \in \mathbb{R}\}$, which has dimension 3, and $\operatorname{range} T = \mathbb{R}^2$, which has dimension 2. ∎

Exercise 3B2

Suppose $S, T \in \mathcal{L}(V)$ are such that $\operatorname{range} S \subseteq \operatorname{null} T$. Prove that $(ST)^2 = 0$.

Solution. Let $v \in V$. Then

$$(ST)^2 v = ST(STv) = S(T(STv)).$$

Since $STv \in \operatorname{range} S$ and $\operatorname{range} S \subseteq \operatorname{null} T$, we have $T(STv) = 0$. Therefore,

$$(ST)^2 v = S(0) = 0.$$

Since v was arbitrary, it follows that $(ST)^2 = 0$. ∎

Exercise 3B3

Suppose $v_1, \dots, v_m$ is a list of vectors in V. Define $T \in \mathcal{L}(\mathbb{F}^m, V)$ by

$$T(z_1, \dots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of T corresponds to $v_1, \dots, v_m$ spanning V?
(b) What property of T corresponds to the list $v_1, \dots, v_m$ being linearly independent?

Solution.
(a) The list $v_1, \dots, v_m$ spans V if and only if $\operatorname{range} T = V$, i.e., T is surjective.
(b) The list $v_1, \dots, v_m$ is linearly independent if and only if $\operatorname{null} T = \{0\}$, i.e., T is injective. ∎

Exercise 3B4

Show that $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) \mid \dim \operatorname{null} T > 2\}$ is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$.

Solution. Let $T_1, T_2 \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$ be defined by

$$T_1(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0)$$

and

$$T_2(x_1, x_2, x_3, x_4, x_5) = (0, 0, x_3, x_4).$$

Then $\operatorname{null} T_1 = \{(0, 0, x_3, x_4, x_5) \mid x_3, x_4, x_5 \in \mathbb{R}\}$ and $\operatorname{null} T_2 = \{(x_1, x_2, 0, 0, x_5) \mid x_1, x_2, x_5 \in \mathbb{R}\}$ both have dimension 3, so T_1 and T_2 are in the set $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) \mid \dim \operatorname{null} T > 2\}$. However,

$$(T_1 + T_2)(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, x_3, x_4),$$

which has null space $\operatorname{null}(T_1 + T_2) = \{(0, 0, 0, 0, x_5) \mid x_5 \in \mathbb{R}\}$ of dimension 1. Therefore,

$$\dim \operatorname{null}(T_1 + T_2) = 1 < 2,$$

so $T_1 + T_2$ is not in the set. Hence, the set is not closed under addition and is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$. ■

Exercise 3B5

Give an example of $T \in \mathcal{L}(\mathbb{R}^4)$ such that $\operatorname{range} T = \operatorname{null} T$.

Solution. Let v_1, v_2, v_3, v_4 be the standard basis of $\mathbb{R}^4$. Define $T \in \mathcal{L}(\mathbb{R}^4)$ by

$$Tv_1 = v_3, \quad Tv_2 = v_4, \quad Tv_3 = 0, \quad Tv_4 = 0.$$

Then for any $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$,

$$Tx = x_1Tv_1 + x_2Tv_2 + x_3Tv_3 + x_4Tv_4 = x_1v_3 + x_2v_4.$$

Then $\operatorname{range} T = \operatorname{span} v_3, v_4$. If $Tx = 0$, then $x_1v_3 + x_2v_4 = 0$, which implies $x_1 = x_2 = 0$. Therefore, $\operatorname{null} T = \{(0, 0, x_3, x_4) \mid x_3, x_4 \in \mathbb{R}\} = \operatorname{span} v_3, v_4$. Hence, $\operatorname{range} T = \operatorname{null} T$. ■

Exercise 3B6

Prove that there does not exist $T \in \mathcal{L}(\mathbb{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$.

Solution. Suppose for contradiction that there exists $T \in \mathcal{L}(\mathbb{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$. Then by the fundamental theorem of linear maps,

$$\dim \mathbb{R}^5 = \dim \operatorname{null} T + \dim \operatorname{range} T = 2 \dim \operatorname{range} T.$$

Therefore, $5 = 2 \dim \operatorname{range} T$, which is impossible since 5 is odd and $2 \dim \operatorname{range} T$ is even. Hence, no such T exists. ■

Exercise 3B7

Suppose V and W are finite-dimensional with $2 \leq \dim V \leq \dim W$. Show that $\{T \in \mathcal{L}(V, W) \mid T \text{ is not injective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

Solution. Let $\dim V = n$ and $\dim W = m$ with $2 \leq n \leq m$. Let $v_1, v_2, \ldots, v_n$ be a basis of V, and let $w_1, w_2, \ldots, w_m$ be a basis of W. Define $T_1, T_2 \in \mathcal{L}(V, W)$ by

$$T_1 v_1 = w_1, \quad T_1 v_k = 0 \text{ for } k \geq 2,$$

and

$$T_2 v_1 = 0, \quad T_2 v_k = w_k \text{ for } k \geq 2.$$

Then $v_2 \in \operatorname{null} T_1$ and $v_1 \in \operatorname{null} T_2$, so both T_1 and T_2 are not injective. However,

$$(T_1 + T_2)v_1 = T_1 v_1 + T_2 v_1 = w_1 + 0 = w_1 \neq 0,$$

and for $k \geq 2$,

$$(T_1 + T_2)v_k = T_1 v_k + T_2 v_k = 0 + w_k = w_k \neq 0.$$

Suppose $v \in V$ is such that $(T_1 + T_2)v = 0$. Then v can be expressed as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n.$$

Then

$$(T_1 + T_2)v = \alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_n w_n = 0.$$

Since $w_1, w_2, \ldots, w_m$ are linearly independent, it follows that $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$, hence $v = 0$. Therefore, $T_1 + T_2$ is injective. Since T_1 and T_2 are not injective, but their sum is injective, the set $\{T \in \mathcal{L}(V, W) \mid T \text{ is not injective}\}$ is not closed under addition and thus is not a subspace of $\mathcal{L}(V, W)$. ■

Exercise 3B8

Suppose V and W are finite-dimensional with $\dim V \geq \dim W \geq 2$. Show that $\{T \in \mathcal{L}(V, W) \mid T \text{ is not surjective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

Solution. We proceed similarly to the previous exercise. Let $\dim V = n$ and $\dim W = m$ with $n \geq m \geq 2$. Let $v_1, v_2, \ldots, v_n$ be a basis of V, and let $w_1, w_2, \ldots, w_m$ be a basis of W. Define $T_1, T_2 \in \mathcal{L}(V, W)$ by

$$T_1 v_1 = w_1, \quad T_1 v_k = 0 \text{ for } k \geq 2,$$

and

$$T_2 v_1 = 0, \quad T_2 v_k = w_k \text{ for } k = 2, \ldots, m, \quad T_2 v_k = 0 \text{ for } k > m.$$

Then $\operatorname{range} T_1 = \operatorname{span} w_1$ and $\operatorname{range} T_2 = \operatorname{span} w_2, \ldots, w_m$, both of which are proper subspaces of W, so T_1 and T_2 are not surjective. However,

$$(T_1 + T_2)v_1 = T_1 v_1 + T_2 v_1 = w_1 + 0 = w_1,$$

and for $k = 2, \ldots, m$,

$$(T_1 + T_2)v_k = T_1 v_k + T_2 v_k = 0 + w_k = w_k.$$

Therefore, $\operatorname{range}(T_1 + T_2) = \operatorname{span} w_1, w_2, \ldots, w_m = W$, so $T_1 + T_2$ is surjective. Since T_1 and T_2 are not surjective, but their sum is surjective, the set $\{T \in \mathcal{L}(V, W) \mid T \text{ is not surjective}\}$ is not closed under addition and thus is not a subspace of $\mathcal{L}(V, W)$. ■

Exercise 3B9

Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \ldots, v_n$ is linearly independent in V. Prove that $Tv_1, \ldots, Tv_n$ is linearly independent in W.

Solution. Suppose $a_1, \ldots, a_n \in \mathbb{F}$ are such that

$$a_1 Tv_1 + \cdots + a_n Tv_n = 0.$$

By linearity of T, we have

$$T(a_1 v_1 + \cdots + a_n v_n) = 0.$$

Injectivity of T implies that $\operatorname{null} T = \{0\}$, so

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Since $v_1, \ldots, v_n$ is linearly independent, it follows that $a_1 = \cdots = a_n = 0$. Therefore, $Tv_1, \ldots, Tv_n$ is linearly independent in W. ■

Exercise 3B10

Suppose $v_1, \ldots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \ldots, Tv_n$ spans $\operatorname{range} T$.

Solution. Let $w \in \operatorname{range} T$. Then there exists $v \in V$ such that $Tv = w$. Since $v_1, \ldots, v_n$ spans V, we can write

$$v = a_1 v_1 + \cdots + a_n v_n$$

for some scalars $a_1, \ldots, a_n \in \mathbb{F}$. By linearity of T, we have

$$w = Tv = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n.$$

Therefore, w is a linear combination of $Tv_1, \ldots, Tv_n$. Since w was arbitrary in $\operatorname{range} T$, it follows that $Tv_1, \ldots, Tv_n$ spans $\operatorname{range} T$. ■

Exercise 3B11

Suppose that V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace U of V such that

$$U \cap \operatorname{null} T = \{0\} \quad \text{and} \quad \operatorname{range} T = \{Tu \mid u \in U\}.$$

Manual Remark. *Note that this exercise proves the existence of a subspace U of V such that $V = U \oplus \operatorname{null} T$.*

Solution. Since $\operatorname{null} T$ is a subspace of V, we can extend a basis of $\operatorname{null} T$ to a basis of V. Let $n_1, \ldots, n_k$ be a basis of $\operatorname{null} T$, and extend this to a basis of V by adding vectors $u_1, \ldots, u_m$. Hence, a basis for V is given by

$$n_1, \ldots, n_k, u_1, \ldots, u_m.$$

Define $U = \operatorname{span}(u_1, \ldots, u_m)$. Suppose $v \in U \cap \operatorname{null} T$. Then v can be expressed as a linear combination of the basis vectors of U:

$$v = \alpha_1 u_1 + \cdots + \alpha_m u_m.$$

Since $v \in \operatorname{null} T$ as well, we can also express v as a linear combination of the basis vectors of $\operatorname{null} T$:

$$v = \beta_1 n_1 + \cdots + \beta_k n_k.$$

Equating the two expressions for v, we have

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = \beta_1 n_1 + \cdots + \beta_k n_k.$$

Since $n_1, \ldots, n_k, u_1, \ldots, u_m$ is a basis for V, the only way this equality can hold is if all coefficients are zero, i.e., $\alpha_1 = \cdots = \alpha_m = 0$ and $\beta_1 = \cdots = \beta_k = 0$. Therefore, $v = 0$, and we conclude that $U \cap \operatorname{null} T = \{0\}$.

Now, let $w \in \operatorname{range} T$. Then there exists $v \in V$ such that $Tv = w$. We can express v as a linear combination of the basis vectors of V:

$$v = \gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 u_1 + \cdots + \delta_m u_m.$$

Applying T to both sides, we have

$$Tv = T(\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 u_1 + \cdots + \delta_m u_m) = \gamma_1 T n_1 + \cdots + \gamma_k T n_k + \delta_1 T u_1 + \cdots + \delta_m T u_m.$$

Since $n_1, \ldots, n_k \in \operatorname{null} T$, we have $Tn_i = 0$ for all $i = 1, \ldots, k$. Therefore,

$$Tv = \delta_1 T u_1 + \cdots + \delta_m T u_m.$$

Let $u = \delta_1 u_1 + \cdots + \delta_m u_m \in U$. Then $Tu = w$. Since w was arbitrary in $\operatorname{range} T$, it follows that $\operatorname{range} T = \{Tu \mid u \in U\}$. Thus, we have found a subspace U of V such that $U \cap \operatorname{null} T = \{0\}$ and $\operatorname{range} T = \{Tu \mid u \in U\}$. ■

Exercise 3B12

Suppose T is a linear map from $\mathbb{F}^4$ to $\mathbb{F}^2$ such that

$$\operatorname{null} T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that T is surjective.

Solution. We may write the null space of T as

$$\operatorname{null} T = \{(5x_2, x_2, 7x_4, x_4) \mid x_2, x_4 \in \mathbb{F}\}.$$

From this, we can see that $\operatorname{null} T$ is spanned by the vectors $(5, 1, 0, 0)$ and $(0, 0, 7, 1)$. Therefore, $\dim \operatorname{null} T = 2$. By the fundamental theorem of linear maps, we have

$$\dim \mathbb{F}^4 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 4 = 2 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 2$. Since $\dim \mathbb{F}^2 = 2$, it follows that $\operatorname{range} T = \mathbb{F}^2$. Therefore, T is surjective. ■

Exercise 3B13

Suppose U is a three-dimensional subspace of $\mathbb{R}^8$ and that T is a linear map from $\mathbb{R}^8$ to $\mathbb{R}^5$ such that $\operatorname{null} T = U$. Prove that T is surjective.

Solution. By the fundamental theorem of linear maps, we have

$$\dim \mathbb{R}^8 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 8 = 3 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 5$. Since $\dim \mathbb{R}^5 = 5$, it follows that $\operatorname{range} T = \mathbb{R}^5$. Therefore, T is surjective. ■

Exercise 3B14

Prove that there does not exist a linear map from $\mathbb{F}^5$ to $\mathbb{F}^2$ whose null space equals $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 \mid x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}$.

Solution. We can express the null space as

$$\operatorname{null} T = \{(3x_2, x_2, x_4, x_4, x_4) \mid x_2, x_4 \in \mathbb{F}\}.$$

From this, we can see that $\operatorname{null} T$ is spanned by the vectors $(3, 1, 0, 0, 0)$ and $(0, 0, 1, 1, 1)$. Therefore, $\dim \operatorname{null} T = 2$. By the fundamental theorem of linear maps, we have

$$\dim \mathbb{F}^5 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 5 = 2 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 3$. However, $\dim \mathbb{F}^2 = 2$, which is less than 3. This is a contradiction, as the dimension of the range cannot exceed the dimension of the target space. Therefore, no such linear map exists. ■

Exercise 3B15

Suppose there exists a linear map on V whose null space and range are both finite-dimensional. Prove that V is finite-dimensional.
Manual Remark. *The spanning list in the solution is actually a basis of V.*

Solution. Let $T \in \mathcal{L}(V)$ be such that $\operatorname{null} T$ and $\operatorname{range} T$ are both finite-dimensional. Let $u_1, \ldots, u_m$ be a basis of $\operatorname{null} T$, and let $w_1, \ldots, w_n$ be a basis of $\operatorname{range} T$. Select $v_j \in V$ be such that $Tv_j = w_j$ for $j = 1, \ldots, n$. We claim that the list $u_1, \ldots, u_m, v_1, \ldots, v_n$ spans V.

Let $v \in V$. Then $Tv \in \operatorname{range} T$, so we may write

$$Tv = \alpha_1 w_1 + \cdots + \alpha_n w_n$$

for some scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$. Consider the vector

$$u = v - (\alpha_1 v_1 + \cdots + \alpha_n v_n).$$

Then

$$Tu = Tv - (\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n) = Tv - (\alpha_1 w_1 + \cdots + \alpha_n w_n) = 0.$$

Therefore, $u \in \operatorname{null} T$, and we can write

$$u = \beta_1 u_1 + \cdots + \beta_m u_m$$

for some scalars $\beta_1, \ldots, \beta_m \in \mathbb{F}$. Thus,

$$v = u + (\alpha_1 v_1 + \cdots + \alpha_n v_n) = \beta_1 u_1 + \cdots + \beta_m u_m + \alpha_1 v_1 + \cdots + \alpha_n v_n,$$

which shows that the list spans V. Hence, V is finite-dimensional. ■

Exercise 3B16

Suppose V and W are both finite-dimensional. Prove that there exists an injective linear map from V to W if and only if $\dim V \leq \dim W$.

Solution. Suppose there exists an injective linear map $T \in \mathcal{L}(V, W)$. By the fundamental theorem of linear maps, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Since T is injective, $\operatorname{null} T = \{0\}$, so $\dim \operatorname{null} T = 0$. Therefore,

$$\dim V = 0 + \dim \operatorname{range} T = \dim \operatorname{range} T.$$

Since $\operatorname{range} T$ is a subspace of W, we have $\dim \operatorname{range} T \leq \dim W$. Thus,

$$\dim V = \dim \operatorname{range} T \leq \dim W.$$

Conversely, suppose $\dim V \leq \dim W$. Let $n = \dim V$ and $m = \dim W$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Let $\{w_1, \ldots, w_m\}$ be a basis of W. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tv_i = w_i \quad \text{for } i = 1, \ldots, n.$$

Since $n \leq m$, we can assign each basis vector of V to a distinct basis vector of W. To show that T is injective, let $v = \alpha_1 v_1 + \cdots + \alpha_n v_n \in V$ such that $Tv = 0$. Then

$$Tv = \alpha_1 Tv_1 + \cdots + \alpha_n Tv_n = \alpha_1 w_1 + \cdots + \alpha_n w_n = 0.$$

Since $w_1, \ldots, w_n$ are linearly independent, it follows that $\alpha_1 = \cdots = \alpha_n = 0$. Therefore, $v = 0$, and T is injective. Hence, there exists an injective linear map from V to W if and only if $\dim V \leq \dim W$. ■

Exercise 3B17

Suppose V and W are both finite-dimensional. Prove that there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$.

Solution. Suppose there exists a surjective linear map $T \in \mathcal{L}(V, W)$. By the fundamental theorem of linear maps, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Since T is surjective, range $T = W$, so $\dim \operatorname{range} T = \dim W$. Therefore,

$$\dim V = \dim \operatorname{null} T + \dim W.$$

Since $\dim \operatorname{null} T \geq 0$, it follows that

$$\dim V \geq \dim W.$$

Conversely, suppose $\dim V \geq \dim W$. Let $n = \dim V$ and $m = \dim W$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Let $\{w_1, \ldots, w_m\}$ be a basis of W. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tv_i = w_i \quad \text{for } i = 1, \ldots, m, \qquad Tv_i = 0 \quad \text{for } i = m+1, \ldots, n.$$

To show that T is surjective, let $w \in W$. Then w can be expressed as a linear combination of the basis vectors of W:

$$w = \beta_1 w_1 + \cdots + \beta_m w_m.$$

Consider the vector $v = \beta_1 v_1 + \cdots + \beta_m v_m \in V$. Then

$$Tv = \beta_1 Tv_1 + \cdots + \beta_m Tv_m = \beta_1 w_1 + \cdots + \beta_m w_m = w.$$

Since w was arbitrary in W, it follows that range $T = W$. Therefore, T is surjective. Hence, there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$. ■

Exercise 3B18

Suppose V and W are finite-dimensional and that U is a subspace of V. Prove that there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$ if and only if $\dim U \geq \dim V - \dim W$.

Solution. Suppose there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$. By the fundamental theorem of linear maps, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim U + \dim \operatorname{range} T.$$

Since range T is a subspace of W, we have $\dim \operatorname{range} T \leq \dim W$. Therefore,

$$\dim V = \dim U + \dim \operatorname{range} T \leq \dim U + \dim W.$$

Rearranging this inequality gives

$$\dim U \geq \dim V - \dim W.$$

Conversely, suppose $\dim U \geq \dim V - \dim W$. Let $n = \dim V$, $m = \dim W$, and $k = \dim U$. Let $u_1, \ldots, u_k$ be a basis of U. Extend this basis to a basis of V by adding vectors $v_1, \ldots, v_{n-k}$. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tu_i = 0 \quad \text{for } i = 1, \ldots, k, \qquad Tv_j = w_j \quad \text{for } j = 1, \ldots, n-k,$$

where $w_1, \ldots, w_{n-k}$ are linearly independent vectors in W. Since $\dim U = k$ and $\dim V - \dim W = n - m$, we have $k \geq n - m$, which implies that $n - k \leq m$. Therefore, we can choose $w_1, \ldots, w_{n-k}$ to be linearly independent in W. To show that null $T = U$, let $v \in \operatorname{null} T$. Then $Tv = 0$. Express v as a linear combination of the basis vectors of V:

$$v = \alpha_1 u_1 + \cdots + \alpha_k u_k + \beta_1 v_1 + \cdots + \beta_{n-k} v_{n-k}.$$

Applying T to both sides, we have

$$Tv = \alpha_1 Tu_1 + \cdots + \alpha_k Tu_k + \beta_1 Tv_1 + \cdots + \beta_{n-k} Tv_{n-k} = 0 + \beta_1 w_1 + \cdots + \beta_{n-k} w_{n-k} = 0.$$

Since $w_1, \ldots, w_{n-k}$ are linearly independent, it follows that $\beta_1 = \cdots = \beta_{n-k} = 0$. Therefore,

$$v = \alpha_1 u_1 + \cdots + \alpha_k u_k \in U.$$

Hence, $\operatorname{null} T \subseteq U$. Conversely, if $v \in U$, then $Tv = 0$ by the definition of T. Therefore, $U \subseteq \operatorname{null} T$. Combining these two inclusions, we have $\operatorname{null} T = U$. Thus, there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \geq \dim V - \dim W$. ■

Exercise 3B19

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V.

Solution. Suppose T is injective. Then $\operatorname{null} T = \{0\}$. Also, $\operatorname{range} T$ is finite dimensional since it is a subspace of W. Since T is injective, it is an isomorphism from V onto $\operatorname{range} T$. Hence, V is finite dimensional. Let $n = \dim V$ and $m = \dim W$. Since T is injective, we have $\dim \operatorname{range} T = n$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Then $\{Tv_1, \ldots, Tv_n\}$ is linearly independent in W. Extend this set to a basis of W by adding vectors $w_1, \ldots, w_{m-n}$. Define a linear map $S \in \mathcal{L}(W, V)$ by

$$S(Tv_i) = v_i \quad \text{for } i = 1, \ldots, n, \quad S(w_j) = 0 \quad \text{for } j = 1, \ldots, m-n.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

Applying T to both sides, we have

$$Tv = \alpha_1 Tv_1 + \cdots + \alpha_n Tv_n.$$

Now, applying S to Tv, we get

$$STv = S(\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n) = \alpha_1 S(Tv_1) + \cdots + \alpha_n S(Tv_n) = \alpha_1 v_1 + \cdots + \alpha_n v_n = v.$$

Therefore, ST is the identity operator on V.

Conversely, suppose there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V. Let $v \in \operatorname{null} T$. Then $Tv = 0$, and applying S gives

$$STv = S(0) = 0.$$

Since ST is the identity operator on V, we have $v = STv = 0$. Therefore, $\operatorname{null} T = \{0\}$, which means T is injective. ■

Exercise 3B20

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that TS is the identity operator on W.

Solution. Suppose T is surjective. Then range $T = W$. Let $m = \dim W$ and choose a basis $\{w_1, \ldots, w_m\}$ of W. Since T is surjective, for each w_i, there exists $v_i \in V$ such that $Tv_i = w_i$. Define a linear map $S \in \mathcal{L}(W, V)$ by

$$S(w_i) = v_i \quad \text{for } i = 1, \ldots, m.$$

Then for any $w \in W$, we can express w as a linear combination of the basis vectors:

$$w = \alpha_1 w_1 + \cdots + \alpha_m w_m.$$

Applying S to both sides, we have

$$Sw = S(\alpha_1 w_1 + \cdots + \alpha_m w_m) = \alpha_1 S(w_1) + \cdots + \alpha_m S(w_m) = \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

Now, applying T to Sw, we get

$$TSw = T(\alpha_1 v_1 + \cdots + \alpha_m v_m) = \alpha_1 Tv_1 + \cdots + \alpha_m Tv_m = \alpha_1 w_1 + \cdots + \alpha_m w_m = w.$$

Therefore, TS is the identity operator on W.

Conversely, suppose there exists $S \in \mathcal{L}(W, V)$ such that TS is the identity operator on W. Let $w \in W$. Then

$$TSw = w.$$

This means that for every $w \in W$, there exists a vector $v = Sw \in V$ such that $Tv = T(Sw) = w$. Therefore, range $T = W$, which means that T is surjective. ■

Exercise 3B21

Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and U is a subspace of W. Prove that $\{v \in V \mid Tv \in U\}$ is a subspace of V and

$$\dim\{v \in V \mid Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

Solution. Let $X = \{v \in V \mid Tv \in U\}$. We first show that X is a subspace of V. Let $v_1, v_2 \in X$ and $\alpha, \beta \in \mathbb{F}$. Then

$$T(\alpha v_1 + \beta v_2) = \alpha T v_1 + \beta T v_2.$$

Since $Tv_1, Tv_2 \in U$ and U is a subspace of W, it follows that $\alpha T v_1 + \beta T v_2 \in U$. Therefore, $\alpha v_1 + \beta v_2 \in X$, and thus X is a subspace of V.

Now, we will compute the dimension of X. Let $Y = U \cap \operatorname{range} T$. Let $\dim T = k$ and $\dim Y = m$. Let $n_1, \ldots, n_k$ be a basis of null T, and let $y_1, \ldots, y_m$ be a basis of Y. Since $y_i \in \operatorname{range} T$, there exist vectors $v_i \in V$ such that $Tv_i = y_i$ for $i = 1, \ldots, m$. We claim that the set $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$ is a basis of X.

First, we show that this set spans X. Let $v \in X$. Then $Tv \in U$, and since $Tv \in \operatorname{range} T$, we have $Tv \in Y$. Therefore, there exist scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$Tv = \alpha_1 y_1 + \cdots + \alpha_m y_m = \alpha_1 T v_1 + \cdots + \alpha_m T v_m.$$

Let $v' = v - (\alpha_1 v_1 + \cdots + \alpha_m v_m)$. Then

$$Tv' = Tv - T(\alpha_1 v_1 + \cdots + \alpha_m v_m) = Tv - (\alpha_1 T v_1 + \cdots + \alpha_m T v_m) = 0.$$

Therefore, $v' \in \operatorname{null} T$, and we can express v' as a linear combination of the basis vectors of null T:

$$v' = \beta_1 n_1 + \cdots + \beta_k n_k$$

for some scalars $\beta_1, \ldots, \beta_k \in \mathbb{F}$. Thus,

$$v = v' + (\alpha_1 v_1 + \cdots + \alpha_m v_m) = \beta_1 n_1 + \cdots + \beta_k n_k + \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

This shows that v is a linear combination of the vectors in $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$, and hence this set spans X.

Next, we show that this set is linearly independent. Suppose

$$\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 v_1 + \cdots + \delta_m v_m = 0$$

for some scalars $\gamma_1, \ldots, \gamma_k, \delta_1, \ldots, \delta_m \in \mathbb{F}$. Observe that $Tn_i = 0$ for all $i = 1, \ldots, k$, and $Tv_j = y_j$ for all $j = 1, \ldots, m$. Therefore,

$$0 = T(\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 v_1 + \cdots + \delta_m v_m) = \delta_1 y_1 + \cdots + \delta_m y_m.$$

Since $y_1, \ldots, y_m$ are linearly independent, it follows that $\delta_1 = \cdots = \delta_m = 0$. Therefore,

$$\gamma_1 n_1 + \cdots + \gamma_k n_k = 0.$$

Since $n_1, \ldots, n_k$ are linearly independent, $\gamma_1 = \cdots = \gamma_k = 0$. Thus, the set $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$ is linearly independent and a basis of X. Therefore,

$$\dim X = k + m = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

■

Exercise 3B22

Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T.$$

Solution. Let $u \in U$ such that $u \in \operatorname{null} ST$. Then $STu = 0$, which implies that $Tu \in \operatorname{null} S$. Therefore, we can express $\operatorname{null} ST$ as

$$\operatorname{null} ST = \{u \in U \mid Tu \in \operatorname{null} S\}.$$

Observe that this looks similar to the set in Exercise 3b21, where T is a linear map from U to V and $\operatorname{null} S$ is a subspace of V. By applying the result from Exercise 3b21, we have

$$\dim \operatorname{null} ST = \dim \operatorname{null} T + \dim(\operatorname{null} S \cap \operatorname{range} T).$$

Since $\operatorname{null} S \cap \operatorname{range} T$ is a subspace of $\operatorname{null} S$, we have

$$\dim(\operatorname{null} S \cap \operatorname{range} T) \leq \dim \operatorname{null} S.$$

Therefore,

$$\dim \operatorname{null} ST = \dim \operatorname{null} T + \dim(\operatorname{null} S \cap \operatorname{range} T) \leq \dim \operatorname{null} T + \dim \operatorname{null} S.$$ ■

Exercise 3B23

Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{range} ST \leq \min(\dim \operatorname{range} S, \dim \operatorname{range} T).$$

Solution. Let $u \in U$ such that $STu \in \operatorname{range} ST$. Then $STu = S(Tu)$, which implies that $Tu \in \operatorname{range} T$. Therefore, we can express $\operatorname{range} ST$ as

$$\operatorname{range} ST = \{S(Tu) \mid u \in U\}.$$

Since $Tu \in \operatorname{range} T$, we can rewrite this as

$$\operatorname{range} ST = \{Sv \mid v \in \operatorname{range} T\}.$$

This shows that $\operatorname{range} ST$ is a subset of $\operatorname{range} S$, and hence

$$\dim \operatorname{range} ST \leq \dim \operatorname{range} S.$$

Now let $v_1, \dots, v_m$ be a basis of $\operatorname{range} T$. Then for any $u \in U$, we can express Tu as a linear combination of the basis vectors:

$$Tu = \alpha_1 v_1 + \cdots + \alpha_m v_m$$

for some scalars $\alpha_1, \dots, \alpha_m \in \mathbb{F}$. Applying S to both sides, we have

$$STu = S(Tu) = S(\alpha_1 v_1 + \cdots + \alpha_m v_m) = \alpha_1 Sv_1 + \cdots + \alpha_m Sv_m.$$

This shows that $\operatorname{range} ST$ is spanned by the vectors $Sv_1, \dots, Sv_m$. Therefore,

$$\dim \operatorname{range} ST \leq m = \dim \operatorname{range} T.$$

Combining the two inequalities, we have

$$\dim \operatorname{range} ST \leq \min(\dim \operatorname{range} S, \dim \operatorname{range} T). \qquad \blacksquare$$

Exercise 3B24

(a) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \operatorname{range} TS \leq 2$.
(b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

Solution.

(a) Since $ST = 0$, then $S(Tv) = 0$ for all $v \in V$. This implies that $\operatorname{range} T \subseteq \operatorname{null} S$. By the fundamental theorem of linear maps, we have

$$5 = \dim \operatorname{null} S + \dim \operatorname{range} S \geq \dim \operatorname{range} T + \dim \operatorname{range} S.$$

Now, by Exercise 3B23, we have

$$\dim \operatorname{range} TS \leq \min(\dim \operatorname{range} T, \dim \operatorname{range} S).$$

Suppose that $\dim \operatorname{range} TS > 2$. Then both $\dim \operatorname{range} T$ and $\dim \operatorname{range} S$ must be greater than 2, or equivalently, $\dim \operatorname{range} T \geq 3$ and $\dim \operatorname{range} S \geq 3$. This implies that

$$\dim \operatorname{range} T + \dim \operatorname{range} S \geq 3 + 3 = 6 > 5,$$

which is a contradiction. Therefore, $\dim \operatorname{range} TS \leq 2$.

(b) Let v_1, v_2, v_3, v_4, v_5 be a basis of $\mathbb{F}^5$. Define $S, T \in \mathcal{L}(\mathbb{F}^5)$ by

$$Sv_1 = v_1, \quad Sv_2 = v_2, \quad Sv_3 = Sv_4 = Sv_5 = 0,$$

and

$$Tv_1 = v_3, \quad Tv_2 = v_4, \quad Tv_3 = Tv_4 = Tv_5 = 0.$$

Then $ST = 0$ since $S(Tv_i) = S(0) = 0$ for all $i = 1, \ldots, 5$. Now, we compute range TS. We have

$$TSv_1 = T(Sv_1) = Tv_1 = v_3, \quad TSv_2 = T(Sv_2) = Tv_2 = v_4, \quad TSv_3 = TSv_4 = TSv_5 = 0.$$

Therefore, range $TS = \operatorname{span}(v_3, v_4)$, which has dimension 2. Thus, we have constructed an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$. ■

Exercise 3B25

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that null $S \subseteq$ null T if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

Manual Remark. *This exercise requires showing that some $R \in \mathcal{L}(\operatorname{range} S, W)$ is well-defined. An issue arises when T is not injective: for some $w \in \operatorname{range} S$, there may be distinct $v_1, v_2 \in V$ such that $Sv_1 = Sv_2 = w$. In this case, we must show that $Tv_1 = Tv_2$.*

Solution. Suppose null $S \subseteq$ null T. Define a linear map $R : \operatorname{range} S \to W$ by $R(Sv) = Tv$ for all $v \in V$. We need to show that R is well-defined. Suppose $Sv_1 = Sv_2$ for some $v_1, v_2 \in V$. Then $S(v_1 - v_2) = 0$, which implies that $v_1 - v_2 \in$ null S. Since null $S \subseteq$ null T, we have $v_1 - v_2 \in$ null T, which means that $T(v_1 - v_2) = 0$. Therefore, $Tv_1 = Tv_2$, and hence $R(Sv_1) = R(Sv_2)$. Thus, R is well-defined.

By Exercise 3A13, we can extend R to some $E \in \mathcal{L}(W)$. Then for any $v \in V$, we have

$$Tv = R(Sv) = E(Sv),$$

and hence $T = ES$.

Conversely, suppose there exists $E \in \mathcal{L}(W)$ such that $T = ES$. Let $v \in$ null S. Then $Sv = 0$, and applying E gives

$$Tv = E(Sv) = E(0) = 0.$$

Therefore, $v \in$ null T, and hence null $S \subseteq$ null T. ■

Exercise 3B26

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that range $S \subseteq$ range T if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

Manual Remark. *From the previous exercise, the reader may be tempted to show that a map is well-defined again. However, this is not necessary here, since we are defining a map on the entire space V, not just on* range S.

Solution. Suppose range $S \subseteq$ range T. Let $v_1, \ldots, v_n$ be a basis of V. For each $i = 1, \ldots, n$, since $Sv_i \in \operatorname{range} S \subseteq \operatorname{range} T$, there exists $u_i \in V$ such that $Tu_i = Sv_i$. Define a linear map $E \in \mathcal{L}(V)$ by

$$Ev_i = u_i \quad \text{for } i = 1, \ldots, n.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

for some scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$. Applying S and TE to both sides, we have

$$Sv = \alpha_1 Sv_1 + \cdots + \alpha_n Sv_n = \alpha_1 Tu_1 + \cdots + \alpha_n Tu_n = T(\alpha_1 u_1 + \cdots + \alpha_n u_n) = TEv.$$

Therefore, $S = TE$.

Conversely, suppose there exists $E \in \mathcal{L}(V)$ such that $S = TE$. Let $w \in \operatorname{range} S$. Then there exists $v \in V$ such that $Sv = w$. Applying the definition of S, we have

$$w = Sv = TE(v) = T(E(v)).$$

Therefore, $w \in \operatorname{range} T$, and hence $\operatorname{range} S \subseteq \operatorname{range} T$. ■

Exercise 3B27

Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

Solution. Let $v \in V$. We can express v as

$$v = (v - Pv) + Pv.$$

Observe that $Pv \in \operatorname{range} P$ by definition. Now, we need to show that $v - Pv \in \operatorname{null} P$. Applying P to $v - Pv$, we have

$$P(v - Pv) = Pv - P^2 v = Pv - Pv = 0.$$

Therefore, $v - Pv \in \operatorname{null} P$. This shows that every vector $v \in V$ can be expressed as a sum of a vector in $\operatorname{null} P$ and a vector in $\operatorname{range} P$, which implies that

$$V = \operatorname{null} P + \operatorname{range} P.$$

Next, we need to show that the sum is direct. Suppose $u \in \operatorname{null} P \cap \operatorname{range} P$. Then there exists some $w \in V$ such that $u = Pw$. Since $u \in \operatorname{null} P$, we have

$$Pu = P(Pw) = P^2 w = Pw = u.$$

But since $u \in \operatorname{null} P$, we also have $Pu = 0$. Therefore, we have

$$u = Pu = 0.$$

This shows that the intersection of $\operatorname{null} P$ and $\operatorname{range} P$ is trivial, i.e., $\operatorname{null} P \cap \operatorname{range} P = \{0\}$. Hence, the sum is direct, and we conclude that

$$V = \operatorname{null} P \oplus \operatorname{range} P.$$ ■

Exercise 3B28

Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ is such that $\deg Dp = (\deg p) - 1$ for every nonconstant polynomial $p \in \mathcal{P}(\mathbb{R})$. Prove that D is surjective.

Manual Remark. *Note that $\mathcal{P}(\mathbb{R})$ is infinite-dimensional, so we cannot use the fundamental theorem of linear maps directly. Instead, one should consider finite-dimensional subspaces of $\mathcal{P}(\mathbb{R})$.*

Solution. If $p = 0$, then $p = D(0) \in \operatorname{range} D$ by linearity. Suppose $p \neq 0$ and let $n = \deg p$. Consider the list $D(x), D(x^2), \dots, D(x^{n+1})$. Since x^k is nonconstant for each $k \in \{1, \dots, n+1\}$, we have $\deg D(x^k) = k-1$. Thus, $D(x), D(x^2), \dots, D(x^{n+1})$ is a list of $n+1$ polynomials with pairwise distinct degrees, which implies linear independence. Since $\dim \mathcal{P}_n(\mathbb{R}) = n + 1$, this list is a basis for $\mathcal{P}_n(\mathbb{R})$. In particular, $p \in \mathcal{P}_n(\mathbb{R})$, so there exist scalars $c_1, \dots, c_{n+1} \in \mathbb{R}$ such that

$$p = \sum_{k=1}^{n+1} c_k D(x^k) = D\left(\sum_{k=1}^{n+1} c_k x^k\right) \in \operatorname{range} D.$$

Since p was arbitrary, D is surjective. ■

Exercise 3B29

Suppose $p \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $5q'' + 3q' = p$.
Appendix. *See* Appendix 3B29 *for an explicit method of determining q given a system of equations.*

Solution. Define $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ by $Dq = 5q'' + 3q'$ for all $q \in \mathcal{P}(\mathbb{R})$. For nonconstant q, we have $\deg Dq = \deg q - 1$. By Exercise 3B28, D is surjective. Therefore, for any polynomial $p \in \mathcal{P}(\mathbb{R})$, there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $Dq = p$, which means that $5q'' + 3q' = p$. ■

Exercise 3B30

Suppose $\varphi \in \mathcal{L}(V, \mathbb{F})$ and $\varphi \neq 0$. Suppose $u \in V$ is not in $\operatorname{null} \varphi$. Prove that

$$V = \operatorname{null} \varphi \oplus \{au \mid a \in \mathbb{F}\}.$$

Solution. Let $v \in V$. We want to show that v can be expressed as a sum of a vector in $\operatorname{null} \varphi$ and a scalar multiple of u. Since $\varphi(u) \neq 0$, we can define a scalar

$$a = \frac{\varphi(v)}{\varphi(u)}.$$

Then consider the vector

$$w = v - au.$$

We have

$$\varphi(w) = \varphi(v - au) = \varphi(v) - a\varphi(u) = \varphi(v) - \frac{\varphi(v)}{\varphi(u)}\varphi(u) = 0.$$

Therefore, $w \in \operatorname{null} \varphi$. Thus, we can express v as

$$v = w + au,$$

where $w \in \operatorname{null} \varphi$ and au is a scalar multiple of u. This shows that every vector in V can be expressed as a sum of a vector in $\operatorname{null} \varphi$ and a scalar multiple of u, which implies that

$$V = \operatorname{null} \varphi + \{au \mid a \in \mathbb{F}\}.$$

Next, we need to show that the sum is direct. Suppose $w \in \operatorname{null}\varphi \cap \{au \mid a \in \mathbb{F}\}$. Then there exists a scalar $a \in \mathbb{F}$ such that $w = au$. Since $w \in \operatorname{null}\varphi$, we have

$$0 = \varphi(w) = \varphi(au) = a\varphi(u).$$

Since $\varphi(u) \neq 0$, it follows that $a = 0$. Therefore, $w = 0$, which shows that the intersection of $\operatorname{null}\varphi$ and $\{au \mid a \in \mathbb{F}\}$ is trivial. Hence, the sum is direct, and we conclude that

$$V = \operatorname{null}\varphi \oplus \{au \mid a \in \mathbb{F}\}. \qquad \blacksquare$$

Exercise 3B31

Suppose V is finite-dimensional, X is a subspace of V, and Y is a finite-dimensional subspace of W. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$ if and only if $\dim X + \dim Y = \dim V$.

Solution. Suppose there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$. By the fundamental theorem of linear maps, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim X + \dim Y.$$

Conversely, suppose $\dim X + \dim Y = \dim V$. Let $n = \dim V$, $k = \dim X$, and $m = \dim Y$. Then $n = k + m$. Choose a basis $\{x_1, \ldots, x_k\}$ of X and extend it to a basis $\{x_1, \ldots, x_k, v_1, \ldots, v_m\}$ of V. Let $\{y_1, \ldots, y_m\}$ be a basis of Y. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tx_i = 0 \quad \text{for } i = 1, \ldots, k,$$

and

$$Tv_j = y_j \quad \text{for } j = 1, \ldots, m.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 x_1 + \cdots + \alpha_k x_k + \beta_1 v_1 + \cdots + \beta_m v_m$$

for some scalars $\alpha_1, \ldots, \alpha_k, \beta_1, \ldots, \beta_m \in \mathbb{F}$. Applying T to both sides gives

$$Tv = \alpha_1 Tx_1 + \cdots + \alpha_k Tx_k + \beta_1 Tv_1 + \cdots + \beta_m Tv_m = 0 + \beta_1 y_1 + \cdots + \beta_m y_m.$$

Therefore, $Tv \in Y$. Now suppose $y = \gamma_1 y_1 + \cdots + \gamma_m y_m \in Y$. Then

$$T(\gamma_1 v_1 + \cdots + \gamma_m v_m) = \gamma_1 y_1 + \cdots + \gamma_m y_m = y.$$

Hence, $\operatorname{range} T = Y$. Moreover, if $Tv = 0$, then $\beta_1 y_1 + \cdots + \beta_m y_m = 0$. Since $y_1, \ldots, y_m$ are linearly independent, it follows that $\beta_1 = \cdots = \beta_m = 0$. Thus, $v \in X$. Additionally, if $v \in X$, then $v = \alpha_1 x_1 + \cdots + \alpha_k x_k$ for some scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, and hence $Tv = 0$. Therefore, $\operatorname{null} T = X$. This shows that there exists a linear map $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$. $\blacksquare$

Exercise 3B32

Suppose V is finite-dimensional with $\dim V > 1$. Show that if $\varphi : \mathcal{L}(V) \to \mathbb{F}$ is a linear map such that $\varphi(ST) = \varphi(S)\varphi(T)$ for all $S, T \in \mathcal{L}(V)$, then $\varphi = 0$.
Axler Note. *The description of the two-sided ideals of $\mathcal{L}(V)$ given by Exercise 3A17 might be useful.*

Solution. Observe that $\operatorname{null}\varphi$ is a two-sided ideal of $\mathcal{L}(V)$ as follows: if $X \in \operatorname{null}\varphi$ and $Y \in \mathcal{L}(V)$, then

$$\varphi(XY) = \varphi(X)\varphi(Y) = 0 \cdot \varphi(Y) = 0, \quad \varphi(YX) = \varphi(Y)\varphi(X) = \varphi(Y) \cdot 0 = 0.$$

Since $\dim V > 1$, then Exercise 3A16 gives the existence of $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$. Then we have

$$\varphi(ST - TS) = \varphi(S)\varphi(T) - \varphi(T)\varphi(S) = 0.$$

Therefore, $ST - TS \in \operatorname{null}\varphi$. Since $\operatorname{null}\varphi$ is a two-sided ideal of $\mathcal{L}(V)$, then by Exercise 3A17, we have $\operatorname{null}\varphi = \mathcal{L}(V)$. Hence, $\varphi = 0$. ■

Exercise 3B33

Suppose that V and W are real vector spaces and $T \in \mathcal{L}(V, W)$. Define $T_{\mathbb{C}} : V_{\mathbb{C}} \to W_{\mathbb{C}}$ by

$$T_{\mathbb{C}}(u + iv) = Tu + iTv$$

for all $u, v \in V$.

(a) Show that $T_{\mathbb{C}}$ is a (complex) linear map from $V_{\mathbb{C}}$ to $W_{\mathbb{C}}$.
(b) Show that $T_{\mathbb{C}}$ is injective if and only if T is injective.
(c) Show that $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$ if and only if $\operatorname{range} T = W$.

Axler Note. *See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbb{C}}$. The linear map $T_{\mathbb{C}}$ is called the **complexification** of the linear map T.*

Solution.

(a) Let $u + vi$ and $w + xi \in V_{\mathbb{C}}$. Then for any $a + bi \in \mathbb{C}$, we have

$$\begin{aligned} T_{\mathbb{C}}((a + bi)(u + vi) + (w + xi)) &= T_{\mathbb{C}}((au - bv + (av + bu)i) + (w + xi)) \\ &= T(au - bv + w) + iT(av + bu + x) \\ &= aTu - bTv + Tw + i(aTv + bTu + Tx) \\ &= (a + bi)(Tu + iTv) + (Tw + iTx) \\ &= (a + bi)T_{\mathbb{C}}(u + vi) + T_{\mathbb{C}}(w + xi). \end{aligned}$$

Therefore, $T_{\mathbb{C}}$ is a complex linear map from $V_{\mathbb{C}}$ to $W_{\mathbb{C}}$.

(b) Suppose $T_{\mathbb{C}}$ is injective. Let $v \in V$ be such that $Tv = 0$. Then

$$T_{\mathbb{C}}(v + i0) = Tv + iT0 = 0 + i0 = 0.$$

Since $T_{\mathbb{C}}$ is injective, we have $v + i0 = 0$, which implies that $v = 0$. Therefore, T is injective.

Conversely, suppose T is injective. Let $u + iv \in V_{\mathbb{C}}$ be such that $T_{\mathbb{C}}(u + iv) = 0$. Then

$$Tu + iTv = 0.$$

This implies that $Tu = 0$ and $Tv = 0$. Since T is injective, we have $u = 0$ and $v = 0$. Therefore, $u + iv = 0$, and hence $T_{\mathbb{C}}$ is injective.

(c) Suppose $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$. Let $w \in W$. Then $w + i0 \in W_{\mathbb{C}}$, and since $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$, there exists $u + iv \in V_{\mathbb{C}}$ such that

$$T_{\mathbb{C}}(u + iv) = Tu + iTv = w + i0.$$

This implies that $Tu = w$ and $Tv = 0$. Therefore, $w \in \operatorname{range} T$, and hence $\operatorname{range} T = W$.

Conversely, suppose $\operatorname{range} T = W$. Let $w_1 + iw_2 \in W_{\mathbb{C}}$. Since $\operatorname{range} T = W$, there exist $u, v \in V$ such that $Tu = w_1$ and $Tv = w_2$. Then

$$T_{\mathbb{C}}(u + iv) = Tu + iTv = w_1 + iw_2.$$

Therefore, $w_1 + iw_2 \in \operatorname{range} T_{\mathbb{C}}$, and hence $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$. ∎

3C Matrices

Exercise 3C1

Suppose $T \in \mathcal{L}(V, W)$. Show that with respect to each choice of bases of V and W, the matrix of T has at least $\dim \operatorname{range} T$ nonzero entries.

Solution. Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W. Let $A = \mathcal{M}(T)$ be the matrix with respect to these bases. Then the j-th column of A is the coordinate vector of Tv_j with respect to the basis for W.

Suppose A had fewer than $\dim \operatorname{range} T$ nonzero columns. Since zero columns do not contribute to the span, the columns of A would then span a space of dimension less than $\dim \operatorname{range} T$. This is a contradiction, since fewer than $\dim \operatorname{range} T$ vectors cannot span $\operatorname{range} T$. Therefore, A must have at least $\dim \operatorname{range} T$ nonzero columns, and hence at least $\dim \operatorname{range} T$ nonzero entries. ■

Exercise 3C2

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \operatorname{range} T = 1$ if and only if there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

Solution. Suppose $\dim \operatorname{range} T = 1$. Then there exists a nonzero vector $y_1 \in W$ such that $\operatorname{range} T = \operatorname{span} y_1$. Let $\dim V = n$ and $\dim W = m$. Extend y_1 to a basis $y_1, \dots, y_m$ of W, and define the following vectors in W:

$$w_1 = y_1 - (y_2 + \cdots + y_m), \quad w_k = y_k \text{ for } k = 2, \dots, m.$$

It is clear that $w_1 + w_2 + \cdots + w_m = y_1$, and hence $w_1, \dots, w_m$ spans W. Now suppose for scalars $\alpha_1, \dots, \alpha_m \in \mathbb{F}$, we have

$$\alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_m w_m = 0.$$

Then we can express this as

$$\alpha_1(y_1 - (y_2 + \cdots + y_m)) + \alpha_2 y_2 + \cdots + \alpha_m y_m = 0.$$

Simplifying, we get

$$(\alpha_1)y_1 + (-\alpha_1 + \alpha_2)y_2 + \cdots + (-\alpha_1 + \alpha_m)y_m = 0.$$

Since $y_1, \dots, y_m$ are linearly independent, we must have $\alpha_1 = \alpha_2 = \cdots = \alpha_m = 0$. Therefore, $w_1, \dots, w_m$ are linearly independent and hence form a basis of W.

We now construct a basis of V. By the fundamental theorem of linear maps, we have

$$\dim V = \dim T + \dim \operatorname{range} T = \dim T + 1.$$

Let $u_2, \dots, u_n$ be a basis of T. Then we can extend this to a basis $u_1, u_2, \dots, u_n$ of V, where $u_1 \notin T$. Since $\operatorname{range} T = \operatorname{span} y_1$, we have

$$Tu_1 = \beta y_1$$

for some nonzero scalar $\beta \in \mathbb{F}$. If $\beta \neq 1$, we may replace u_1 with $\beta^{-1}u_1$ to ensure that $Tu_1 = y_1$. We construct the following vectors in V:

$$v_1 = u_1, \quad v_k = u_1 + u_k \text{ for } k = 2, \dots, n.$$

Note that for every $j = 1, \dots, n$, we have $Tv_j = Tu_1 + Tu_j = y_1 + 0 = y_1$. We claim that $v_1, \dots, v_n$ form a basis of V. Suppose for scalars $\alpha_1, \dots, \alpha_n \in \mathbb{F}$, we have

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0.$$

Then we can express this as

$$(\alpha_1 + \alpha_2 + \cdots + \alpha_n)u_1 + \alpha_2 u_2 + \cdots + \alpha_n u_n = 0.$$

Since $u_1, \dots, u_n$ are linearly independent, we must have $\alpha_1 + \alpha_2 + \cdots + \alpha_n = 0$ and $\alpha_2 = \cdots = \alpha_n = 0$. This implies that $\alpha_1 = 0$, and hence $v_1, \dots, v_n$ are linearly independent and form a basis of V. Then with respect to the bases $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W, we have that $Tv_j = y_1 = w_1 + w_2 + \cdots + w_m$ for each $j = 1, \dots, n$. Therefore, the matrix of T with respect to these bases has all entries equal to 1.

Conversely, suppose there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1. Then for each basis vector v_j of V, we have

$$Tv_j = w$$

for some fixed nonzero vector $w \in W$, where $w = w_1 + \cdots + w_m$ is the sum of the basis vectors of W. Since $\operatorname{range} T$ is spanned by the images of the basis vectors of V, then any sum of Tv_j is a scalar multiple of w. Therefore, $\operatorname{range} T = \operatorname{span} w$, and hence $\dim \operatorname{range} T = 1$. ■

Exercise 3C3

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W.

(a) Show that if $S, T \in \mathcal{L}(V, W)$, then $\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$.

(b) Show that if $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$, then $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

Solution.

(a) Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W. Then the matrix of $S + T$ with respect to these bases is given by

$$\begin{aligned}
\mathcal{M}(S+T) &= \begin{pmatrix} (S+T)v_1 & (S+T)v_2 & \cdots & (S+T)v_n \end{pmatrix} \\
&= \begin{pmatrix} Sv_1 + Tv_1 & Sv_2 + Tv_2 & \cdots & Sv_n + Tv_n \end{pmatrix} \\
&= \begin{pmatrix} Sv_1 & Sv_2 & \cdots & Sv_n \end{pmatrix} + \begin{pmatrix} Tv_1 & Tv_2 & \cdots & Tv_n \end{pmatrix} \\
&= \mathcal{M}(S) + \mathcal{M}(T).
\end{aligned}$$

(b) Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W. Then the matrix of λT with respect to these bases is given by

$$\begin{aligned}
\mathcal{M}(\lambda T) &= \begin{pmatrix} (\lambda T)v_1 & (\lambda T)v_2 & \cdots & (\lambda T)v_n \end{pmatrix} \\
&= \begin{pmatrix} \lambda Tv_1 & \lambda Tv_2 & \cdots & \lambda Tv_n \end{pmatrix} \\
&= \lambda \begin{pmatrix} Tv_1 & Tv_2 & \cdots & Tv_n \end{pmatrix} \\
&= \lambda \mathcal{M}(T).
\end{aligned}$$

■

Exercise 3C4

Suppose that $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ is the differentiation map defined by $Dp = p'$. Find a basis of $\mathcal{P}_3(\mathbb{R})$ and a basis of $\mathcal{P}_2(\mathbb{R})$ such that the matrix of D with respect to these bases is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Solution. Consider the standard basis of $\mathcal{P}_3(\mathbb{R})$. Reorder it to obtain the basis $x^3, x^2, x, 1$. Consider the standard basis of $\mathcal{P}_2(\mathbb{R})$. Reorder it to obtain the basis $x^2, x, 1$. Observe that $D(x^3) = 3x^2$, $D(x^2) = 2x$, $D(x) = 1$, and $D(1) = 0$. However, we need the matrix of D to have the desired form. To achieve this, we can scale the basis vectors of $\mathcal{P}_3(\mathbb{R})$ and $\mathcal{P}_2(\mathbb{R})$ appropriately. Then we may define the basis for $\mathcal{P}_3(\mathbb{R})$ as

$$\frac{1}{3}x^3, \frac{1}{2}x^2, x, 1$$

while maintaining the standard basis for $\mathcal{P}_2(\mathbb{R})$ as $x^2, x, 1$. Then we have

$$D\left(\frac{1}{3}x^3\right) = x^2, \quad D\left(\frac{1}{2}x^2\right) = x, \quad D(x) = 1, \quad D(1) = 0.$$

Therefore, with respect to the bases, we get the desired matrix representation of D. ■

Exercise 3C5

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that there exist a basis of V and a basis of W such that with respect to these bases, all entries of $\mathcal{M}(T)$ are 0 except that the entries in row k, column k, equal 1 if $1 \le k \le \dim \operatorname{range} T$.

Solution. Let $n = \dim V$ and $m = \dim W$. By the fundamental theorem of linear maps, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Let $r = \dim \operatorname{range} T$. Let $u_1, \dots, u_{n-r}$ be a basis of $\operatorname{null} T$. Extend this to a basis $v_1, \dots, v_r, u_1, \dots, u_{n-r}$ of V. We claim that $Tv_1, \dots, Tv_r$ are linearly independent. Suppose for scalars $\alpha_1, \dots, \alpha_r \in \mathbb{F}$, we have

$$\alpha_1 Tv_1 + \cdots + \alpha_r Tv_r = 0.$$

Then

$$T(\alpha_1 v_1 + \cdots + \alpha_r v_r) = 0,$$

which implies that $\alpha_1 v_1 + \cdots + \alpha_r v_r \in \operatorname{null} T$. Since $v_1, \dots, v_r, u_1, \dots, u_{n-r}$ is a basis of V, it follows that $\alpha_1 = \cdots = \alpha_r = 0$. Therefore, $Tv_1, \dots, Tv_r$ are linearly independent and form a basis of $\operatorname{range} T$. Extend this to a basis $Tv_1, \dots, Tv_r, w_1, \dots, w_{m-r}$ of W. Then with respect to these bases, Tv_k is the k-th basis vector of W for each $k = 1, \dots, r$, and

$$Tu_j = 0, \quad \text{for } j = 1, \dots, n-r.$$

Therefore, the matrix of T with respect to these bases has 1's in the diagonal entries corresponding to Tv_k and 0's elsewhere, as desired. ■

Exercise 3C6

Suppose $v_1, \ldots, v_m$ is a basis of V and W is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \ldots, w_n$ of W such that all entries in the first column of $\mathcal{M}(T)$ [with respect to the bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$] are 0 except for possibly a 1 in the first row, first column.

Solution. Let $n = \dim W$. If $Tv_1 = 0$, then we can choose any basis of W and the first column of $\mathcal{M}(T)$ will be all zeros. If $Tv_1 \neq 0$, then we can extend Tv_1 to a basis of W. Let $w_1 = Tv_1$ and extend it to a basis $w_1, w_2, \ldots, w_n$ of W. Then with respect to the bases $v_1, \ldots, v_m$ of V and $w_1, \ldots, w_n$ of W, we have

$$Tv_1 = w_1$$

and for each $j = 2, \ldots, m$, we can express

$$Tv_j = \alpha_{j1}w_1 + \alpha_{j2}w_2 + \cdots + \alpha_{jn}w_n$$

for some scalars $\alpha_{j1}, \alpha_{j2}, \ldots, \alpha_{jn} \in \mathbb{F}$. Therefore, the first column of $\mathcal{M}(T)$ has a 1 in the first row and zeros in all other rows, as desired. ■

Exercise 3C7

Suppose $w_1, \ldots, w_n$ is a basis of W and V is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $v_1, \ldots, v_m$ of V such that all entries in the first row of $\mathcal{M}(T)$ [with respect to the bases $v_1, \ldots, v_m$ and $w_1, \ldots, w_n$] are 0 except for possibly a 1 in the first row, first column.

Solution. Let $m = \dim V$ and $u_1, \ldots, u_m$ be a basis for V. If T has trivial w_1 component, meaning that the coefficient of w_1 in Tv is zero for all $v \in V$, then choose any basis of V and the first row of $\mathcal{M}(T)$ will be all zeros. Otherwise, reorder the basis so that Tu_1 has non-trivial w_1 component. Then we can express

$$Tu_1 = \alpha w_1 + \beta_2 w_2 + \cdots + \beta_n w_n$$

for some scalars $\alpha, \beta_2, \ldots, \beta_n \in \mathbb{F}$ such that $\alpha \neq 0$. We can scale u_1 by α^{-1} to obtain a new vector $v_1 = \alpha^{-1}u_1$ such that

$$Tv_1 = w_1 + \frac{\beta_2}{\alpha}w_2 + \cdots + \frac{\beta_n}{\alpha}w_n.$$

With the remaining $u_2, \ldots, u_m$ basis vectors of V, note that they can be expressed as

$$Tu_j = \alpha_{j1}w_1 + \alpha_{j2}w_2 + \cdots + \alpha_{jn}w_n$$

for some scalars $\alpha_{j1}, \alpha_{j2}, \ldots, \alpha_{jn} \in \mathbb{F}$. Consider the set of vectors of V given by

$$v_1 = \alpha^{-1}u_1, \quad v_j = u_j - \alpha_{j1}v_1 \text{ for } j = 2, \ldots, m.$$

Note that $Tv_j = Tu_j - \alpha_{j1}Tv_1$ is just a combination of $w_2, \ldots, w_n$. We now claim that $v_1, v_2, \ldots, v_m$ are linearly independent. Suppose for scalars $\gamma_1, \ldots, \gamma_m \in \mathbb{F}$, we have

$$\gamma_1 v_1 + \gamma_2 v_2 + \cdots + \gamma_m v_m = 0.$$

Then we can express this as

$$\gamma_1 v_1 + \gamma_2(u_2 - \alpha_{21}v_1) + \cdots + \gamma_m(u_m - \alpha_{m1}v_1) = 0.$$

Simplifying, we get

$$(\gamma_1 - \gamma_2\alpha_{21} - \cdots - \gamma_m\alpha_{m1})v_1 + \gamma_2 u_2 + \cdots + \gamma_m u_m = 0.$$

Since $v_1, u_2, \ldots, u_m$ are linearly independent, we must have $\gamma_1 - \gamma_2\alpha_{21} - \cdots - \gamma_m\alpha_{m1} = 0$ and $\gamma_2 = \cdots = \gamma_m = 0$. This implies that $\gamma_1 = 0$, and hence $v_1, v_2, \ldots, v_m$ are linearly independent. Then $v_1, v_2, \ldots, v_m$ form a basis of V. It follows that Tv_1 has a 1 in the first row, while Tv_j for $j = 2, \ldots, m$ have zeros in the first row. ■

Exercise 3C8

Suppose A is an m-by-n matrix and B is an n-by-p matrix. Prove that $(AB)_{j,\cdot} = A_{j,\cdot}B$ for each $1 \leq j \leq m$. In other words, show that row j of AB equals (row j of A) times B.

Solution. Let A be an m-by-n matrix and B be an n-by-p matrix. The entry in row j, column k, of the product AB is given by

$$(AB)_{j,k} = \sum_{r=1}^{n} A_{j,r}B_{r,k}.$$

The row vector $A_{j,\cdot}$ is the j-th row of A, which can be expressed as

$$A_{j,\cdot} = (A_{j,1}, A_{j,2}, \ldots, A_{j,n}).$$

When we multiply this row vector by the matrix B, we get a new row vector whose entries are given by

$$(A_{j,\cdot}B)_k = \sum_{r=1}^{n} A_{j,r}B_{r,k}.$$

Comparing this with the expression for $(AB)_{j,k}$, we see that

$$(AB)_{j,k} = (A_{j,\cdot}B)_k$$

for each column index k. Therefore, the entire row vector $(AB)_{j,\cdot}$ is equal to the row vector $A_{j,\cdot}B$. This proves that row j of AB equals (row j of A) times B. ■

Exercise 3C9

Suppose $a = (a_1 \quad \cdots \quad a_n)$ is a 1-by-n matrix and B is an n-by-p matrix. Prove that

$$aB = a_1 B_{1,\cdot} + \cdots + a_n B_{n,\cdot}.$$

In other words, show that aB is a linear combination of the rows of B, with the scalars that multiply the rows coming from a.

Solution. Let $a = (a_1 \quad \cdots \quad a_n)$ be a 1-by-n matrix and B be an n-by-p matrix. The product aB is a 1-by-p matrix, and its entries are given by

$$(aB)_{1,k} = \sum_{j=1}^{n} a_j B_{j,k}$$

for each column index k.

Now, consider the linear combination of the rows of B given by

$$a_1 B_{1,\cdot} + a_2 B_{2,\cdot} + \cdots + a_n B_{n,\cdot}.$$

The entry in row 1, column k, of this linear combination is

$$(a_1 B_{1,\cdot} + a_2 B_{2,\cdot} + \cdots + a_n B_{n,\cdot})_k = a_1 B_{1,k} + a_2 B_{2,k} + \cdots + a_n B_{n,k} = \sum_{j=1}^{n} a_j B_{j,k}.$$

Comparing this with the expression for $(aB)_{1,k}$, we see that

$$(aB)_{1,k} = (a_1 B_{1,\cdot} + a_2 B_{2,\cdot} + \cdots + a_n B_{n,\cdot})_k$$

for each column index k. Therefore, we conclude that

$$aB = a_1 B_{1,\cdot} + a_2 B_{2,\cdot} + \cdots + a_n B_{n,\cdot},$$

which shows that aB is indeed a linear combination of the rows of B, with the scalars coming from the entries of a. ■

Exercise 3C10

Give an example of 2-by-2 matrices A and B such that $AB \neq BA$.

Solution. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then we compute the products AB and BA:

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 \cdot 0 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 0 \\ 3 \cdot 0 + 4 \cdot 1 & 3 \cdot 1 + 4 \cdot 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix},$$

and

$$BA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 3 & 0 \cdot 2 + 1 \cdot 4 \\ 1 \cdot 1 + 0 \cdot 3 & 1 \cdot 2 + 0 \cdot 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

By comparing the two products, we see that $AB \neq BA$. ■

Exercise 3C11

Prove that the distributive property holds for matrix addition and matrix multiplication. In other words, suppose A, B, C, D, E, and F are matrices whose sizes are such that $A(B + C)$ and $(D + E)F$ make sense. Explain why $AB + AC$ and $DF + EF$ both make sense and prove that

$$A(B + C) = AB + AC \quad \text{and} \quad (D + E)F = DF + EF.$$

Solution. Let A be an m-by-n matrix, and let B and C be n-by-p matrices. Then $B + C$ is an n-by-p matrix, and hence $A(B+C)$ is an m-by-p matrix. The products AB and AC are also defined, and both are m-by-p matrices. Therefore, the sum $AB + AC$ is defined and is also an m-by-p matrix. By Exercise 3C8 and Exercise 3C9, we have

$$A(B+C)_{j,\cdot} = A_{j,\cdot}(B+C) = A_{j,\cdot}B + A_{j,\cdot}C = (AB)_{j,\cdot} + (AC)_{j,\cdot} = (AB+AC)_{j,\cdot}$$

for all $j = 1, \ldots, m$. Therefore, $A(B+C) = AB + AC$.

Similarly, let D and E be m-by-n matrices, and let F be an n-by-p matrix. Then $D + E$ is an m-by-n matrix, and hence $(D+E)F$ is an m-by-p matrix. The products DF and EF are also defined, and both are m-by-p matrices. Therefore, the sum $DF + EF$ is defined and is also an m-by-p matrix. Then the product $(D+E)F$ is defined. By Exercise 3C8 and Exercise 3C9, we have

$$(D+E)F_{j,\cdot} = (D+E)_{j,\cdot}F = D_{j,\cdot}F + E_{j,\cdot}F = (DF)_{j,\cdot} + (EF)_{j,\cdot} = (DF+EF)_{j,\cdot}$$

for all $j = 1, \ldots, m$. Therefore, $(D+E)F = DF + EF$. ■

Exercise 3C12

Prove that matrix multiplication is associative. In other words, suppose A, B, and C are matrices whose sizes are such that $(AB)C$ makes sense. Explain why $A(BC)$ makes sense and prove that

$$(AB)C = A(BC).$$

Manual Remark. *Unfortunately, we cannot use the same trick as in Exercise 3C11 to prove this result.*

Solution. Let A be an m-by-n matrix, B be an n-by-p matrix, and C be a p-by-q matrix. Then the product AB is defined and is an m-by-p matrix. Therefore, the product $(AB)C$ is also defined and is an m-by-q matrix. Going by the definition of matrix multiplication, we have

$$((AB)C)_{j,k} = \sum_{r=1}^{p} (AB)_{j,r}C_{r,k} = \sum_{r=1}^{p} \left(\sum_{s=1}^{n} A_{j,s}B_{s,r} \right) C_{r,k} = \sum_{r=1}^{p} \sum_{s=1}^{n} A_{j,s}B_{s,r}C_{r,k}.$$

On the other hand, the product BC is defined and is an n-by-q matrix. Therefore, the product $A(BC)$ is also defined and is an m-by-q matrix. Again, by the definition of matrix multiplication, we have

$$(A(BC))_{j,k} = \sum_{s=1}^{n} A_{j,s}(BC)_{s,k} = \sum_{s=1}^{n} A_{j,s} \left(\sum_{r=1}^{p} B_{s,r}C_{r,k} \right) = \sum_{s=1}^{n} \sum_{r=1}^{p} A_{j,s}B_{s,r}C_{r,k}.$$

Since the order of summation does not matter, we can interchange the order of the sums in the last expression to get

$$(A(BC))_{j,k} = \sum_{r=1}^{p} \sum_{s=1}^{n} A_{j,s}B_{s,r}C_{r,k}.$$

Comparing the expressions for $((AB)C)_{j,k}$ and $(A(BC))_{j,k}$, we see that they are equal for all j and k. Therefore, $(AB)C = A(BC)$. ■

Exercise 3C13

Suppose A is an n-by-n matrix and $1 \le j,k \le n$. Show that the entry in row j, column k, of A^3 (which is defined to mean AAA) is

$$\sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k}.$$

Solution. Note that $A^3 = (AA)A = A^2A$. By the definition of matrix multiplication, we have

$$(A^3)_{j,k} = (A^2A)_{j,k} = \sum_{p=1}^{n}(A^2)_{j,p}A_{p,k}.$$

Now, we can express $(A^2)_{j,p}$ as

$$(A^2)_{j,p} = (AA)_{j,p} = \sum_{r=1}^{n} A_{j,r}A_{r,p}.$$

Substituting this back into the expression for $(A^3)_{j,k}$, we get

$$(A^3)_{j,k} = \sum_{p=1}^{n}\left(\sum_{r=1}^{n} A_{j,r}A_{r,p}\right)A_{p,k} = \sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,r}A_{r,p}A_{p,k}.$$

By renaming the indices r and p, we can rewrite this as

$$(A^3)_{j,k} = \sum_{p=1}^{n}\sum_{r=1}^{n} A_{j,p}A_{p,r}A_{r,k},$$

which is the desired expression. ■

Exercise 3C14

Suppose m and n are positive integers. Prove that the function $A \mapsto A^t$ is a linear map from $\mathbb{F}^{m,n}$ to $\mathbb{F}^{n,m}$.

Solution. Let $A, B \in \mathbb{F}^{m,n}$ and let $\lambda \in \mathbb{F}$. We need to show that $(A+B)^t = A^t + B^t$ and $(\lambda A)^t = \lambda A^t$.

First, consider the sum $A + B$. The entry in row j, column k, of $A + B$ is given by

$$(A+B)_{j,k} = A_{j,k} + B_{j,k}.$$

Taking the transpose, we have

$$((A+B)^t)_{k,j} = (A+B)_{j,k} = A_{j,k} + B_{j,k} = (A^t)_{k,j} + (B^t)_{k,j} = (A^t + B^t)_{k,j}.$$

Therefore, $(A+B)^t = A^t + B^t$.

Next, consider the scalar multiplication λA. The entry in row j, column k, of λA is given by

$$(\lambda A)_{j,k} = \lambda A_{j,k}.$$

Taking the transpose, we have

$$((\lambda A)^t)_{k,j} = (\lambda A)_{j,k} = \lambda A_{j,k} = \lambda (A^t)_{k,j} = (\lambda A^t)_{k,j}.$$

Therefore, $(\lambda A)^t = \lambda A^t$.

Since both properties hold, we conclude that the function $A \mapsto A^t$ is a linear map from $\mathbb{F}^{m,n}$ to $\mathbb{F}^{n,m}$. ■

Exercise 3C15

Prove that if A is an m-by-n matrix and C is an n-by-p matrix, then

$$(AC)^t = C^t A^t.$$

Solution. Let A be an m-by-n matrix and C be an n-by-p matrix. The entry in row j, column k, of the product AC is given by

$$(AC)_{j,k} = \sum_{r=1}^{n} A_{j,r} C_{r,k}.$$

Taking the transpose, we have

$$((AC)^t)_{k,j} = (AC)_{j,k} = \sum_{r=1}^{n} A_{j,r} C_{r,k}.$$

Now, consider the product $C^t A^t$. The entry in row k, column j, of this product is given by

$$(C^t A^t)_{k,j} = \sum_{r=1}^{n} (C^t)_{k,r} (A^t)_{r,j} = \sum_{r=1}^{n} C_{r,k} A_{j,r}.$$

Since multiplication in the field $\mathbb{F}$ is commutative, we can rewrite this as

$$(C^t A^t)_{k,j} = \sum_{r=1}^{n} A_{j,r} C_{r,k},$$

which is exactly the same as the expression for $((AC)^t)_{k,j}$.

Therefore, we conclude that

$$(AC)^t = C^t A^t.$$

■

Exercise 3C16

Suppose A is an m-by-n matrix with $A \neq 0$. Prove that the rank of A is 1 if and only if there exist $(c_1, \ldots, c_m) \in \mathbb{F}^m$ and $(d_1, \ldots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \ldots, m$ and every $k = 1, \ldots, n$.

Solution. Suppose the rank of A is 1. Then the column rank of A is 1, which means that all columns of A are scalar multiples of a single non-zero column vector. Let $c = (c_1, \ldots, c_m)^t$ be this non-zero column vector. Then for each column k of A, there exists a scalar $d_k \in \mathbb{F}$ such that the k-th column of A is given by $d_k c$. Therefore, we can express the entries of A as

$$A_{j,k} = c_j d_k$$

for all $j = 1, \ldots, m$ and $k = 1, \ldots, n$.

Conversely, suppose there exist $(c_1, \ldots, c_m) \in \mathbb{F}^m$ and $(d_1, \ldots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for all j and k. Since $A \neq 0$, then $(c_1, \ldots, c_m) \neq 0$ and $(d_1, \ldots, d_n) \neq 0$. This means that all columns of A are scalar multiples of the column vector c, and hence the column rank of A is 1. Therefore, the rank of A is 1. ■

Exercise 3C17

Suppose $T \in \mathcal{L}(V)$, and $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$ are bases of V. Prove that the following are equivalent.

(a) T is injective.
(b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$.
(c) The columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n,1}$.
(d) The rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1,n}$.
(e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1,n}$.

Solution.

- (a) $\iff$ (b): (a) $\iff$ $\dim \operatorname{null} T = 0$ $\iff$ $\dim \operatorname{range} T = n$ $\iff$ $\operatorname{column rank} \mathcal{M}(T) = n$. For n column vectors, rank n implies linear independence $\iff$ (b).
- (b) $\iff$ (c): n columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$ $\iff$ n columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n,1}$.
- (c) $\iff$ (d): columns span $\mathbb{F}^{n,1}$ $\iff$ $\operatorname{rank} \mathcal{M}(T) = n$ by (b) $\iff$ (c). By Theorem 3.57, $\operatorname{rank} \mathcal{M}(T) = \operatorname{row rank} \mathcal{M}(T)$ $\iff$ rows span $\mathbb{F}^{1,n}$.
- (d) $\iff$ (e): n rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1,n}$ $\iff$ n rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1,n}$. ■

3D Invertibility and Isomorphisms

Exercise 3D1

Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and

$$(T^{-1})^{-1} = T.$$

Solution. Since T is invertible, then $T^{-1} \in \mathcal{L}(W, V)$. By $TT^{-1} = I$ and $T^{-1}T = I$, we have that T^{-1} is invertible and $(T^{-1})^{-1} = T$. ■

Exercise 3D2

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

Solution. Since T and S are invertible, then $T^{-1} \in \mathcal{L}(V, U)$ and $S^{-1} \in \mathcal{L}(W, V)$. By the definition of the inverse, we have that

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SIS^{-1} = SS^{-1} = I,$$

and

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}IT = T^{-1}T = I.$$

Thus, ST is invertible and $(ST)^{-1} = T^{-1}S^{-1}$. ■

Exercise 3D3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) T is invertible.

(b) $Tv_1, \ldots, Tv_n$ is a basis of V for every basis $v_1, \ldots, v_n$ of V.

(c) $Tv_1, \ldots, Tv_n$ is a basis of V for some basis $v_1, \ldots, v_n$ of V.

Solution.

- (a) $\implies$ (b): Suppose T is invertible. Consider scalars $\alpha_1, \ldots, \alpha_n$ such that

  $$\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n = 0.$$

  Applying T^{-1} to both sides, we get

  $$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0.$$

  Since $v_1, \ldots, v_n$ is a basis of V, it follows that $\alpha_1 = \cdots = \alpha_n = 0$. Hence, $Tv_1, \ldots, Tv_n$ is linearly independent. Since $Tv_1, \ldots, Tv_n$ has n elements and V is n-dimensional, it follows that $Tv_1, \ldots, Tv_n$ is a basis of V.
- (b) $\implies$ (c): This is trivial.
- (c) $\implies$ (a): Suppose $Tv_1, \ldots, Tv_n$ is a basis of V for some basis $v_1, \ldots, v_n$ of V. Then range $T = V$. By the fundamental theorem of linear maps, we have that $\dim \operatorname{null} T = 0$. Thus, T is injective. Since T is injective and range $T = V$, then T is invertible. ■

Exercise 3D4

Suppose V is finite-dimensional and $\dim V > 1$. Prove that the set of non-invertible linear maps from V to itself is not a subspace of $\mathcal{L}(V)$.

Solution. Let $v_1, \dots, v_n$ be a basis of V. Define $T, S \in \mathcal{L}(V)$ such that

$$Tv_1 = 0, \quad Tv_k = v_k \text{ for } k = 2, \dots, n,$$

and

$$Sv_1 = v_1, \quad Sv_k = 0 \text{ for } k = 2, \dots, n.$$

Then T and S are non-invertible linear maps, since $\operatorname{null} T = \operatorname{span} v_1$ and $\operatorname{null} S = \operatorname{span}(v_2, \dots, v_n)$. However, $T + S$ is invertible, since

$$(T+S)v_1 = v_1, \quad (T+S)v_k = v_k \text{ for } k = 2, \dots, n.$$

Hence, $T + S = I$ is invertible. Therefore, the set of non-invertible linear maps from V to itself is not a subspace of $\mathcal{L}(V)$. ■

Exercise 3D5

Suppose V is finite-dimensional, U is a subspace of V, and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$ if and only if S is injective.

Solution. Suppose there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$. If $Su = 0$, then $Tu = 0$, and since T is invertible, $u = 0$. Hence, S is injective.

Conversely, suppose S is injective. Let $u_1, \dots, u_m$ be a basis of U. Since S is injective, then $Su_1, \dots, Su_m$ is linearly independent. Extend $u_1, \dots, u_m$ to a basis $u_1, \dots, u_n$ of V. Extend $Su_1, \dots, Su_m$ to a basis $v_1, \dots, v_n$ of V. Define $T \in \mathcal{L}(V)$ such that

$$Tu_k = Su_k \text{ for } k = 1, \dots, m, \quad Tu_k = v_k \text{ for } k = m+1, \dots, n.$$

Then T is invertible, since $Tu_1, \dots, Tu_n$ is a basis of V. Thus, there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$. ■

Exercise 3D6

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S = \operatorname{null} T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

Solution. Suppose $\operatorname{null} S = \operatorname{null} T$. Define a linear map E_0 from $\operatorname{range} T$ to $\operatorname{range} S$ by setting $E_0(Tv) = Sv$ for all $v \in V$. This is well-defined because if $Tv = Tv'$ for some $v, v' \in V$, then $T(v - v') = 0$, so $v - v' \in \operatorname{null} T = \operatorname{null} S$, and thus $S(v - v') = 0$, which implies $Sv = Sv'$. Moreover, if $E_0(Tv) = 0$ for some $v \in V$, then $Sv = 0$, which implies $v \in \operatorname{null} S = \operatorname{null} T$, and thus $Tv = 0$. Hence, E_0 is injective. By Exercise 3D5, we can extend E_0 to an invertible linear map E on all of W, and then we have $S = ET$.

Conversely, suppose there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$. If $v \in \operatorname{null} S$, then $Sv = 0$, and since $S = ET$, we have $ETv = 0$. Since E is invertible, $Tv = 0$ so that $v \in \operatorname{null} T$. Hence, $\operatorname{null} S \subseteq \operatorname{null} T$. Similarly, if $v \in \operatorname{null} T$, then $Tv = 0$, and since $S = ET$, we have $Sv = ETv = E0 = 0$. Thus, $v \in \operatorname{null} S$, and $\operatorname{null} T \subseteq \operatorname{null} S$. Therefore, $\operatorname{null} S = \operatorname{null} T$. ■

Exercise 3D7

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S = \operatorname{range} T$ if and only if there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$.

Solution. Suppose $\operatorname{range} S = \operatorname{range} T$. Then $\dim \operatorname{null} S = \dim \operatorname{null} T$. Let $u_1, \ldots, u_k$ be a basis of $\operatorname{null} S$ and $v_1, \ldots, v_k$ be a basis of $\operatorname{null} T$. Extend both bases to a basis of V, say $u_1, \ldots, u_n$ and $v_1, \ldots, v_n$. Observe that for $i > k$, we have that $Su_i \in \operatorname{range} S = \operatorname{range} T$. Then there must exist $y_i \in V$ such that $Ty_i = Su_i$. Define a linear map $E \in \mathcal{L}(V)$ by setting $Eu_i = v_i$ for $i \leq k$ and $Eu_i = y_i$ for $i > k$. To see that $TE = S$, we have for $i \leq k$ that $TEu_i = Tv_i = 0 = Su_i$. Moreover, for $i > k$, we have $TEu_i = Ty_i = Su_i$. To see that E is invertible, suppose we have scalars $\alpha_1, \ldots, \alpha_n$ such that

$$\alpha_1 Eu_1 + \alpha_2 Eu_2 + \cdots + \alpha_n Eu_n = 0.$$

Applying T, we get

$$\alpha_{k+1} Su_{k+1} + \cdots + \alpha_n Su_n = 0,$$

where we used the fact that $TE = S$ and $Su_1 = \cdots = Su_k = 0$ because $u_1, \ldots, u_k \in \operatorname{null} S$. Since $u_1, \ldots, u_n$ is a basis of V, the vectors $u_{k+1}, \ldots, u_n$ map to a spanning list of $\operatorname{range} S$. Since $\dim \operatorname{range} S = n - k$, it follows that $Su_{k+1}, \ldots, Su_n$ are linearly independent. Then from the previous equation, $\alpha_{k+1} = \cdots = \alpha_n = 0$. Substituting this back into the original equation, we get $\alpha_1 v_1 + \cdots + \alpha_k v_k = 0$, and since $v_1, \ldots, v_k$ are linearly independent, we must have $\alpha_1 = \cdots = \alpha_k = 0$. Hence, all the scalars are zero, proving that E is invertible.

Conversely, suppose there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$. We have that $\operatorname{range} S \subseteq \operatorname{range} T$ by Exercise 3B26. Because E is invertible, we also have that $T = SE^{-1}$, so $\operatorname{range} T \subseteq \operatorname{range} S$ by the same reasoning. Hence, $\operatorname{range} S = \operatorname{range} T$. Similarly, if $w \in \operatorname{range} T$, then there exists some $v \in V$ such that $Tv = w$. Since E is invertible, there exists some $u \in V$ such that $Eu = v$. Then we have

$$Su = TEu = Tv = w,$$

and thus $w \in \operatorname{range} S$. Hence, $\operatorname{range} T \subseteq \operatorname{range} S$. Therefore, $\operatorname{range} S = \operatorname{range} T$. ■

Exercise 3D8

Suppose V and W are finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that there exist invertible $E_1 \in \mathcal{L}(V)$ and $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ if and only if $\dim \operatorname{null} S = \dim \operatorname{null} T$.

Solution. Suppose there exist invertible $E_1 \in \mathcal{L}(V)$ and invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$. We claim that E_1 is an invertible map from $\operatorname{null} S$ to $\operatorname{null} T$. To see this, suppose $v, w \in \operatorname{null} S$ and $E_1 v = E_1 w$. Then $v = w$ since E_1 is invertible, proving injectivity. To see surjectivity, let $u \in \operatorname{null} T$. Since E_1 is invertible, there exists some $v \in V$ such that $E_1 v = u$. Then

$$Sv = E_2 T E_1 v = E_2 T u = E_2 0 = 0,$$

and thus $v \in \operatorname{null} S$. Hence, E_1 is an invertible map from $\operatorname{null} S$ to $\operatorname{null} T$, and thus $\dim \operatorname{null} S = \dim \operatorname{null} T$.

Conversely, suppose $\dim \operatorname{null} S = \dim \operatorname{null} T$. Then we may construct an invertible linear map $E_1 \in \mathcal{L}(V)$ that maps $\operatorname{null} S$ onto $\operatorname{null} T$. We now claim that $\operatorname{null} TE_1 = \operatorname{null} S$. To see this, let $v \in \operatorname{null} TE_1$. This is only true when $TE_1 v = 0$, or $E_1 v \in \operatorname{null} T$. By invertibility of E_1, we have $v \in \operatorname{null} S$. By Exercise 3D6 with $\operatorname{null} S = \operatorname{null} TE_1$, there exists an invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2 TE_1$. ■

Exercise 3D9

Suppose V is finite-dimensional and $T : V \to W$ is a surjective linear map of V onto W. Prove that there is a subspace U of V such that $T|_U$ is an isomorphism of U onto W.

Axler Note. *Here $T|_U$ means the function T restricted to U. Thus, $T|_U$ is the function whose domain is U, with $T|_U(u) = Tu$ for every $u \in U$.*

Solution. Since T is surjective, then $\operatorname{range} T = W$. By Exercise 3B11, there exists a subspace U of V such that $V = \operatorname{null} T \oplus U$. We show that $T|_U$ is an isomorphism of U onto W.

Suppose $u \in U$ such that $T|_U u = 0$. Then $Tu = 0$, and thus $u \in \operatorname{null} T \cap U$. Since $V = \operatorname{null} T \oplus U$, we have $\operatorname{null} T \cap U = \{0\}$, and thus $u = 0$. This shows that $T|_U$ is injective.

To see surjectivity, let $w \in W$. Since T is surjective, there exists $v \in V$ such that $Tv = w$. Write $v = n + u$ with $n \in \operatorname{null} T$ and $u \in U$. Then $Tu = T(v - n) = Tv - Tn = w - 0 = w$. Hence, $T|_U$ is surjective onto W, and therefore an isomorphism of U onto W. ■

Exercise 3D10

Suppose V and W are finite-dimensional and U is a subspace of V. Let

$$\mathcal{E} = \{T \in \mathcal{L}(V, W) \mid U \subseteq \operatorname{null} T\}.$$

(a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.
(b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

Axler Hint. *Define $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$. What is $\operatorname{null} \Phi$. What is $\operatorname{range} \Phi$?*

Solution.

(a) Consider the zero map $0 \in \mathcal{L}(V, W)$. For this map, $U \subseteq \operatorname{null} 0$ since $\operatorname{null} 0 = V$. Hence, $0 \in \mathcal{E}$, showing that $\mathcal{E}$ is non-empty. To see that $\mathcal{E}$ is closed under addition and scalar multiplication, let $S, T \in \mathcal{E}$ and $\alpha \in \mathbb{F}$. For any $u \in U$, we have $(S + T)u = Su + Tu = 0 + 0 = 0$, so $S + T \in \mathcal{E}$. Similarly, $(\alpha T)u = \alpha(Tu) = \alpha \cdot 0 = 0$, so $\alpha T \in \mathcal{E}$. Therefore, $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Consider the linear map $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ defined by $\Phi(T) = T|_U$. Let $T \in \mathcal{L}(V, W)$ such that $\Phi(T) = T|_U = 0$. Then $T|_U = 0$ implies that for every $u \in U$, we have $Tu = 0$, i.e., $U \subseteq \operatorname{null} T$. Hence, $T \in \mathcal{E}$, showing that $\operatorname{null} \Phi \subseteq \mathcal{E}$. Conversely, if $T \in \mathcal{E}$, then $U \subseteq \operatorname{null} T$, which means that $T|_U = 0$, i.e., $T \in \operatorname{null} \Phi$. Therefore, $\operatorname{null} \Phi = \mathcal{E}$, and $\dim \mathcal{E} = \dim \operatorname{null} \Phi$.

We now show that Φ is surjective. Let $S \in \mathcal{L}(U, W)$, and let $u_1, \ldots, u_m$ be a basis for U. Extend this to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ for V. Define a linear map $T \in \mathcal{L}(V, W)$ by setting $Tu_i = Su_i$ for $i = 1, \ldots, m$ and $Tv_j = 0$ for $j = 1, \ldots, n$. Then $T|_U = S$, so $\Phi(T) = S$. Hence, Φ is surjective, and $\operatorname{range} \Phi = \mathcal{L}(U, W)$. Therefore, $\dim \operatorname{range} \Phi = \dim \mathcal{L}(U, W) = (\dim U)(\dim W)$. By the fundamental theorem of linear maps,

$$\dim \mathcal{L}(V, W) = \dim \operatorname{null} \Phi + \dim \operatorname{range} \Phi = \dim \mathcal{E} + (\dim U)(\dim W),$$

Observing that $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$, we get

$$\dim \mathcal{E} = (\dim V)(\dim W) - (\dim U)(\dim W) = (\dim V - \dim U)(\dim W).$$ ■

Exercise 3D11

Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that

$$ST \text{ is invertible} \iff S \text{ and } T \text{ are invertible.}$$

Solution. Suppose ST is invertible. Then $\operatorname{null} ST = \{0\}$. If $v \in \operatorname{null} T$, then $STv = S0 = 0$, and thus $v \in \operatorname{null} ST$. Since $\operatorname{null} ST = \{0\}$, we have $v = 0$, and thus $\operatorname{null} T = \{0\}$. Hence, T is injective, and since V is finite-dimensional, T is invertible. Then $S = (ST)T^{-1}$ is a product of invertible linear maps, and thus S is invertible by Exercise 3D2.

Conversely, suppose S and T are invertible. Then ST is a product of invertible linear maps, and thus ST is invertible by Exercise 3D2. ■

Exercise 3D12

Suppose V is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ and $STU = I$. Show that T is invertible and that $T^{-1} = US$.

Solution. Since $STU = I$, then we have that $S(TU) = I$. By Exercise 3B20, then S is surjective, hence invertible with $S^{-1} = TU$. Similarly, $(ST)U = I$. By Exercise 3B19, then U is injective, hence invertible with $U^{-1} = ST$.

$$T(US) = (TU)S = S^{-1}S = I, \quad (US)T = U(ST) = UU^{-1} = I.$$

Hence, T is invertible with $T^{-1} = US$. ■

Exercise 3D13

Show that the result in Exercise 3D12 can fail without the hypothesis that V is finite-dimensional.

Solution. If V were not finite-dimensional, let $v_1, v_2, \ldots$ be an infinite basis for V. Let $S, T, U \in \mathcal{L}(V)$ be defined as follows: let $S = I$,

$$Tv_1 = 0, \quad Tv_k = v_{k-1} \text{ for } k = 2, 3, \ldots$$

and

$$Uv_k = v_{k+1} \text{ for } k = 1, 2, \ldots$$

It is clear that $TUv_k = v_k$ for all $k = 1, 2, \ldots$, so $STU = I$. However, T is not injective since $\operatorname{null} T = \operatorname{span} v_1$, and thus T is not invertible. ■

Exercise 3D14

Prove or give a counterexample: If V is a finite-dimensional vector space and $R, S, T \in \mathcal{L}(V)$ are such that RST is surjective, then S is injective.

Solution. Since RST is surjective, and V is finite dimensional, it follows that RST is injective, hence invertible. By Exercise 3D11, then $RST = (RS)T$ is invertible if and only if RS and T are invertible. In particular, RS is invertible, and applying Exercise 3D11 again, we have that R and S are invertible. Hence, S is injective. ∎

Exercise 3D15

Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_m$ is a list in V such that $Tv_1, \ldots, Tv_m$ spans V. Prove that $v_1, \ldots, v_m$ spans V.

Solution. Since $Tv_1, \ldots, Tv_m$ spans V, we have T surjective. Moreover, V is finite-dimensional, so a surjective linear map on V is also injective, hence invertible. Therefore, for every $v \in V$, there exist scalars $\alpha_1, \ldots, \alpha_m$ such that

$$v = \alpha_1 Tv_1 + \cdots + \alpha_m Tv_m.$$

Applying T^{-1} to both sides, we get

$$T^{-1}v = \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

Since T^{-1} is surjective, every vector $w \in V$ can be written as $T^{-1}v$ for some $v \in V$, and thus $v_1, \ldots, v_m$ spans V. ∎

Exercise 3D16

Prove that every linear map from $\mathbb{F}^{n,1}$ to $\mathbb{F}^{m,1}$ is given by a matrix multiplication. In other words, prove that if $T \in \mathcal{L}(\mathbb{F}^{n,1}, \mathbb{F}^{m,1})$, then there exists an m-by-n matrix A such that $Tx = Ax$ for every $x \in \mathbb{F}^{n,1}$.

Solution. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{F}^{n,1}$. Define the k-th column of A to be $Te_k \in \mathbb{F}^{m,1}$ for $k = 1, \ldots, n$. Then for every $x \in \mathbb{F}^{n,1}$, there exist scalars $\alpha_1, \ldots, \alpha_n$ such that

$$x = \alpha_1 e_1 + \cdots + \alpha_n e_n.$$

Applying T to both sides, we have

$$Tx = \alpha_1 Te_1 + \cdots + \alpha_n Te_n = Ax,$$

where the last equality follows from the definition of matrix multiplication. Hence, $Tx = Ax$ for every $x \in \mathbb{F}^{n,1}$. ∎

Exercise 3D17

Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(T) = ST$$

for $T \in \mathcal{L}(V)$.

(a) Show that $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.
(b) Show that $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$.

Solution.

(a) Consider $T \in \operatorname{null} \mathcal{A}$. Then $\mathcal{A}(T) = ST = 0$, so T is a linear map from V to $\operatorname{null} S$. Conversely, if T is a linear map from V to $\operatorname{null} S$, then $ST = 0$, so $T \in \operatorname{null} \mathcal{A}$. Hence, $\operatorname{null} \mathcal{A}$ is the set of linear maps from V to $\operatorname{null} S$, and thus $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.

(b) By fundamental theorem of linear maps, we have

$$\dim \mathcal{L}(V) = \dim \operatorname{null} \mathcal{A} + \dim \operatorname{range} \mathcal{A}.$$

Since $\dim \mathcal{L}(V) = (\dim V)^2$ and $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$, we get

$$(\dim V)^2 = (\dim V)(\dim \operatorname{null} S) + \dim \operatorname{range} \mathcal{A},$$

which implies

$$\dim \operatorname{range} \mathcal{A} = (\dim V)^2 - (\dim V)(\dim \operatorname{null} S) = (\dim V)(\dim \operatorname{range} S). \quad \blacksquare$$

Exercise 3D18

Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces.

Solution. Let $T : \mathcal{L}(\mathbb{F}, V) \to V$ be defined by $T(\alpha) = \alpha(1)$ for $\alpha \in \mathcal{L}(\mathbb{F}, V)$. We first show that T is a linear map. Let $\alpha, \beta \in \mathcal{L}(\mathbb{F}, V)$ and $\lambda \in \mathbb{F}$. Then

$$T(\alpha + \beta) = (\alpha + \beta)(1) = \alpha(1) + \beta(1) = T\alpha + T\beta,$$

and

$$T(\lambda\alpha) = (\lambda\alpha)(1) = \lambda\alpha(1) = \lambda T\alpha.$$

We now show that T is an isomorphism. To see injectivity, let $\alpha, \beta \in \mathcal{L}(\mathbb{F}, V)$ such that $T\alpha = T\beta$. Then $\alpha(1) = \beta(1)$. For any $x \in \mathbb{F}$, we have $\alpha(x) = x\alpha(1) = x\beta(1) = \beta(x)$, so $\alpha = \beta$. Hence, T is injective.

To see surjectivity, let $v \in V$. Define $\alpha \in \mathcal{L}(\mathbb{F}, V)$ by $\alpha(x) = xv$ for $x \in \mathbb{F}$. Then $T\alpha = \alpha(1) = v$, so T is surjective. Therefore, T is an isomorphism, and thus V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces. $\blacksquare$

Exercise 3D19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has the same matrix with respect to every basis of V if and only if T is a scalar multiple of the identity operator.

Manual Remark. *See* Appendix 3D19 *for other constructions of bases to eliminate off-diagonal entries.*

Solution. Suppose T has the same matrix with respect to every basis of V. Then for any set of bases for V, its matrix must be the same. If $\dim V \leq 1$, then every operator on V is a scalar multiple of the identity operator. Now suppose $\dim V \geq 2$. For any bases $v_1, \ldots, v_n$ and $u_1, \ldots, u_n$ of V, we have

$$\mathcal{M}(T, (v_1, \ldots, v_n)) = \mathcal{M}(T, (u_1, \ldots, u_n)).$$

Hence, we can declare that their matrix is just some A. We now consider the basis $v_1 + v_2, v_2, \ldots, v_n$, which yields the same matrix A. Let $w_1 = v_1 + v_2$ and $w_i = v_i$ for $i \geq 2$. In the new basis, we have the expression

$$Tw_1 = A_{1,1}w_1 + A_{2,1}w_2 + \cdots + A_{n,1}w_n = A_{1,1}v_1 + (A_{1,1} + A_{2,1})v_2 + A_{3,1}v_3 + \cdots + A_{n,1}v_n.$$

This must equal the expression for $Tw_1 = Tv_1 + Tv_2$ in the original basis:

$$Tv_1 + Tv_2 = (A_{1,1} + A_{1,2})v_1 + (A_{2,1} + A_{2,2})v_2 + \cdots + (A_{n,1} + A_{n,2})v_n.$$

Comparing coefficients of $v_1, \ldots, v_n$ in the two expressions, we get

$$\begin{aligned} A_{1,1} &= A_{1,1} + A_{1,2}, \\ A_{1,1} + A_{2,1} &= A_{2,1} + A_{2,2}, \\ A_{j,1} &= A_{j,1} + A_{j,2} \text{ for } j \geq 3. \end{aligned}$$

From this, we obtain that $A_{1,2} = 0, A_{2,2} = A_{1,1}$, and $A_{j,2} = 0$ for $j \geq 3$, i.e., the diagonal elements $A_{1,1}$ and $A_{2,2}$ are equal, while off-diagonal entries in column 2 are 0. Now consider the basis $v_1, v_1 + v_2, \ldots, v_n$. Let $u_1 = v_1, u_2 = v_1 + v_2$, and $u_i = v_i$ for $i \geq 3$. In the new basis, we have

$$\begin{aligned} Tu_2 &= A_{1,2}u_1 + \cdots + A_{n,2}u_n \\ &= A_{1,2}v_1 + A_{2,2}(v_1 + v_2) + A_{3,2}v_3 + \cdots + A_{n,2}v_n \\ &= (A_{1,2} + A_{2,2})v_1 + A_{2,2}v_2 + A_{3,2}v_3 + \cdots + A_{n,2}v_n. \\ &= A_{1,1}v_1 + A_{2,2}v_2. \end{aligned}$$

This must equal the expression for $Tu_2 = Tv_1 + Tv_2$ in the original basis, whose expression is listed above. Comparing coefficients of v_1 and v_2, we get

$$\begin{aligned} A_{1,1} &= A_{1,1} + A_{1,2}, \\ A_{2,2} &= A_{2,1} + A_{2,2}, \\ A_{j,2} &= A_{j,1} + A_{j,2} \quad \text{for } j \geq 3. \end{aligned}$$

From this, we obtain similar results about diagonal entries, but the off-diagonal entries in column 1 must also be 0. We may repeat the first process for $v_1 + v_k$ for $k \geq 3$ to get that entries off the diagonal in column k must also be 0. Continuing this process, we conclude all off-diagonal entries are 0, and all diagonal entries are equal, i.e., A is a scalar multiple of the identity matrix.

Conversely, suppose T is a scalar multiple of the identity operator. Then there exists some $\lambda \in \mathbb{F}$ such that $T = \lambda I$. For any basis of V, the matrix representation of T with respect to that basis is the diagonal matrix with λ on the diagonal. Hence, T has the same matrix with respect to every basis of V. ■

Exercise 3D20

Suppose $q \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{R})$ such that

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$$

for all $x \in \mathbb{R}$.

Solution. Let $T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$ be defined by

$$Tp = (x^2 + x)p'' + 2xp' + p(3)$$

for $p \in \mathcal{P}_n(\mathbb{R})$. We first show that T is a linear map. Let $p, q \in \mathcal{P}_n(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$T(p+q) = (x^2 + x)(p+q)'' + 2x(p+q)' + (p+q)(3) = Tp + Tq,$$

and

$$T(\alpha p) = (x^2 + x)(\alpha p)'' + 2x(\alpha p)' + (\alpha p)(3) = \alpha Tp.$$

Then $T \in \mathcal{L}(\mathcal{P}_n(\mathbb{R}))$. We now show that T is invertible. To see injectivity, let $p \in \mathcal{P}_n(\mathbb{R})$ such that $Tp = 0$. Then

$$(x^2 + x)p'' + 2xp' + p(3) = 0.$$

Then

$$(x^2 + x)p'' + 2xp' = -p(3).$$

We analyze the degree of p as follows:

- If $\deg p \geq 2$, consider $(x^2 + x)p''$. In particular, $(a_m x^m)'' = a_m m(m-1)x^{m-2}$, hence the coefficient of x^m is $a_m m(m-1)$. Moreover, $2xp'$ has coefficient $2ma_m$. Since $\deg p \geq 2$, then $m \geq 2$, so the total coefficient of x^m is $a_m m(m-1) + 2ma_m = a_m m(m-1+2) = a_m m(m+1)$. Since this is nonzero for $a_m \neq 0$, we get a contradiction with the right-hand side being a constant.
- If $\deg p = 1$, then $p(x) = ax + b$ for some $a, b \in \mathbb{R}$. Then $p''(x) = 0$ and $p'(x) = a$, so the left-hand side is $2ax$. Since the right-hand side is a constant, this is a contradiction.
- If $\deg p = 0$, then $p(x) = c$ for some $c \in \mathbb{R}$. Then $p''(x) = 0$ and $p'(x) = 0$, so the left-hand side is 0, and the right-hand side is $p(3) = c$. Hence, $c = 0$, and thus $p = 0$. Therefore, T is injective.

Since T is injective and $\mathcal{P}_n(\mathbb{R})$ is finite-dimensional, then T is surjective, hence invertible. Then there does exist some $p \in \mathcal{P}_n(\mathbb{R})$ such that $Tp = q$. ■

Exercise 3D21

Suppose n is a positive integer and $A_{j,k} \in \mathbb{F}$ for all $j,k = 1,\ldots,n$. Prove that the following are equivalent (note that in both parts below, the number of equations equals the number of variables).

(a) The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = 0$$
$$\vdots$$
$$\sum_{k=1}^{n} A_{n,k}x_k = 0$$

(b) For every $c_1,\ldots,c_n \in \mathbb{F}$, there exists a solution to the system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = c_1$$
$$\vdots$$
$$\sum_{k=1}^{n} A_{n,k}x_k = c_n$$

Solution.

(a) Suppose the trivial solution $0 \in \mathbb{F}^{n,1}$ is the only solution to the homogeneous system of equations. Then $\operatorname{null} T = \{0\}$, where $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ is defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$. Since $\operatorname{null} T = \{0\}$, then T is injective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, then T is surjective, hence invertible. Then for every $c_1,\ldots,c_n \in \mathbb{F}$, there exists some $x \in \mathbb{F}^{n,1}$ such that $Tx = c$, where c is the column vector with entries $c_1,\ldots,c_n$. Hence, there exists a solution to the system of equations.

(b) Suppose for every $c_1,\ldots,c_n \in \mathbb{F}$, there exists a solution to the system of equations. Then $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$ is surjective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, then T is injective. Hence, the trivial solution is the only solution to the homogeneous system of equations. ■

Exercise 3D22

Suppose $T \in \mathcal{L}(V)$ and $v_1,\ldots,v_n$ is a basis of V. Prove that

$$\mathcal{M}(T,(v_1,\ldots,v_n)) \text{ is invertible} \iff T \text{ is invertible.}$$

Solution. Suppose $\mathcal{M}(T,(v_1,\ldots,v_n))$ is invertible. Then there exists some $A \in \mathbb{F}^{n,n}$ such that $A\mathcal{M}(T,(v_1,\ldots,v_n)) = I$. Let U be the linear map from V to V whose matrix with respect to the basis $v_1,\ldots,v_n$ is A. Then

$$\mathcal{M}(UT,(v_1,\ldots,v_n)) = \mathcal{M}(U,(v_1,\ldots,v_n))\mathcal{M}(T,(v_1,\ldots,v_n)) = A\mathcal{M}(T,(v_1,\ldots,v_n)) = I.$$

Then UT is the identity operator, and thus T is injective. Since V is finite-dimensional, then T is surjective, hence invertible.

Suppose T is invertible. Then T^{-1} exists, and we have

$$\mathcal{M}(T^{-1}T,(v_1,\ldots,v_n)) = \mathcal{M}(T^{-1},(v_1,\ldots,v_n))\mathcal{M}(T,(v_1,\ldots,v_n)) = I.$$

Hence, $\mathcal{M}(T,(v_1,\ldots,v_n))$ is invertible. ■

Exercise 3D23

Suppose that $u_1,\ldots,u_n$ and $v_1,\ldots,v_n$ are bases of V. Let $T \in \mathcal{L}(V)$ be such that $Tv_k = u_k$ for each $k = 1,\ldots,n$. Prove that

$$\mathcal{M}(T,(v_1,\ldots,v_n)) = \mathcal{M}(I,(u_1,\ldots,u_n),(v_1,\ldots,v_n)).$$

Solution. By definition, column k of $\mathcal{M}(T,(v_1,\ldots,v_n))$ is $Tv_k = u_k$ with respect to the basis $v_1,\ldots,v_n$. Similarly, column k of $\mathcal{M}(I,(u_1,\ldots,u_n),(v_1,\ldots,v_n))$ is also u_k with respect to the basis $v_1,\ldots,v_n$. ■

Exercise 3D24

Suppose A and B are square matrices of the same size and $AB = I$. Prove that $BA = I$.

Solution. Let $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ be defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$. Since $AB = I$, then T is surjective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, surjectivity implies injectivity. Hence, T is invertible, and $T(Bx) = x = T(T^{-1}x)$. Then $Bx = T^{-1}x$ for every $x \in \mathbb{F}^{n,1}$. Therefore,

$$BAx = B(Tx) = T^{-1}(Tx) = x$$

for every $x \in \mathbb{F}^{n,1}$. Hence, $BA = I$. ■

3E Products and Quotients of Vector Spaces

Exercise 3E1

Suppose T is a function from V to W. The *graph of T* is the subset of $V \times W$ defined by

$$\text{graph of } T = \{(v, Tv) \in V \times W \mid v \in V\}.$$

Prove that T is a linear map if and only if the graph of T is a subspace of $V \times W$.

Solution. Let $G(T)$ denote the graph of T. Now, suppose T is a linear map. Then $G(T)$ is nonempty since $(0, T0) = (0, 0) \in G(T)$. Now, let $(v_1, Tv_1), (v_2, Tv_2) \in G(T)$. Then

$$(v_1 - v_2, Tv_1 - Tv_2) = (v_1, Tv_1) - (v_2, Tv_2) \in G(T)$$

since T is linear. Now, let $(v, Tv) \in G(T)$ and $\alpha \in \mathbb{F}$. Then

$$(\alpha v, \alpha Tv) = \alpha(v, Tv) \in G(T)$$

since T is linear. Thus, $G(T)$ is a subspace of $V \times W$.

Conversely, suppose $G(T)$ is a subspace of $V \times W$. By definition, we have $(v_1, Tv_1), (v_2, Tv_2) \in G(T)$ for all $v_1, v_2 \in V$. Since $G(T)$ is a subspace of $V \times W$, for all $\alpha, \beta \in \mathbb{F}$, we have

$$(\alpha v_1 + \beta v_2, \alpha Tv_1 + \beta Tv_2) = \alpha(v_1, Tv_1) + \beta(v_2, Tv_2) \in G(T).$$

By definition of $G(T)$, we have $\alpha Tv_1 + \beta Tv_2 = T(\alpha v_1 + \beta v_2)$. Thus, T is a linear map. ■

Exercise 3E2

Suppose that $V_1, \ldots, V_m$ are vector spaces such that $V_1 \times \cdots \times V_m$ is finite-dimensional. Prove that V_k is finite-dimensional for each $k = 1, \ldots, m$.

Solution. Let $V = V_1 \times \cdots \times V_m$, and consider $T \in \mathcal{L}(V, V_k)$ defined by $T(v_1, \ldots, v_m) = v_k$. Then T is a linear map since for all $\alpha, \beta \in \mathbb{F}$ and all $(v_1, \ldots, v_m), (w_1, \ldots, w_m) \in V$, we have

$$\begin{aligned} T(\alpha(v_1, \ldots, v_m) + \beta(w_1, \ldots, w_m)) &= T(\alpha v_1 + \beta w_1, \ldots, \alpha v_m + \beta w_m) \\ &= \alpha v_k + \beta w_k \\ &= \alpha T(v_1, \ldots, v_m) + \beta T(w_1, \ldots, w_m). \end{aligned}$$

Moreover, T is surjective since for any $v_k \in V_k$, we have $v_k = T(0, \ldots, 0, v_k, 0, \ldots, 0)$ where v_k is in the k-th coordinate. Since V is finite-dimensional and T is a linear map that is surjective, it follows that V_k is finite-dimensional. ■

Exercise 3E3

Suppose $V_1, \ldots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Solution. Let $V = V_1 \times \cdots \times V_m$. Let $L = \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$. Define $T \in \mathcal{L}(\mathcal{L}(V, W), L)$ by

$$T(S) = (S \circ \iota_1, \ldots, S \circ \iota_m)$$

where $\iota_k \in \mathcal{L}(V_k, V)$ is the map defined as

$$\iota_k(v_k) = (0, \ldots, 0, v_k, 0, \ldots, 0)$$

with $\iota_k(v_k)$ having v_k in the k-th coordinate and 0 in all other coordinates. Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $S_1, S_2 \in \mathcal{L}(V, W)$, we have

$$\begin{aligned} T(\alpha S_1 + \beta S_2) &= ((\alpha S_1 + \beta S_2) \circ \iota_1, \ldots, (\alpha S_1 + \beta S_2) \circ \iota_m) \\ &= (\alpha(S_1 \circ \iota_1) + \beta(S_2 \circ \iota_1), \ldots, \alpha(S_1 \circ \iota_m) + \beta(S_2 \circ \iota_m)) \\ &= \alpha T(S_1) + \beta T(S_2), \end{aligned}$$

hence T is a linear map. Now suppose $T(S) = 0$. Then $S \circ \iota_k = 0$ for all $k = 1, \ldots, m$. Moreover, for all $(v_1, \ldots, v_m) \in V$, we have

$$S(v_1, \ldots, v_m) = S(\iota_1(v_1) + \cdots + \iota_m(v_m)) = S \circ \iota_1(v_1) + \cdots + S \circ \iota_m(v_m) = 0.$$

It follows that $S(v_1, \ldots, v_m) = 0$ for all $(v_1, \ldots, v_m) \in V$. Thus, $S = 0$, and hence T is injective.

Now, let $(S_1, \ldots, S_m) \in L$. Define $S \in \mathcal{L}(V, W)$ by $S(v_1, \ldots, v_m) = S_1 v_1 + \cdots + S_m v_m$. Then $T(S) = (S_1, \ldots, S_m)$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E4

Suppose $W_1, \ldots, W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1 \times \cdots \times W_m)$ and $\mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$ are isomorphic vector spaces.
Manual Remark. *Note the difference between this exercise and Exercise 3E3. Here, the domain is decomposed, while in Exercise 3E3, the codomain is decomposed. Hence, the use of the maps ι_k and π_k in the solutions to these exercises are reversed.*

Solution. Let $W = W_1 \times \cdots \times W_m$. Let $L = \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$. Define $T \in \mathcal{L}(\mathcal{L}(V, W), L)$ by

$$T(S) = (\pi_1 \circ S, \ldots, \pi_m \circ S)$$

where $\pi_k \in \mathcal{L}(W, W_k)$ is the map defined as

$$\pi_k(w_1, \ldots, w_m) = w_k$$

Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $S_1, S_2 \in \mathcal{L}(V, W)$, we have

$$\begin{aligned} T(\alpha S_1 + \beta S_2) &= (\pi_1 \circ (\alpha S_1 + \beta S_2), \ldots, \pi_m \circ (\alpha S_1 + \beta S_2)) \\ &= (\alpha(\pi_1 \circ S_1) + \beta(\pi_1 \circ S_2), \ldots, \alpha(\pi_m \circ S_1) + \beta(\pi_m \circ S_2)) \\ &= \alpha T(S_1) + \beta T(S_2), \end{aligned}$$

hence T is a linear map. Now suppose $T(S) = 0$. Then $\pi_k \circ S = 0$ for all $k = 1, \ldots, m$. Moreover, for all $v \in V$, we have

$$Sv = (\pi_1(Sv), \ldots, \pi_m(Sv)) = (0, \ldots, 0) = 0.$$

It follows that $Sv = 0$ for all v, hence $S = 0$, and thus T is injective.

Now let $(S_1, \ldots, S_m) \in L$. Define $S \in \mathcal{L}(V, W)$ by $Sv = (S_1 v, \ldots, S_m v)$. Then $T(S) = (S_1, \ldots, S_m)$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E5

For m a positive integer, define V^m by

$$V^m = \underbrace{V \times \cdots \times V}_{m \text{ times}}.$$

Prove that V^m and $\mathcal{L}(\mathbb{F}^m, V)$ are isomorphic vector spaces.

Solution. Let $e_1, \ldots, e_m$ be the standard basis of $\mathbb{F}^m$. Define $T \in \mathcal{L}(V^m, \mathcal{L}(\mathbb{F}^m, V))$ by

$$T(v_1, \ldots, v_m)(\lambda_1, \ldots, \lambda_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m.$$

Then T is linear since for all $\alpha, \beta \in \mathbb{F}$ and all $v = (v_1, \ldots, v_m), w = (w_1, \ldots, w_m) \in V^m$, we have

$$\begin{aligned} T(\alpha v + \beta w)(\lambda_1, \ldots, \lambda_m) &= T(\alpha v_1 + \beta w_1, \ldots, \alpha v_m + \beta w_m)(\lambda_1, \ldots, \lambda_m) \\ &= \lambda_1(\alpha v_1 + \beta w_1) + \cdots + \lambda_m(\alpha v_m + \beta w_m) \\ &= \alpha(\lambda_1 v_1 + \cdots + \lambda_m v_m) + \beta(\lambda_1 w_1 + \cdots + \lambda_m w_m) \\ &= \alpha T(v)(\lambda_1, \ldots, \lambda_m) + \beta T(w)(\lambda_1, \ldots, \lambda_m), \end{aligned}$$

hence T is a linear map. Now suppose $T(v) = 0$. Then $T(v)(e_k) = 0$ for all $k = 1, \ldots, m$. It follows that $v_k = 0$ for all $k = 1, \ldots, m$, hence $v = 0$, and thus T is injective.

Now let $S \in \mathcal{L}(\mathbb{F}^m, V)$. Define $v \in V^m$ by $v_k = Se_k$ for all $k = 1, \ldots, m$. Then $T(v) = S$. Thus, T is surjective. Therefore, T is an isomorphism. ■

Exercise 3E6

Suppose that v, x are vectors in V and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$.

Solution. Let $u \in U$. Then $v + u \in v + U = x + W$. Thus, there exists $w \in W$ such that $v + u = x + w$. Note that $v \in x + W$ implies the existence of some $w' \in W$ such that $v = x + w'$, or $x - v = -w' \in W$. Similarly, $x \in v + U$ implies the existence of some $u' \in U$ such that $x = v + u'$, or $v - x = -u' \in U$. It follows that $u = (x - v) + w \in W$, hence $U \subseteq W$. Now let $w \in W$. Then $x + w \in x + W = v + U$. Thus, there exists $u \in U$ such that $x + w = v + u$. It follows that $w = (v - x) + u \in U$, hence $W \subseteq U$. Therefore, $U = W$. ■

Exercise 3E7

Let $U = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\}$. Suppose $A \subseteq \mathbb{R}^3$. Prove that A is a translate of U if and only if there exists $c \in \mathbb{R}$ such that

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = c\}.$$

Solution. Suppose A is a translate of U. Then there exists $v \in \mathbb{R}^3$ such that $A = v + U$. Let $v = (x_0, y_0, z_0)$.

Then

$$\begin{aligned} A &= v + U \\ &= \{(x_0, y_0, z_0) + (x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\} \\ &= \{(x', y', z') \in \mathbb{R}^3 \mid 2(x' - x_0) + 3(y' - y_0) + 5(z' - z_0) = 0\} \\ &= \{(x', y', z') \in \mathbb{R}^3 \mid 2x' + 3y' + 5z' = 2x_0 + 3y_0 + 5z_0\}. \end{aligned}$$

Conversely, suppose there exists $c \in \mathbb{R}$ such that

$$A = \{(x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = c\}.$$

Let $v = (x_0, y_0, z_0)$ where x_0, y_0, z_0 are any real numbers such that $2x_0 + 3y_0 + 5z_0 = c$. Then

$$\begin{aligned} v + U &= (x_0, y_0, z_0) + U \\ &= \{(x_0, y_0, z_0) + (x, y, z) \in \mathbb{R}^3 \mid 2x + 3y + 5z = 0\} \\ &= A. \end{aligned}$$

■

Exercise 3E8

(a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Prove that $\{x \in V \mid Tx = c\}$ is either the empty set or is a translate of $\operatorname{null} T$.
(b) Explain why the set of solutions to a system of linear equations such as 3.27 is either the empty set or is a translate of some subspace of $\mathbb{F}^n$.

Solution.

(a) Suppose $T \in \mathcal{L}(V, W)$ and $c \in W$. Let $A = \{x \in V \mid Tx = c\}$. If A is nonempty, then there exists $v \in V$ such that $Tv = c$. We claim that $A = v + \operatorname{null} T$. Let $x \in A$. Then $Tx = c = Tv$, hence $T(x - v) = 0$, and thus $x - v \in \operatorname{null} T$. It follows that $x \in v + \operatorname{null} T$, hence $A \subseteq v + \operatorname{null} T$. Now let $y \in v + \operatorname{null} T$. Then there exists $u \in \operatorname{null} T$ such that $y = v + u$. It follows that $Ty = T(v + u) = Tv + Tu = c + 0 = c$, hence $y \in A$, and thus $v + \operatorname{null} T \subseteq A$. Therefore, $A = v + \operatorname{null} T$.
(b) A system of linear equations such as 3.27 can be written in the form $Tx = c$ for some linear map T and some vector c. By part (a), the set of solutions to this system of linear equations is either the empty set or is a translate of $\operatorname{null} T$, which is a subspace of $\mathbb{F}^n$. ■

Exercise 3E9

Prove that a nonempty subset A of V is a translate of some subspace of V if and only if $\lambda v + (1 - \lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$.

Solution. Suppose A is a translate of some subspace of V. Then there exists $v \in V$ and a subspace U of V such that $A = v + U$. Let $v + u, v + u' \in A$. Then for any $\lambda \in \mathbb{F}$,

$$\lambda(v + u) + (1 - \lambda)(v + u') = v + \underbrace{(\lambda u + (1 - \lambda)u')}_{\in U} \in A.$$

Conversely, suppose $\lambda v + (1-\lambda)w \in A$ for all $v, w \in A$ and all $\lambda \in \mathbb{F}$. Fix some $v_0 \in A$, and let $U = \{u \in V \mid v_0 + u \in A\}$. We first claim that U is a subspace of V. Since $v_0 = v_0 + 0 \in A$, we have $0 \in U$. Let $u_1, u_2 \in U$. Then $v_0 + u_1, v_0 + u_2 \in A$. Now let $\lambda \in \mathbb{F}$. Then

$$\begin{aligned}\lambda(v_0 + u_1) + (1-\lambda)(v_0 + u_2) &= v_0 + \lambda u_1 + (1-\lambda)u_2 \\ &= v_0 + (\lambda u_1 + (1-\lambda)u_2) \in A.\end{aligned}$$

It follows that $\lambda u_1 + (1-\lambda)u_2 \in U$, hence U is a subspace of V. Now we claim that $A = v_0 + U$. Let $w \in A$. By definition, $v_0 + (w - v_0) = w \in A$, hence $w - v_0 \in U$, and thus $w \in v_0 + U$, so $A \subseteq v_0 + U$. Now let $v_0 + u \in v_0 + U$. Then $v_0 + u \in A$ by definition of U, hence $v_0 + U \subseteq A$. Therefore, $A = v_0 + U$. ■

Exercise 3E10

Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces U_1, U_2 of V. Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of V or is the empty set.

Solution. If $A_1 \cap A_2$ is the empty set, then we are done. Suppose that $A_1 \cap A_2$ is nonempty. Let $x, y \in A_1 \cap A_2$ and $\lambda \in \mathbb{F}$. Then $x, y \in A_1$, hence $\lambda x + (1-\lambda)y \in A_1$. Moreover, $x, y \in A_2$, hence $\lambda x + (1-\lambda)y \in A_2$. It follows that $\lambda x + (1-\lambda)y \in A_1 \cap A_2$. Thus, by Exercise 3E9, $A_1 \cap A_2$ is a translate of some subspace of V. ■

Exercise 3E11

Suppose $U = \{(x_1, x_2, \ldots) \in \mathbb{F}^\infty \mid x_k \neq 0 \text{ for only finitely many } k\}$.

(a) Show that U is a subspace of $\mathbb{F}^\infty$.

(b) Prove that $\mathbb{F}^\infty / U$ is infinite-dimensional.

Appendix. *See Appendix 3E11 for a generalization of this treatment, so long as you are comfortable with the idea $\mathbb{N}$ can be partitioned into infinitely many disjoint infinite sets.*

Solution.

(a) We claim that U can be reformulated as the set of vectors $(x_1, x_2, \ldots) \in \mathbb{F}^\infty$ such that there exists an index N such that $x_k = 0$ for all $k > N$. To see this, suppose $(x_1, x_2, \ldots) \in U$. Then there exists a finite set of indices K such that $x_k = 0$ for all $k \notin K$. If K is empty, let $N = 0$. Otherwise, let $N = \max K$. Then $x_k = 0$ for all $k > N$.

Conversely, suppose there exists an index N such that $x_k = 0$ for all $k > N$. Then the set of indices $\{1, 2, \ldots, N\}$ is finite and contains all indices where $x_k \neq 0$, hence $(x_1, x_2, \ldots) \in U$. Thus, the two formulations of U are equivalent.

We now show that U is a subspace of $\mathbb{F}^\infty$. First, note that the zero vector $(0, 0, \ldots) \in U$ since it has no nonzero coordinates. Now let $u = (u_1, u_2, \ldots), v = (v_1, v_2, \ldots) \in U$. Then there exist indices N_u and N_v such that $u_k = 0$ for all $k > N_u$ and $v_k = 0$ for all $k > N_v$. Let $N = \max(N_u, N_v)$. Then for all $k > N$, we have $u_k = 0$ and $v_k = 0$, hence $(u+v)_k = u_k + v_k = 0 + 0 = 0$. Thus, $u + v \in U$. Now let $\alpha \in \mathbb{F}$. Then for all $k > N_u$, we have $(\alpha u)_k = \alpha u_k = \alpha \cdot 0 = 0$, hence $\alpha u \in U$. Therefore, U is a subspace of $\mathbb{F}^\infty$.

(b) For any given positive integer m, define the sets B_i for $1 \le i \le m$ as follows:

$$B_i = \{i + km \mid k \in \mathbb{N}\} = \{i, i+m, i+2m, \ldots\}.$$

Then $B_1, \ldots, B_m$ are pairwise disjoint subsets of $\mathbb{N}$. For each $i = 1, \ldots, m$, define

$$v_i = (v_{i,1}, v_{i,2}, \ldots) \in \mathbb{F}^\infty$$

such that $v_{i,j} = 1$ if $j \in B_i$ and $v_{i,j} = 0$ otherwise. We claim that $v_1 + U, \ldots, v_m + U$ are linearly independent in $\mathbb{F}^\infty/U$.

Let $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ such that $\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = U$. Then $\lambda_1 v_1 + \cdots + \lambda_m v_m \in U$. Then we can show that $\lambda_i = 0$ for all $i = 1, \ldots, m$ as follows. For each i, consider the j-th coordinate of v_i for $j \in B_i$. Since v_i has 1's in all coordinates indexed by B_i and 0's elsewhere, and B_i is disjoint from B_k for $k \neq i$, the j-th coordinate of $\lambda_1 v_1 + \cdots + \lambda_m v_m$ is λ_i. But this sum is in U, which means it has only finitely many nonzero coordinates. Since B_i is infinite, it follows that $\lambda_i = 0$. Hence, all $\lambda_i = 0$, proving that $v_1 + U, \ldots, v_m + U$ are linearly independent. Since m was arbitrary, $\mathbb{F}^\infty/U$ is infinite-dimensional. ■

Exercise 3E12

Suppose $v_1, \ldots, v_m \in V$. Let

$$A = \{\lambda_1 v_1 + \cdots + \lambda_m v_m \mid \lambda_1, \ldots, \lambda_m \in \mathbb{F} \text{ and } \lambda_1 + \cdots + \lambda_m = 1\}.$$

(a) Prove that A is a translate of some subspace of V.
(b) Prove that if B is a translate of some subspace of V and $\{v_1, \ldots, v_m\} \subseteq B$, then $A \subseteq B$.
(c) Prove that A is a translate of some subspace of V of dimension less than m.

Solution.

(a) For any element $\lambda_1 v_1 + \cdots + \lambda_m v_m \in A$, we can write

$$\lambda_1 v_1 + \cdots + \lambda_m v_m = v_1 + \lambda_2(v_2 - v_1) + \cdots + \lambda_m(v_m - v_1) = v_1 + \sum_{i=2}^{m} \lambda_i(v_i - v_1),$$

where $\lambda_1 = 1 - (\lambda_2 + \cdots + \lambda_m)$. Let $U = \operatorname{span}(v_2 - v_1, \ldots, v_m - v_1)$. Then $A = v_1 + U$, hence A is a translate of the subspace U of V.

(b) By Exercise 3E9, since B is a translate of some subspace of V, we have $\lambda v + (1 - \lambda)w \in B$ for all $v, w \in B$ and all $\lambda \in \mathbb{F}$. Since $\{v_1, \ldots, v_m\} \subseteq B$, it follows that any linear combination of $v_1, \ldots, v_m$ with coefficients summing to 1 is in B. Hence, $A \subseteq B$.

(c) The subspace $U = \operatorname{span}(v_2 - v_1, \ldots, v_m - v_1)$ has dimension at most $m - 1$, since it is spanned by $m - 1$ vectors. Therefore, $A = v_1 + U$ is a translate of a subspace of V of dimension less than m. ■

Exercise 3E13

Suppose U is a subspace of V such that V/U is finite-dimensional. Prove that V is isomorphic to $U \times (V/U)$.

Solution. Since V/U is finite-dimensional, let $v_1 + U, \ldots, v_m + U$ be a basis of V/U. Let $W = \operatorname{span}(v_1, \ldots, v_m)$. We show that $V = U \oplus W$. First, note that any $v \in V$ can be written as $v = u + w$ for some $u \in U$ and $w \in W$. Indeed, since $v + U \in V/U$, we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$$

for some scalars $\lambda_1, \ldots, \lambda_m$. This gives

$$v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U.$$

So setting $u = v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U$ and $w = \lambda_1 v_1 + \cdots + \lambda_m v_m \in W$, we have $v = u + w$. To see that the sum is direct, suppose $u + w = 0$ with $u \in U$ and $w \in W$. Then $w = -u \in W \cap U$. Since $w \in W$, we can write $w = \lambda_1 v_1 + \cdots + \lambda_m v_m$ for some scalars $\lambda_1, \ldots, \lambda_m$. Since we established that $w \in U$, we have $w + U = 0 + U$. But $v_1 + U, \ldots, v_m + U$ are linearly independent in V/U, hence $\lambda_1 = \cdots = \lambda_m = 0$, and thus $w = 0$. It follows that $u = 0$ as well, proving that the sum is direct. Therefore, $V = U \oplus W$.

Consider the map $T : V \to U \times (V/U)$ given by $Tv = (u, v + U)$, where $v = u + w$ with $u \in U$ and $w \in W$. This map is well-defined because the decomposition $v = u + w$ is unique. It is linear, injective, and surjective, hence an isomorphism. Therefore, $V \cong U \times (V/U)$. ■

Exercise 3E14

Suppose U and W are subspaces of V and $V = U \oplus W$. Suppose $w_1, \ldots, w_m$ is a basis of W. Prove that $w_1 + U, \ldots, w_m + U$ is a basis of V/U.

Solution. Since $V = U \oplus W$, every $v \in V$ can be written uniquely as $v = u + w$ with $u \in U$ and $w \in W$. We claim that $w_1 + U, \ldots, w_m + U$ is a basis of V/U. First, we show that they span V/U. Let $v + U \in V/U$. Write $v = u + w$ with $u \in U$ and $w \in W$. Then $v + U = (u + w) + U = w + U$. Since $w \in W$, we can write $w = \lambda_1 w_1 + \cdots + \lambda_m w_m$ for some scalars $\lambda_1, \ldots, \lambda_m$. Hence, $v + U = \lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U)$, showing that $w_1 + U, \ldots, w_m + U$ span V/U.

Next, we show linear independence. Suppose $\lambda_1(w_1 + U) + \cdots + \lambda_m(w_m + U) = 0 + U$. Then $\lambda_1 w_1 + \cdots + \lambda_m w_m \in U$. But $\lambda_1 w_1 + \cdots + \lambda_m w_m \in W$ and $U \cap W = \{0\}$, so $\lambda_1 w_1 + \cdots + \lambda_m w_m = 0$. Since $w_1, \ldots, w_m$ are linearly independent, we have $\lambda_1 = \cdots = \lambda_m = 0$. Therefore, $w_1 + U, \ldots, w_m + U$ are linearly independent, and hence form a basis of V/U. ■

Exercise 3E15

Suppose U is a subspace of V and $v_1 + U, \ldots, v_m + U$ is a basis of V/U and $u_1, \ldots, u_n$ is a basis of U. Prove that $v_1, \ldots, v_m, u_1, \ldots, u_n$ is a basis of V.

Solution. First, we show that $v_1, \ldots, v_m, u_1, \ldots, u_n$ span V. Let $v \in V$. Then $v + U \in V/U$, so we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U)$$

for some scalars $\lambda_1, \ldots, \lambda_m$. This gives $v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U$. Since $u_1, \ldots, u_n$ is a basis of U, we can write $v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) = \mu_1 u_1 + \cdots + \mu_n u_n$ for some scalars $\mu_1, \ldots, \mu_n$. Hence, $v = \lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n$, showing that $v_1, \ldots, v_m, u_1, \ldots, u_n$ span V.

Next, we show linear independence. Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_n u_n = 0$. Then $\lambda_1 v_1 + \cdots + \lambda_m v_m \in U$. Then

$$\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0 + U.$$

Since $v_1 + U, \ldots, v_m + U$ are linearly independent in V/U, we must have $\lambda_1 = \cdots = \lambda_m = 0$. Then the original equation reduces to $\mu_1 u_1 + \cdots + \mu_n u_n = 0$, and since $u_1, \ldots, u_n$ are linearly independent, we have $\mu_1 = \cdots = \mu_n = 0$. Therefore, $v_1, \ldots, v_m, u_1, \ldots, u_n$ are linearly independent, and hence form a basis of V. ■

Exercise 3E16

Suppose $\varphi \in \mathcal{L}(V, \mathbb{F})$ and $\varphi \neq 0$. Prove that $\dim V/(\operatorname{null} \varphi) = 1$.

Solution. From 3.107 in the text, we know that $V/\operatorname{null}\varphi \cong \operatorname{range}\varphi$. Since $\varphi \neq 0$, we have $\operatorname{range}\varphi \neq \{0\}$, and since $\operatorname{range}\varphi$ is a subspace of $\mathbb{F}$, it follows that $\dim\operatorname{range}\varphi = 1$. This implies $\operatorname{range}\varphi = \mathbb{F}$, and hence $\dim V/(\operatorname{null}\varphi) = \dim\operatorname{range}\varphi = 1$. ■

Exercise 3E17

Suppose U is a subspace of V such that $\dim V/U = 1$. Prove that there exists $\varphi \in \mathcal{L}(V, \mathbb{F})$ such that $\operatorname{null}\varphi = U$.

Solution. Since $\dim V/U = \dim \mathbb{F} = 1$, there exists some isomorphism $\psi \in \mathcal{L}(V/U, \mathbb{F})$. Let $\pi : V \to V/U$ be the quotient map. Define $\varphi = \psi \circ \pi \in \mathcal{L}(V, \mathbb{F})$. Suppose $\varphi(v) = 0$. Then $\psi(\pi(v)) = 0$, which implies $\pi(v) = 0$ since ψ is an isomorphism. Hence, $v \in U$, and we have $\operatorname{null}\varphi \subseteq U$. Conversely, if $v \in U$, then $\pi(v) = 0$, so $\varphi(v) = \psi(\pi(v)) = 0$. Therefore, $\operatorname{null}\varphi = U$. ■

Exercise 3E18

Suppose that U is a subspace of V such that V/U is finite-dimensional.

(a) Show that if W is a finite-dimensional subspace of V and $V = U + W$, then $\dim W \geq \dim V/U$.

(b) Prove that there exists a finite-dimensional subspace W of V such that $\dim W = \dim V/U$ and $V = U \oplus W$.

Solution.

(a) Suppose W is a finite-dimensional subspace of V and $V = U + W$. Let $\pi : V \to V/U$ be the quotient map, and consider the restriction $\pi|_W : W \to V/U$. To see that $\operatorname{range}(\pi|_W) = V/U$, note that for any $v + U \in V/U$, we can write $v = u + w$ for some $u \in U$ and $w \in W$. Then $\pi(v) = \pi(u + w) = \pi(u) + \pi(w) = 0 + \pi(w) = \pi(w)$, so $v + U = \pi(w) \in \operatorname{range}(\pi|_W)$. Hence, $\operatorname{range}(\pi|_W) = V/U$. From the fundamental theorem of linear maps, we have

$$\dim W = \dim\operatorname{null}\pi|_W + \dim\operatorname{range}(\pi|_W) = \dim\operatorname{null}\pi|_W + \dim V/U \geq \dim V/U.$$

(b) Let $v_1 + U, \dots, v_m + U$ be a basis of V/U. We claim that $v_1, \dots, v_m$ are linearly independent. Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ for some scalars $\lambda_1, \dots, \lambda_m$. Then

$$\lambda_1 v_1 + \cdots + \lambda_m v_m \in U.$$

But $\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0$ in V/U, and since $v_1 + U, \dots, v_m + U$ are linearly independent, it follows that $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $v_1, \dots, v_m$ are linearly independent.

Let $W = \operatorname{span}\{v_1, \dots, v_m\}$. Then $\dim W = m = \dim V/U$. For any $v \in V$, we can write

$$v + U = \lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U).$$

This implies that

$$v - (\lambda_1 v_1 + \cdots + \lambda_m v_m) \in U,$$

so $v \in U + W$. Hence, $V = U + W$. To see that $U \cap W = \{0\}$, let $w \in U \cap W$. Then $w \in W$, so $w = \lambda_1 v_1 + \cdots + \lambda_m v_m$ for some scalars $\lambda_1, \ldots, \lambda_m$. But $w \in U$, so $w + U = 0$ in V/U. Hence,

$$\lambda_1(v_1 + U) + \cdots + \lambda_m(v_m + U) = 0$$

in V/U. Since $v_1 + U, \ldots, v_m + U$ are linearly independent, it follows that $\lambda_1 = \cdots = \lambda_m = 0$. Hence, $w = 0$, and we have $U \cap W = \{0\}$. Therefore, $V = U \oplus W$. ■

Exercise 3E19

Suppose $T \in \mathcal{L}(V, W)$ and U is a subspace of V. Let π denote the quotient map from V onto V/U. Prove that there exists $S \in \mathcal{L}(V/U, W)$ such that $T = S \circ \pi$ if and only if $U \subseteq \operatorname{null} T$.

Solution. Suppose $S \in \mathcal{L}(V/U, W)$ is such that $T = S \circ \pi$. Then for any $u \in U$, we have $T(u) = S(\pi(u)) = S(0) = 0$, so $U \subseteq \operatorname{null} T$.

Conversely, suppose $U \subseteq \operatorname{null} T$. Define $S : V/U \to W$ by $S(v + U) = T(v)$. This is well-defined because if $v + U = v' + U$, then $v - v' \in U \subseteq \operatorname{null} T$, so $T(v) = T(v')$. It is straightforward to check that S is linear and that $T = S \circ \pi$. ■

3F Duality

Manual Remark. *For the rest of the solutions manual, unless otherwise stated, we will be using the Kronecker delta notation, i.e., we define $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.*

Exercise 3F1

Explain why each linear functional is surjective or is the zero map.

Solution. Let $\varphi \in V'$. If $\varphi = 0$, then it is the zero map. Otherwise, there exists $v \in V$ such that $\varphi(v) \neq 0$, say $\varphi(v) = c \neq 0$. Then for any $x \in \mathbb{F}$, we have

$$\varphi\left(\frac{x}{c}v\right) = \frac{x}{c}\varphi(v) = \frac{x}{c}c = x.$$

Hence, φ is surjective. ■

Exercise 3F2

Give three distinct examples of linear functionals on $\mathbb{R}^{[0,1]}$.

Solution. Consider the linear functionals on $\mathbb{R}^{[0,1]}$ defined by

$$\varphi_1(f) = f(0), \quad \varphi_2(f) = f(1), \quad \varphi_3(f) = f\left(\frac{1}{2}\right).$$ ■

Exercise 3F3

Suppose V is finite-dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

Solution. Since V is finite-dimensional, we can extend v to a basis of V, say $v, v_2, \ldots, v_n$. Define $\varphi \in V'$ by setting $\varphi(v) = 1$ and $\varphi(v_i) = 0$ for $i = 2, \ldots, n$. This defines a linear functional because it is defined on a basis and extended linearly. Clearly, $\varphi(v) = 1$. ■

Exercise 3F4

Suppose V is finite-dimensional and U is a subspace of V such that $U \neq V$. Prove that there exists $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

Solution. Since $U \neq V$, there exists $v_1 \in V$ such that $v_1 \notin U$. Extend a basis of U to a basis of V, say $u_1, \ldots, u_k, v_1, \ldots, v_m$. Define $\varphi \in V'$ by setting $\varphi(u_i) = 0$ for all $i = 1, \ldots, k$ and $\varphi(v_j) = 0$ for all $j = 2, \ldots, m$ but $\varphi(v_1) = 1$. This defines a linear functional because it is defined on a basis and extended linearly. Clearly, $\varphi(u) = 0$ for all $u \in U$ and $\varphi \neq 0$. ■

Exercise 3F5

Suppose $T \in \mathcal{L}(V, W)$ and $w_1, \ldots, w_m$ is a basis of range T. Hence for each $v \in V$, there exist unique numbers $\varphi_1(v), \ldots, \varphi_m(v)$ such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m,$$

thus defining functions $\varphi_1, \ldots, \varphi_m$ from V to $\mathbb{F}$. Show that each of the functions $\varphi_1, \ldots, \varphi_m$ is a linear functional on V.

Solution. Let $v \in V$. Since $w_1, \ldots, w_m$ is a basis of range T, there exist unique scalars $\varphi_1(v), \ldots, \varphi_m(v)$ such that

$$Tv = \varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m.$$

To show that each φ_i is a linear functional, let $v, u \in V$ and $a, b \in \mathbb{F}$. Then

$$T(av + bu) = aTv + bTu = a(\varphi_1(v)w_1 + \cdots + \varphi_m(v)w_m) + b(\varphi_1(u)w_1 + \cdots + \varphi_m(u)w_m).$$

By the uniqueness of the representation in terms of the basis $w_1, \ldots, w_m$, we must have

$$\varphi_i(av + bu) = a\varphi_i(v) + b\varphi_i(u) \quad \text{for each } i = 1, \ldots, m.$$

Hence, each φ_i is a linear functional on V. ■

Exercise 3F6

Suppose $\varphi, \beta \in V'$. Prove that $\operatorname{null} \varphi \subseteq \operatorname{null} \beta$ if and only if there exists $c \in \mathbb{F}$ such that $\beta = c\varphi$.

Solution. Suppose $\operatorname{null} \varphi \subseteq \operatorname{null} \beta$. If $\varphi = 0$, then $\operatorname{null} \varphi = V$, and hence $\operatorname{null} \beta = V$, which implies $\beta = 0$. In this case, we can take $c = 0$ and have $\beta = c\varphi$. Now assume $\varphi \neq 0$. Pick $v_0 \in V$ such that $\varphi(v_0) \neq 0$. Define

$$c = \frac{\beta(v_0)}{\varphi(v_0)}.$$

For any $v \in V$, consider

$$v - \frac{\varphi(v)}{\varphi(v_0)}v_0.$$

This vector is in $\operatorname{null} \varphi$, so it is also in $\operatorname{null} \beta$. Hence,

$$\beta\left(v - \frac{\varphi(v)}{\varphi(v_0)}v_0\right) = 0 \implies \beta(v) = \frac{\varphi(v)}{\varphi(v_0)}\beta(v_0) = c\varphi(v).$$

Therefore, $\beta = c\varphi$.

Conversely, if $\beta = c\varphi$, then clearly $\operatorname{null} \varphi \subseteq \operatorname{null} \beta$ because if $\varphi(v) = 0$, then $\beta(v) = c\varphi(v) = 0$. ■

Exercise 3F7

Suppose that $V_1, \ldots, V_m$ are vector spaces. Prove that $(V_1 \times \cdots \times V_m)'$ and $V_1' \times \cdots \times V_m'$ are isomorphic vector spaces.

Solution. Define a map $\Phi : (V_1 \times \cdots \times V_m)' \to V_1' \times \cdots \times V_m'$ by

$$\Phi(\varphi) = (\varphi_1, \ldots, \varphi_m),$$

where $\varphi_i \in V_i'$ is defined by $\varphi_i(v_i) = \varphi(0, \ldots, 0, v_i, 0, \ldots, 0)$, with v_i in the ith position.

First, we show that Φ is linear. Let $\varphi, \psi \in (V_1 \times \cdots \times V_m)'$ and $a, b \in \mathbb{F}$. Then

$$\Phi(a\varphi + b\psi) = ((a\varphi + b\psi)_1, \ldots, (a\varphi + b\psi)_m) = (a\varphi_1 + b\psi_1, \ldots, a\varphi_m + b\psi_m) = a\Phi(\varphi) + b\Phi(\psi).$$

Next, we show that Φ is bijective. For injectivity, suppose $\Phi(\varphi) = 0$. Then $\varphi_i = 0$ for all i, which implies $\varphi(0, \ldots, 0, v_i, 0, \ldots, 0) = 0$ for all $v_i \in V_i$. By linearity, φ must be zero on all of $V_1 \times \cdots \times V_m$, so $\varphi = 0$. For surjectivity, given $(\varphi_1, \ldots, \varphi_m) \in V_1' \times \cdots \times V_m'$, define $\varphi \in (V_1 \times \cdots \times V_m)'$ by

$$\varphi(v_1, \ldots, v_m) = \varphi_1(v_1) + \cdots + \varphi_m(v_m),$$

which is linear by linearity of each φ_i. Then $\Phi(\varphi) = (\varphi_1, \ldots, \varphi_m)$, so Φ is surjective.

Therefore, Φ is a linear isomorphism, and $(V_1 \times \cdots \times V_m)'$ and $V_1' \times \cdots \times V_m'$ are isomorphic vector spaces. ■

Exercise 3F8

Suppose $v_1, \ldots, v_n$ is a basis of V and $\varphi_1, \ldots, \varphi_n$ is the dual basis of V'. Define $\Gamma : V \to \mathbb{F}^n$ and $\Lambda : \mathbb{F}^n \to V$ by

$$\Gamma(v) = (\varphi_1(v), \ldots, \varphi_n(v)) \quad \text{and} \quad \Lambda(a_1, \ldots, a_n) = a_1 v_1 + \cdots + a_n v_n.$$

Explain why Γ and Λ are inverses of each other.

Solution. To show that Γ and Λ are inverses of each other, we need to check that $\Gamma(\Lambda(a_1, \ldots, a_n)) = (a_1, \ldots, a_n)$ for all $(a_1, \ldots, a_n) \in \mathbb{F}^n$ and $\Lambda(\Gamma(v)) = v$ for all $v \in V$.

First, let $(a_1, \ldots, a_n) \in \mathbb{F}^n$. Then

$$\Gamma(\Lambda(a_1, \ldots, a_n)) = \Gamma(a_1 v_1 + \cdots + a_n v_n) = (\varphi_1(a_1 v_1 + \cdots + a_n v_n), \ldots, \varphi_n(a_1 v_1 + \cdots + a_n v_n)).$$

By the definition of the dual basis, we have $\varphi_i(v_j) = \delta_{ij}$ for all i, j. Therefore,

$$\varphi_i(a_1 v_1 + \cdots + a_n v_n) = a_i.$$

Therefore,

$$\Gamma(\Lambda(a_1, \ldots, a_n)) = (a_1, \ldots, a_n).$$

Next, let $v \in V$. Write $v = b_1 v_1 + \cdots + b_n v_n$ for some scalars $b_1, \ldots, b_n$. Then

$$\Lambda(\Gamma(v)) = \Lambda(\varphi_1(v), \ldots, \varphi_n(v)) = \Lambda(b_1, \ldots, b_n) = b_1 v_1 + \cdots + b_n v_n = v.$$

Hence, Γ and Λ are inverses of each other. ■

Exercise 3F9

Suppose m is a positive integer. Show that the dual basis of the basis $1, x, \ldots, x^m$ of $\mathcal{P}_m(\mathbb{R})$ is $\varphi_0, \varphi_1, \ldots, \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!}.$$

Axler Note. *Here $p^{(k)}$ denotes the kth derivative of p, with the understanding that the 0th derivative of p is p.*

Solution. Let $p \in \mathcal{P}_m(\mathbb{R})$. We can write $p(x) = a_0 + a_1 x + \cdots + a_m x^m$ for some scalars $a_0, \ldots, a_m$. Then for each $k = 0, 1, \ldots, m$, we have

$$\begin{aligned}
p(x) &= a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m \\
p'(x) &= a_1 + 2a_2 x + \cdots + m a_m x^{m-1} \\
p''(x) &= 2a_2 + \cdots + m(m-1) a_m x^{m-2} \\
&\vdots \\
p^{(k)}(x) &= k! a_k + \cdots + \frac{m!}{(m-k)!} a_m x^{m-k}
\end{aligned}$$

Then

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!} = a_k.$$

Therefore, $\varphi_k(x^j) = \delta_{kj}$ for all $0 \le k, j \le m$, which shows that $\varphi_0, \varphi_1, \ldots, \varphi_m$ is indeed the dual basis of $1, x, \ldots, x^m$. ■

Exercise 3F10

Suppose m is a positive integer.

(a) Show that $1, x-5, \ldots, (x-5)^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$.

(b) What is the dual basis of the basis in (a)?

Solution.

(a) Note that $1, x-5, \ldots, (x-5)^m$ is a list of $m+1$ polynomials in $\mathcal{P}_m(\mathbb{R})$ with differing degrees, and since $\mathcal{P}_m(\mathbb{R})$ has dimension $m+1$, this list forms a basis of $\mathcal{P}_m(\mathbb{R})$.

(b) Substitute $y = x - 5$. Then the basis $1, x-5, \ldots, (x-5)^m$ becomes $1, y, \ldots, y^m$. By Exercise 3F9, the dual basis of $1, y, \ldots, y^m$ is $\varphi_0, \varphi_1, \ldots, \varphi_m$, where

$$\varphi_k(p) = \frac{p^{(k)}(0)}{k!}.$$

Since $y = x - 5$, then

$$\varphi_k(p) = \frac{p^{(k)}(5)}{k!}.$$ ■

Exercise 3F11

Suppose $v_1, \ldots, v_n$ is a basis of V and $\varphi_1, \ldots, \varphi_n$ is the corresponding dual basis of V'. Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$

Solution. Let $v \in V$. Since $\varphi_1, \ldots, \varphi_n$ is the dual basis of $v_1, \ldots, v_n$, we have $\varphi_i(v_j) = \delta_{ij}$ for all $1 \le i, j \le n$. Let $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ for some scalars $\alpha_1, \ldots, \alpha_n$. Then

$$\begin{aligned}(\psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n)(v) &= \psi(v_1)\varphi_1(v) + \cdots + \psi(v_n)\varphi_n(v)\\ &= \psi(v_1)\alpha_1 + \cdots + \psi(v_n)\alpha_n\\ &= \psi(\alpha_1 v_1 + \cdots + \alpha_n v_n)\\ &= \psi(v)\end{aligned}$$

Since this holds for all $v \in V$, we conclude that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$ ■

Exercise 3F12

Suppose $S, T \in \mathcal{L}(V, W)$.

(a) Prove that $(S + T)' = S' + T'$.

(b) Prove that $(\lambda T)' = \lambda T'$ for all $\lambda \in \mathbb{F}$.

Solution.

(a) Let $\psi \in W'$. Then for any $v \in V$, we have

$$\begin{aligned}((S+T)'(\psi))(v) &= \psi((S+T)(v))\\ &= \psi(S(v) + T(v))\\ &= \psi(S(v)) + \psi(T(v))\\ &= (S'(\psi))(v) + (T'(\psi))(v)\\ &= (S'(\psi) + T'(\psi))(v).\end{aligned}$$

Since this holds for all $v \in V$, we conclude that

$$(S+T)'(\psi) = S'(\psi) + T'(\psi).$$

Since this holds for all $\psi \in W'$, we have

$$(S+T)' = S' + T'.$$

(b) Let $\psi \in W'$. Then for any $v \in V$, we have

$$\begin{aligned}((\lambda T)'(\psi))(v) &= \psi((\lambda T)(v))\\ &= \psi(\lambda T(v))\\ &= \lambda\psi(T(v))\\ &= (\lambda T'(\psi))(v).\end{aligned}$$

Since this holds for all $v \in V$, we conclude that

$$(\lambda T)'(\psi) = \lambda T'(\psi).$$

Since this holds for all $\psi \in W'$, we have

$$(\lambda T)' = \lambda T'.$$

■

Exercise 3F13

Show that the dual map of the identity operator on V is the identity operator on V'.

Solution. Let $\psi \in V'$. Let I' denote the dual map of the identity operator I on V. Then for any $v \in V$, we have

$$\begin{aligned}(I'(\psi))(v) &= \psi(I(v)) \\ &= \psi(v).\end{aligned}$$

Since this holds for all $v \in V$, we conclude that

$$I'(\psi) = \psi.$$

Since this holds for all $\psi \in V'$, then I' must be the identity operator on V'. ■

Exercise 3F14

Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose φ_1, φ_2 denotes the dual basis of the standard basis of $\mathbb{R}^2$ and ψ_1, ψ_2, ψ_3 denotes the dual basis of the standard basis of $\mathbb{R}^3$.

(a) Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

(b) Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as linear combinations of ψ_1, ψ_2, ψ_3.

Solution.

(a) Let $(x, y, z) \in \mathbb{R}^3$. Then we have

$$\begin{aligned}(T'(\varphi_1))((x, y, z)) &= \varphi_1(T(x, y, z)) \\ &= \varphi_1(4x + 5y + 6z, 7x + 8y + 9z) \\ &= 4x + 5y + 6z, \\ (T'(\varphi_2))((x, y, z)) &= \varphi_2(T(x, y, z)) \\ &= \varphi_2(4x + 5y + 6z, 7x + 8y + 9z) \\ &= 7x + 8y + 9z.\end{aligned}$$

(b) We can write $T'(\varphi_1)$ and $T'(\varphi_2)$ as linear combinations of ψ_1, ψ_2, ψ_3 as follows:

$$\begin{aligned}T'(\varphi_1) &= 4\psi_1 + 5\psi_2 + 6\psi_3, \\ T'(\varphi_2) &= 7\psi_1 + 8\psi_2 + 9\psi_3.\end{aligned}$$

■

Exercise 3F15

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by

$$(Tp)(x) = x^2 p(x) + p''(x)$$

for each $x \in \mathbb{R}$.

(a) Suppose $\varphi \in \mathcal{P}(\mathbb{R})'$ is defined by $\varphi(p) = p'(4)$. Describe the linear functional $T'(\varphi)$ on $\mathcal{P}(\mathbb{R})$.

(b) Suppose $\varphi \in \mathcal{P}(\mathbb{R})'$ is defined by

$$\varphi(p) = \int_0^1 p.$$

Evaluate $(T'(\varphi))(x^3)$.

Solution.

(a) Let $p \in \mathcal{P}(\mathbb{R})$. Then we have

$$\begin{aligned}(T'(\varphi))(p) &= \varphi(T(p)) \\ &= (Tp)'(4) \\ &= (2xp(x) + x^2 p'(x) + p'''(x))\big|_{x=4}. \\ &= 8p(4) + 16p'(4) + p'''(4).\end{aligned}$$

(b) Let $p(x) = x^3$. Then we have

$$\begin{aligned}(T'(\varphi))(x^3) &= \varphi(T(x^3)) \\ &= \int_0^1 T(x^3)\,\mathrm{d}x \\ &= \int_0^1 (x^2 \cdot x^3 + (x^3)'')\,\mathrm{d}x \\ &= \int_0^1 (x^5 + 6x)\,\mathrm{d}x \\ &= \frac{1}{6} + \frac{18}{6} \\ &= \frac{19}{6}.\end{aligned}$$

■

Exercise 3F16

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T' = 0 \iff T = 0.$$

Solution. Suppose $T' = 0$. Then for every $\varphi \in W'$ and every $v \in V$, we have

$$(T'(\varphi))(v) = \varphi(T(v)) = 0.$$

If $T(v) \neq 0$ for some $v \in V$, then $T(v) \in W$. Since W is finite-dimensional, then by Exercise 3F3, there exists $\psi \in W'$ such that $\psi(T(v)) \neq 0$, which contradicts the fact that $(T'(\psi))(v) = \psi(T(v)) = 0$. Hence, $T(v) = 0$ for every $v \in V$, and so $T = 0$.

Conversely, suppose $T = 0$. Then for every $\varphi \in W'$ and every $v \in V$, we have

$$(T'(\varphi))(v) = \varphi(T(v)) = \varphi(0) = 0.$$

Hence, $T' = 0$. ■

Exercise 3F17

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is invertible if and only if $T' \in \mathcal{L}(W', V')$ is invertible.

Solution. Suppose T is invertible. Then there exists $S \in \mathcal{L}(W, V)$ such that $ST = I_V$ and $TS = I_W$. Taking the dual of both sides, we get $T'S' = I_{W'}$ and $S'T' = I_{V'}$, which shows that T' is invertible with inverse S'.

Conversely, suppose T' is invertible. Then T' is an isomorphism. Then

$$\dim V = \dim V' = \dim W' = \dim W.$$

By injectivity of T', we have $\operatorname{null} T' = \{0\}$. Since $\operatorname{null} T' = (\operatorname{range} T)^0$, we have $(\operatorname{range} T)^0 = \{0\}$ so that $\operatorname{range} T = W$. Hence, T is surjective. Since $\dim V = \dim W$, we conclude that T is injective as well. Therefore, T is invertible. ■

Exercise 3F18

Suppose V and W are finite-dimensional. Prove that the map that takes $T \in \mathcal{L}(V, W)$ to $T' \in \mathcal{L}(W', V')$ is an isomorphism of $\mathcal{L}(V, W)$ onto $\mathcal{L}(W', V')$.

Solution. Let $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(W', V')$ be the map defined by $\Phi(T) = T'$. We need to show that Φ is linear, injective, and surjective.

Linearity: For $T_1, T_2 \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, we have

$$\Phi(T_1 + T_2) = (T_1 + T_2)' = T_1' + T_2' = \Phi(T_1) + \Phi(T_2),$$

and

$$\Phi(aT_1) = (aT_1)' = aT_1' = a\Phi(T_1).$$

Hence, Φ is linear.

To see that Φ is injective, suppose $\Phi(T) = T' = 0$. By Exercise 3F16, this implies $T = 0$. Hence, Φ is injective.

To see that Φ is surjective, note that since V and W are finite-dimensional, $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W = \dim W' \cdot \dim V' = \dim \mathcal{L}(W', V')$. A linear map between finite-dimensional vector spaces of the same dimension that is injective is also surjective. Hence, Φ is surjective and therefore an isomorphism. ■

Exercise 3F19

Suppose $U \subseteq V$. Explain why

$$U^0 = \{\varphi \in V' \mid U \subseteq \operatorname{null} \varphi\}.$$

Solution. By definition, the annihilator U^0 of U is the set of all linear functionals $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$. Equivalently, this means that $U \subseteq \operatorname{null} \varphi$. ∎

Exercise 3F20

Suppose V is finite-dimensional and U is a subspace of V. Show that

$$U = \{v \in V \mid \varphi(v) = 0 \text{ for every } \varphi \in U^0\}.$$

Manual Remark. *This says that $U = (U^0)^0$.*

Solution. Let $S = \{v \in V \mid \varphi(v) = 0 \text{ for every } \varphi \in U^0\}$. We need to show that $S = U$.

First, if $v \in U$, then for any $\varphi \in U^0$, we have $\varphi(v) = 0$ by definition of the annihilator. Hence, $v \in S$, so $U \subseteq S$.

Conversely, suppose $v \in S$. If $v \notin U$, then by the construction done in Exercise 3F4 with $v_1 = v$, there exists a linear functional $\varphi \in V'$ such that $\varphi(u) = 0$ for all $u \in U$ and $\varphi(v) \neq 0$. But this φ is in U^0, which contradicts $v \in S$. Hence, $v \in U$, and therefore $S \subseteq U$.

Combining both inclusions, we get $S = U$, as desired. ∎

Exercise 3F21

Suppose V is finite-dimensional and U and W are subspaces of V.

(a) Prove that $W^0 \subseteq U^0$ if and only if $U \subseteq W$.

(b) Prove that $W^0 = U^0$ if and only if $U = W$.

Solution.

(a) Suppose $W^0 \subseteq U^0$. If $u \in U$, then for any $\varphi \in W^0$, we have $\varphi(u) = 0$ because $\varphi \in U^0$. Hence, by Exercise 3F20, we have $u \in W$, so $U \subseteq W$. Conversely, if $U \subseteq W$ and $\varphi \in W^0$, then $\varphi(w) = 0$ for all $w \in W$, and in particular for all $u \in U$. Hence, $\varphi \in U^0$, so $W^0 \subseteq U^0$.

(b) Suppose $W^0 = U^0$. Then $W^0 \subseteq U^0$ and $U^0 \subseteq W^0$. By part (a), this implies that $U \subseteq W$ and $W \subseteq U$. Hence, $U = W$. Conversely, if $U = W$, then clearly $U^0 = W^0$. ∎

Exercise 3F22

Suppose V is finite-dimensional and U and W are subspaces of V.

(a) Show that $(U + W)^0 = U^0 \cap W^0$.

(b) Show that $(U \cap W)^0 = U^0 + W^0$.

Solution.

(a) Let $\varphi \in (U + W)^0$. Then for any $u \in U$ and $w \in W$, we have $\varphi(u + w) = 0$. In particular, taking $w = 0$ gives $\varphi(u) = 0$ for all $u \in U$, so $\varphi \in U^0$. Similarly, taking $u = 0$ gives $\varphi(w) = 0$ for all $w \in W$, so $\varphi \in W^0$. Hence, $\varphi \in U^0 \cap W^0$, and we have shown that $(U + W)^0 \subseteq U^0 \cap W^0$.

Conversely, let $\varphi \in U^0 \cap W^0$. Then for any $u \in U$ and $w \in W$, we have $\varphi(u) = 0$ and $\varphi(w) = 0$, so $\varphi(u + w) = \varphi(u) + \varphi(w) = 0$. Hence, $\varphi \in (U + W)^0$, and we have shown that $U^0 \cap W^0 \subseteq (U + W)^0$. Combining both inclusions, we get $(U + W)^0 = U^0 \cap W^0$.

(b) From part (a), we know that $(U+W)^0 = U^0 \cap W^0$. Then

$$\dim(U+W)^0 = \dim V - \dim(U+W).$$

Applying the dimension formula for a sum of subspaces, we have

$$\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0).$$

Since $(U+W)^0 = U^0 \cap W^0$, we have $\dim(U^0 \cap W^0) = \dim(U+W)^0 = \dim V - \dim(U+W)$. Therefore,

$$\dim(U^0 + W^0) = \dim U^0 + \dim W^0 - (\dim V - \dim(U+W)).$$

Using the fact that $\dim U^0 = \dim V - \dim U$ and $\dim W^0 = \dim V - \dim W$, we get

$$\begin{aligned}\dim(U^0 + W^0) &= \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0)\\ &= (\dim V - \dim U) + (\dim V - \dim W) - (\dim V - \dim(U+W))\\ &= \dim V - \dim U + \dim V - \dim W - \dim V + \dim(U+W)\\ &= \dim V - (\dim U + \dim W - \dim(U+W))\\ &= \dim V - \dim(U \cap W).\end{aligned}$$

Hence, $\dim(U^0 + W^0) = \dim(U \cap W)^0$. To see equality, we just need to prove one inclusion. If $\varphi = \varphi_U + \varphi_W \in U^0 + W^0$, then for any $v \in U \cap W$, we have $\varphi(v) = \varphi_U(v) + \varphi_W(v) = 0 + 0 = 0$, so $\varphi \in (U \cap W)^0$. Therefore, $U^0 + W^0 \subseteq (U \cap W)^0$. Since the dimensions are equal, we conclude that $(U \cap W)^0 = U^0 + W^0$. ■

Exercise 3F23

Suppose V is finite-dimensional and $\varphi_1, \ldots, \varphi_m \in V'$. Prove that the following three sets are equal to each other.

(a) $\operatorname{span}(\varphi_1, \ldots, \varphi_m)$

(b) $((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0$

(c) $\{\varphi \in V' \mid (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m) \subseteq \operatorname{null} \varphi\}$

Solution. For ease of notation, let A, B, C be the sets in the exercise, i.e.,

$$\begin{aligned}A &= \operatorname{span}(\varphi_1, \ldots, \varphi_m),\\ B &= ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0,\\ C &= \{\varphi \in V' \mid (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m) \subseteq \operatorname{null} \varphi\}.\end{aligned}$$

We will show that $A = B$ and $B = C$. Starting off with $B = C$, note that by definition,

$$\begin{aligned}B &= ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0\\ &= \{\varphi \in V' \mid (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m) \subseteq \operatorname{null} \varphi\} = C,\end{aligned}$$

where the reformulation of the annihilator subspace was shown in Exercise 3F19.

To show that $A = B$, we may use Exercise 3F22 (b) repeatedly to rewrite B as

$$B = (\operatorname{null} \varphi_1)^0 + \cdots + (\operatorname{null} \varphi_m)^0.$$

By Exercise 3F6, we have $(\operatorname{null} \varphi_i)^0 = \operatorname{span}(\varphi_i)$ for each $i = 1, \ldots, m$. Therefore,

$$B = (\operatorname{null} \varphi_1)^0 + \cdots + (\operatorname{null} \varphi_m)^0 = \operatorname{span}(\varphi_1) + \cdots + \operatorname{span}(\varphi_m) = \operatorname{span}(\varphi_1, \ldots, \varphi_m) = A.$$

We conclude that $A = B$. ■

Exercise 3F24

Suppose V is finite-dimensional and $v_1, \ldots, v_m \in V$. Define a linear map $\Gamma : V' \to \mathbb{F}^m$ by $\Gamma(\varphi) = (\varphi(v_1), \ldots, \varphi(v_m))$.

(a) Prove that $v_1, \ldots, v_m$ spans V if and only if Γ is injective.
(b) Prove that $v_1, \ldots, v_m$ is linearly independent if and only if Γ is surjective.

Solution.

(a) Suppose $v_1, \ldots, v_m$ spans V. If $\Gamma(\varphi) = 0$, then $\varphi(v_i) = 0$ for all $i = 1, \ldots, m$. Since $v_1, \ldots, v_m$ spans V, it follows that $\varphi(v) = 0$ for all $v \in V$, i.e., $\varphi = 0$. Hence, Γ is injective.

Conversely, suppose Γ is injective. If $v_1, \ldots, v_m$ does not span V, then by Exercise 3F4, there exists a nonzero $\varphi \in V'$ that vanishes on $v_1, \ldots, v_m$. But then $\Gamma(\varphi) = 0$, contradicting the injectivity of Γ. Hence, $v_1, \ldots, v_m$ must span V.

(b) Suppose $v_1, \ldots, v_m$ is linearly independent. Extend this list to a basis $v_1, \ldots, v_m, v_{m+1}, \ldots, v_n$ of V. Define linear functionals $\varphi_1, \ldots, \varphi_m \in V'$ by setting $\varphi_i(v_j) = \delta_{ij}$ for $i, j = 1, \ldots, m$ and $\varphi_i(v_j) = 0$ for $j = m+1, \ldots, n$. Then for any $(a_1, \ldots, a_m) \in \mathbb{F}^m$, the linear functional $\varphi = a_1\varphi_1 + \cdots + a_m\varphi_m$ satisfies $\Gamma(\varphi) = (a_1, \ldots, a_m)$. Hence, Γ is surjective.

Conversely, suppose Γ is surjective. If $v_1, \ldots, v_m$ is not linearly independent, then there exist scalars $a_1, \ldots, a_m$, not all zero, such that $a_1v_1 + \cdots + a_mv_m = 0$. Consider some index j such that $a_j \neq 0$. Since Γ is surjective, there exists $\varphi \in V'$ such that $\varphi(v_j) = 1$ and $\varphi(v_i) = 0$ for all $i \neq j$. Then

$$0 = \varphi(a_1v_1 + \cdots + a_mv_m) = a_1\varphi(v_1) + \cdots + a_m\varphi(v_m) = a_j \neq 0,$$

a contradiction. Hence, $v_1, \ldots, v_m$ must be linearly independent. ■

Exercise 3F25

Suppose V is finite-dimensional and $\varphi_1, \ldots, \varphi_m \in V'$. Define a linear map $\Gamma : V \to \mathbb{F}^m$ by $\Gamma(v) = (\varphi_1(v), \ldots, \varphi_m(v))$.

(a) Prove that $\varphi_1, \ldots, \varphi_m$ spans V' if and only if Γ is injective.
(b) Prove that $\varphi_1, \ldots, \varphi_m$ is linearly independent if and only if Γ is surjective.

Solution.

(a) Suppose $\varphi_1, \ldots, \varphi_m$ spans V'. If $\Gamma(v) = 0$, then $\varphi_i(v) = 0$ for all $i = 1, \ldots, m$. Since $\varphi_1, \ldots, \varphi_m$ spans V', then by Exercise 3F3, it follows that $\psi(v) = 0$ for all $\psi \in V'$, i.e., $v = 0$. Hence, Γ is injective.

Conversely, suppose Γ is injective. Then $\operatorname{null}\Gamma = \{0\}$. By definition, we have

$$\operatorname{null}\Gamma = \operatorname{null}\varphi_1 \cap \cdots \cap \operatorname{null}\varphi_m.$$

By Exercise 3F23, we have

$$\operatorname{span}(\varphi_1, \ldots, \varphi_m) = ((\operatorname{null}\varphi_1) \cap \cdots \cap (\operatorname{null}\varphi_m))^0 = (\operatorname{null}\Gamma)^0 = \{\psi \in V' \mid \operatorname{null}\Gamma \subseteq \operatorname{null}\psi\} = V'.$$

Hence, $\varphi_1, \ldots, \varphi_m$ spans V'.

(b) Suppose $\varphi_1, \ldots, \varphi_m$ is linearly independent. Then $\dim\operatorname{span}(\varphi_1, \ldots, \varphi_m) = m$. By the fundamental theorem of linear maps and Exercise 3F23, we have

$$\dim\operatorname{range}\Gamma = \dim V - \dim\operatorname{null}\Gamma = \dim V - \dim((\operatorname{null}\varphi_1) \cap \cdots \cap (\operatorname{null}\varphi_m)) = m.$$

Since range Γ is a subspace of $\mathbb{F}^m$ and has dimension m, we conclude that range $\Gamma = \mathbb{F}^m$, so Γ is surjective.

Conversely, suppose Γ is surjective. Then range $\Gamma = \mathbb{F}^m$, so $\dim \operatorname{range} \Gamma = m$. By the fundamental theorem of linear maps, we have

$$\dim \operatorname{range} \Gamma = \dim V - \dim \operatorname{null} \Gamma = \dim V - \dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)).$$

By Exercise 3F23, we have

$$\operatorname{span}(\varphi_1, \ldots, \varphi_m) = ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0,$$

so $\dim \operatorname{span}(\varphi_1, \ldots, \varphi_m) = \dim V - \dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)) = m$. Hence, $\varphi_1, \ldots, \varphi_m$ is linearly independent. ■

Exercise 3F26

Suppose V is finite-dimensional and Ω is a subspace of V'. Prove that

$$\Omega = \{v \in V \mid \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}^0.$$

Manual Remark. *This says* $\Omega = (\Omega^0)^0$.

Solution. Let $S = \{v \in V \mid \varphi(v) = 0 \text{ for every } \varphi \in \Omega\}$. We need to show that $\Omega = S^0$. Let $\varphi_1, \ldots, \varphi_m$ be a basis of Ω. Then $S = (\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)$. By Exercise 3F23, we have

$$\Omega = \operatorname{span}(\varphi_1, \ldots, \varphi_m) = ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0 = S^0.$$ ■

Exercise 3F27

Suppose $T \in \mathcal{L}(\mathcal{P}_5(\mathbb{R}))$ and $\operatorname{null} T' = \operatorname{span}(\varphi)$, where φ is the linear functional on $\mathcal{P}_5(\mathbb{R})$ defined by $\varphi(p) = p(8)$. Prove that

$$\operatorname{range} T = \{p \in \mathcal{P}_5(\mathbb{R}) \mid p(8) = 0\}.$$

Solution. Let $W = \{p \in \mathcal{P}_5(\mathbb{R}) \mid p(8) = 0\}$. We need to show that range $T = W$.

First, if $p \in \operatorname{range} T$, then there exists $q \in \mathcal{P}_5(\mathbb{R})$ such that $T(q) = p$. Since $\varphi \in \operatorname{null} T'$, we have

$$\varphi(T(q)) = (T'(\varphi))(q) = 0.$$

But $\varphi(T(q)) = T(q)(8) = p(8)$, so $p(8) = 0$. Hence, $p \in W$, and we have shown that range $T \subseteq W$.

Now we compare the dimensions of range T and W. Recall that $(\operatorname{range} T)^0 = \operatorname{null} T' = \operatorname{span}(\varphi)$, so $\dim(\operatorname{range} T)^0 = 1$. Since $\mathcal{P}_5(\mathbb{R})$ has dimension 6, we have $\dim \operatorname{range} T = 6 - 1 = 5$. On the other hand, W is the set of all polynomials in $\mathcal{P}_5(\mathbb{R})$ that vanish at 8. Using the fundamental theorem of linear maps and the fact that $\varphi \neq 0$ is surjective by Exercise 3F1, we have $\dim W = \dim \mathcal{P}_5(\mathbb{R}) - 1 = 5$. Hence, range T and W are subspaces of $\mathcal{P}_5(\mathbb{R})$ with the same dimension, and range $T \subseteq W$. We conclude that range $T = W$, as desired. ■

Exercise 3F28

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_m$ is a linearly independent list in V'. Prove that

$$\dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)) = (\dim V) - m.$$

Solution. Since $\varphi_1, \dots, \varphi_m$ is linearly independent, we have $\dim \operatorname{span}(\varphi_1, \dots, \varphi_m) = m$. By Exercise 3F23, we have

$$\operatorname{span}(\varphi_1, \dots, \varphi_m) = ((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0.$$

Hence, $\dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0 = m$. Since V is finite-dimensional, we have

$$\dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m)) = \dim V - \dim((\operatorname{null} \varphi_1) \cap \cdots \cap (\operatorname{null} \varphi_m))^0 = \dim V - m,$$

as desired. ■

Exercise 3F29

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$.

(a) Prove that if $\varphi \in W'$ and $\operatorname{null} T' = \operatorname{span}(\varphi)$, then $\operatorname{range} T = \operatorname{null} \varphi$.

(b) Prove that if $\psi \in V'$ and $\operatorname{range} T' = \operatorname{span}(\psi)$, then $\operatorname{null} T = \operatorname{null} \psi$.

Solution.

(a) From $\operatorname{null} T' = \operatorname{span}(\varphi)$ and $\operatorname{null} T' = (\operatorname{range} T)^0$, we have by Exercise 3F20 that

$$\operatorname{range} T = ((\operatorname{range} T)^0)^0 = (\operatorname{null} T')^0 = (\operatorname{span}(\varphi))^0 = \operatorname{null} \varphi,$$

where the last equality follows as such:

$$v \in \operatorname{null} \varphi \iff \varphi(v) = 0 \iff \alpha\varphi(v) = 0 \text{ for all } \alpha \in \mathbb{F} \iff v \in (\operatorname{span}(\varphi))^0.$$

(b) Note that $\operatorname{range} T' = \operatorname{span}(\psi)$ and $\operatorname{range} T' = (\operatorname{null} T)^0$. Then $(\operatorname{null} T)^0 = \operatorname{span}(\psi)$. By Exercise 3F20, we have

$$\operatorname{null} T = ((\operatorname{null} T)^0)^0 = (\operatorname{span}(\psi))^0 = \operatorname{null} \psi,$$

where the last equality follows as such:

$$v \in \operatorname{null} \psi \iff \psi(v) = 0 \iff \alpha\psi(v) = 0 \text{ for all } \alpha \in \mathbb{F} \iff v \in (\operatorname{span}(\psi))^0.$$

■

Exercise 3F30

Suppose V is finite-dimensional and $\varphi_1, \dots, \varphi_n$ is a basis of V'. Show that there exists a basis of V whose dual basis is $\varphi_1, \dots, \varphi_n$.

Solution. Let $\varphi_1, \dots, \varphi_n$ be a basis of V'. From Exercise 3F25, there exists a map $\Gamma : V \to \mathbb{F}^n$ that is an isomorphism, since $\varphi_1, \dots, \varphi_n$ is a basis for V'. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{F}^n$. Since Γ

is an isomorphism, there exist $v_1, \ldots, v_n \in V$ such that $\Gamma(v_i) = e_i$ for each $i = 1, \ldots, n$. We claim that $v_1, \ldots, v_n$ is a basis of V whose dual basis is $\varphi_1, \ldots, \varphi_n$.

To see that $v_1, \ldots, v_n$ is a basis of V, we show linear independence. Suppose $a_1, \ldots, a_n$ are scalars such that $a_1 v_1 + \cdots + a_n v_n = 0$. Applying Γ to both sides gives

$$a_1 e_1 + \cdots + a_n e_n = \Gamma(a_1 v_1 + \cdots + a_n v_n) = \Gamma(0) = 0.$$

Since $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$, it follows that $a_1 = \cdots = a_n = 0$. Hence, $v_1, \ldots, v_n$ is linearly independent. Since $\dim V = n$, we conclude that $v_1, \ldots, v_n$ is a basis of V. To see that the dual basis of $v_1, \ldots, v_n$ is $\varphi_1, \ldots, \varphi_n$, we need to show that $\varphi_i(v_j) = \delta_{ij}$ for each $i, j = 1, \ldots, n$. Since $\Gamma(v_j) = e_j$, we have

$$\varphi_i(v_j) = (\Gamma(v_j))_i = (e_j)_i = \delta_{ij},$$

as desired. ■

Exercise 3F31

Suppose U is a subspace of V. Let $i : U \to V$ be the inclusion map defined by $i(u) = u$. Thus, $i' \in \mathcal{L}(V', U')$.

(a) Show that $\operatorname{null} i' = U^0$.

(b) Prove that if V is finite-dimensional, then $\operatorname{range} i' = U'$.

(c) Prove that if V is finite-dimensional, then i' is an isomorphism from V'/U^0 onto U'.

Solution.

(a) Note that if $\varphi \in \operatorname{null} i'$, then $i'(\varphi) = 0$, so $\varphi(i(u)) = 0$ for all $u \in U$. Since $i(u) = u$, we have $\varphi(u) = 0$ for all $u \in U$, so $\varphi \in U^0$. Hence, $\operatorname{null} i' \subseteq U^0$. Conversely, if $\varphi \in U^0$, then $\varphi(u) = 0$ for all $u \in U$, so $i'(\varphi) = 0$. Hence, $U^0 \subseteq \operatorname{null} i'$, and we conclude that $\operatorname{null} i' = U^0$.

(b) To see $\operatorname{range} i' \subseteq U'$, note that if $\varphi \in V'$, then $i'(\varphi) \in U'$ since $i'(\varphi)(u) = \varphi(i(u)) = \varphi(u)$ for all $u \in U$. Hence, $\operatorname{range} i' \subseteq U'$.

Since V is finite-dimensional, we have $\dim V' = \dim V$. By the fundamental theorem of linear maps, we have

$$\dim \operatorname{range} i' = \dim V' - \dim \operatorname{null} i' = \dim V - \dim U^0.$$

By the annihilator dimension formula, we have $\dim U^0 = \dim V - \dim U$. Hence,

$$\dim \operatorname{range} i' = \dim V - (\dim V - \dim U) = \dim U = \dim U'.$$

Since $\operatorname{range} i'$ is a subspace of U' and has the same dimension, we conclude that $\operatorname{range} i' = U'$, as desired.

(c) From part (a), we have $\operatorname{null} i' = U^0$. Then we know that V'/U^0 is isomorphic to $\operatorname{range} i'$. Then the induced map $\tilde{i}' : V'/U^0 \to \operatorname{range} i'$ defined by $\tilde{i}'(\varphi + U^0) = i'(\varphi)$ is an isomorphism. Since $\operatorname{range} i' = U'$ by part (b), we conclude that i' induces an isomorphism from V'/U^0 onto U', as desired. ■

Exercise 3F32

The **double dual space** of V, denoted by V'', is defined to be the dual space of V'. In other words, $V'' = (V')'$. Define $\Lambda : V \to V''$ by

$$(\Lambda v)(\varphi) = \varphi(v)$$

for each $v \in V$ and each $\varphi \in V'$.

(a) Show that Λ is a linear map from V to V''.
(b) Show that if $T \in \mathcal{L}(V)$, then $T'' \circ \Lambda = \Lambda \circ T$, where $T'' = (T')'$.
(c) Show that if V is finite-dimensional, then Λ is an isomorphism from V onto V''.

Axler Note. *Suppose V is finite-dimensional. Then V and V' are isomorphic, but finding an isomorphism from V onto V' generally requires choosing a basis of V. In contrast, the isomorphism Λ from V onto V'' does not require a choice of basis and thus is considered more natural.*

Solution.

(a) To see that Λ is linear, let $v, w \in V$ and $a \in \mathbb{F}$. Then for any $\varphi \in V'$, we have

$$\begin{aligned}(\Lambda(v+w))(\varphi) &= \varphi(v+w) = \varphi(v) + \varphi(w) = (\Lambda v)(\varphi) + (\Lambda w)(\varphi),\\ (\Lambda(av))(\varphi) &= \varphi(av) = a\varphi(v) = a(\Lambda v)(\varphi).\end{aligned}$$

Hence, Λ is linear.

(b) Let $v \in V$ and $\varphi \in V'$. Then

$$\begin{aligned}((T'' \circ \Lambda)(v))(\varphi) &= (T''(\Lambda v))(\varphi)\\ &= (\Lambda v)(T'(\varphi))\\ &= T'(\varphi)(v)\\ &= \varphi(T(v))\\ &= (\Lambda(T(v)))(\varphi) = ((\Lambda \circ T)(v))(\varphi).\end{aligned}$$

Hence, $T'' \circ \Lambda = \Lambda \circ T$.

(c) Since V is finite-dimensional, we have $\dim V = \dim V'$. Then $\dim V'' = \dim(V')' = \dim V' = \dim V$. By the fundamental theorem of linear maps, we have

$$\dim \operatorname{range} \Lambda = \dim V - \dim \operatorname{null} \Lambda.$$

To see that Λ is injective, suppose $\Lambda v = 0$ for some $v \in V$. Then for any $\varphi \in V'$, we have $\varphi(v) = (\Lambda v)(\varphi) = 0$. Since this holds for all $\varphi \in V'$ by Exercise 3F3, it follows that $v = 0$. Hence, Λ is injective, so $\operatorname{null} \Lambda = \{0\}$. Therefore, $\dim \operatorname{range} \Lambda = \dim V - 0 = \dim V$. Since $\operatorname{range} \Lambda$ is a subspace of V'' and has the same dimension as V'', we conclude that $\operatorname{range} \Lambda = V''$, so Λ is surjective. Combining injectivity and surjectivity, we conclude that Λ is an isomorphism from V onto V'', as desired. ■

Exercise 3F33

Suppose U is a subspace of V. Let $\pi : V \to V/U$ be the usual quotient map. Thus, $\pi' \in \mathcal{L}((V/U)', V')$.

(a) Show that π' is injective.
(b) Show that $\operatorname{range} \pi' = U^0$.
(c) Conclude that π' is an isomorphism from $(V/U)'$ onto U^0.

Solution.

(a) Suppose $\pi'(\varphi) = 0$ for some $\varphi \in (V/U)'$. Then $\varphi(\pi(v)) = 0$ for all $v \in V$. Since $\pi(v) = v + U$, we have $\varphi(v + U) = 0$ for all $v \in V$. Since this holds for all $v \in V$ by surjectivity of π, it follows that $\varphi = 0$. Hence, π' is injective.

(b) To see $\operatorname{range} \pi' \subseteq U^0$, note that if $\varphi \in (V/U)'$, then for any $u \in U$, we have $\pi'(\varphi)(u) = \varphi(\pi(u)) = \varphi(0 + U) = 0$. Hence, $\operatorname{range} \pi' \subseteq U^0$.

To see the other inclusion, let $\psi \in U^0$. We want to find $\varphi \in (V/U)'$ such that $\pi'(\varphi) = \psi$. Define $\varphi : V/U \to \mathbb{F}$ by $\varphi(v + U) = \psi(v)$ for each $v \in V$. We need to check that φ is well-defined. If $v + U = w + U$, then $v - w \in U$, so $\psi(v - w) = 0$, which implies that $\psi(v) = \psi(w)$. Hence, φ is well-defined. It is also linear since for any $a, b \in \mathbb{F}$ and $v, w \in V$, we have

$$\begin{aligned}\varphi(a(v + U) + b(w + U)) &= \varphi((av + bw) + U) = \psi(av + bw)\\ &= a\psi(v) + b\psi(w) = a\varphi(v + U) + b\varphi(w + U).\end{aligned}$$

Finally, we have $\pi'(\varphi)(v) = \varphi(\pi(v)) = \varphi(v + U) = \psi(v)$ for all $v \in V$, so $\pi'(\varphi) = \psi$. Hence, $U^0 \subseteq \operatorname{range} \pi'$, and we conclude that $\operatorname{range} \pi' = U^0$.

(c) From part (a), we have π' is injective. From part (b), we have $\operatorname{range} \pi' = U^0$. Then the induced map $\tilde{\pi}' : (V/U)' \to U^0$ defined by $\tilde{\pi}'(\varphi) = \pi'(\varphi)$ is an isomorphism. Hence, π' is an isomorphism from $(V/U)'$ onto U^0, as desired. ■

4 Polynomials

Exercise 4-1

Suppose $w, z \in \mathbb{C}$. Verify the following equalities and inequalities.

(a) $z + \bar{z} = 2\Re z$

(b) $z - \bar{z} = 2(\Im z)i$

(c) $z\bar{z} = |z|^2$

(d) $\overline{w + z} = \bar{w} + \bar{z}$ and $\overline{wz} = \bar{w}\bar{z}$

(e) $\bar{\bar{z}} = z$

(f) $|\Re z| \le |z|$ and $|\Im z| \le |z|$

(g) $|\bar{z}| = |z|$

(h) $|wz| = |w||z|$

Solution. In this problem, let $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$. Then

(a) $z + \bar{z} = (a + bi) + (a - bi) = 2a = 2\Re z$.

(b) $z - \bar{z} = (a + bi) - (a - bi) = 2bi = 2(\Im z)i$.

(c) $z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$.

(d) $\overline{w + z} = \overline{(c + di) + (a + bi)} = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = (a - bi) + (c - di) = \bar{w} + \bar{z}$ and $\overline{wz} = \overline{(c + di)(a + bi)} = \overline{(ac - bd) + (ad + bc)i} = (ac - bd) - (ad + bc)i = (c - di)(a - bi) = \bar{w}\bar{z}$.

(e) $\bar{\bar{z}} = \overline{a - bi} = a + bi = z$.

(f) $|\Re z| = |a| \le \sqrt{a^2 + b^2} = |z|$ and $|\Im z| = |b| \le \sqrt{a^2 + b^2} = |z|$.

(g) $|\bar{z}| = |a - bi| = \sqrt{a^2 + b^2} = |z|$.

(h) $|wz| = |(c + di)(a + bi)| = |(ac - bd) + (ad + bc)i| = \sqrt{(ac - bd)^2 + (ad + bc)^2} = \sqrt{(a^2 + b^2)(c^2 + d^2)} = |a + bi||c + di| = |z||w|$. ■

Exercise 4-2

Prove that if $w, z \in \mathbb{C}$, then $||w| - |z|| \le |w - z|$.

Axler Note. *The inequality above is called the* **reverse triangle inequality**.

Solution. Let $w, z \in \mathbb{C}$. By the triangle inequality, we have

$$|w| = |(w - z) + z| \le |w - z| + |z|.$$

Rearranging gives $|w| - |z| \le |w - z|$. Similarly,

$$|z| = |(z - w) + w| \le |z - w| + |w| = |w - z| + |w|,$$

which gives $|z| - |w| \le |w - z|$, or equivalently $-|w - z| \le |w| - |z|$. Combining these inequalities, we get

$$||w| - |z|| \le |w - z|.$$ ■

Exercise 4-3

Suppose V is a complex vector space and $\varphi \in V'$. Define $\sigma : V \to \mathbb{R}$ by $\sigma(v) = \Re\varphi(v)$ for each $v \in V$. Show that

$$\varphi(v) = \sigma(v) - i\sigma(iv)$$

for all $v \in V$.

Solution. Let $v \in V$. Then there exist $s, t \in \mathbb{R}$ such that

$$\varphi(v) = s + ti.$$

Then

$$\sigma(v) = \Re\varphi(v) = s \quad \text{and} \quad \sigma(iv) = \Re\varphi(iv) = \Re(i(s+ti)) = \Re(-t+si) = -t.$$

Therefore,

$$\sigma(v) - i\sigma(iv) = s - i(-t) = s + ti = \varphi(v).$$

■

Exercise 4-4

Suppose m is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) \mid \deg p = m\}$$

a subspace of $\mathcal{P}(\mathbb{F})$?

Solution. No, the set is not a subspace of $\mathcal{P}(\mathbb{F})$. To see this, consider two polynomials $p, q \in \mathcal{P}(\mathbb{F})$ with $p(x) = x^m$ and $q(x) = -x^m + x^{m-1}$. Then $p + q = x^{m-1}$, which has degree $m - 1$ and is not in the set. Therefore, the set is not closed under addition and hence not a subspace. ■

Exercise 4-5

Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) \mid \deg p \text{ is even}\}$$

a subspace of $\mathcal{P}(\mathbb{F})$?

Solution. No, the set is not a subspace of $\mathcal{P}(\mathbb{F})$. To see this, consider two polynomials $p, q \in \mathcal{P}(\mathbb{F})$ with $p(x) = x^2$ and $q(x) = x - x^2$. Then $p + q = x$, which has degree 1 and is not even. Therefore, the set is not closed under addition and hence not a subspace. ■

Exercise 4-6

Suppose that m and n are positive integers with $m \leq n$, and suppose $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \cdots = p(\lambda_m)$ and such that p has no other zeros.

Solution. Consider the polynomial

$$p(x) = (x - \lambda_1)^{n-m+1}(x - \lambda_2) \cdots (x - \lambda_m).$$

This polynomial has degree n, vanishes at $\lambda_1, \ldots, \lambda_m$, and has no other zeros. Therefore, it satisfies the required conditions. ■

Exercise 4-7

Suppose that m is a nonnegative integer, $z_1, \ldots, z_{m+1}$ are distinct elements of $\mathbb{F}$, and $w_1, \ldots, w_{m+1} \in \mathbb{F}$. Prove that there exists a unique polynomial $p \in \mathcal{P}_m(\mathbb{F})$ such that

$$p(z_k) = w_k$$

for each $k = 1, \ldots, m + 1$.
Axler Note. *This result can be proved without using linear algebra. However, try to find the clearer, shorter proof that uses some linear algebra.*

Solution. Consider the linear map $T : \mathcal{P}_m(\mathbb{F}) \to \mathbb{F}^{m+1}$ defined by

$$T(p) = (p(z_1), \ldots, p(z_{m+1})).$$

Since $\dim \mathcal{P}_m(\mathbb{F}) = m + 1$ and $\dim \mathbb{F}^{m+1} = m + 1$, it suffices to show that T is injective to conclude that it is also surjective. Suppose $T(p) = 0$, i.e., $p(z_k) = 0$ for all $k = 1, \ldots, m + 1$. Since p has degree at most m and has $m + 1$ distinct zeros, it must be the zero polynomial. Therefore, T is injective, and hence bijective. This shows that for any $(w_1, \ldots, w_{m+1}) \in \mathbb{F}^{m+1}$, there exists a unique $p \in \mathcal{P}_m(\mathbb{F})$ such that $T(p) = (w_1, \ldots, w_{m+1})$, which is exactly the desired polynomial. ■

Exercise 4-8

Suppose $p \in \mathcal{P}(\mathbb{C})$ has degree m. Prove that p has m distinct zeros if and only if p and its derivative p' have no zeros in common.

Solution. Suppose p has m distinct zeros, say $z_1, \ldots, z_m$. If p and p' have a common zero z_k, then z_k would be a multiple zero of p, contradicting the assumption that all zeros are distinct. Hence, p and p' have no zeros in common.

Conversely, suppose p and p' have no zeros in common. If p had a multiple zero z, then $p(z) = 0$ and $p'(z) = 0$, which would contradict the assumption. Since p has degree m, it can have at most m zeros by the fundamental theorem of algebra. Therefore, p must have exactly m distinct zeros. ■

Exercise 4-9

Prove that every polynomial of odd degree with real coefficients has a real zero.
Appendix. *See* Appendix 4A9 *provides an alternate solution using the intermediate value theorem.*

Solution. Let $p(x)$ be a polynomial of odd degree with real coefficients. Since non-real zeros come in pairs, there are an even number of non-real zeros. Therefore, there must be at least one real zero. ■

Exercise 4-10

For $p \in \mathcal{P}(\mathbb{R})$, define $Tp : \mathbb{R} \to \mathbb{R}$ by

$$(Tp)(x) = \begin{cases} \frac{p(x)-p(3)}{x-3} & \text{if } x \neq 3, \\ p'(3) & \text{if } x = 3 \end{cases}$$

for each $x \in \mathbb{R}$. Show that $Tp \in \mathcal{P}(\mathbb{R})$ for every polynomial $p \in \mathcal{P}(\mathbb{R})$ and also show that $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is a linear map.

Solution. Let $p \in \mathcal{P}(\mathbb{R})$. Since $p(x) - p(3)$ has a zero at $x = 3$, we can factor it as $(x-3)q(x)$ for some polynomial $q \in \mathcal{P}(\mathbb{R})$. Then for $x \neq 3$,

$$(Tp)(x) = \frac{p(x) - p(3)}{x - 3} = q(x),$$

and differentiating p gives $p'(3) = q(3)$. Since $Tp(3) = p'(3) = q(3)$, we have $Tp(x) = q(x)$ for all $x \in \mathbb{R}$. Therefore, $Tp \in \mathcal{P}(\mathbb{R})$.

Let $p_1, p_2 \in \mathcal{P}(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$. Let $p_1(x) - p_1(3) = (x-3)q_1(x)$ and $p_2(x) - p_2(3) = (x-3)q_2(x)$ for some $q_1, q_2 \in \mathcal{P}(\mathbb{R})$. Then

$$(\alpha p_1 + \beta p_2)(x) = (\alpha p_1(3) + \beta p_2(3)) + (x-3)(\alpha q_1(x) + \beta q_2(x)).$$

Hence, $T(\alpha p_1 + \beta p_2)(x) = \alpha q_1(x) + \beta q_2(x) = \alpha T p_1(x) + \beta T p_2(x)$ for all $x \in \mathbb{R}$. Therefore, T is a linear map. ■

Exercise 4-11

Suppose $p \in \mathcal{P}(\mathbb{C})$. Define $q : \mathbb{C} \to \mathbb{C}$ by

$$q(z) = p(z)\overline{p(\bar{z})}.$$

Prove that q is a polynomial with real coefficients.

Appendix. *See* Appendix 4A11 *provides an alternate solution by looking at the roots of p.*

Solution. Let $p \in \mathcal{P}(\mathbb{C})$ have the form $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ with $a_k \in \mathbb{C}$. Then

$$\overline{p(\bar{z})} = \overline{\sum_{i=0}^{n} a_i \bar{z}^i} = \sum_{i=0}^{n} \bar{a}_i z^i.$$

Therefore,

$$q(z) = p(z)\overline{p(\bar{z})} = \left(\sum_{i=0}^{n} a_i z^i\right)\left(\sum_{j=0}^{n} \bar{a}_j z^j\right) = \sum_{k=0}^{2n}\left(\sum_{i+j=k} a_i \bar{a}_j\right) z^k.$$

Let b_m denote the coefficient of z^m in $q(z)$. Then

$$b_m = \sum_{i+j=m} a_i \bar{a}_j = \sum_{i+j=m} \bar{a}_j a_i = \overline{b_m}.$$

Hence, $b_m \in \mathbb{R}$ for each $m = 0, 1, \ldots, 2n$. Therefore, q is a polynomial with real coefficients. ■

Exercise 4-12

Suppose m is a nonnegative integer and $p \in \mathcal{P}_m(\mathbb{C})$ is such that there are distinct real numbers $x_0, x_1, \ldots, x_m$ with $p(x_k) \in \mathbb{R}$ for each $k = 0, 1, \ldots, m$. Prove that all coefficients of p are real.

Solution. Let $p(x_k) = y_k \in \mathbb{R}$ for each $k = 0, 1, \ldots, m$. By Exercise 4-7 with $\mathbb{F} = \mathbb{R}$, there exists a polynomial $q \in \mathcal{P}_m(\mathbb{R})$ that satisfies $q(x_k) = y_k$ for each $k = 0, 1, \ldots, m$. Note that this polynomial must be the same as the original $p \in \mathcal{P}_m(\mathbb{C})$ because a polynomial of degree at most m is uniquely determined by its values at $m + 1$ distinct points. Therefore, all coefficients of p are real. ■

Exercise 4-13

Suppose $p \in \mathcal{P}(\mathbb{F})$ with $p \neq 0$. Let $U = \{pq \mid q \in \mathcal{P}(\mathbb{F})\}$.

(a) Show that $\dim \mathcal{P}(\mathbb{F})/U = \deg p$.

(b) Find a basis of $\mathcal{P}(\mathbb{F})/U$.

Solution.

(a) We first show that U is a subspace of $\mathcal{P}(\mathbb{F})$. Clearly, $0 \in U$ since $0 = p \cdot 0$. If $u_1, u_2 \in U$, then $u_1 = pq_1$ and $u_2 = pq_2$ for some $q_1, q_2 \in \mathcal{P}(\mathbb{F})$. For any scalars $\alpha, \beta \in \mathbb{F}$, we have $\alpha u_1 + \beta u_2 = \alpha p q_1 + \beta p q_2 = p(\alpha q_1 + \beta q_2) \in U$. Therefore, U is closed under addition and scalar multiplication, and hence U is a subspace of $\mathcal{P}(\mathbb{F})$.

Now let $r \in \mathcal{P}(\mathbb{F})$. By the division algorithm for polynomials, there exist unique polynomials $q_0 \in \mathcal{P}(\mathbb{F})$ and $s \in \mathcal{P}(\mathbb{F})$ with $\deg s < \deg p$ such that $r = pq_0 + s$. This shows that every polynomial in $\mathcal{P}(\mathbb{F})$ can be written as an element of U plus a polynomial of degree less than $\deg p$. Consider the map $T : \mathcal{P}_{\deg p - 1}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})/U$ defined by $T(s) = s + U$. This map is linear and surjective, and null T is trivial since if $s + U = U$, then $s \in U$, which implies $s = pq$ for some $q \in \mathcal{P}(\mathbb{F})$. Then

$$\deg s = \deg p + \deg q < \deg p.$$

Therefore, $q = 0$, hence $s = 0$, and T is an isomorphism. We conclude that $\dim \mathcal{P}(\mathbb{F})/U = \dim \mathcal{P}_{\deg p - 1}(\mathbb{F}) = \deg p$.

(b) A basis of $\mathcal{P}(\mathbb{F})/U$ is given by the images of the standard basis of $\mathcal{P}_{\deg p - 1}(\mathbb{F})$ under T, namely $\{1 + U, x + U, \ldots, x^{\deg p - 1} + U\}$. ■

Exercise 4-14

Suppose $p, q \in \mathcal{P}(\mathbb{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)–(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that

$$rp + sq = 1.$$

(a) Define $T : \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) \to \mathcal{P}_{m+n-1}(\mathbb{C})$ by

$$T(r, s) = rp + sq.$$

Show that the linear map T is injective.

(b) Show that the linear map T in (a) is surjective.

(c) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $rp + sq = 1$.

Manual Remark. *In part* (a), *one may also observe that a multiple of p and q must have* $\deg p + \deg q$, *but* $\deg rp = \deg r + \deg p \leq n - 1 + m$.

Solution.

(a) To show that T is injective, suppose that $T(r, s) = rp + sq = 0$ for some $(r, s) \in \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C})$. Then $rp = -sq$ implies rp is a polynomial multiple of q, so every zero of q must be a zero of rp. Because p and q have no zeros in common, every zero of q must be a zero of r. But $\deg r < \deg q$, so it must be that $r = 0$. Because $r = 0$, we have $-sq = 0$. Then since q is nonconstant, it must be that $s = 0$. Therefore, if $(r, s) \in \operatorname{null} T$, we have $(r, s) = 0$, so T is injective.

(b) To show that T is surjective, note that $\dim(\mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C})) = n + m = \dim \mathcal{P}_{m+n-1}(\mathbb{C})$. Since T is a linear map between vector spaces of the same finite dimension, and we have already shown that T is injective, it follows that T is also surjective.

(c) Since T is surjective, for the polynomial $1 \in \mathcal{P}_{m+n-1}(\mathbb{C})$, there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $T(r, s) = rp + sq = 1$. ■

5 Eigenvalues and Eigenvectors

5A Invariant Subspaces

Exercise 5A1

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V.

(a) Prove that if $U \subseteq \operatorname{null} T$, then U is invariant under T.

(b) Prove that if $\operatorname{range} T \subseteq U$, then U is invariant under T.

Solution.

(a) If $U \subseteq \operatorname{null} T$, then for any $u \in U$, we have $T(u) = 0 \in U$, so U is invariant under T.

(b) If $\operatorname{range} T \subseteq U$, then for any $u \in U$, $T(u) \in \operatorname{range} T \subseteq U$, so U is invariant under T. ■

Exercise 5A2

Suppose that $T \in \mathcal{L}(V)$ and $V_1, \ldots, V_m$ are subspaces of V invariant under T. Prove that $V_1 + \cdots + V_m$ is invariant under T.

Solution. Let $v \in V_1 + \cdots + V_m$. Then $v = v_1 + \cdots + v_m$ for some $v_i \in V_i$. Since each V_i is invariant under T, we have $T(v_i) \in V_i$ for each i. Therefore,

$$T(v) = T(v_1 + \cdots + v_m) = T(v_1) + \cdots + T(v_m) \in V_1 + \cdots + V_m,$$

so $V_1 + \cdots + V_m$ is invariant under T. ■

Exercise 5A3

Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of V invariant under T is invariant under T.

Solution. Let $\{U_i\}_{i \in I}$ be a collection of subspaces of V invariant under T, and let

$$U = \bigcap_{i \in I} U_i.$$

Let $u \in U$. Then $u \in U_i$ for all $i \in I$. Since each U_i is invariant under T, we have $T(u) \in U_i$ for all $i \in I$. Therefore, $T(u) \in U$, so U is invariant under T. ■

Exercise 5A4

Prove or give a counterexample: If V is finite-dimensional and U is a subspace of V that is invariant under every operator on V, then $U = \{0\}$ or $U = V$.

Solution. If $U = \{0\}$, then we are done. Now suppose $U \neq \{0\}$ so that there exists nonzero $v_1 \in U$. Extend v_1 to a basis $v_1, \ldots, v_m$ of V, and let $v \in V$. Define the linear operator $T \in \mathcal{L}(V)$ by $T(v_1) = v$

and $T(v_i) = 0$ for $i = 2, \ldots, m$. Since U is invariant under T, we have $T(v_1) = v \in U$. But v was an arbitrary element of V, so $U = V$. Therefore, $U = \{0\}$ or $U = V$. ■

Exercise 5A5

Suppose $T \in \mathcal{L}(\mathbb{R}^2)$ is defined by $T(x, y) = (-3y, x)$. Find the eigenvalues of T.

Solution. If (x, y) is an eigenvector of T with eigenvalue λ, then we have the system of equations

$$\begin{cases} -3y = \lambda x, \\ x = \lambda y \end{cases}$$

Solving the system, we get $\lambda^2 = -3$ which has no real solutions. Therefore, T has no real eigenvalues. ■

Exercise 5A6

Define $T \in \mathcal{L}(\mathbb{F}^2)$ by $T(w, z) = (z, w)$. Find all eigenvalues and eigenvectors of T.

Solution. If (w, z) is an eigenvector of T with eigenvalue λ, then we have

$$(z, w) = \lambda(w, z),$$

which gives the system

$$\begin{cases} z = \lambda w, \\ w = \lambda z. \end{cases}$$

Solving, we get $\lambda^2 = 1$, so $\lambda = 1$ or $\lambda = -1$. If $\lambda = 1$, then $z = w$, so the eigenvectors are of the form (w, w). If $\lambda = -1$, then $z = -w$, so the eigenvectors are of the form $(w, -w)$. ■

Exercise 5A7

Define $T \in \mathcal{L}(\mathbb{F}^3)$ by $T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$. Find all eigenvalues and eigenvectors of T.

Solution. If (z_1, z_2, z_3) is an eigenvector of T with eigenvalue λ, then we have

$$(2z_2, 0, 5z_3) = \lambda(z_1, z_2, z_3),$$

which gives the system

$$\begin{cases} 2z_2 = \lambda z_1, \\ 0 = \lambda z_2, \\ 5z_3 = \lambda z_3. \end{cases}$$

Solving, we get the eigenvalues $\lambda = 0$ and $\lambda = 5$. For $\lambda = 0$, the eigenvectors are of the form $(z_1, 0, 0)$. For $\lambda = 5$, the eigenvectors are of the form $(0, 0, z_3)$. ■

Exercise 5A8

Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that if λ is an eigenvalue of P, then $\lambda = 0$ or $\lambda = 1$.

Solution. If v is an eigenvector of P with eigenvalue λ, then $Pv = \lambda v$. Applying P again, we get $P^2 v = P(\lambda v) = \lambda P v = \lambda^2 v$. But $P^2 = P$, so $P^2 v = Pv = \lambda v$. Therefore, $\lambda^2 v = \lambda v$, which gives $(\lambda^2 - \lambda)v = 0$. Since $v \neq 0$, we must have $\lambda^2 - \lambda = 0$, so $\lambda(\lambda - 1) = 0$. Hence, $\lambda = 0$ or $\lambda = 1$. ■

Exercise 5A9

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of T.

Solution. Let $p \in \mathcal{P}(\mathbb{R})$ such that $p' = \lambda p$ for some $\lambda \in \mathbb{R}$.

- If $\lambda \neq 0$, suppose $\deg p = 0$. Then p is a constant polynomial, and $p' = 0 \neq \lambda p$, so $\deg p \geq 1$. If $\deg p = n \geq 1$, then $\deg p' = n - 1$, so $\deg p' \neq \deg p$, which contradicts $p' = \lambda p$.
- If $\lambda = 0$, then $p' = 0$, which implies that p is a constant polynomial. Therefore, the eigenvectors corresponding to the eigenvalue $\lambda = 0$ are all nonzero constant polynomials. ■

Exercise 5A10

Define $T \in \mathcal{L}(\mathcal{P}_4(\mathbb{R}))$ by $(Tp)(x) = xp'(x)$ for all $x \in \mathbb{R}$. Find all eigenvalues and eigenvectors of T.

Solution. Let $p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4$ be an eigenvector of T with eigenvalue λ. Then

$$(Tp)(x) = xp'(x) = x(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3) = a_1 x + 2a_2 x^2 + 3a_3 x^3 + 4a_4 x^4.$$

Equating coefficients with $\lambda p(x) = \lambda a_0 + \lambda a_1 x + \lambda a_2 x^2 + \lambda a_3 x^3 + \lambda a_4 x^4$ yields the system:

$$\lambda a_0 = 0, \quad \lambda a_1 = a_1, \quad \lambda a_2 = 2a_2, \quad \lambda a_3 = 3a_3, \quad \lambda a_4 = 4a_4.$$

For each $n \in \{0, 1, 2, 3, 4\}$, the equation $\lambda a_n = n a_n$ is satisfied with $a_n \neq 0$ if and only if $\lambda = n$. When $\lambda = n$, all other coefficients must be zero, so the eigenvector is a nonzero scalar multiple of x^n. Therefore, the eigenvalues of T are $0, 1, 2, 3, 4$, with corresponding eigenvectors being nonzero scalar multiples of $1, x, x^2, x^3, x^4$, respectively. ■

Exercise 5A11

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\alpha \in \mathbb{F}$. Prove that there exists $\delta > 0$ such that $T - \lambda I$ is invertible for all $\lambda \in \mathbb{F}$ such that $0 < |\alpha - \lambda| < \delta$.

Solution. Since V is finite-dimensional, the set of eigenvalues of T is finite. If T has no eigenvalues, then $T - \lambda I$ is invertible for all $\lambda \in \mathbb{F}$, and we can choose any $\delta > 0$. Otherwise, let $\lambda_1, \ldots, \lambda_m$ be the eigenvalues of T. Define

$$\delta = \min_{\lambda_j \neq \alpha} |\alpha - \lambda_j|.$$

If α is the only eigenvalue, then we can select any $\delta > 0$ since any λ satisfying $0 < |\alpha - \lambda| < \delta$ will not be equal to α, and hence $T - \lambda I$ will be invertible. Otherwise, we have for any $\lambda \in \mathbb{F}$ such that

$0 < |\alpha - \lambda| < \delta$, we have $\lambda \neq \lambda_j$ for all $j = 1, \dots, m$, so λ is not an eigenvalue of T. Hence, $T - \lambda I$ is invertible for all such λ. ■

Exercise 5A12

Suppose $V = U \oplus W$, where U and W are nonzero subspaces of V. Define $P \in \mathcal{L}(V)$ by $P(u + w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of P.

Solution. Let $v = u + w \in V$ be an eigenvector of P with eigenvalue λ. Then $P(v) = P(u + w) = u = \lambda v = \lambda(u + w)$. Comparing the U and W components, we get $u = \lambda u$ and $0 = \lambda w$. If $\lambda \neq 0$, then $w = 0$ and $u = \lambda u$ implies $\lambda = 1$. If $\lambda = 0$, then $u = 0$ and w can be any nonzero vector in W. Therefore, the eigenvalues are $\lambda = 1$ and $\lambda = 0$, with corresponding eigenvectors being the nonzero vectors in U for $\lambda = 1$ and the nonzero vectors in W for $\lambda = 0$. ■

Exercise 5A13

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible.

(a) Prove that T and $S^{-1}TS$ have the same eigenvalues.

(b) What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?

Solution.

(a) Let λ be an eigenvalue of T with eigenvector $v \in V$, so $T(v) = \lambda v$. Then

$$(S^{-1}TS)(S^{-1}v) = S^{-1}T(v) = S^{-1}(\lambda v) = \lambda(S^{-1}v).$$

Thus, $S^{-1}v$ is an eigenvector of $S^{-1}TS$ with eigenvalue λ. This shows that every eigenvalue of T is an eigenvalue of $S^{-1}TS$. Conversely, if λ is an eigenvalue of $S^{-1}TS$ with eigenvector u, then Su is an eigenvector of T with eigenvalue λ. Therefore, T and $S^{-1}TS$ have the same eigenvalues.

(b) The relation between the eigenvectors of T and the eigenvectors of $S^{-1}TS$ is given by $v \mapsto S^{-1}v$. Specifically, if v is an eigenvector of T with eigenvalue λ, then $S^{-1}v$ is an eigenvector of $S^{-1}TS$ with the same eigenvalue λ. Conversely, if u is an eigenvector of $S^{-1}TS$ with eigenvalue λ, then Su is an eigenvector of T with eigenvalue λ. ■

Exercise 5A14

Give an example of an operator on $\mathbb{R}^4$ that has no (real) eigenvalues.

Solution. Consider the linear operator on $\mathbb{R}^4$ given by the matrix

$$T = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Letting T act on $(x_1, x_2, x_3, x_4) \in \mathbb{R}^4$, we have

$$T(x_1, x_2, x_3, x_4) = (-x_2, x_1, -x_4, x_3).$$

Then any eigenvector $(x_1, x_2, x_3, x_4) \in \mathbb{R}^4$ with eigenvalue λ must satisfy

$$(-x_2, x_1, -x_4, x_3) = \lambda(x_1, x_2, x_3, x_4).$$

This yields $\lambda^2 + 1 = 0$, which has no solution in $\mathbb{R}$. ■

Exercise 5A15

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that λ is an eigenvalue of T if and only if λ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

Solution.

$$\begin{aligned} \lambda \text{ is an eigenvalue of } T &\iff T - \lambda I \text{ is not invertible} \\ &\iff (T - \lambda I)' \text{ is not invertible} \\ &\iff T' - \lambda I \text{ is not invertible} \\ &\iff \lambda \text{ is an eigenvalue of } T' \end{aligned}$$

■

Exercise 5A16

Suppose $v_1, \ldots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T, then

$$|\lambda| \le n \max\{|\mathcal{M}(T)_{j,k}| : 1 \le j, k \le n\},$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j, column k of the matrix of T with respect to the basis $v_1, \ldots, v_n$.
Axler Note. *See* Exercise 6A19 *for a different bound on* $|\lambda|$.

Solution. Let λ be an eigenvalue of T with eigenvector $u \in V$, so $Tu = \lambda u$. Let $\mathcal{M}(T)$ be the matrix of T with respect to the basis $v_1, \ldots, v_n$. Then

$$\lambda u_j = (\mathcal{M}(T)u)_j = \sum_{k=1}^{n} \mathcal{M}(T)_{j,k} u_k \quad \text{for each } j = 1, \ldots, n.$$

Taking absolute values, we get

$$|\lambda||u_j| = \left| \sum_{k=1}^{n} \mathcal{M}(T)_{j,k} u_k \right| \le \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}||u_k| \le n \max\{|\mathcal{M}(T)_{j,k}| : 1 \le j, k \le n\} \max\{|u_k| : 1 \le k \le n\}.$$

Choosing j such that $|u_j| = \max\{|u_k| : 1 \le k \le n\}$, we obtain

$$|\lambda| \le n \max\{|\mathcal{M}(T)_{j,k}| : 1 \le j, k \le n\}.$$

■

Exercise 5A17

Suppose $\mathbb{F} = \mathbb{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{R}$. Prove that λ is an eigenvalue of T if and only if λ is an eigenvalue of the complexification $T_{\mathbb{C}}$.
Axler Note. *See* Exercise 3B33 *for the definition of* $T_{\mathbb{C}}$.

Solution. Suppose λ is an eigenvalue of T with eigenvector $v \in V$, so $Tv = \lambda v$. Then in the complexification, we have $T_{\mathbb{C}}(v + i0) = Tv + iT0 = \lambda v + i0 = \lambda(v + i0)$, showing that λ is an eigenvalue of $T_{\mathbb{C}}$ with eigenvector $v + i0$.

Conversely, suppose λ is an eigenvalue of $T_{\mathbb{C}}$ with eigenvector $v + iw \in V_{\mathbb{C}}$, where $v, w \in V$. Then

$$T_{\mathbb{C}}(v + iw) = Tv + iTw = \lambda(v + iw) = \lambda v + i\lambda w.$$

Equating real and imaginary parts, we get

$$Tv = \lambda v \quad \text{and} \quad Tw = \lambda w.$$

Since $\lambda \in \mathbb{R}$, if $v \neq 0$, then v is an eigenvector of T corresponding to λ. If $v = 0$, then $w \neq 0$ and $Tw = \lambda w$, so w is an eigenvector of T corresponding to λ. This shows that λ is an eigenvalue of T. ■

Exercise 5A18

Suppose $\mathbb{F} = \mathbb{R}$, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{C}$. Prove that λ is an eigenvalue of the complexification $T_{\mathbb{C}}$ if and only if $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$.

Solution. Suppose λ is an eigenvalue of $T_{\mathbb{C}}$ with eigenvector $v + iw \in V_{\mathbb{C}}$, where $v, w \in V$. Then

$$T_{\mathbb{C}}(v + iw) = Tv + iTw = \lambda(v + iw) = \lambda v + i\lambda w.$$

Moreover, we have

$$T_{\mathbb{C}}(v - iw) = Tv - iTw.$$

Note that $v + iw = 0$ if and only if both $v = w = 0$. Then $v + iw \neq 0$ implies $v - iw \neq 0$. Let $\lambda = a + bi$ with $a, b \in \mathbb{R}$. Then equating real and imaginary parts, we have

$$Tv = av - bw \quad \text{and} \quad Tw = bv + aw.$$

Then

$$\begin{aligned}
T_{\mathbb{C}}(v - iw) &= Tv - iTw \\
&= (av - bw) - i(bv + aw) \\
&= av - ibv - bw - iaw \\
&= (a - ib)(v - iw) \\
&= \overline{\lambda}(v - iw).
\end{aligned}$$

Since $v - iw \neq 0$, this shows that $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$ with eigenvector $v - iw$.

Conversely, if $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$, then by the same argument, λ is an eigenvalue of $T_{\mathbb{C}}$. This completes the proof. ■

Exercise 5A19

Show that the forward shift operator $T \in \mathcal{L}(\mathbb{F}^{\infty})$ defined by

$$T(z_1, z_2, \ldots) = (0, z_1, z_2, \ldots)$$

has no eigenvalues.

Solution. Suppose $\lambda \in \mathbb{F}$ is an eigenvalue of T with eigenvector $(z_1, z_2, \ldots) \in \mathbb{F}^\infty$. Then

$$T(z_1, z_2, \ldots) = (0, z_1, z_2, \ldots) = \lambda(z_1, z_2, \ldots).$$

Equating the first component, we get $0 = \lambda z_1$, so either $\lambda = 0$ or $z_1 = 0$. If $\lambda = 0$, then the second component gives $z_1 = 0$, and the third component gives $z_2 = 0$, and so on. By induction, all $z_k = 0$, contradicting that the eigenvector is nonzero. If $\lambda \neq 0$, then $z_1 = 0$, and the second component gives $z_2 = 0$, and so on. Again, all $z_k = 0$, a contradiction. Therefore, T has no eigenvalues. ■

Exercise 5A20

Define the backward shift operator $S \in \mathcal{L}(\mathbb{F}^\infty)$ by

$$S(z_1, z_2, z_3, \ldots) = (z_2, z_3, \ldots).$$

(a) Show that every element of $\mathbb{F}$ is an eigenvalue of S.
(b) Find all eigenvectors of S.

Solution.

(a) Let $\alpha \in \mathbb{F}$. Consider the vector $(1, \alpha, \alpha^2, \ldots) \in \mathbb{F}^\infty$. Then

$$S(1, \alpha, \alpha^2, \ldots) = (\alpha, \alpha^2, \alpha^3, \ldots) = \alpha(1, \alpha, \alpha^2, \ldots),$$

showing that α is an eigenvalue of S with eigenvector $(1, \alpha, \alpha^2, \ldots)$. Since $\alpha \in \mathbb{F}$ was arbitrary, every element of $\mathbb{F}$ is an eigenvalue of S.

(b) Let $\alpha \in \mathbb{F}$ and consider the eigenvalue α of S. Any eigenvector corresponding to α must satisfy

$$S(z_1, z_2, z_3, \ldots) = \alpha(z_1, z_2, z_3, \ldots),$$

which gives

$$(z_2, z_3, z_4, \ldots) = (\alpha z_1, \alpha z_2, \alpha z_3, \ldots).$$

Equating components, we get $z_2 = \alpha z_1$, $z_3 = \alpha z_2 = \alpha^2 z_1$, and in general $z_k = \alpha^{k-1} z_1$ for $k \geq 1$. Therefore, all eigenvectors corresponding to α are of the form $(z_1, \alpha z_1, \alpha^2 z_1, \ldots) = z_1(1, \alpha, \alpha^2, \ldots)$ with $z_1 \neq 0$. ■

Exercise 5A21

Suppose $T \in \mathcal{L}(V)$ is invertible.

(a) Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$. Prove that λ is an eigenvalue of T if and only if $\frac{1}{\lambda}$ is an eigenvalue of T^{-1}.
(b) Prove that T and T^{-1} have the same eigenvectors.

Solution.

(a) Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$ is an eigenvalue of T with eigenvector $v \in V$. Then

$$Tv = \lambda v.$$

Since T is invertible, we can apply T^{-1} to both sides to get

$$v = T^{-1}(\lambda v) = \lambda T^{-1} v.$$

Dividing both sides by λ (which is nonzero), we obtain

$$T^{-1}v = \frac{1}{\lambda}v.$$

This shows that $\frac{1}{\lambda}$ is an eigenvalue of T^{-1} with the same eigenvector v.

Conversely, suppose $\mu \in \mathbb{F}$ with $\mu \neq 0$ is an eigenvalue of T^{-1} with eigenvector $v \in V$. Then

$$T^{-1}v = \mu v.$$

Applying T to both sides gives

$$v = \mu Tv \implies Tv = \frac{1}{\mu}v.$$

This shows that $\frac{1}{\mu}$ is an eigenvalue of T with the same eigenvector v.

Therefore, λ is an eigenvalue of T if and only if $\frac{1}{\lambda}$ is an eigenvalue of T^{-1}.

(b) Part (a) shows that if λ is an eigenvalue of T, then $\frac{1}{\lambda}$ is an eigenvalue of T^{-1} with the same eigenvector. Conversely, if μ is an eigenvalue of T^{-1}, then $\frac{1}{\mu}$ is an eigenvalue of T with the same eigenvector. Therefore, T and T^{-1} have the same eigenvectors. ∎

Exercise 5A22

Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors u and w in V such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u.$$

Prove that 3 or -3 is an eigenvalue of T.

Solution. Consider the vectors $u + w$ and $u - w$. Since u and w are both nonzero, at least one of $u + w$ or $u - w$ is nonzero. We have

$$T(u + w) = Tu + Tw = 3w + 3u = 3(u + w),$$

and

$$T(u - w) = Tu - Tw = 3w - 3u = -3(u - w).$$

Hence, one of $u + w$ or $u - w$ is an eigenvector with eigenvalue 3 or -3, respectively. ∎

Exercise 5A23

Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that ST and TS have the same eigenvalues.

Solution. Let λ be an eigenvalue of ST with eigenvector $v \in V$, so that

$$STv = \lambda v.$$

If $\lambda = 0$, then ST is not invertible. Since V is finite-dimensional, TS is also not invertible, so 0 is an eigenvalue of TS. Otherwise, if $\lambda \neq 0$, let $w = Tv$. Then

$$Sw = S(Tv) = STv = \lambda v \neq 0 \implies w \neq 0.$$

Now, we have

$$TSw = T(STv) = T(\lambda v) = \lambda Tv = \lambda w.$$

Therefore, w is an eigenvector of TS with eigenvalue λ. Then every eigenvalue of ST is also an eigenvalue of TS. By symmetry, the same argument shows that if λ is an eigenvalue of TS, then it is also an eigenvalue of ST. Hence, ST and TS have the same eigenvalues. ■

Exercise 5A24

Suppose A is an n-by-n matrix with entries in $\mathbb{F}$. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $Tx = Ax$, where elements of $\mathbb{F}^n$ are thought of as n-by-1 column vectors.

(a) Suppose the sum of the entries in each row of A equals 1. Prove that 1 is an eigenvalue of T.

(b) Suppose the sum of the entries in each column of A equals 1. Prove that 1 is an eigenvalue of T.

Solution.

(a) Suppose the sum of the entries in each row of A equals 1. Let $v = (1, 1, \ldots, 1)^t \in \mathbb{F}^n$. Then

$$Tv = Av = v,$$

showing that v is an eigenvector of T corresponding to the eigenvalue 1.

(b) Suppose the sum of the entries in each column of A equals 1. By part (a), this is equivalent to saying that the sum of the entries in each row of A^t equals 1, hence 1 is an eigenvalue of the linear operator defined by A^t. But the eigenvalues of A^t are the same as the eigenvalues of A because $A - \lambda I$ is not invertible if and only if $A^t - \lambda I$ is not invertible. Therefore, 1 is an eigenvalue of T. ■

Exercise 5A25

Suppose $T \in \mathcal{L}(V)$ and u, w are eigenvectors of T such that $u + w$ is also an eigenvector of T. Prove that u and w are eigenvectors of T corresponding to the same eigenvalue.

Solution. Let $Tu = \lambda u$ and $Tw = \mu w$ for some scalars $\lambda, \mu \in \mathbb{F}$. Since $u + w$ is also an eigenvector of T, we have $T(u + w) = \nu(u + w)$ for some scalar $\nu \in \mathbb{F}$. Then

$$\lambda u + \mu w = Tu + Tw = T(u + w) = \nu(u + w) = \nu u + \nu w.$$

Suppose u and w correspond to different eigenvalues. Then the list u, w is linearly independent, and we can equate coefficients of u and w above to obtain $\lambda = \nu = \mu$, which is a contradiction. Therefore, u and w must correspond to the same eigenvalue. ■

Exercise 5A26

Suppose $T \in \mathcal{L}(V)$ is such that every nonzero vector in V is an eigenvector of T. Prove that T is a scalar multiple of the identity operator.

Solution. Let $v \in V$ be any nonzero vector. By assumption, v is an eigenvector of T, so $Tv = \lambda_v v$ for some scalar $\lambda_v \in \mathbb{F}$. Now, let $u \in V$ be another nonzero vector. Consider the vector $u + v$. If u and v are linearly dependent, then without loss of generality, let $u = \alpha v$ for some scalar $\alpha \in \mathbb{F}$. Then $Tu = \lambda_u u = \lambda_u \alpha v$, and $Tu = T(\alpha v) = \alpha Tv = \alpha\lambda_v v$. Equating these, we get $\lambda_u \alpha v = \alpha\lambda_v v$, and since $\alpha \neq 0$ and $v \neq 0$, it follows that $\lambda_u = \lambda_v$.

The remaining case is when $u + v$ are linearly independent. In this case, $u + v$ is an eigenvector of T, so $T(u+v) = \lambda_{u+v}(u+v)$ for some scalar $\lambda_{u+v} \in \mathbb{F}$. On the other hand, $T(u+v) = Tu + Tv = \lambda_u u + \lambda_v v$. Equating these, we get $\lambda_{u+v} u + \lambda_{u+v} v = \lambda_u u + \lambda_v v$. Since u and v are linearly independent, we can equate coefficients to get $\lambda_{u+v} = \lambda_u = \lambda_v$.

Since u and v were arbitrary nonzero vectors, it follows that there exists a scalar $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all nonzero $v \in V$. Hence, $T = \lambda I$. ■

Exercise 5A27

Suppose that V is finite-dimensional and $k \in \{1, \ldots, \dim V - 1\}$. Suppose $T \in \mathcal{L}(V)$ is such that every subspace of V of dimension k is invariant under T. Prove that T is a scalar multiple of the identity operator.

Appendix. *See* Appendix 5A27 *for a more elegant proof that doesn't require construction of subspaces.*

Solution. Let $v_1 \in V$ be nonzero. Our goal is to intersect enough k-dimensional subspaces to isolate $\operatorname{span} v_1$. We proceed as follows. Let $\dim V = n$, and extend v_1 to a basis $v_1, \ldots, v_n$ of V. For each $i = 2, \ldots, n$, let

$$U_i = \operatorname{span}(v_1, w_2, \ldots, w_k), \quad w_2, \ldots, w_k \in \{v_2, \ldots, v_n\} \setminus \{v_i\},$$

i.e., each U_i must contain v_1 as well as $k-1$ vectors from the basis except v_i. This choice is possible, since $k \leq n-1$, so there are enough vectors in $\{v_2, \ldots, v_n\} \setminus \{v_i\}$ to choose $k-1$ of them. The intersection of all such U_i will be exactly $\operatorname{span} v_1$. Since each U_i is invariant under T, we have $Tv_1 \in U_i$ for all i. Therefore, $Tv_1 \in \operatorname{span} v_1$, which means Tv_1 is a scalar multiple of v_1. Since v_1 was arbitrary, it follows that T is a scalar multiple of the identity operator. ■

Exercise 5A28

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has at most $1 + \dim \operatorname{range} T$ distinct eigenvalues.

Solution. Let $\lambda_1, \ldots, \lambda_m$ be distinct eigenvalues of T with corresponding eigenvectors $v_1, \ldots, v_m$. Since the eigenvalues are distinct, at most one eigenvalue must be 0. Without loss of generality, assume $\lambda_1 = 0$ if 0 is an eigenvalue. Then $\lambda_2, \ldots, \lambda_m$ are nonzero eigenvalues of T. For $j = 2, \ldots, m$, we have $Tv_j = \lambda_j v_j$, so $T(\lambda_j^{-1} v_j) = v_j$, which implies $v_j \in \operatorname{range} T$. Therefore, $v_2, \ldots, v_m$ are linearly independent vectors in $\operatorname{range} T$, so $m - 1 \leq \dim \operatorname{range} T$. Hence, $m \leq 1 + \dim \operatorname{range} T$. ■

Exercise 5A29

Suppose $T \in \mathcal{L}(\mathbb{R}^3)$ and $-4, 5$, and $\sqrt{7}$ are eigenvalues of T. Prove that there exists $x \in \mathbb{R}^3$ such that $Tx - 9x = (-4, 5, \sqrt{7})$.

Solution. Since $\dim \mathbb{R}^3 = 3$, the eigenvalues of T are exactly $-4, 5$, and $\sqrt{7}$. It follows that 9 is not an eigenvalue of T, and $\operatorname{null}(T-9I) = \{0\}$. Then $T-9I$ is injective, and since $\mathbb{R}^3$ is finite-dimensional, $T-9I$ is also surjective. Therefore, for any vector in $\mathbb{R}^3$, there exists $x \in \mathbb{R}^3$ such that $(T-9I)x = (-4, 5, \sqrt{7})$. ■

Exercise 5A30

Suppose $T \in \mathcal{L}(V)$ and $(T-2I)(T-3I)(T-4I) = 0$. Suppose λ is an eigenvalue of T. Prove that $\lambda = 2$ or $\lambda = 3$ or $\lambda = 4$.

Solution. Let v be an eigenvector of T corresponding to the eigenvalue λ. Then

$$(T-2I)(T-3I)(T-4I)v = (T-2I)(T-3I)(\lambda-4)v = (T-2I)(\lambda-3)(\lambda-4)v = (\lambda-2)(\lambda-3)(\lambda-4)v = 0.$$

Since $v \neq 0$, it must be that $(\lambda-2)(\lambda-3)(\lambda-4) = 0$. Therefore, $\lambda = 2$, $\lambda = 3$, or $\lambda = 4$. ■

Exercise 5A31

Give an example of $T \in \mathcal{L}(\mathbb{R}^2)$ such that $T^4 = -I$.

Solution. Consider the rotation matrix by $\pi/4$ radians:

$$T = \begin{pmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{pmatrix}.$$

Then

$$T^4 = \begin{pmatrix} \cos(\pi) & -\sin(\pi) \\ \sin(\pi) & \cos(\pi) \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

■

Exercise 5A32

Suppose $T \in \mathcal{L}(V)$ has no eigenvalues and $T^4 = I$. Prove that $T^2 = -I$.
Manual Remark. *The proof of this is a little easier if we may use minimal polynomials.*

Solution. Note that

$$T^4 - I = (T^2 - I)(T^2 + I) = (T - I)(T + I)(T^2 + I) = 0.$$

Since T has no eigenvalues, $T - I$ and $T + I$ are both injective, hence $T^2 - I$ is injective. Then for any $v \in V$, we have

$$(T^2 - I)(T^2 + I)v = 0,$$

which implies $(T^2 + I)v = 0$ since $T^2 - I$ is injective. Therefore, $T^2 + I = 0$, or $T^2 = -I$. ■

Exercise 5A33

Suppose $T \in \mathcal{L}(V)$ and m is a positive integer.

(a) Prove that T is injective if and only if T^m is injective.

(b) Prove that T is surjective if and only if T^m is surjective.

Solution.

(a) Suppose T is injective. Then if $T^m v = 0$, we have $T(T^{m-1}v) = 0$, which implies $T^{m-1}v = 0$ since T is injective. Repeating this argument, we eventually get $v = 0$. Hence, T^m is injective.

Conversely, suppose T^m is injective. If $Tv = 0$, then $T^m v = 0$, which implies $v = 0$. Hence, T is injective.

(b) Suppose T is surjective. Then for any $v \in V$, there exists $w \in V$ such that $Tw = v$. Similarly, there exists $w_2 \in V$ such that $Tw_2 = w$, and so on. Repeating this m times, we find $u \in V$ such that $T^m u = v$. Hence, T^m is surjective.

Conversely, suppose T^m is surjective. Then for any $v \in V$, there exists $u \in V$ such that $T^m u = v$. Let $w = T^{m-1}u$. Then $Tw = T(T^{m-1}u) = T^m u = v$. Hence, T is surjective. ■

Exercise 5A34

Suppose V is finite-dimensional and $v_1, \dots, v_m \in V$. Prove that the list $v_1, \dots, v_m$ is linearly independent if and only if there exists $T \in \mathcal{L}(V)$ such that $v_1, \dots, v_m$ are eigenvectors of T corresponding to distinct eigenvalues.

Solution. Suppose $v_1, \dots, v_m$ is linearly independent. Since V is finite-dimensional, we can extend this list to a basis $v_1, \dots, v_m, v_{m+1}, \dots, v_n$ of V. Define $T \in \mathcal{L}(V)$ by setting $Tv_i = \lambda_i v_i$ for $i = 1, \dots, m$, where $\lambda_1, \dots, \lambda_m$ are distinct scalars, and choose $\lambda_{m+1}, \dots, \lambda_n$ arbitrarily (distinct from each other and from $\lambda_1, \dots, \lambda_m$). Then $v_1, \dots, v_m$ are eigenvectors of T corresponding to distinct eigenvalues.

Conversely, suppose there exists $T \in \mathcal{L}(V)$ such that $v_1, \dots, v_m$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_m$. By 5.11, eigenvectors corresponding to distinct eigenvalues are linearly independent. Hence, $v_1, \dots, v_m$ is linearly independent. ■

Exercise 5A35

Suppose that $\lambda_1, \dots, \lambda_n$ is a list of distinct real numbers. Prove that the list $e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ is linearly independent in the vector space of real-valued functions on $\mathbb{R}$.

Axler Hint. *Let $V = \operatorname{span}(e^{\lambda_1 x}, \dots, e^{\lambda_n x})$, and define an operator $D \in \mathcal{L}(V)$ by $Df = f'$. Find eigenvalues and eigenvectors of D.*

Solution. Suppose there exists a linear combination

$$a_1 e^{\lambda_1 x} + \cdots + a_n e^{\lambda_n x} = 0$$

for all $x \in \mathbb{R}$. Let $V = \operatorname{span}(e^{\lambda_1 x}, \dots, e^{\lambda_n x})$ and define $D \in \mathcal{L}(V)$ by $Df = f'$. Then $De^{\lambda_i x} = \lambda_i e^{\lambda_i x}$, so $e^{\lambda_i x}$ is an eigenvector of D corresponding to the eigenvalue λ_i. Since the λ_i are distinct, by 5.11, the eigenvectors corresponding to distinct eigenvalues are linearly independent. Hence, $a_1 = \cdots = a_n = 0$, proving that $e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ are linearly independent. ■

Exercise 5A36

Suppose that $\lambda_1, \ldots, \lambda_n$ is a list of distinct positive numbers. Prove that the list $\cos(\lambda_1 x), \ldots,$ $\cos(\lambda_n x)$ is linearly independent in the vector space of real-valued functions on $\mathbb{R}$.

Solution. Suppose there exists a linear combination

$$a_1 \cos(\lambda_1 x) + \cdots + a_n \cos(\lambda_n x) = 0$$

for all $x \in \mathbb{R}$. Let $V = \operatorname{span}(\cos(\lambda_1 x), \sin(\lambda_1 x), \ldots, \cos(\lambda_n x), \sin(\lambda_n x))$ and define $D \in \mathcal{L}(V)$ by $Df = f'$. Then $D^2 \cos(\lambda_i x) = -\lambda_i^2 \cos(\lambda_i x)$, so $\cos(\lambda_i x)$ is an eigenvector of D^2 corresponding to the eigenvalue $-\lambda_i^2$. Since the λ_i are distinct and positive, the $-\lambda_i^2$ are distinct, and by 5.11, the eigenvectors corresponding to distinct eigenvalues are linearly independent. Hence, $a_1 = \cdots = a_n = 0$, proving that $\cos(\lambda_1 x), \ldots, \cos(\lambda_n x)$ are linearly independent. ∎

Exercise 5A37

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(S) = TS$$

for each $S \in \mathcal{L}(V)$. Prove that the set of eigenvalues of T equals the set of eigenvalues of $\mathcal{A}$.

Solution. Suppose λ is an eigenvalue of T with corresponding eigenvector $v \in V$, so $Tv = \lambda v$. Consider a nonzero $\varphi \in V'$ and consider $S \in \mathcal{L}(V)$ defined by $Sw = \varphi(w)v$ for all $w \in V$. Then $S \neq 0$ and for all $w \in V$,

$$TSw = T(\varphi(w)v) = \varphi(w)Tv = \varphi(w)\lambda v = \lambda\varphi(w)v = \lambda Sw.$$

Hence, λ is an eigenvalue of $\mathcal{A}$.

Conversely, if λ is an eigenvalue of $\mathcal{A}$ with corresponding eigenvector $S \in \mathcal{L}(V)$, then $\mathcal{A}(S) = TS = \lambda S$. Since $S \neq 0$, there exists $v \in V$ such that $Sv \neq 0$. Then $TSv = \lambda Sv$, so λ is an eigenvalue of T. Therefore, the set of eigenvalues of T equals the set of eigenvalues of $\mathcal{A}$. ∎

Exercise 5A38

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V invariant under T. The **quotient operator** $T/U \in \mathcal{L}(V/U)$ is defined by

$$(T/U)(v + U) = Tv + U$$

for each $v \in V$.

(a) Show that the definition of T/U makes sense (which requires using the condition that U is invariant under T) and show that T/U is an operator on V/U.

(b) Show that each eigenvalue of T/U is an eigenvalue of T.

Solution.

(a) To show that the definition of T/U makes sense, suppose $v + U = v' + U$ for some $v, v' \in V$. Then $v - v' \in U$, and since U is invariant under T, we have $T(v - v') \in U$, so $Tv - Tv' \in U$, which implies $Tv + U = Tv' + U$. Hence, the definition of T/U is well-defined.

To show that T/U is an operator on V/U, note that for any $v + U \in V/U$, $(T/U)(v + U) = Tv + U \in V/U$. Linearity follows from the linearity of T: for any $\alpha, \beta \in \mathbb{F}$ and $v + U, w + U \in V/U$,

$$\begin{aligned}(T/U)(\alpha(v + U) + \beta(w + U)) &= (T/U)(\alpha v + \beta w + U)\\ &= T(\alpha v + \beta w) + U\\ &= \alpha Tv + \beta Tw + U\\ &= \alpha(T/U)(v + U) + \beta(T/U)(w + U)\end{aligned}$$

(b) Let λ be an eigenvalue of T/U with eigenvector $v + U \neq 0$. Then $v \notin U$ and $(T - \lambda I)v \in U$. Consider $(T - \lambda I)|_U \in \mathcal{L}(U)$.
- If $(T - \lambda I)|_U$ is not injective, then λ is an eigenvalue of $T|_U$ and hence of T.
- If $(T - \lambda I)|_U$ is injective, then by finite-dimensionality it is surjective onto U. Thus, there exists $u \in U$ with $(T - \lambda I)u = (T - \lambda I)v$. Consequently, $(T - \lambda I)(v - u) = 0$ with $v - u \neq 0$ (since $v \notin U$ and $u \in U$), showing λ is an eigenvalue of T.

In either case, λ is an eigenvalue of T. ■

Exercise 5A39

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an eigenvalue if and only if there exists a subspace of V of dimension $\dim V - 1$ that is invariant under T.

Solution. Suppose T has an eigenvalue λ with corresponding eigenvector $v \in V$. Consider $\operatorname{range}(T - \lambda I)$. Then $\dim \operatorname{range}(T - \lambda I) \leq \dim V - 1$. Extend $\operatorname{range}(T - \lambda I)$ to a subspace U of V of dimension $\dim V - 1$. Then U is invariant under T because for any $u \in U$, $Tu = (T - \lambda I)u + \lambda u \in U$. Hence, there exists a subspace of V of dimension $\dim V - 1$ that is invariant under T.

Conversely, suppose there exists a subspace U of V of dimension $\dim V - 1$ that is invariant under T. Then V/U is one-dimensional, and the induced operator T/U on V/U must have an eigenvalue λ. By Exercise 5A38 (b), any eigenvalue of T/U is also an eigenvalue of T. Hence, T has an eigenvalue. ■

Exercise 5A40

Suppose $S, T \in \mathcal{L}(V)$ and S is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

Solution. We first prove by induction that $(STS^{-1})^k = ST^kS^{-1}$ for all non-negative integers k. For $k = 0$, we have $(STS^{-1})^0 = I = ST^0S^{-1}$. Suppose the result holds for some $k = n$, so that $(STS^{-1})^n = ST^nS^{-1}$.

Then

$$\begin{aligned}(STS^{-1})^{n+1} &= (STS^{-1})^n(STS^{-1})\\ &= (ST^nS^{-1})(STS^{-1})\\ &= ST^n(S^{-1}S)TS^{-1}\\ &= ST^nTS^{-1}\\ &= ST^{n+1}S^{-1}.\end{aligned}$$

By induction, the result holds for all non-negative integers k.

Now let $p(z) = a_0 + a_1z + \cdots + a_mz^m$ be any polynomial. Then

$$\begin{aligned}p(STS^{-1}) &= a_0I + a_1(STS^{-1}) + \cdots + a_m(STS^{-1})^m\\ &= a_0I + a_1(STS^{-1}) + \cdots + a_m(ST^mS^{-1})\\ &= S(a_0I + a_1T + \cdots + a_mT^m)S^{-1}\\ &= Sp(T)S^{-1}.\end{aligned}$$

Hence, the result holds for all polynomials p. ■

Exercise 5A41

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V invariant under T. Prove that U is invariant under $p(T)$ for every polynomial $p \in \mathcal{P}(\mathbb{F})$.

Solution. Let $u \in U$. Since U is invariant under T, we have $T(u) \in U$. By induction, suppose $T^k(u) \in U$ for some non-negative integer k. Then $T^{k+1}(u) = T(T^k(u)) \in U$ since U is invariant under T. Hence, $T^k(u) \in U$ for all non-negative integers k. Now let $p(z) = a_0 + a_1z + \cdots + a_mz^m$ be any polynomial. Then

$$p(T)(u) = a_0u + a_1T(u) + \cdots + a_mT^m(u) \in U,$$

since each term is in U. Therefore, U is invariant under $p(T)$ for every polynomial $p \in \mathcal{P}(\mathbb{F})$. ■

Exercise 5A42

Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \ldots, x_n) = (x_1, 2x_2, 3x_3, \ldots, nx_n)$.

(a) Find all eigenvalues and eigenvectors of T.

(b) Find all subspaces of $\mathbb{F}^n$ that are invariant under T.

Appendix. *See* Appendix 5A42 *for a generalization to any diagonal operator. The solution here remains as to show the student the mechanism.*

Solution.

(a) Let λ be an eigenvalue of T with corresponding eigenvector $v = (x_1, x_2, \ldots, x_n) \in \mathbb{F}^n$. Then $T(v) = \lambda v$, which means

$$(x_1, 2x_2, 3x_3, \ldots, nx_n) = (\lambda x_1, \lambda x_2, \ldots, \lambda x_n).$$

Comparing components, we get $\lambda x_i = ix_i$ for each $i = 1, 2, \ldots, n$. Hence, if $x_i \neq 0$, then $\lambda = i$. Therefore, the eigenvalues of T are $1, 2, \ldots, n$, and the corresponding eigenvectors are nonzero multiples of the standard basis vectors $e_1, e_2, \ldots, e_n$.

(b) Let U be an invariant subspace of $\mathbb{F}^n$ under T. Let

$$u = \sum_{i=1}^{n} \alpha_i e_i \in U,$$

where e_i are the standard basis vectors of $\mathbb{F}^n$ and $\alpha_i \in \mathbb{F}$. Since U is invariant under T, we have

$$Tu = \sum_{i=1}^{n} \alpha_i T(e_i) = \sum_{i=1}^{n} \alpha_i i e_i \in U.$$

Consider some index i such that $\alpha_i \neq 0$. Then

$$u_1 = Tu - iu = \sum_{j=1}^{n} \alpha_j j e_j - i \sum_{j=1}^{n} \alpha_j e_j = \sum_{j=1}^{n} \alpha_j (j-i) e_j \in U.$$

The i-th term in this sum is $\alpha_i(i-i)e_i = 0$, hence the e_i term is eliminated from u_1. Consider another index j of u_1 such that $\alpha_j \neq 0$. Then we can form

$$u_2 = Tu_1 - ju_1 = \sum_{k=1}^{n} \alpha_k (k-i) k e_k - j \sum_{k=1}^{n} \alpha_k (k-i) e_k = \sum_{k=1}^{n} \alpha_k (k-i)(k-j) e_k \in U.$$

The j-th term in this sum is $\alpha_j(j-i)(j-j)e_j = 0$, and u_2 now has all terms except the i-th and j-th terms. By repeating this process for each index with a nonzero coefficient, we can isolate each $\alpha_i e_i$ in U whenever $\alpha_i \neq 0$. This shows that for any $u \in U$, all the standard basis vectors e_i corresponding to nonzero coefficients in u must also be in U. Therefore, U must be the span of some subset of the standard basis vectors $\{e_1, e_2, \ldots, e_n\}$. ■

Exercise 5A43

Suppose that V is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that $\{p(T) \mid p \in \mathcal{P}(\mathbb{F})\} \neq \mathcal{L}(V)$.

Appendix. *See* Appendix 5A43 *for a proof that utilizes commutativity of operators in* $\{p(T) \mid p \in \mathcal{P}(\mathbb{F})\}$.

Solution. If T is the zero transformation or a scalar multiple of the identity, the result trivially holds. Otherwise, because $\dim V > 1$, there is at least one nonzero vector $v_1 \in V$ that is not an eigenvector of T. Let $Tv_1 = v_2$, so that v_1 and v_2 are linearly independent, and extend v_1, v_2 to a basis $v_1, \ldots, v_n$ for V.

Now suppose that $\{p(T) \mid p \in \mathcal{P}(\mathbb{F})\} = \mathcal{L}(V)$ and consider the map $S \in \mathcal{L}(V)$ defined by $Sv_1 = 0$ and $Sv_i = v_i$ for $i = 2, \ldots, n$. Because $S \in \mathcal{L}(V)$, there must be a polynomial $p \in \mathcal{P}(\mathbb{F})$ such that $p(T) = S$. Because the null space of $p(T)$ is invariant under T, it must be that the null space of S is invariant under T. This is impossible, though, as $Sv_1 = 0$ and $STv_1 = Sv_2 = v_2 \neq 0$. Therefore, there is no $T \in \mathcal{L}(V)$ such that $\{p(T) \mid p \in \mathcal{P}(\mathbb{F})\} = \mathcal{L}(V)$. ■

5B The Minimal Polynomial

Exercise 5B1

Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of T^2 if and only if 3 or -3 is an eigenvalue of T.

Solution. Suppose 9 is an eigenvalue of T^2. Then there exists a nonzero vector $v \in V$ such that $T^2v = 9v$. This implies $(T - 3I)(T + 3I)v = 0$. Let $w = (T + 3I)v$, and suppose $w \neq 0$. Then

$$(T - 3I)w = Tw - 3w = 0 \implies Tw = 3w,$$

showing that 3 is an eigenvalue of T. If $w = 0$, then $(T + 3I)v = 0$, which implies $Tv = -3v$, showing that -3 is an eigenvalue of T.

Conversely, suppose 3 is an eigenvalue of T. Then there exists a nonzero vector $v \in V$ such that $Tv = 3v$. Applying T again, we get $T^2v = T(3v) = 3Tv = 3 \cdot 3v = 9v$, showing that 9 is an eigenvalue of T^2. Similarly, if -3 is an eigenvalue of T, then 9 is an eigenvalue of T^2. ■

Exercise 5B2

Suppose V is a complex vector space and $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that every subspace of V invariant under T is either $\{0\}$ or infinite-dimensional.

Solution. Suppose U is a subspace of V invariant under T. If $U = \{0\}$, we are done. Suppose U is nonzero and finite-dimensional. By invariance of U under T, $T|_U \in \mathcal{L}(U)$, and $T|_U$ has an eigenvalue $\lambda \in \mathbb{C}$ with corresponding eigenvector $v \in U$, $v \neq 0$. But then v is also an eigenvector of T with eigenvalue λ, contradicting the assumption that T has no eigenvalues. Hence, any nonzero invariant subspace U must be infinite-dimensional. ■

Exercise 5B3

Suppose n is a positive integer and $T \in \mathcal{L}(\mathbb{F}^n)$ is defined by

$$T(x_1, \dots, x_n) = (x_1 + \cdots + x_n, \dots, x_1 + \cdots + x_n).$$

(a) Find all eigenvalues and eigenvectors of T.
(b) Find the minimal polynomial of T.

Solution.

(a) If $n = 1$, then $T(x_1) = (x_1)$, and the only eigenvalue is 1 with corresponding eigenvectors being all nonzero vectors in $\mathbb{F}^1$. Otherwise, suppose $T(x_1, \dots, x_n) = \lambda(x_1, \dots, x_n)$ for some $\lambda \in \mathbb{F}$ and $(x_1, \dots, x_n) \neq 0$. Then

$$\lambda x_j = x_1 + \cdots + x_n \quad \text{for all } j = 1, \dots, n.$$

This implies that $\lambda x_1 = \lambda x_2 = \cdots = \lambda x_n$. If $\lambda \neq 0$, then $x_1 = x_2 = \cdots = x_n$, and we can write $(x_1, \dots, x_n) = (x_1, \dots, x_1) = x_1(1, \dots, 1)$. Substituting this into the eigenvalue equation gives

$$\lambda(x_1, \dots, x_1) = T(x_1, \dots, x_1) = (nx_1, \dots, nx_1),$$

which implies $\lambda = n$. If $\lambda = 0$, then $x_1 + \cdots + x_n = 0$, and the eigenvectors are all nonzero vectors $(x_1, \ldots, x_n)$ such that $x_1 + \cdots + x_n = 0$. Hence, the eigenvalues of T are 0 and n, with corresponding eigenvectors as described.

(b) If $n = 1$, the minimal polynomial is $z - 1$.

For $n > 1$, we know from (a) that the eigenvalues are 0 and n. Then the minimal polynomial is a polynomial multiple of $z(z - n)$. Letting $x = x_1 + \cdots + x_n$, then

$$T(x_1, \ldots, x_n) = (x, \ldots, x), \quad (T - nI)(x_1, \ldots, x_n) = (x - nx_1, \ldots, x - nx_n).$$

Observe that $(x - nx_1) + \cdots + (x - nx_n) = nx - n(x_1 + \cdots + x_n) = 0$. Therefore, $(T - nI)(x_1, \ldots, x_n)$ lies in $\operatorname{null} T$, and applying T again gives

$$T(T - nI)(x_1, \ldots, x_n) = 0.$$

Hence, $z(z - n)$ is a polynomial multiple of the minimal polynomial, and since it must be monic, the minimal polynomial of T is $z(z - n)$. ■

Exercise 5B4

Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T)$ if and only if $\alpha = p(\lambda)$ for some eigenvalue λ of T.

Solution. Suppose α is an eigenvalue of $p(T)$. Then there exists a nonzero vector $v \in V$ such that $p(T)v = \alpha v$. Then $p(T) - \alpha I$ is not injective. If $p(z) - \alpha = 0$, then $\alpha = p(\lambda)$ for any $\lambda \in \mathbb{C}$, and we are done. If $p(z) - \alpha = \beta$ for some constant $\beta \in \mathbb{F}$, then $p(T) - \alpha I = \beta I$ is injective, which contradicts the fact that $p(T) - \alpha I$ is not injective. We may now assume that $p(z) - \alpha$ is a nonconstant polynomial. By the Fundamental Theorem of Algebra, we can factor $p(z) - \alpha$ as

$$p(z) - \alpha = c(z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_m)$$

for some $c \in \mathbb{C} \setminus \{0\}$ and $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$. Then

$$p(T) - \alpha I = c(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_m I).$$

Since $p(T) - \alpha I$ is not injective, at least one of the factors on the right-hand side is not injective. Therefore, $T - \lambda_j I$ is not injective for some j, which means λ_j is an eigenvalue of T. Then $p(\lambda_j) - \alpha = 0$, so $\alpha = p(\lambda_j)$ for some eigenvalue λ_j of T.

Conversely, suppose $\alpha = p(\lambda)$ for some eigenvalue λ of T with corresponding eigenvector v. Let $p(z) = a_m z^m + \cdots + a_1 z + a_0$. Since $T^m v = \lambda^m v$, we have

$$p(T)v = a_m T^m v + \cdots + a_1 Tv + a_0 v = a_m \lambda^m v + \cdots + a_1 \lambda v + a_0 v = p(\lambda)v = \alpha v.$$

Hence, α is an eigenvalue of $p(T)$. ■

Exercise 5B5

Give an example of an operator on $\mathbb{R}^2$ that shows the result in Exercise 4 does not hold if $\mathbb{C}$ is replaced with $\mathbb{R}$.

Solution. Consider the operator $T \in \mathcal{L}(\mathbb{R}^2)$ defined by $T(x,y) = (-y,x)$, which represents a counterclockwise rotation by 90°. Observe that T has no real eigenvalues as follows: If (x,y) is an eigenvector of T with eigenvalue $\lambda \in \mathbb{R}$, then $T(x,y) = \lambda(x,y)$ implies $(-y,x) = (\lambda x, \lambda y)$. This gives the system of equations:

$$-y = \lambda x \quad \text{and} \quad x = \lambda y.$$

Since x and y cannot both be zero, one can work out that $\lambda^2 = -1$, which has no real solutions. Therefore, T has no real eigenvalues. Now consider the polynomial $p(z) = z^2 + 1$. Then

$$p(T)(x,y) = T^2(x,y) + (x,y) = (-x,-y) + (x,y) = (0,0).$$

Thus, 0 is an eigenvalue of $p(T)$ with corresponding eigenvector $(1,0)$ (or any nonzero vector in $\mathbb{R}^2$). However, there is no real eigenvalue λ of T such that $p(\lambda) = 0$, since the eigenvalues of T are complex. This example demonstrates that the result in Exercise 5B4 does not hold when $\mathbb{C}$ is replaced with $\mathbb{R}$. ■

Exercise 5B6

Suppose $T \in \mathcal{L}(\mathbb{F}^2)$ is defined by $T(w,z) = (-z,w)$. Find the minimal polynomial of T.

Solution. By the computation done in Exercise 5B5, we have $T^2(w,z) = (-w,-z)$, so $T^2 + I = 0$. Therefore, the polynomial $p(z) = z^2 + 1$ annihilates T. To see that this is the minimal polynomial, suppose $T = \alpha I$. Then $T(1,0) = (0,1) \neq \alpha(1,0)$ for any $\alpha \in \mathbb{F}$, so $T \neq \alpha I$. Hence, the minimal polynomial of T is $p(z) = z^2 + 1$. ■

Exercise 5B7

(a) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^2)$ such that the minimal polynomial of ST does not equal the minimal polynomial of TS.

(b) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that if at least one of S, T is invertible, then the minimal polynomial of ST equals the minimal polynomial of TS.

Axler Hint. *Show that if S is invertible and $p \in \mathcal{P}(\mathbb{F})$, then $p(TS) = S^{-1}p(ST)S$.*

Solution.

(a) Consider the operators $S, T \in \mathcal{L}(\mathbb{F}^2)$ defined by $S(w,z) = (0,w)$ and $T(w,z) = (w,0)$. Then

$$ST(w,z) = S(w,0) = (0,w) \quad \text{and} \quad TS(w,z) = T(0,w) = (0,0).$$

The minimal polynomial of ST is $p(z) = z^2$, since $ST \neq 0$ but $(ST)^2 = 0$, while the minimal polynomial of TS is $q(z) = z$. Hence, the minimal polynomials of ST and TS are not equal.

(b) Without loss of generality, suppose S is invertible, and note that $ST = STSS^{-1}$. Let p be the minimal polynomial of TS. By Exercise 5A40, we have

$$p(ST) = p(S(TS)S^{-1}) = Sp(TS)S^{-1} = 0,$$

so p annihilates ST. Similarly, let q be the minimal polynomial of ST. Then

$$q(TS) = q(S^{-1}(ST)S) = S^{-1}q(ST)S = 0$$

as $TS = S^{-1}(ST)S$. Therefore, since p annihilates ST, p divides q, and since q annihilates TS, q divides p, so $p = q$. ■

Exercise 5B8

Suppose $T \in \mathcal{L}(\mathbb{R}^2)$ is the operator of counterclockwise rotation by $1°$. Find the minimal polynomial of T.
Axler Note. *Because* $\dim \mathbb{R}^2 = 2$*, the degree of the minimal polynomial of T is at most* 2*. Thus, the minimal polynomial of T is not the tempting polynomial* $x^{180} + 1$*, even though* $T^{180} = -I$*.*

Solution. Since T is not a scalar multiple of the identity operator, the minimal polynomial of T cannot have degree 1. Moreover, $\dim \mathbb{R}^2 = 2$ implies that the degree of the minimal polynomial of T must be 2. Therefore, the minimal polynomial of T is of the form

$$p(z) = z^2 + bz + c.$$

Consider the point $(1, 0) \in \mathbb{R}^2$. Note that $T(x, y) = (x \cos 1° - y \sin 1°, x \sin 1° + y \cos 1°)$. Then

$$T(1, 0) = (\cos 1°, \sin 1°), \quad T^2(1, 0) = (\cos 2°, \sin 2°).$$

Since $p(T) = 0$, we have

$$p(T)(1, 0) = T^2(1, 0) + bT(1, 0) + c(1, 0) = (\cos 2° + b \cos 1° + c, \sin 2° + b \sin 1°) = (0, 0).$$

This gives the system of equations:

$$\begin{cases} \cos 2° + b \cos 1° + c = 0, \\ \sin 2° + b \sin 1° = 0. \end{cases}$$

From this, we get the values of b and c:

$$b = -\frac{\sin 2°}{\sin 1°} = -2 \cos 1°, \quad c = -\cos 2° - b \cos 1° = 1.$$

Therefore, the minimal polynomial of T is $p(z) = z^2 - 2 \cos 1° z + 1$. ■

Exercise 5B9

Suppose $T \in \mathcal{L}(V)$ is such that with respect to some basis of V, all entries of the matrix of T are rational numbers. Explain why all coefficients of the minimal polynomial of T are rational numbers.

Solution. Let A be the matrix of T with respect to some basis of V. Since all entries of A are rational numbers, A^k also has rational entries for any positive integer k. Then the minimal polynomial $p(z) = z^m + a_{m-1} z^{m-1} + \cdots + a_1 z + a_0$ of T in terms of the matrix A satisfies

$$p(A) = A^m + a_{m-1} A^{m-1} + \cdots + a_1 A + a_0 I = 0.$$

Then this gives a system of $(\dim V)^2$ linear equations in the coefficients $a_0, a_1, \ldots, a_{m-1}$ with rational coefficients. By uniqueness of the minimal polynomial, this system has a unique solution. Since the system has rational coefficients and a unique solution, the solution must also be rational. Therefore, all coefficients of the minimal polynomial of T are rational numbers. ■

Exercise 5B10

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$. Prove that

$$\operatorname{span}(v, Tv, \dots, T^m v) = \operatorname{span}(v, Tv, \dots, T^{\dim V - 1} v)$$

for all integers $m \geq \dim V - 1$.

Solution. Let $n = \dim V$ and let $p(z) = z^d + a_{d-1} z^{d-1} + \cdots + a_0$ be the minimal polynomial of T. Then $d \leq n$ and $p(T) = 0$. For any $v \in V$,

$$p(T)v = 0 \implies T^d v = -a_{d-1} T^{d-1} v - \cdots - a_1 T v - a_0 v \implies T^d v \in \operatorname{span}\left(v, Tv, \dots, T^{d-1} v\right).$$

Let $U = \operatorname{span}\left(v, Tv, \dots, T^{d-1} v\right)$.

We claim that U is invariant under T. Take any $u = c_0 v + c_1 T v + \cdots + c_{d-1} T^{d-1} v \in U$. Then,

$$Tu = c_0 T v + c_1 T^2 v + \cdots + c_{d-2} T^{d-1} v + c_{d-1} T^d v.$$

Every term on the right-hand side except $T^d v$ is automatically in the spanning set of U. We also showed that $T^d v \in U$ earlier, so $Tu \in U$. Hence, $T(U) \subseteq U$.

Because $v \in U$ and U is invariant under T, we see that $T^m v \in U$ for every $m \geq 0$ (this is immediate by induction). So, for any $m \geq n - 1$, we have that

$$\operatorname{span}(v, Tv, \dots, T^m v) \subseteq U \subseteq \operatorname{span}(v, Tv, \dots, T^{n-1} v)$$

by using $d - 1 \leq n - 1$ to obtain the second inclusion. Since $\{v, Tv, \dots, T^{n-1} v\} \subseteq \{v, Tv, \dots, T^m v\}$, the reverse inclusion is immediate, and so

$$\operatorname{span}(v, Tv, \dots, T^m v) = \operatorname{span}(v, Tv, \dots, T^{n-1} v). \qquad \blacksquare$$

Exercise 5B11

Suppose V is a two-dimensional vector space, $T \in \mathcal{L}(V)$, and the matrix of T with respect to some basis of V is $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$.

(a) Show that $T^2 - (a + d)T + (ad - bc)I = 0$.

(b) Show that the minimal polynomial of T equals

$$\begin{cases} z - a & \text{if } b = c = 0 \text{ and } a = d, \\ z^2 - (a + d)z + (ad - bc) & \text{otherwise.} \end{cases}$$

Solution.

(a) We compute T^2:

$$T^2 = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ac + cd \\ ab + bd & bc + d^2 \end{pmatrix}.$$

Then

$$\begin{aligned}
T^2 - (a+d)T + (ad-bc)I &= \begin{pmatrix} a^2+bc & ac+cd \\ ab+bd & bc+d^2 \end{pmatrix} - (a+d)\begin{pmatrix} a & c \\ b & d \end{pmatrix} + (ad-bc)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} a^2+bc-(a+d)a+ad-bc & ac+cd-(a+d)c \\ ab+bd-(a+d)b & bc+d^2-(a+d)d+ad-bc \end{pmatrix} \\
&= \begin{pmatrix} a^2+bc-a^2-ad+ad-bc & ac+cd-ac-cd \\ ab+bd-ab-bd & bc+d^2-d^2-ad+ad-bc \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.
\end{aligned}$$

(b) If $b = c = 0$ and $a = d$, then the matrix of T is

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix},$$

which is a scalar multiple of the identity matrix. In this case, the minimal polynomial of T is $z - a$.

Otherwise, the polynomial $z^2 - (a+d)z + (ad-bc)$ annihilates T by part (a). To see that this is the minimal polynomial, we need to check that T is not a scalar multiple of the identity matrix. If T were a scalar multiple of the identity matrix, then we would have $b = c = 0$ and $a = d$, which contradicts our assumption. Therefore, the minimal polynomial of T has degree 2 and is $z^2 - (a+d)z + (ad-bc)$ in this case. ■

Exercise 5B12

Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \ldots, x_n) = (x_1, 2x_2, 3x_3, \ldots, nx_n)$. Find the minimal polynomial of T.

Solution. For each $k \in \{1, \ldots, n\}$, the standard basis vector e_k satisfies $Te_k = ke_k$, so each k is an eigenvalue. Let $p(z) = (z-1)(z-2)\cdots(z-n)$. So,

$$p(T)e_k = p(k)e_k = 0 \cdot e_k = 0 \text{ for all } k \in \{1, \ldots, n\},$$

and since the e_k form a basis of V, $p(T) = 0$. Therefore, the minimal polynomial $m_T(z)$ divides $p(z)$.

Now suppose $m_T(z)$ is the minimal polynomial of T. For each eigenvalue k,

$$0 = m_T(T)e_k = m_T(k)e_k \implies m_T(k) = 0.$$

Thus, $(z-k)$ divides $m_T(z)$ for every $k \in \{1, \ldots, n\}$. Because the factors are distinct, $p(z)$ divides $m_T(z)$.

Since both $p(z)$ and $m_T(z)$ are monic, $p(z) = m_T(z)$. Hence, the minimal polynomial is $p(z) = (z-1)\cdots(z-n)$. ■

Exercise 5B13

Suppose $T \in \mathcal{L}(V)$ and $p \in \mathcal{P}(\mathbb{F})$. Prove that there exists a unique $r \in \mathcal{P}(\mathbb{F})$ such that $p(T) = r(T)$ and $\deg r$ is less than the degree of the minimal polynomial of T.

Solution. Let m be the minimal polynomial of T with degree d. By the Division Algorithm for polynomials, we can write

$$p(z) = q(z)m(z) + r(z),$$

where $q, r \in \mathcal{P}(\mathbb{F})$ and $\deg r < d$. Applying this to T, we have

$$p(T) = q(T)m(T) + r(T).$$

Since $m(T) = 0$, it follows that

$$p(T) = r(T).$$

Thus, there exists a polynomial r such that $p(T) = r(T)$ and $\deg r < d$.

To show uniqueness, suppose there exists another polynomial $s \in \mathcal{P}(\mathbb{F})$ such that $p(T) = s(T)$ and $\deg s < d$. Then we have

$$r(T) = p(T) = s(T).$$

This implies that $(r - s)(T) = 0$. Since $\deg(r - s) < d$, it follows that $r - s$ must be the zero polynomial (otherwise, it would contradict the minimality of m). Hence, $r = s$, proving uniqueness. ■

Exercise 5B14

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$ has minimal polynomial $4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$. Find the minimal polynomial of T^{-1}.

Solution. Let $p(z) = 4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$ be the minimal polynomial of T. We want to find the minimal polynomial of T^{-1}. Since $p(z)$ is the minimal polynomial of T, we have

$$p(T) = 4I + 5T - 6T^2 - 7T^3 + 2T^4 + T^5 = 0.$$

To find the minimal polynomial of T^{-1}, apply T^{-1} to both sides of the equation until we express it in terms of T^{-1}:

$$4T^{-5} + 5T^{-4} - 6T^{-3} - 7T^{-2} + 2T^{-1} + I = 0.$$

Then the polynomial $q(z) = 4z^5 + 5z^4 - 6z^3 - 7z^2 + 2z + 1$ annihilates T^{-1}. To make it monic, we can divide by the leading coefficient:

$$q(z) = z^5 + \frac{5}{4}z^4 - \frac{3}{2}z^3 - \frac{7}{4}z^2 + \frac{1}{2}z + \frac{1}{4}.$$

Suppose T^{-1} has a minimal polynomial $m(z)$ with $\deg m < 5$, hence $m(T^{-1}) = 0$. Applying $T^{\deg m}$ to both sides gives $n(T) = 0$ for some polynomial $n(z)$ of degree less than 5, which contradicts the minimality of $p(z)$. Therefore, the minimal polynomial of T^{-1} is $q(z)$. ■

Exercise 5B15

Suppose V is a finite-dimensional complex vector space with $\dim V > 0$ and $T \in \mathcal{L}(V)$. Define $f : \mathbb{C} \to \mathbb{R}$ by

$$f(\lambda) = \dim \operatorname{range}(T - \lambda I).$$

Prove that f is not a continuous function.

Solution. Since V is a finite-dimensional complex vector space, the operator T has at least one eigenvalue $\lambda_0 \in \mathbb{C}$. By Exercise 5A11, there exists $\delta > 0$ such that $T - \lambda I$ is invertible for all $\lambda \in \mathbb{C}$ with $0 < |\lambda - \lambda_0| < \delta$. This implies that for all such λ, we have $\dim \operatorname{range}(T - \lambda I) = \dim V$. However, at the eigenvalue λ_0, we have $\dim \operatorname{range}(T - \lambda_0 I) < \dim V$ since $\dim \operatorname{null}(T - \lambda_0 I) > 0$. Therefore,

$$\lim_{\lambda \to \lambda_0} f(\lambda) \neq f(\lambda_0),$$

which shows that f is not continuous at λ_0. ■

Exercise 5B16

Suppose $a_0, \dots, a_{n-1} \in \mathbb{F}$. Let T be the operator on $\mathbb{F}^n$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} 0 & & & & & -a_0 \\ 1 & 0 & & & & -a_1 \\ & 1 & \ddots & & & -a_2 \\ & & \ddots & & & \vdots \\ & & & & 0 & -a_{n-2} \\ & & & & 1 & -a_{n-1} \end{pmatrix}.$$

Here all entries of the matrix are 0 except for all 1's on the line under the diagonal and the entries in the last column (some of which might also be 0). Show that the minimal polynomial of T is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

Solution. Let $p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$. Let $e_1, \dots, e_n$ be the standard basis of $\mathbb{F}^n$. We will show that $p(T) = 0$ and that no polynomial of lower degree annihilates T. First, we compute $T^k e_1$ for $k = 0, 1, \dots, n$:

$$T^0 e_1 = e_1, \quad T^1 e_1 = e_2, \quad T^2 e_1 = e_3, \quad \dots, \quad T^{n-1} e_1 = e_n, \quad T^n e_1 = -a_0 e_1 - a_1 e_2 - \cdots - a_{n-1} e_n.$$

Then we have

$$\begin{aligned} p(T)e_1 &= a_0 T^0 e_1 + a_1 T^1 e_1 + \cdots + a_{n-1} T^{n-1} e_1 + T^n e_1 \\ &= a_0 e_1 + a_1 e_2 + \cdots + a_{n-1} e_n - a_0 e_1 - a_1 e_2 - \cdots - a_{n-1} e_n \\ &= 0. \end{aligned}$$

Since $T^{j-1} e_1 = e_j$ for $j = 1, 2, \dots, n$, and $p(T)$ commutes with T, we have

$$p(T)e_j = p(T)T^{j-1} e_1 = T^{j-1} p(T) e_1 = T^{j-1} 0 = 0$$

for all $j = 1, 2, \dots, n$. Since $\{e_1, e_2, \dots, e_n\}$ is a basis for $\mathbb{F}^n$, it follows that $p(T) = 0$.

To show that p is the minimal polynomial, suppose there exists a polynomial $q(z) = b_0 + b_1 z + \cdots + b_{m-1} z^{m-1} + z^m$ of degree $m < n$ such that $q(T) = 0$. Then we would have

$$q(T)e_1 = b_0 e_1 + b_1 e_2 + \cdots + b_{m-1} e_m + T^m e_1 = 0.$$

However, $T^m e_1 = e_{m+1}$, which is not in the span of $\{e_1, e_2, \dots, e_m\}$ since $m < n$. This leads to a contradiction, as e_{m+1} cannot be expressed as a linear combination of $e_1, e_2, \dots, e_m$. Therefore, no polynomial of degree less than n annihilates T, and the minimal polynomial of T is indeed $p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$. ■

Exercise 5B17

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T. Suppose $\lambda \in \mathbb{F}$. Show that the minimal polynomial of $T - \lambda I$ is the polynomial q defined by $q(z) = p(z + \lambda)$.

Solution. Let $p(z)$ be the minimal polynomial of T. Then we have

$$p(T) = 0.$$

Consider the polynomial $q(z) = p(z + \lambda)$. Then we have

$$q(T - \lambda I) = p((T - \lambda I) + \lambda I) = p(T) = 0.$$

Thus, q annihilates $T - \lambda I$.

To show that q is the minimal polynomial, suppose there exists a polynomial $r(z)$ of degree less than $\deg q$ such that $r(T - \lambda I) = 0$. Then we can write

$$s(z) = r(z - \lambda)$$

for some polynomial $s(z)$ of degree less than $\deg p$. Then we have

$$s(T) = r(T - \lambda I) = 0.$$

However, this contradicts the minimality of p, since $\deg s < \deg p$ and $s(T) = 0$. Therefore, no polynomial of degree less than $\deg q$ annihilates $T - \lambda I$, and the minimal polynomial of $T - \lambda I$ is indeed $q(z) = p(z + \lambda)$. ■

Exercise 5B18

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T. Suppose $\lambda \in \mathbb{F} \setminus \{0\}$. Show that the minimal polynomial of λT is the polynomial q defined by $q(z) = \lambda^{\deg p} p(\frac{z}{\lambda})$.

Solution. Let $p(z)$ be the minimal polynomial of T. Then we have

$$p(T) = 0.$$

Consider the polynomial $q(z) = \lambda^{\deg p} p(\frac{z}{\lambda})$. Then we have

$$q(\lambda T) = \lambda^{\deg p} p\left(\frac{\lambda T}{\lambda}\right) = \lambda^{\deg p} p(T) = 0.$$

Moreover, q has leading term $\lambda^{\deg p} (\frac{z}{\lambda})^{\deg p} = z^{\deg p}$, so q is monic and has the same degree as p. To show that q is the minimal polynomial, suppose there exists a polynomial $r(z)$ of degree less than $\deg q$ such that $r(\lambda T) = 0$. Define $s(z) = r(\lambda z)$. Then we have

$$s(T) = r(\lambda T) = 0.$$

However, $\deg s = \deg r < \deg q = \deg p$, which contradicts the minimality of p. Therefore, no polynomial of degree less than $\deg q$ annihilates λT, and the minimal polynomial of λT is indeed $q(z) = \lambda^{\deg p} p(\frac{z}{\lambda})$. ■

Exercise 5B19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\mathcal{E}$ be the subspace of $\mathcal{L}(V)$ defined by

$$\mathcal{E} = \{q(T) \mid q \in \mathcal{P}(\mathbb{F})\}.$$

Prove that $\dim \mathcal{E}$ equals the degree of the minimal polynomial of T.

Solution. Let $p(z)$ be the minimal polynomial of T with degree d. Let $\mathcal{I} = \{I, T, T^2, \ldots, T^{d-1}\}$. We will show that $\mathcal{I}$ is a basis for $\mathcal{E}$.

First, we show that $\dim \mathcal{E} \leq d$. By the Division Algorithm for polynomials, for any polynomial $q \in \mathcal{P}(\mathbb{F})$, we can write

$$q(z) = s(z)p(z) + r(z),$$

where $s, r \in \mathcal{P}(\mathbb{F})$ and $\deg r < d$. Then we have

$$q(T) = s(T)p(T) + r(T) = 0 + r(T) = r(T).$$

Since $\deg r < d$, the set $\mathcal{I}$ spans $\mathcal{E}$. Hence, $\dim \mathcal{E} \leq d$.

Next, we show that $\dim \mathcal{E} \geq d$. Suppose

$$a_0 I + a_1 T + a_2 T^2 + \cdots + a_{d-1} T^{d-1} = 0$$

for scalars $a_0, a_1, \ldots, a_{d-1} \in \mathbb{F}$ not all zero. Then the polynomial

$$q(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_{d-1} z^{d-1}$$

annihilates T, contradicting the minimality of $p(z)$ since $\deg q < d$. Therefore, the set $\mathcal{I}$ is linearly independent in $\mathcal{E}$, which implies that $\dim \mathcal{E} \geq d$.

Combining both inequalities, we conclude that $\dim \mathcal{E} = d$, which is the degree of the minimal polynomial of T. ■

Exercise 5B20

Suppose $T \in \mathcal{L}(\mathbb{F}^4)$ is such that the eigenvalues of T are $3, 5, 8$. Prove that $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$.

Solution. Let p be the minimal polynomial of T. The eigenvalues of T are 3, 5, and 8, so each is a root of p. Hence,

$$p(z) = (z - 3)^{m_1}(z - 5)^{m_2}(z - 8)^{m_3}$$

where m_1, m_2, $m_3 \geq 1$. Because T acts on a 4-dimensional space, $\deg p \leq 4$, and so $m_1 + m_2 + m_3 \leq 4$. Since m_2, $m_3 \geq 1$, we have $m_1 \leq 4 - 1 - 1 = 2$. By similar logic, m_1, m_2, $m_3 \leq 2$. This means $p(z)$ divides $(z - 3)^2(z - 5)^2(z - 8)^2$. Applying this polynomial to T yields

$$(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0.$$ ■

Exercise 5B21

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of T has degree at most $1 + \dim \operatorname{range} T$.
Axler Note. *If* $\dim \operatorname{range} T < \dim V - 1$, *then this exercise gives a better upper bound than 5.22 for the degree of the minimal polynomial of* T.

Solution. Let $r = \dim \operatorname{range} T$. Observe that $\operatorname{range} T$ is invariant under T, and we can consider the restriction of T to $\operatorname{range} T$, denoted by $T|_{\operatorname{range} T}$. Let $p(z)$ be the minimal polynomial of T and let $q(z)$ be the minimal polynomial of $T|_{\operatorname{range} T}$. Then $\deg q \le r$ since $T|_{\operatorname{range} T}$ is an operator on a space of dimension r.

We now show that the polynomial $zq(z)$ annihilates T. For any $v \in V$, we know that $Tv \in \operatorname{range} T$. By invariance, we know that $w \in \operatorname{range} T$ implies $(T|_{\operatorname{range} T})^k w \in \operatorname{range} T$ for any $k \ge 0$. Then

$$q(T)w = q(T|_{\operatorname{range} T})w = 0.$$

Therefore, for any $v \in V$, we have

$$zq(z)(T)v = Tq(T)v = q(T)Tv = 0,$$

where we used the fact that T commutes with $q(T)$ since $q(T)$ is a polynomial in T. This shows that $zq(z)$ annihilates T, and hence the minimal polynomial $p(z)$ of T divides $zq(z)$. Since $\deg q \le r$, we have $\deg p \le 1 + r = 1 + \dim \operatorname{range} T$. ∎

Exercise 5B22

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if $I \in \operatorname{span}(T, T^2, \dots, T^{\dim V})$.

Solution. Suppose T is invertible. Let $q(z)$ be the minimal polynomial of T. Since T is invertible, 0 is not an eigenvalue of T, and thus $q(0) \ne 0$. We may write $q(z)$ as

$$q(z) = z^m + a_{m-1}z^{m-1} + \cdots + a_1 z + a_0,$$

where $a_0 \ne 0$ and $m \le \dim V$. Then we have

$$0 = q(T) = T^m + a_{m-1}T^{m-1} + \cdots + a_1 T + a_0 I.$$

Rearranging gives

$$I = -\frac{1}{a_0}(T^m + a_{m-1}T^{m-1} + \cdots + a_1 T) \in \operatorname{span}(T, T^2, \dots, T^m) \subseteq \operatorname{span}(T, T^2, \dots, T^{\dim V}).$$

Hence, if T is invertible, then $I \in \operatorname{span}(T, T^2, \dots, T^{\dim V})$.

Conversely, suppose $I \in \operatorname{span}(T, T^2, \dots, T^{\dim V})$. Then there exist scalars $a_1, a_2, \dots, a_{\dim V}$ such that

$$I = a_1 T + a_2 T^2 + \cdots + a_{\dim V} T^{\dim V}.$$

Rearranging gives

$$0 = -I + a_1 T + a_2 T^2 + \cdots + a_{\dim V} T^{\dim V} = p(T),$$

where $p(z) = -1 + a_1 z + a_2 z^2 + \cdots + a_{\dim V} z^{\dim V}$. Since the minimal polynomial of T divides any polynomial that annihilates T, the minimal polynomial of T must also have a nonzero constant term. Therefore, 0 is not an eigenvalue of T, and hence T is invertible. ■

Exercise 5B23

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $n = \dim V$. Prove that if $v \in V$, then $\operatorname{span}(v, Tv, \dots, T^{n-1}v)$ is invariant under T.

Solution. Let $W = \operatorname{span}(v, Tv, \dots, T^{n-1}v)$. To show that W is invariant under T, we may show that $Tv, T^2v, \dots, T^n v \in W$. Observe that $Tv, T(Tv) = T^2v, T(T^2v) = T^3v, \dots, T(T^{n-2}v) = T^{n-1}v$ are all in W by definition. It remains to show that $T^n v \in W$.

Let $p(z)$ be the minimal polynomial of T. Let $m = \deg p$. Since $m \leq n$, we have

$$p(T)v = T^m v + a_{m-1}T^{m-1}v + \cdots + a_1 Tv + a_0 v = 0.$$

Rearranging gives

$$T^m v = -a_{m-1}T^{m-1}v - \cdots - a_1 Tv - a_0 v \in W.$$

If $m = n$, then we are done. If $m < n$, then we can apply T^{n-m} to both sides to get

$$T^n v = -a_{m-1}T^{n-1}v - \cdots - a_1 T^{n-m+1}v - a_0 T^{n-m}v \in W,$$

where we used the fact that $T^{n-m}v, T^{n-m+1}v, \dots, T^{n-1}v \in W$ since $n - m < n$. Therefore, $T^n v \in W$, and hence W is invariant under T. ■

Exercise 5B24

Suppose V is a finite-dimensional complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that 5 and 6 are eigenvalues of T and that T has no other eigenvalues. Prove that $(T - 5I)^{\dim V - 1}(T - 6I)^{\dim V - 1} = 0$.

Solution. Let p be the minimal polynomial of T. The eigenvalues of T are 5 and 6, so each is a root of p. Hence,

$$p(z) = (z-5)^{m_1}(z-6)^{m_2}$$

where $m_1, m_2 \geq 1$. Because $\deg p \leq \dim V$, we have $m_1 + m_2 \leq \dim V$. Since $m_2 \geq 1$, we get $m_1 \leq \dim V - 1$, and similarly $m_2 \leq \dim V - 1$. This means $p(z)$ divides $(z-5)^{\dim V - 1}(z-6)^{\dim V - 1}$, so

$$(T - 5I)^{\dim V - 1}(T - 6I)^{\dim V - 1} = 0.$$ ■

Exercise 5B25

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T.

(a) Prove that the minimal polynomial of T is a polynomial multiple of the minimal polynomial of the quotient operator T/U.

(b) Prove that

$$(\text{minimal polynomial of } T|_U) \times (\text{minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of T.

Axler Note. *The quotient operator T/U was defined* Exercise 5A38.

Solution.

(a) Let $p(z)$ be the minimal polynomial of T and let $q(z)$ be the minimal polynomial of T/U. Recall that T/U is defined by $(T/U)(v+U) = Tv+U$ for all $v \in V$. Inductively, we have $(T/U)^k(v+U) = T^k v+U$ for all $k \geq 0$. Then we have

$$p(T/U)(v+U) = p(T)v + U = 0 + U = U,$$

where we used the fact that $p(T) = 0$. Therefore, $p(T/U) = 0$, which implies that $q(z)$ divides $p(z)$. Hence, the minimal polynomial of T is a polynomial multiple of the minimal polynomial of the quotient operator T/U.

(b) Let $p(z)$ be the minimal polynomial of T, let $q(z)$ be the minimal polynomial of $T|_U$, and let $r(z)$ be the minimal polynomial of T/U. By definition,

$$r(T/U)(v+U) = r(T)v + U = U,$$

where we used the fact that $r(T/U) = 0$. Then $r(T)v \in U$ for all $v \in V$. Moreover, invariance of U under T implies that $q(T)u = q(T|_U)u = 0$ for all $u \in U$. Since $r(T)v \in U$, and $q(T) = q(T|_U)$ on U, we have

$$q(T)r(T)v = q(T|_U)r(T)v = 0$$

for all $v \in V$. Therefore, $q(T)r(T) = 0$, which implies $p(z)$ divides $q(z)r(z)$. Hence, the product of the minimal polynomials of $T|_U$ and T/U is a polynomial multiple of the minimal polynomial of T. ■

Exercise 5B26

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V that is invariant under T. Prove that the set of eigenvalues of T equals the union of the set of eigenvalues of $T|_U$ and the set of eigenvalues of T/U.

Solution. Let λ be an eigenvalue of T with corresponding eigenvector $v \in V \setminus \{0\}$. Then we have

$$Tv = \lambda v.$$

If $v \in U$, then λ is an eigenvalue of $T|_U$. If $v \notin U$, then consider the coset $v + U \in V/U$. We have

$$(T/U)(v+U) = Tv + U = \lambda v + U = \lambda(v+U),$$

which shows that λ is an eigenvalue of T/U.

Conversely, suppose λ is an eigenvalue of $T|_U$ with corresponding eigenvector $u \in U \setminus \{0\}$. Then we have

$$T|_U u = Tu = \lambda u,$$

which shows that λ is also an eigenvalue of T.

Now suppose λ is an eigenvalue of T/U with corresponding eigenvector $v + U \in V/U \setminus \{0\}$. From Exercise 5B25 (a), we know that the minimal polynomial of T is a multiple of the minimal polynomial of T/U. Since λ is an eigenvalue of T/U, it must be a root of the minimal polynomial of T/U. Therefore, λ is also a root of the minimal polynomial of T, which implies that λ is an eigenvalue of T. ■

Exercise 5B27

Suppose $\mathbb{F} = \mathbb{R}$, V is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T_{\mathbb{C}}$ equals the minimal polynomial of T.

Axler Note. *The complexification $T_{\mathbb{C}}$ was defined in Exercise 3B33.*

Solution. Let $p(z)$ be the minimal polynomial of T. Then we have

$$p(T) = 0.$$

Consider the complexification $T_{\mathbb{C}}$ of T. By definition, $T_{\mathbb{C}}$ applied on $v + iw \in V_{\mathbb{C}}$ is given by

$$T_{\mathbb{C}}(v + iw) = T(v) + iT(w)$$

for all $v, w \in V$. By induction, we can extend the polynomial $p(z)$ to act on $V_{\mathbb{C}}$ by defining

$$p(T_{\mathbb{C}})(v + iw) = p(T)v + ip(T)w.$$

Since $p(T) = 0$, it follows that

$$p(T_{\mathbb{C}})(v + iw) = 0 + i0 = 0$$

for all $v, w \in V$. Therefore, $p(T_{\mathbb{C}}) = 0$, which shows that the minimal polynomial of $T_{\mathbb{C}}$ divides $p(z)$.

Conversely, let $q(z)$ be the minimal polynomial of $T_{\mathbb{C}}$. We can split $q(z)$ into its real and imaginary parts, say $q(z) = r(z) + is(z)$, where $r(z), s(z) \in \mathcal{P}(\mathbb{R})$. Then for any $v \in V$ which can be viewed as $v + i0 \in V_{\mathbb{C}}$, we have

$$q(T_{\mathbb{C}})(v + i0) = r(T)(v + i0) + is(T)(v + i0) = r(T)v + is(T)v.$$

Since $q(T_{\mathbb{C}}) = 0$, it follows that $r(T)v = 0$ and $s(T)v = 0$ for all $v \in V$. Therefore, both $r(T)$ and $s(T)$ must be 0, which implies that the minimal polynomial of T divides $r(z)$ and $s(z)$. Since $q(z) = r(z) + is(z)$, it follows that the minimal polynomial of T divides $q(z)$. Combining both directions, we conclude that the minimal polynomial of $T_{\mathbb{C}}$ equals the minimal polynomial of T. ■

Exercise 5B28

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the minimal polynomial of $T' \in \mathcal{L}(V')$ equals the minimal polynomial of T.

Axler Note. *The dual map T' was defined in Section 3F, 3.118.*

Solution. Let p be the minimal polynomial of T. For any $\varphi \in V'$, we have

$$p(T')(\varphi) = \varphi \circ p(T) = \varphi \circ 0 = 0 \implies p(T') = 0.$$

Hence, the minimal polynomial of T' divides p.

Conversely, let q be the minimal polynomial of T', so $q(T') = 0$. Then for all $\varphi \in V'$ and all $v \in V$,

$$0 = (q(T')\varphi)(v) = \varphi(q(T)v).$$

Thus, for each fixed v, $\varphi(q(T)v) = 0$ for all $\varphi \in V'$. Since $\varphi(w) = 0$ for all $\varphi \in V'$ implies $w = 0$, this forces $q(T)v = 0$ for all v; i.e., $q(T) = 0$. Therefore, p divides q.

Both p and q divide each other and are monic, so they are equal. ■

Exercise 5B29

Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

Axler Note. Exercise 5C6 *will give an improvement of this result when* $\mathbb{F} = \mathbb{C}$.

Appendix. *See* Appendix 5B29 *for an alternate proof using upper-triangular matrices, which appear in the next section.*

Solution. Let V be a finite-dimensional vector space with $\dim V \geq 2$, and let $T \in \mathcal{L}(V)$. Let $p(z)$ be the minimal polynomial of T. Over $\mathbb{F}$, every nonconstant polynomial factors into linear and irreducible quadratic factors. We split the proof into two cases based on the factorization of $p(z)$.

(1) Suppose $p(z)$ has a linear factor $z - \lambda$ for some $\lambda \in \mathbb{F}$. If $\deg p(z) = 1$, then $p(z) = z - \lambda$, which implies $T - \lambda I = 0$, or simply $T = \lambda I$. Since T is just a scalar multiple of the identity, every single subspace of V is invariant under T. Because $\dim V \geq 2$, we can choose any two linearly independent vectors $v, w \in V$, and $\operatorname{span}(v, w)$ will form a 2-dimensional invariant subspace.

If $\deg p(z) \geq 2$, then $p(z)/(z - \lambda)$ has degree at least 1 and must itself contain either a linear factor or an irreducible quadratic factor. The latter is handled by Case 2 below, so we consider the two linear cases:

- Repeated linear factor: $p(z) = (z - \lambda)^2 q(z)$. Since $p(z)$ is the minimal polynomial, there exists $v \in V$ such that $(T - \lambda I)q(T)v \neq 0$ but $(T - \lambda I)^2 q(T)v = 0$. Let $w = q(T)v$ and $u = (T - \lambda I)w$. Then $u \neq 0$ and $(T - \lambda I)u = (T - \lambda I)^2 w = 0$, which implies $Tu = \lambda u$. Thus, u is an eigenvector corresponding to the eigenvalue λ. Moreover, $Tw = (T - \lambda I)w + \lambda w = u + \lambda w$, which shows that $Tw \in \operatorname{span}(u, w)$.

  To confirm linear independence, suppose $u = \alpha w$ for some $\alpha \in \mathbb{F}$. Then $(T - \lambda I)w = \alpha w$, so w is an eigenvector of T with eigenvalue $\lambda + \alpha$. But then $(T - \lambda I)^2 w = \alpha^2 w$. Since $(T - \lambda I)^2 w = 0$ and $w \neq 0$, this forces $\alpha = 0$, contradicting our deduction that $\alpha \neq 0$ (since $u \neq 0$). Therefore, u and w are linearly independent, and $\operatorname{span}(u, w)$ is a 2-dimensional invariant subspace of V under T.
- Distinct linear factor: $p(z) = (z - \lambda)(z - \mu)g(z)$ with $\lambda \neq \mu$. Then T has two distinct eigenvalues, yielding two linearly independent eigenvectors v and u. The subspace $\operatorname{span}(v, u)$ is 2-dimensional, and since $Tv = \lambda v \in \operatorname{span}(v, u)$ and $Tu = \mu u \in \operatorname{span}(v, u)$, it is explicitly invariant.

(2) Suppose $p(z)$ has an irreducible quadratic factor, say $z^2 + az + b$ for some $a, b \in \mathbb{F}$. This means we can write $p(z) = (z^2 + az + b)q(z)$ for some polynomial $q(z)$. Since $p(z)$ is the minimal polynomial, $q(T)$ is not the zero operator, meaning there exists some vector $x \in V$ such that $q(T)x \neq 0$.

Let $u_1 = q(T)x$. Evaluating the minimal polynomial identity gives:

$$0 = p(T)x = (T^2 + aT + bI)q(T)x = (T^2 + aT + bI)u_1.$$

Rearranging this provides the operator relation $T^2 u_1 = -bu_1 - au_2$, where we define $u_2 = Tu_1$.

The vectors u_1 and u_2 must be linearly independent. If $Tu_1 = cu_1$ for some $c \in \mathbb{F}$, then $(T^2 + aT + bI)u_1 = (c^2 + ac + b)u_1 = 0$. Since $u_1 \neq 0$, this implies $c^2 + ac + b = 0$, meaning c is a root of $z^2 + az + b$ over $\mathbb{F}$, contradicting its irreducibility.

Let $U = \operatorname{span}(u_1, u_2)$, which has dimension two. Checking the images of our basis elements under T yields:

$$\begin{aligned} T(u_1) &= u_2 \in U \\ T(u_2) &= T^2 u_1 = -bu_1 - au_2 \in U. \end{aligned}$$

Since T maps the generating elements of U back into U, the subspace U is a 2-dimensional invariant subspace of V under T.

■

5C Upper-Triangular Matrices

Exercise 5C1

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and T^2 has an upper-triangular matrix with respect to some basis of V, then T has an upper-triangular matrix with respect to some basis of V.
Manual Remark. *The matrix in this exercise represents the* $90°$ *counterclockwise rotation of* $\mathbb{R}^2$.

Solution. We claim that the statement is false. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ with matrix representation

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

with respect to the standard basis. Then T^2 is given by

$$A^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix},$$

which is an upper-triangular matrix with respect to the standard basis of $\mathbb{R}^2$. However, T does not have an upper-triangular matrix with respect to any basis of $\mathbb{R}^2$, since it has no real eigenvalues and thus cannot be represented in upper-triangular form over $\mathbb{R}$. ■

Exercise 5C2

Suppose A and B are upper-triangular matrices of the same size, with $\alpha_1, \ldots, \alpha_n$ on the diagonal of A and $\beta_1, \ldots, \beta_n$ on the diagonal of B.

(a) Show that $A + B$ is an upper-triangular matrix with $\alpha_1 + \beta_1, \ldots, \alpha_n + \beta_n$ on the diagonal.

(b) Show that AB is an upper-triangular matrix with $\alpha_1\beta_1, \ldots, \alpha_n\beta_n$ on the diagonal.

Axler Note. *The results in this exercise are used in the proof of 5.81.*

Solution.

(a) Let $C = A + B$. Since A and B are upper-triangular, for any entry c_{ij} of C where $i > j$, we have

$$c_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0,$$

which shows that C is upper-triangular. The diagonal entries of C are given by

$$c_{ii} = a_{ii} + b_{ii} = \alpha_i + \beta_i,$$

for $i = 1, \ldots, n$.

(b) Let $D = AB$. Since A and B are upper-triangular, for any entry d_{ij} of D where $i > j$, we have the following sum for the (i, j)-th entry of D:

$$d_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Consider a_{ik} when $k < i$. Since A is upper-triangular, $a_{ik} = 0$. Now consider b_{kj} when $k > j$. Since B is upper-triangular, $b_{kj} = 0$. Therefore, for $i > j$, all terms in the sum are zero, and we have $d_{ij} = 0$ for $i > j$, showing that D is upper-triangular.

To see that the diagonal entries of D are $\alpha_1\beta_1, \dots, \alpha_n\beta_n$, we compute the diagonal entries:

$$d_{ii} = \sum_{k=1}^{n} a_{ik} b_{ki},$$

for the i-th diagonal entry. Since A and B are upper-triangular, then $k < i$ implies $a_{ik} = 0$ and $k > i$ implies $b_{ki} = 0$. Therefore, the only nonzero term in the sum occurs when $k = i$, giving

$$d_{ii} = a_{ii} b_{ii} = \alpha_i \beta_i,$$

for $i = 1, \dots, n$. ■

Exercise 5C3

Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \dots, v_n$ is a basis of V with respect to which the matrix of T is upper triangular, with $\lambda_1, \dots, \lambda_n$ on the diagonal. Show that the matrix of T^{-1} is also upper triangular with respect to the basis $v_1, \dots, v_n$, with

$$\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$$

on the diagonal.
Appendix. *See* Appendix 5C3 *for a direct computation of the entries of* T^{-1}.

Solution. Let $V_k = \operatorname{span}(v_1, \dots, v_k)$ for each $k \in \{1, \dots, n\}$. Because $\mathcal{M}(T, (v_1, \dots, v_n))$ is upper-triangular, $T(V_k) \subseteq V_k$ for every k; thus each V_k is invariant under T. Since T is invertible, its restriction to the (finite-dimensional) V_k is injective, hence surjective onto V_k. Therefore, $T(V_k) = V_k$. Applying T^{-1} yields $V_k = T^{-1}(V_k)$, meaning each V_k is also invariant under T^{-1}. So, $\mathcal{M}(T^{-1}, (v_1, \dots, v_n))$ is upper-triangular.

Let $\mu_1, \dots, \mu_n$ be the diagonal entries of $\mathcal{M}(T^{-1}, (v_1, \dots, v_n))$. Since $TT^{-1} = I$ is upper-triangular, we have from Exercise 5C2 (b) that $\lambda_k \mu_k = 1$ for each k, giving $\mu_k = \frac{1}{\lambda_k}$. ■

Exercise 5C4

Give an example of an operator whose matrix with respect to some basis contains only 0's on the diagonal, but the operator is invertible.
Axler Note. *This exercise and the exercise below show that 5.41 fails without the hypothesis that an upper-triangular matrix is under consideration.*

Solution. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ with matrix representation

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

with respect to the standard basis. A is invertible with inverse

$$A^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

■

Exercise 5C5

Give an example of an operator whose matrix with respect to some basis contains only nonzero numbers on the diagonal, but the operator is not invertible.

Solution. Consider the linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ with matrix representation

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

with respect to the standard basis. Since $T(1, -1) = (0, 0)$, T is not invertible. ∎

Exercise 5C6

Suppose $\mathbb{F} = \mathbb{C}$, V is finite-dimensional, and $T \in \mathcal{L}(V)$. Prove that if $k \in \{1, \dots, \dim V\}$, then V has a k-dimensional subspace invariant under T.

Solution. Recall that every operator on a finite-dimensional complex vector space has an upper-triangular matrix with respect to some basis. Let v_1, …, v_n be such a basis. Then, for each $k \in \{1, \dots, n\}$,

$$Tv_j \in \operatorname{span}(v_1, \dots, v_j) \subseteq \operatorname{span}(v_1, \dots, v_k)$$

for all $j \leq k$. Hence, $\operatorname{span}(v_1, \dots, v_k)$ is a k-dimensional subspace that is invariant under T. ∎

Exercise 5C7

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $v \in V$.

(a) Prove that there exists a unique monic polynomial p_v of smallest degree such that $p_v(T)v = 0$.

(b) Prove that the minimal polynomial of T is a polynomial multiple of p_v.

Solution.

(a) If $v = 0$, then $p(T)v = 0$ for any polynomial p, so we can take $p_v(x) = 1$. Otherwise, suppose $v \neq 0$. Consider the list $v, Tv, T^2v, \dots, T^{\dim V}v$. Since V is finite-dimensional, this list is linearly dependent. Then there exist scalars $a_0, \dots, a_m$ not all zero such that

$$a_0 v + a_1 Tv + \cdots + a_m T^m v = 0.$$

Here, m denotes the smallest integer such that there exist scalars $a_0, \dots, a_m$ not all zero satisfying the above equation. Since m is minimal, we have $a_m \neq 0$. Let $p_v(x)$ be the monic polynomial after dividing by a_m:

$$p_v(x) = x^m + \frac{a_{m-1}}{a_m} x^{m-1} + \cdots + \frac{a_0}{a_m}.$$

Then $p_v(T)v = 0$. Suppose q is another monic polynomial of smallest degree such that $q(T)v = 0$. Then $p_v - q$ is a polynomial of smaller degree than p_v such that $(p_v - q)(T)v = 0$, which contradicts the minimality of p_v.

(b) Let m_T be the minimal polynomial of T. Then $m_T(T) = 0$, so $m_T(T)v = 0$. By the division algorithm for polynomials, there exist unique polynomials q_0 and r such that

$$m_T = p_v q_0 + r,$$

where $r = 0$ or $\deg r < \deg p_v$. Then

$$0 = m_T(T)v = (p_v q_0 + r)(T)v = p_v(T)q_0(T)v + r(T)v = q_0(T)p_v(T)v + r(T)v = r(T)v,$$

since $p_v(T)v = 0$. If $r \neq 0$, then r is a polynomial of smaller degree than p_v such that $r(T)v = 0$, which contradicts the minimality of p_v. Therefore, $r = 0$, and we have $m_T = p_v q_0$, which shows that m_T is a polynomial multiple of p_v. ■

Exercise 5C8

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and there exists a nonzero vector $v \in V$ such that $T^2 v + 2Tv = -2v$.

(a) Prove that if $\mathbb{F} = \mathbb{R}$, then there does not exist a basis of V with respect to which T has an upper-triangular matrix.

(b) Prove that if $\mathbb{F} = \mathbb{C}$ and A is an upper-triangular matrix that equals the matrix of T with respect to some basis of V, then $-1 + i$ or $-1 - i$ appears on the diagonal of A.

Solution.

(a) Suppose there exists a basis of V with respect to which T has an upper-triangular matrix. Then the diagonal entries of this matrix are the eigenvalues of T. Let $p(z) = z^2 + 2z + 2$. Then $p(T)$ also has an upper-triangular matrix with respect to the same basis, and the diagonal entries of this matrix are $p(\lambda)$, where λ is an eigenvalue of T. Since $p(T)v = 0$ for some nonzero vector v, $p(T)$ is not injective. Then $p(T)$ is not invertible, so $p(\lambda) = 0$ for some eigenvalue $\lambda \in \mathbb{R}$ of T. However, $p(z)$ has no real roots, contradicting that $p(\lambda) = 0$ for some real eigenvalue λ.

(b) Utilizing the same deductions as in part (a), we conclude that there exists an eigenvalue λ of T such that $p(\lambda) = 0$. The roots of $p(z)$ are $-1 + i$ and $-1 - i$, so either $-1 + i$ or $-1 - i$ must be an eigenvalue of T. Since the diagonal entries of an upper-triangular matrix are the eigenvalues of the corresponding operator, either $-1 + i$ or $-1 - i$ must appear on the diagonal of A. ■

Exercise 5C9

Suppose B is a square matrix with complex entries. Prove that there exists an invertible square matrix A with complex entries such that $A^{-1}BA$ is an upper-triangular matrix.

Solution. Since B represents a linear operator on a finite-dimensional complex vector space, let T be the linear operator corresponding to B, i.e., $B = \mathcal{M}(T, (e_1, \ldots, e_n))$, where $(e_1, \ldots, e_n)$ is the standard basis of $\mathbb{C}^n$. Since T has an upper-triangular matrix with respect to some basis of the vector space, let A be the change of basis matrix from the new basis to the standard basis. Then $A^{-1}BA$ is the matrix representation of T with respect to this new basis, which is upper-triangular. ■

Exercise 5C10

Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V. Show that the following are equivalent.

(a) The matrix of T with respect to $v_1, \ldots, v_n$ is lower triangular.
(b) $\operatorname{span}(v_k, \ldots, v_n)$ is invariant under T for each $k = 1, \ldots, n$.
(c) $Tv_k \in \operatorname{span}(v_k, \ldots, v_n)$ for each $k = 1, \ldots, n$.

Axler Note. *A square matrix is called* ***lower triangular*** *if all entries above the diagonal are* 0.

Solution.

- Suppose the matrix of T with respect to $v_1, \ldots, v_n$ is lower triangular. Fix some $k \in \{1, \ldots, n\}$. For $j \in \{1, \ldots, n\}$,
$$Tv_j \in \operatorname{span}(v_j, \ldots, v_n)$$
because the matrix is lower triangular. If $j \geq k$, then $\operatorname{span}(v_j, \ldots, v_n) \subseteq \operatorname{span}(v_k, \ldots, v_n)$. Therefore, $Tv_j \in \operatorname{span}(v_k, \ldots, v_n)$ for all $j \geq k$, which shows that $\operatorname{span}(v_k, \ldots, v_n)$ is invariant under T.
- Suppose $\operatorname{span}(v_k, \ldots, v_n)$ is invariant under T for each $k = 1, \ldots, n$. Then for each k, we have $Tv_k \in \operatorname{span}(v_k, \ldots, v_n)$, which shows that Tv_k can be expressed as a linear combination of $v_k, \ldots, v_n$.
- Suppose $Tv_k \in \operatorname{span}(v_k, \ldots, v_n)$ for each $k = 1, \ldots, n$. Then
$$Tv_k = \sum_{j=k}^{n} A_{j,k} v_j$$
for some scalars $A_{j,k}$. Then $A_{i,k} = 0$ for all $i < k$, which shows that the matrix of T with respect to $v_1, \ldots, v_n$ is lower triangular. ■

Exercise 5C11

Suppose $\mathbb{F} = \mathbb{C}$ and V is finite-dimensional. Prove that if $T \in \mathcal{L}(V)$, then there exists a basis of V with respect to which T has a lower-triangular matrix.

Solution. Since T has an upper-triangular matrix with respect to some basis of V, let $v_1, \ldots, v_n$ be such a basis. Then $\operatorname{span}(v_1, \ldots, v_k)$ is invariant under T for each $k = 1, \ldots, n$. Consider the reverse basis $w_1 = v_n, w_2 = v_{n-1}, \ldots, w_n = v_1$. Then $\operatorname{span}(w_k, \ldots, w_n) = \operatorname{span}(v_1, \ldots, v_{n-k+1})$ is invariant under T for each $k = 1, \ldots, n$. By Exercise 5C10, the matrix of T with respect to the basis $w_1, \ldots, w_n$ is lower triangular. ■

Exercise 5C12

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V, and U is a subspace of V that is invariant under T.

(a) Prove that $T|_U$ has an upper-triangular matrix with respect to some basis of U.
(b) Prove that the quotient operator T/U has an upper-triangular matrix with respect to some basis of V/U.

Axler Note. *The quotient operator T/U was defined in Exercise 5A38.*

Solution.

(a) Since T has an upper-triangular matrix with respect to some basis of V, then its minimal polynomial is a product of linear factors. Since U is invariant under T, the minimal polynomial of $T|_U$ divides the minimal polynomial of T, and thus is also a product of linear factors. Therefore, $T|_U$ has an upper-triangular matrix with respect to some basis of U.

(b) Since the minimal polynomial of T is a polynomial multiple of the minimal polynomial of T/U, the same argument as in part (a) shows that T/U has an upper-triangular matrix with respect to some basis of V/U. ■

Exercise 5C13

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Suppose there exists a subspace U of V that is invariant under T such that $T|_U$ has an upper-triangular matrix with respect to some basis of U and also T/U has an upper-triangular matrix with respect to some basis of V/U. Prove that T has an upper-triangular matrix with respect to some basis of V.

Solution. Let $u_1, \ldots, u_m$ be a basis of U with respect to which $T|_U$ has an upper-triangular matrix. Let $v_1 + U, \ldots, v_n + U$ be a basis of V/U with respect to which T/U has an upper-triangular matrix. Then the list $u_1, \ldots, u_m, v_1, \ldots, v_n$ is a basis of V. Since $T|_U$ has an upper-triangular matrix with respect to the basis $u_1, \ldots, u_m$, we have

$$Tu_j = \sum_{k=1}^{j} A_{k,j} u_k$$

for some scalars $A_{k,j}$ and for each $j = 1, \ldots, m$. Since T/U has an upper-triangular matrix with respect to the basis $v_1 + U, \ldots, v_n + U$, we have

$$(T/U)(v_j + U) = \sum_{k=1}^{j} B_{k,j}(v_k + U)$$

for some scalars $B_{k,j}$ and for each $j = 1, \ldots, n$. This implies that

$$Tv_j - \sum_{k=1}^{j} B_{k,j} v_k \in U,$$

so there exist scalars $C_{l,j}$ such that

$$Tv_j - \sum_{k=1}^{j} B_{k,j} v_k = \sum_{l=1}^{m} C_{l,j} u_l.$$

Therefore,

$$Tv_j = \sum_{l=1}^{m} C_{l,j} u_l + \sum_{k=1}^{j} B_{k,j} v_k,$$

which shows that the matrix of T with respect to the basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ is upper-triangular. ■

Exercise 5C14

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an upper-triangular matrix with respect to some basis of V if and only if the dual operator T' has an upper-triangular matrix with respect to some basis of the dual space V'.

Solution. By Exercise 5B28, the minimal polynomials of T and T' are the same. Therefore, the following are equivalent:

- T has an upper-triangular matrix with respect to some basis of V.
- The minimal polynomial of T is a product of linear factors.
- The minimal polynomial of T' is a product of linear factors.
- T' has an upper-triangular matrix with respect to some basis of V'. ■

5D Diagonalizable Operators

Exercise 5D1

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$.

(a) Prove that if $T^4 = I$, then T is diagonalizable.

(b) Prove that if $T^4 = T$, then T is diagonalizable.

(c) Give an example of an operator $T \in \mathcal{L}(\mathbf{C}^2)$ such that $T^4 = T^2$ and T is not diagonalizable.

Manual Remark. *The roots ω in part (b) are the roots obtained in Exercise 1C6.*

Solution.

(a) If $T^4 = I$, then the minimal polynomial of T divides $z^4 - 1 = (z-1)(z+1)(z-i)(z+i)$, which has no repeated roots. Hence, the minimal polynomial of T has no repeated roots, so T is diagonalizable.

(b) If $T^4 = T$, then the minimal polynomial of T divides $z^4 - z = z(z-1)(z-\omega)(z-\omega^2)$. This polynomial has no repeated roots, so the minimal polynomial of T has no repeated roots, and hence T is diagonalizable.

(c) Let $T \in \mathcal{L}(\mathbf{C}^2)$ be defined by

$$T(z_1, z_2) = (z_2, 0).$$

Then $T^2(z_1, z_2) = (0, 0)$, so $T^4 = T^2$. However, T is not diagonalizable because the only eigenvalue of T is 0, and $\dim E(0, T) = 1 < 2 = \dim \mathbf{C}^2$. ■

Exercise 5D2

Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix A with respect to some basis of V. Prove that if $\lambda \in \mathbf{F}$, then λ appears on the diagonal of A precisely $\dim E(\lambda, T)$ times.

Solution. Suppose A is a diagonal matrix of T with respect to some basis of V. Let $\lambda \in \mathbf{F}$. Let m denote the number of times λ appears on the diagonal of A. Then $A - \lambda I$ is a diagonal matrix of $T - \lambda I$ with respect to the same basis, with exactly m zeros on the diagonal. A diagonal matrix with m zeros on the diagonal has null space of dimension m, so $\dim \operatorname{null}(T - \lambda I) = m$. Since $E(\lambda, T) = \operatorname{null}(T - \lambda I)$, we have $\dim E(\lambda, T) = m$. ■

Exercise 5D3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that if the operator T is diagonalizable, then $V = \operatorname{null} T \oplus \operatorname{range} T$.

Solution. Suppose T is diagonalizable. Let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T. Then

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T).$$

If 0 is not an eigenvalue of T, then $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = V$, so $V = \operatorname{null} T \oplus \operatorname{range} T$. If 0 is an eigenvalue of T, then we may assume $\lambda_1 = 0$. Let

$$U = E(\lambda_2, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Since $\operatorname{null} T = E(0, T)$, we have $V = \operatorname{null} T \oplus U$. It remains to show that $U = \operatorname{range} T$.

Let $u \in U$. Then $u = u_2 + \cdots + u_m$ for some $u_k \in E(\lambda_k, T)$. Since $\lambda_k \neq 0$ for $k = 2, \ldots, m$, we have $u_k = \lambda_k^{-1} T(u_k)$ for each $k = 2, \ldots, m$. Hence,

$$u = u_2 + \cdots + u_m = \lambda_2^{-1} T(u_2) + \cdots + \lambda_m^{-1} T(u_m) = T(\lambda_2^{-1} u_2 + \cdots + \lambda_m^{-1} u_m) \in \operatorname{range} T.$$

Thus, $U \subseteq \operatorname{range} T$.

Now let $v \in \operatorname{range} T$. Note that eigenspaces are invariant as follows: If $u \in E(\lambda_k, T)$, then $T(u) = \lambda_k u \in E(\lambda_k, T)$. Since U is a direct sum of eigenspaces, it is invariant under T. Then $v = T(w)$ for some $w \in V$. Since $V = \operatorname{null} T \oplus U$, we may write $w = w_1 + w_2$ for some $w_1 \in \operatorname{null} T$ and $w_2 \in U$. Then

$$v = T(w) = T(w_1 + w_2) = T(w_1) + T(w_2) = 0 + T(w_2) = T(w_2) \in U.$$

Thus, $\operatorname{range} T \subseteq U$. Since $U \subseteq \operatorname{range} T$ and $\operatorname{range} T \subseteq U$, we have $U = \operatorname{range} T$. Therefore, $V = \operatorname{null} T \oplus U = \operatorname{null} T \oplus \operatorname{range} T$. ■

Exercise 5D4

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) $V = \operatorname{null} T \oplus \operatorname{range} T$.
(b) $V = \operatorname{null} T + \operatorname{range} T$.
(c) $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.

Solution.

(a) $\implies$ (b): If $V = \operatorname{null} T \oplus \operatorname{range} T$, then $V = \operatorname{null} T + \operatorname{range} T$.
(b) $\implies$ (c): If $V = \operatorname{null} T + \operatorname{range} T$, then $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T - \dim(\operatorname{null} T \cap \operatorname{range} T)$. Since $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ by the fundamental theorem of linear maps, we have $\dim(\operatorname{null} T \cap \operatorname{range} T) = 0$, so $\operatorname{null} T \cap \operatorname{range} T = \{0\}$.
(c) $\implies$ (a): If $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, then

$$\begin{aligned} \dim(\operatorname{null} T + \operatorname{range} T) &= \dim \operatorname{null} T + \dim \operatorname{range} T - \dim(\operatorname{null} T \cap \operatorname{range} T) \\ &= \dim \operatorname{null} T + \dim \operatorname{range} T. \end{aligned}$$

By the fundamental theorem of linear maps, we have $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$. Hence,

$$\dim(\operatorname{null} T + \operatorname{range} T) = \dim V.$$

Since $\operatorname{null} T + \operatorname{range} T \subseteq V$ and $\dim(\operatorname{null} T + \operatorname{range} T) = \dim V$, we have $\operatorname{null} T + \operatorname{range} T = V$. Along with the assumption that $\operatorname{null} T \cap \operatorname{range} T = \{0\}$, we have $V = \operatorname{null} T \oplus \operatorname{range} T$. ■

Exercise 5D5

Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$$

for every $\lambda \in \mathbb{C}$.

Solution. Suppose T is diagonalizable. Let $\lambda \in \mathbb{C}$. Since T is diagonalizable, we have

$$V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$$

for some distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ of T. If λ is not an eigenvalue of T, then $\operatorname{null}(T - \lambda I) = \{0\}$ and $\operatorname{range}(T - \lambda I) = V$, so

$$V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I).$$

If λ is an eigenvalue of T, then we may assume $\lambda_1 = \lambda$. Let

$$U = E(\lambda_2, T) \oplus \cdots \oplus E(\lambda_m, T).$$

Then $V = E(\lambda, T) \oplus U$. It remains to show that $U = \operatorname{range}(T - \lambda I)$.

By dimension counting, we have

$$\dim U = \dim V - \dim E(\lambda, T) = \dim V - \dim \operatorname{null}(T - \lambda I) = \dim \operatorname{range}(T - \lambda I).$$

Since each $u_k \in E(\lambda_k, T)$ satisfies $T(u_k) = \lambda_k u_k$, we have

$$(T - \lambda I)(u_k) = (\lambda_k - \lambda)u_k \in E(\lambda_k, T).$$

To see that $T - \lambda I$ is injective on U, suppose $u \in U$ and $(T - \lambda I)(u) = 0$. Then $u = u_2 + \cdots + u_m$ for some $u_k \in E(\lambda_k, T)$, and

$$0 = (T - \lambda I)(u) = (T - \lambda I)(u_2 + \cdots + u_m) = (T - \lambda I)(u_2) + \cdots + (T - \lambda I)(u_m) = (\lambda_2 - \lambda)u_2 + \cdots + (\lambda_m - \lambda)u_m.$$

Since $\lambda_2, \ldots, \lambda_m$ are distinct from λ, we have $\lambda_k - \lambda \neq 0$ for each $k = 2, \ldots, m$. Since the sum above is a direct sum, we have $(\lambda_k - \lambda)u_k = 0$ for each $k = 2, \ldots, m$, so $u_k = 0$ for each $k = 2, \ldots, m$. Hence, $u = u_2 + \cdots + u_m = 0$, so $T - \lambda I$ is injective on U. Since $T - \lambda I$ is injective on U and $\dim U = \dim \operatorname{range}(T - \lambda I)$, we have $U = \operatorname{range}(T - \lambda I)$.

Conversely, suppose $V = \operatorname{null}(T - \lambda I) \oplus \operatorname{range}(T - \lambda I)$ for every $\lambda \in \mathbb{C}$. Then $\operatorname{null}(T - \lambda I) \cap \operatorname{range}(T - \lambda I) = \{0\}$ for every $\lambda \in \mathbb{C}$. Note that $T - \lambda I$ restricted to $\operatorname{range}(T - \lambda I)$ is injective, so λ is not an eigenvalue of $T|_{\operatorname{range}(T - \lambda I)}$. Then the minimal polynomial of $T|_{\operatorname{range}(T - \lambda I)}$ does not have λ as a root. Suppose $(z - \lambda)^2$ divides the minimal polynomial of T. Then there is some $v \in V$ such that $(T - \lambda I)^2 v = 0$, but $(T - \lambda I)v \neq 0$. Then $(T - \lambda I)v \in \operatorname{range}(T - \lambda I)$, and $v \in \operatorname{null}(T - \lambda I) \cap \operatorname{range}(T - \lambda I) = \{0\}$, which is a contradiction. Hence, $(z - \lambda)^2$ does not divide the minimal polynomial of T for any $\lambda \in \mathbb{C}$, so the minimal polynomial of T has no repeated roots. Since the minimal polynomial of T has no repeated roots, T is diagonalizable. ■

Exercise 5D6

Suppose $T \in \mathcal{L}(\mathbb{F}^5)$ and $\dim E(8, T) = 4$. Prove that $T - 2I$ or $T - 6I$ is invertible.

Solution. Suppose $T - 2I$ and $T - 6I$ are not invertible. Then 2 and 6 are eigenvalues of T, so $\dim E(2, T) \geq 1$ and $\dim E(6, T) \geq 1$. Since 8 is also an eigenvalue of T, we have

$$\dim E(2, T) + \dim E(6, T) + \dim E(8, T) \geq 1 + 1 + 4 = 6 > 5 = \dim \mathbb{F}^5,$$

which is a contradiction. Hence, $T - 2I$ or $T - 6I$ is invertible. ■

Exercise 5D7

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

$$E(\lambda, T) = E\left(\frac{1}{\lambda}, T^{-1}\right)$$

for every $\lambda \in \mathbb{F}$ with $\lambda \neq 0$.

Solution. Suppose $v \in E(\lambda, T)$. Then $T(v) = \lambda v$. Since T is invertible, we have $v = T^{-1}(\lambda v) = \lambda T^{-1}(v)$, so $T^{-1}(v) = \frac{1}{\lambda} v$. Hence,

$$v \in E(\lambda^{-1}, T^{-1}),$$

so $E(\lambda, T) \subseteq E(\lambda^{-1}, T^{-1})$.

Now suppose $v \in E(\lambda^{-1}, T^{-1})$. Then

$$T^{-1}(v) = \lambda^{-1} v \implies v = T(\lambda^{-1} v) \implies v = \lambda^{-1} T(v) \implies T(v) = \lambda v.$$

Hence, $v \in E(\lambda, T)$, so $E(\lambda^{-1}, T^{-1}) \subseteq E(\lambda, T)$. Then $E(\lambda, T) = E(\lambda^{-1}, T^{-1})$. ∎

Exercise 5D8

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct nonzero eigenvalues of T. Prove that

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim \operatorname{range} T.$$

Solution. For every nonzero eigenvalue λ of T and every $v \in E(\lambda, T)$, we have $v = T(\lambda^{-1} v) \in \operatorname{range} T$, so $E(\lambda, T) \subseteq \operatorname{range} T$. Since eigenspaces corresponding to distinct eigenvalues are linearly independent, we have

$$E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T) \subseteq \operatorname{range} T.$$

It follows that

$$\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) = \dim(E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)) \leq \dim \operatorname{range} T.$$ ∎

Exercise 5D9

Suppose $R, T \in \mathcal{L}(\mathbb{F}^3)$ each have $2, 6, 7$ as eigenvalues. Prove that there exists an invertible operator $S \in \mathcal{L}(\mathbb{F}^3)$ such that $R = S^{-1} T S$.

Solution. Since R and T each have 3 distinct eigenvalues, they are both diagonalizable. Let r_2, r_6, r_7 be eigenvectors of R with $R r_k = k r_k$ for $k = 2, 6, 7$, and let t_2, t_6, t_7 be eigenvectors of T with $T t_k = k t_k$ for $k = 2, 6, 7$. Each list of 3 eigenvectors is linearly independent, so they each form a basis of $\mathbb{F}^3$. Define $S \in \mathcal{L}(\mathbb{F}^3)$ by $S(r_k) = t_k$ for $k = 2, 6, 7$. Then S is invertible because it maps a basis of $\mathbb{F}^3$ to a basis of $\mathbb{F}^3$. For each $k = 2, 6, 7$, we have

$$S^{-1} T S(r_k) = S^{-1} T(t_k) = S^{-1}(k t_k) = k S^{-1}(t_k) = k r_k = R(r_k).$$

Since $S^{-1} T S$ and R agree on a basis of $\mathbb{F}^3$, we have $R = S^{-1} T S$. ∎

Exercise 5D10

Find $R, T \in \mathcal{L}(\mathbb{F}^4)$ such that R and T each have $2, 6, 7$ as eigenvalues, R and T have no other eigenvalues, and there does not exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$.

Solution. Define R to be the operator whose matrix with respect to the standard basis of $\mathbb{F}^4$ has diagonal entries $2, 6, 7, 7$ and all other entries 0. Then R has eigenvalues $2, 6, 7$ and no other eigenvalues. Define T to be the operator whose matrix with respect to the standard basis of $\mathbb{F}^4$ has diagonal entries $2, 6, 7, 2$ and all other entries 0. Then T has eigenvalues $2, 6, 7$ and no other eigenvalues. If there did exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$, then

$$Rv = S^{-1}TSv = \lambda v \implies T(Sv) = \lambda Sv$$

for every eigenvalue λ of R and every eigenvector $v \in E(\lambda, R)$. Hence, the map $S|_{E(\lambda,R)}$ maps $E(\lambda, R)$ to $E(\lambda, T)$. Since S is invertible, $S|_{E(\lambda,R)}$ is injective, so $\dim E(\lambda, R) \leq \dim E(\lambda, T)$. However, $\dim E(7, R) = 2 > 1 = \dim E(7, T)$, so there does not exist an invertible operator $S \in \mathcal{L}(\mathbb{F}^4)$ such that $R = S^{-1}TS$. ■

Exercise 5D11

Find $T \in \mathcal{L}(\mathbb{C}^3)$ such that 6 and 7 are eigenvalues of T and such that T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$.

Solution. Consider the operator $T \in \mathcal{L}(\mathbb{C}^3)$ whose matrix with respect to the standard basis of $\mathbb{C}^3$ is

$$\begin{pmatrix} 6 & 1 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{pmatrix}.$$

Observe that 6 and 7 are the only eigenvalues of T since the matrix is upper triangular with diagonal entries 6, 6, 7. If T had a diagonal matrix with respect to some basis of $\mathbb{C}^3$, then $\dim E(6, T) + \dim E(7, T) = 3$. However, $E(6, T)$ is the null space of $T - 6I$, which has matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Observe that the null space of this matrix is spanned by $(1, 0, 0)$, so $\dim E(6, T) = 1$. Similarly, $E(7, T)$ is the null space of $T - 7I$, which has matrix

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Observe that the null space of this matrix is spanned by $(0, 0, 1)$, so $\dim E(7, T) = 1$. Hence, $\dim E(6, T) + \dim E(7, T) = 1 + 1 = 2 < 3$, so T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$. ■

Exercise 5D12

Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is such that 6 and 7 are eigenvalues of T. Furthermore, suppose T does not have a diagonal matrix with respect to any basis of $\mathbb{C}^3$. Prove that there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that

$$T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3).$$

Solution. Since T cannot be diagonalized, it cannot have 3 distinct eigenvalues. Then 6 and 7 must be the only eigenvalues of T. In particular, 8 is not an eigenvalue of T, so $T - 8I$ is surjective. Then there exists $(z_1, z_2, z_3) \in \mathbb{C}^3$ such that

$$(T - 8I)(z_1, z_2, z_3) = (6, 7, 13) \implies T(z_1, z_2, z_3) = (6 + 8z_1, 7 + 8z_2, 13 + 8z_3). \qquad \blacksquare$$

Exercise 5D13

Suppose A is a diagonal matrix with distinct entries on the diagonal and B is a matrix of the same size as A. Show that $AB = BA$ if and only if B is a diagonal matrix.

Solution. Let A be a diagonal matrix with distinct entries on the diagonal, and let B be a matrix of the same size as A. Suppose $AB = BA$. Let $A_{j,j}$ denote the j-th diagonal entry of A, and let $B_{j,k}$ denote the entry of B in row j and column k. Then

$$(AB)_{j,k} = A_{j,j}B_{j,k}, \quad (BA)_{j,k} = B_{j,k}A_{k,k}.$$

Since $AB = BA$, we have

$$A_{j,j}B_{j,k} = B_{j,k}A_{k,k}.$$

If $j \neq k$, then $A_{j,j} - A_{k,k} \neq 0$ because the diagonal entries of A are distinct. Hence,

$$(A_{j,j} - A_{k,k})B_{j,k} = 0 \implies B_{j,k} = 0.$$

Thus, all off-diagonal entries of B are zero, so B is a diagonal matrix.

Conversely, if B is a diagonal matrix, then it is straightforward to verify that $AB = BA$. $\blacksquare$

Exercise 5D14

(a) Give an example of a finite-dimensional complex vector space and an operator T on that vector space such that T^2 is diagonalizable but T is not diagonalizable.

(b) Suppose $\mathbb{F} = \mathbb{C}$, k is a positive integer, and $T \in \mathcal{L}(V)$ is invertible. Prove that T is diagonalizable if and only if T^k is diagonalizable.

Solution.

(a) Consider the operator $T \in \mathcal{L}(\mathbb{C}^2)$ defined by

$$T(z_1, z_2) = (z_2, 0).$$

Then $T^2(z_1, z_2) = (0, 0)$, so T^2 is diagonalizable. However, T is not diagonalizable because the only eigenvalue of T is 0, and $\dim E(0, T) = 1 < 2 = \dim \mathbb{C}^2$.

(b) Suppose T is diagonalizable. Then a basis of V consists of eigenvectors of T, and each eigenvector of T is also an eigenvector of T^k. Hence, T^k is diagonalizable.

Conversely, suppose T^k is diagonalizable. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T^k. Then the minimal polynomial of T^k has no repeated roots, so we may write

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m).$$

Consider the polynomial

$$q(z) = (z^k - \lambda_1) \cdots (z^k - \lambda_m).$$

Note that $q(T) = p(T^k) = 0$, so the minimal polynomial of T divides q. Since T is invertible, T^k is also invertible. Then $\lambda_j \neq 0$ for each $j = 1, \ldots, m$, so q has no repeated roots. Hence, the minimal polynomial of T has no repeated roots, so T is diagonalizable. ■

Exercise 5D15

Suppose V is a finite-dimensional complex vector space, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T. Prove that the following are equivalent.

(a) T is diagonalizable.
(b) There does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$.
(c) p and its derivative p' have no zeros in common.
(d) The greatest common divisor of p and p' is the constant polynomial 1.

Axler Note. *The* **greatest common divisor** *of p and p' is the monic polynomial q of largest degree such that p and p' are both polynomial multiples of q. The Euclidean algorithm for polynomials* (*look it up*) *can quickly determine the greatest common divisor of two polynomials, without requiring any information about the zeros of the polynomials. Thus, the equivalence of* (a) *and* (d) *above shows that we can determine whether T is diagonalizable without knowing anything about the zeros of p.*

Solution.

(a) $\implies$ (b) Suppose T is diagonalizable. Then the minimal polynomial of T has no repeated roots, so there does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$.

(b) $\implies$ (c) Suppose there does not exist $\lambda \in \mathbb{C}$ such that p is a polynomial multiple of $(z - \lambda)^2$. Then p has no repeated roots, so we may let

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m)$$

for some distinct $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$. Then evaluating the derivative p' at λ_k gives

$$p'(\lambda_k) = \prod_{j \neq k} (\lambda_k - \lambda_j) \neq 0$$

by the distinctness of $\lambda_1, \ldots, \lambda_m$. Hence, p and p' have no zeros in common.

(c) $\implies$ (d) Suppose p and its derivative p' have no zeros in common. Then the greatest common divisor of p and p' has no zeros, so it is a constant polynomial.

(d) $\implies$ (a) Suppose the greatest common divisor of p and p' is the constant polynomial 1. If p had a repeated root $\lambda \in \mathbb{C}$, then p would be a polynomial multiple of $(z - \lambda)^2$, so p' would be a polynomial multiple of $(z - \lambda)$. Hence, p and p' would have a zero in common, which is a contradiction. Thus, p has no repeated roots, so T is diagonalizable. ■

Exercise 5D16

Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Let $\lambda_1, \ldots, \lambda_m$ denote the distinct eigenvalues of T. Prove that a subspace U of V is invariant under T if and only if there exist subspaces $U_1, \ldots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and $U = U_1 \oplus \cdots \oplus U_m$.

Solution. Suppose U is invariant under T. Let $U_k = U \cap E(\lambda_k, T)$ for each $k = 1, \ldots, m$. Then $U_k \subseteq E(\lambda_k, T)$ for each k. For each k, define the polynomial p_k by

$$p_k(z) = \prod_{j \neq k} \frac{z - \lambda_j}{\lambda_k - \lambda_j}.$$

Then $p_k(\lambda_j) = 0$ for $j \neq k$ and $p_k(\lambda_k) = 1$. Let $u \in U$ with

$$u = u_1 + \cdots + u_m, \quad u_k \in E(\lambda_k, T).$$

Then $p_k(T)(u) = u_k \in U$ for each k, so $u_k \in U_k$ for each k. Hence, $U = U_1 \oplus \cdots \oplus U_m$.

Conversely, suppose there exist subspaces $U_1, \ldots, U_m$ of V such that $U_k \subseteq E(\lambda_k, T)$ for each k and $U = U_1 \oplus \cdots \oplus U_m$. Then for each $u \in U$, we have

$$u = u_1 + \cdots + u_m, \quad u_k \in U_k.$$

Then

$$T(u) = T(u_1 + \cdots + u_m) = T(u_1) + \cdots + T(u_m) = \lambda_1 u_1 + \cdots + \lambda_m u_m \in U,$$

so U is invariant under T. ■

Exercise 5D17

Suppose V is finite-dimensional. Prove that $\mathcal{L}(V)$ has a basis consisting of diagonalizable operators.

Solution. Let $n = \dim V$. Let $\{v_1, \ldots, v_n\}$ be a basis of V. For each $j, k \in \{1, \ldots, n\}$, define $T_{j,k} \in \mathcal{L}(V)$ by

$$T_{j,k}(v_\ell) = \begin{cases} v_j & \text{if } \ell = k, \\ 0 & \text{if } \ell \neq k. \end{cases}$$

If $j = k$, let $D_j = T_{j,j}$. Then D_j is diagonalizable because $D_j(v_j) = v_j$ and $D_j(v_\ell) = 0$ for $\ell \neq j$. If $j \neq k$, let $N_{j,k} = D_j + T_{j,k}$. Then $N_{j,k}$ is diagonalizable because $N_{j,k}(v_j) = v_j$, $N_{j,k}(v_k) = v_j$, and $N_{j,k}(v_\ell) = 0$ for $\ell \neq j, k$. Then the eigenvalues of $N_{j,k}$ are 0 and 1 with corresponding eigenvectors $v_j - v_k$ ($n - 2$ of these) and v_j (1 of these), so $N_{j,k}$ is diagonalizable.

Note that the set

$$\{D_1, \ldots, D_n\} \cup \{N_{j,k} : j, k \in \{1, \ldots, n\}, j \neq k\}$$

spans $\mathcal{L}(V)$. For any $T \in \mathcal{L}(V)$, we have

$$T = \sum_{j,k=1}^{n} \alpha_{j,k} T_{j,k} = \sum_{j \neq k} \alpha_{j,k}(N_{j,k} - D_j) + \sum_{j=1}^{n} \alpha_{j,j} D_j,$$

which shows that the set above spans $\mathcal{L}(V)$.

Moreover, note that $\dim \mathcal{L}(V) = n^2$ and the set above has $n + n(n-1) = n^2$ elements, so it is a basis of $\mathcal{L}(V)$ consisting of diagonalizable operators. ■

Exercise 5D18

Suppose that $T \in \mathcal{L}(V)$ is diagonalizable and U is a subspace of V that is invariant under T. Prove that the quotient operator T/U is a diagonalizable operator on V/U.

Solution. Since T is diagonalizable, V has a basis $v_1, \ldots, v_n$ of eigenvectors of T. Let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues of T. Note that $v_1 + U, \ldots, v_n + U$ span V/U. Moreover, for each j, we have

$$(T/U)(v_j + U) = T(v_j) + U = \lambda_j v_j + U = \lambda_j(v_j + U),$$

so V/U is spanned by eigenvectors of T/U. We may reduce this spanning set to a basis of V/U consisting of eigenvectors of T/U, so T/U is diagonalizable. ■

Exercise 5D19

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there exists a subspace U of V that is invariant under T such that $T|_U$ and T/U are both diagonalizable, then T is diagonalizable.

Axler Note. *See* Exercise 5C13 *for an analogous statement about upper-triangular matrices.*

Solution. This statement is false. Consider the operator $T \in \mathcal{L}(\mathbb{F}^2)$ whose matrix with respect to the standard basis of $\mathbb{F}^2$ is

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Let $U = \operatorname{span}\{(1,0)\}$. Then U is invariant under T since $T(1,0) = (0,0) \in U$. The operator $T|_U$ is diagonalizable because it is the zero operator on a one-dimensional space. The quotient space V/U is one-dimensional, so T/U is diagonalizable. However, T is not diagonalizable because the only eigenvalue of T is 0, and $\dim E(0,T) = 1 < 2 = \dim \mathbb{F}^2$. ■

Exercise 5D20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if the dual operator T' is diagonalizable.

Solution. Suppose T is diagonalizable. Then there exists a basis of V consisting of eigenvectors of T. Let $\{v_1, \ldots, v_n\}$ be a basis of V consisting of eigenvectors of T, and let $\lambda_1, \ldots, \lambda_n$ be the corresponding eigenvalues. Let $\{\varphi_1, \ldots, \varphi_n\}$ be the dual basis of V' corresponding to $\{v_1, \ldots, v_n\}$. Then for each $k = 1, \ldots, n$, we have

$$T'(\varphi_k)(v_j) = \varphi_k(T(v_j)) = \varphi_k(\lambda_j v_j) = \lambda_j \varphi_k(v_j) = \begin{cases} \lambda_k & j = k, \\ 0 & j \neq k, \end{cases}$$

so $T'(\varphi_k) = \lambda_k \varphi_k$. Hence, $\{\varphi_1, \ldots, \varphi_n\}$ is a basis of V' consisting of eigenvectors of T', so T' is diagonalizable.

Conversely, suppose T' is diagonalizable. Then there exists a basis of V' consisting of eigenvectors of T'. Using the same reasoning as above, $(T')'$ is diagonalizable with the same eigenvalues as T'. From Exercise 3F32, we know that V is isomorphic to V'' with the map $\Lambda : V \to V''$ defined by

$$\Lambda(v)(\varphi) = \varphi(v)$$

for each $v \in V$ and $\varphi \in V'$. Then $(T')'(\Lambda(v)) = \Lambda(T(v))$ for each $v \in V$, so T has a basis of eigenvectors corresponding to the same eigenvalues as T', so T is diagonalizable. ■

Exercise 5D21

The ***Fibonacci sequence*** $F_0, F_1, F_2, \ldots$ is defined by

$$F_0 = 0, \quad F_1 = 1, \quad \text{and } F_n = F_{n-2} + F_{n-1} \text{ for } n \geq 2.$$

Define $T \in \mathcal{L}(\mathbb{R}^2)$ by $T(x, y) = (y, x + y)$.

(a) Show that $T^n(0, 1) = (F_n, F_{n+1})$ for each nonnegative integer n.
(b) Find the eigenvalues of T.
(c) Find a basis of $\mathbb{R}^2$ consisting of eigenvectors of T.
(d) Use the solution to (c) to compute $T^n(0, 1)$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

for each nonnegative integer n.
(e) Use (d) to conclude that if n is a nonnegative integer, then the Fibonacci number F_n is the integer that is closest to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

Axler Note. *Each F_n is a nonnegative integer, even though the right side of the formula in* (d) *does not look like an integer. The number*

$$\frac{1+\sqrt{5}}{2}$$

is called the ***golden ratio****.*
Manual Remark. *Part* (e) *uses the triangle inequality and creating "zeros" in the right places to show that the second term in the formula for F_n is less than $1/2$ in absolute value.*

Solution.

(a) We prove the statement by induction on n. For the base case $n = 0$, we have $T^0(0, 1) = (0, 1) = (F_0, F_1)$. Now suppose the statement holds for some nonnegative integer n. Then

$$\begin{aligned} T^{n+1}(0,1) &= T(T^n(0,1)) \\ &= T(F_n, F_{n+1}) \\ &= (F_{n+1}, F_n + F_{n+1}) \\ &= (F_{n+1}, F_{n+2}), \end{aligned}$$

so the statement holds for $n + 1$.

(b) Suppose $T(x,y) = \lambda(x,y)$ for some $\lambda \in \mathbb{R}$ and $(x,y) \in \mathbb{R}^2$ with $(x,y) \neq (0,0)$. Then

$$(y, x+y) = (\lambda x, \lambda y),$$

so $y = \lambda x$ and $x + y = \lambda y$. We then have

$$x + \lambda x = \lambda^2 x \implies (1 + \lambda - \lambda^2)x = 0.$$

If $x = 0$, then $y = \lambda x = 0$, which is a contradiction. Hence, $x \neq 0$, so we have

$$\lambda^2 - \lambda - 1 = 0 \implies \lambda = \frac{1 \pm \sqrt{5}}{2}.$$

(c) Let

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}.$$

Using $y = \lambda x$, we have the basis $\{(1,\lambda_1),(1,\lambda_2)\}$ of $\mathbb{R}^2$ consisting of eigenvectors of T.

(d) Let v_1 and v_2 be the eigenvectors of T corresponding to the eigenvalues λ_1 and λ_2, respectively. Since v_1, v_2 form a basis of $\mathbb{R}^2$, we may write $(0,1) = \alpha v_1 + \beta v_2$ for some $\alpha, \beta \in \mathbb{R}$:

$$(0,1) = \alpha\left(1, \frac{1+\sqrt{5}}{2}\right) + \beta\left(1, \frac{1-\sqrt{5}}{2}\right) = (\alpha+\beta, \alpha\lambda_1 + \beta\lambda_2).$$

Then $\alpha + \beta = 0$ and $\alpha\lambda_1 + \beta\lambda_2 = 1$, so $\beta = -\alpha$ and $\alpha(\lambda_1 - \lambda_2) = 1$. Note that $\lambda_1 - \lambda_2 = \sqrt{5}$, so $\alpha = 1/\sqrt{5}$ and $\beta = -1/\sqrt{5}$. Then

$$\begin{aligned} T^n(0,1) &= T^n\left(\frac{1}{\sqrt{5}}v_1 - \frac{1}{\sqrt{5}}v_2\right) \\ &= \frac{1}{\sqrt{5}}T^n(v_1) - \frac{1}{\sqrt{5}}T^n(v_2) \\ &= \frac{1}{\sqrt{5}}\lambda_1^n v_1 - \frac{1}{\sqrt{5}}\lambda_2^n v_2 \\ &= \frac{1}{\sqrt{5}}\left(\lambda_1^n - \lambda_2^n, \lambda_1^{n+1} - \lambda_2^{n+1}\right). \end{aligned}$$

Hence, we have

$$F_n = \frac{1}{\sqrt{5}}\left(\lambda_1^n - \lambda_2^n\right) = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right].$$

(e) Observe that $\frac{1}{\sqrt{5}} < \frac{1}{2}$. Moreover,

$$|\lambda_2| = \frac{\sqrt{5}-1}{2} < 1$$

since $\sqrt{5} < 3$. Then for all $n \geq 0$, we have $|\lambda_2^n| \leq 1$, so

$$\left|\frac{1}{\sqrt{5}}\lambda_2^n\right| < \frac{1}{2} \implies \left|\frac{1}{\sqrt{5}}\lambda_1^n - F_n\right| < \frac{1}{2}.$$

Let m be an integer such that $m \neq F_n$. Then $|m - F_n| \geq 1$, so

$$\left|\frac{1}{\sqrt{5}}\lambda_1^n - m\right| = \left|\frac{1}{\sqrt{5}}\lambda_1^n - F_n + F_n - m\right| \geq |F_n - m| - \left|\frac{1}{\sqrt{5}}\lambda_1^n - F_n\right| > 1 - \frac{1}{2} = \frac{1}{2}.$$

Since the distance from $\frac{1}{\sqrt{5}}\lambda_1^n$ to any integer other than F_n is greater than $1/2$, we conclude that F_n is the integer closest to $\frac{1}{\sqrt{5}}\lambda_1^n$. ■

Exercise 5D22

Suppose $T \in \mathcal{L}(V)$ and A is an n-by-n matrix that is the matrix of T with respect to some basis of V. Prove that if

$$|A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|$$

for each $j \in \{1, \dots, n\}$, then T is invertible.

Axler Note. *This exercise states that if the diagonal entries of the matrix of T are large compared to the nondiagonal entries, then T is invertible.*

Solution. By Theorem 5.67, each eigenvalue λ of T lies in at least one of the Gershgorin disks

$$D_j = \left\{ z \in \mathbb{C} \;\middle|\; |z - A_{j,j}| \leq \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}| \right\}.$$

If $\lambda = 0$ is an eigenvalue of T, then 0 lies in at least one of the Gershgorin disks D_j. However, by the given inequality, we have

$$|0 - A_{j,j}| = |A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^{n} |A_{j,k}|,$$

so $0 \notin D_j$ for each $j \in \{1, \dots, n\}$. Hence, 0 is not an eigenvalue of T, so T is invertible. ■

Exercise 5D23

Suppose the definition of the Gershgorin disks is changed so that the radius of the k^{th} disk is the sum of the absolute values of the entries in column (instead of row) k of A, excluding the diagonal entry. Show that the Gershgorin disk theorem (5.67) still holds with this changed definition.

Solution. Let $A = \mathcal{M}(T)$ be the matrix of T with respect to a basis $v_1, \dots, v_n$ for V. Let λ be an eigenvalue of T. By Exercise 5B28, we know that T and T' have the same minimal polynomial, so they have the same eigenvalues. Moreover, we know that the matrix of T' with respect to the dual basis is just A^t. Using Theorem 5.67 on T', there must be some $k \in \{1, \dots, n\}$ such that

$$|\lambda - A^t_{k,k}| \leq \sum_{\substack{j=1 \\ j \neq k}}^{n} |A^t_{k,j}|.$$

Since $A^t_{k,k} = A_{k,k}$ and $A^t_{k,j} = A_{j,k}$, we have

$$|\lambda - A_{k,k}| \leq \sum_{\substack{j=1 \\ j \neq k}}^{n} |A_{j,k}|,$$

so λ lies in the k^{th} Gershgorin disk with the changed definition. ■

5E Commuting Operators

Exercise 5E1

Give an example of two commuting operators S, T on $\mathbb{F}^4$ such that there is a subspace of $\mathbb{F}^4$ that is invariant under S but not under T and there is a subspace of $\mathbb{F}^4$ that is invariant under T but not under S.

Solution. Let e_1, e_2, e_3, e_4 be the standard basis of $\mathbb{F}^4$. Consider the operators $S, T \in \mathcal{L}(\mathbb{F}^4)$ defined by

$$S(x, y, z, w) = (0, x, 0, 0) \quad \text{and} \quad T(x, y, z, w) = (0, 0, x, 0).$$

Then

$$\begin{aligned} ST(x, y, z, w) &= S(0, 0, x, 0) = (0, 0, 0, 0), \\ TS(x, y, z, w) &= T(0, x, 0, 0) = (0, 0, 0, 0), \end{aligned}$$

so $ST = TS$. The subspace $U = \operatorname{span}(e_1, e_2)$ is invariant under S, since $Se_1 = e_2 \in U$ and $Se_2 = 0 \in U$, but not under T, since $Te_1 = e_3 \notin U$. The subspace $W = \operatorname{span}(e_1, e_3)$ is invariant under T, since $Te_1 = e_3 \in W$ and $Te_3 = 0 \in W$, but not under S, since $Se_1 = e_2 \notin W$. ■

Exercise 5E2

Suppose $\mathcal{E}$ is a subset of $\mathcal{L}(V)$ and every element of $\mathcal{E}$ is diagonalizable. Prove that there exists a basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix if and only if every pair of elements of $\mathcal{E}$ commutes.

Axler Note. *This exercise extends 5.76, which considers the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

Solution. Suppose there exists a basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix. Then every pair of elements of $\mathcal{E}$ commutes, since diagonal matrices commute.

Conversely, suppose every pair of elements of $\mathcal{E}$ commutes. We proceed by induction on $\dim V$. If $\dim V = 0$, then $\varnothing$ is a basis, every element of $\mathcal{E}$ has empty matrix representation, and every pair of elements of $\mathcal{E}$ commutes. If $\dim V = 1$, then every operator on V is diagonalizable and has a diagonal matrix with respect to any basis of V. Now suppose $\dim V > 1$ and the statement holds for all vector spaces of dimension less than $\dim V$. If every element of $\mathcal{E}$ is a scalar multiple of the identity operator, then every element of $\mathcal{E}$ has a diagonal matrix with respect to any basis of V. Otherwise, there exists an operator $S \in \mathcal{E}$ that is not a scalar multiple of the identity operator. Decompose V into a direct sum of eigenspaces of S:

$$V = E(\lambda_1, S) \oplus \cdots \oplus E(\lambda_m, S),$$

where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of S. Since every element of $\mathcal{E}$ commutes with S, each eigenspace $E(\lambda_i, S)$ is invariant under every element of $\mathcal{E}$. Since S is not a scalar multiple of the identity, $m \geq 2$, so $\dim E(\lambda_i, S) < \dim V$ for each i. For any $T \in \mathcal{E}$, Theorem 5.62 implies that the minimal polynomial of T is a product of distinct linear factors. Since the minimal polynomial of $T|_{E(\lambda_i, S)}$ divides the minimal polynomial of T, it follows that $T|_{E(\lambda_i, S)}$ is diagonalizable. Furthermore, any pair of elements of $\mathcal{E}$ commutes on $E(\lambda_i, S)$ since they commute on V and $E(\lambda_i, S)$ is invariant. By the induction hypothesis, for each i there exists a basis of $E(\lambda_i, S)$ with respect to which every restricted element of $\mathcal{E}$ has a diagonal matrix. Concatenating these bases gives a basis of V with respect to which every element of $\mathcal{E}$ has a diagonal matrix. ■

Exercise 5E3

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Suppose $p \in \mathcal{P}(\mathbf{F})$.

(a) Prove that $\operatorname{null} p(S)$ is invariant under T.

(b) Prove that $\operatorname{range} p(S)$ is invariant under T.

Axler Note. *See 5.18 for the special case $S = T$.*

Solution. We first show that $ST = TS$ implies $S^nT = TS^n$ for every nonnegative integer n by induction on n. The base Now suppose $S^nT = TS^n$ for some positive integer n. Then

$$S^{n+1}T = S(S^nT) = S(TS^n) = (ST)S^n = (TS)S^n = T(S^{n+1}),$$

so $S^{n+1}T = TS^{n+1}$. This completes the induction, so $S^nT = TS^n$ for all nonnegative integers n. Using this result, it follows that $p(S)T = Tp(S)$ for any polynomial $p \in \mathcal{P}(\mathbf{F})$.

(a) Suppose $v \in \operatorname{null} p(S)$. Then $p(S)v = 0$, so

$$p(S)(Tv) = Tp(S)v = T(0) = 0,$$

which shows that $Tv \in \operatorname{null} p(S)$. Thus, $\operatorname{null} p(S)$ is invariant under T.

(b) Suppose $v \in \operatorname{range} p(S)$. Then there exists $u \in V$ such that $p(S)u = v$, so

$$p(S)(Tu) = Tp(S)u = T(v),$$

which shows that $Tv \in \operatorname{range} p(S)$. Thus, $\operatorname{range} p(S)$ is invariant under T. ■

Exercise 5E4

Prove or give a counterexample: If A is a diagonal matrix and B is an upper-triangular matrix of the same size as A, then A and B commute.

Solution. This statement is false. Consider the diagonal matrix

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

and the upper-triangular matrix

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

but

$$BA = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Since $AB \neq BA$, A and B do not commute. ■

Exercise 5E5

Prove that a pair of operators on a finite-dimensional vector space commute if and only if their dual operators commute.
Axler Note. *See 3.118 for the definition of the dual of an operator.*

Solution. Suppose $S, T \in \mathcal{L}(V)$ commute. Then for any $\varphi \in V'$ and $v \in V$,

$$(S'T'\varphi)(v) = (T'\varphi)(Sv) = \varphi(TSv) = \varphi(STv) = (S'\varphi)(Tv) = (T'S'\varphi)(v),$$

so $S'T' = T'S'$.

Conversely, suppose $S', T' \in \mathcal{L}(V')$ commute. Applying the result we just proved to S' and T', we have that $(S')'(T')' = (T')'(S')'$, or equivalently $S''T'' = T''S''$. By Exercise 3F32, we know that S and T are isomorphic to S'' and T'', respectively, via the isomorphism Λ. Then for any $v \in V$,

$$\Lambda(STv) = S''T''\Lambda(v) = T''S''\Lambda(v) = \Lambda(TSv),$$

and injectivity of Λ implies that $ST = TS$. ■

Exercise 5E6

Suppose V is a finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute. Prove that there exist $\alpha, \lambda \in \mathbb{C}$ such that

$$\operatorname{range}(S - \alpha I) + \operatorname{range}(T - \lambda I) \neq V.$$

Solution. Assume $V \neq \{0\}$. Since V is finite-dimensional and over $\mathbb{C}$, S has an eigenvalue $\alpha \in \mathbb{C}$. Let $W = \operatorname{range}(S - \alpha I)$. Since $\operatorname{null}(S - \alpha I) \neq \{0\}$, the fundamental theorem of linear maps gives $W \neq V$, so V/W is a nonzero complex finite-dimensional space. Since $ST = TS$, the subspace W is invariant under T: for $w = (S - \alpha I)u \in W$,

$$Tw = T(S - \alpha I)u = (S - \alpha I)Tu \in W.$$

Thus, the quotient operator $T/W \in \mathcal{L}(V/W)$ is well-defined and has an eigenvalue $\lambda \in \mathbb{C}$.

Suppose for contradiction that $W + \operatorname{range}(T - \lambda I) = V$. For every $v \in V$, write $v = w + (T - \lambda I)u$ with $w \in W$ and $u \in V$. Then

$$v + W = (T - \lambda I)u + W = (T/W - \lambda I)(u + W),$$

so $T/W - \lambda I$ is surjective on V/W. Since V/W is finite-dimensional, $T/W - \lambda I$ is invertible, contradicting λ being an eigenvalue of T/W. ■

Exercise 5E7

Suppose V is a complex vector space, $S \in \mathcal{L}(V)$ is diagonalizable, and $T \in \mathcal{L}(V)$ commutes with S. Prove that there is a basis of V such that S has a diagonal matrix with respect to this basis and T has an upper-triangular matrix with respect to this basis.

Solution. Since S is diagonalizable, we can decompose V into a direct sum of eigenspaces of S:

$$V = E(\lambda_1, S) \oplus \cdots \oplus E(\lambda_m, S),$$

where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of S. Since $ST = TS$, each eigenspace $E(\lambda_i, S)$ is invariant under T. Since V is complex and each $E(\lambda_i, S)$ is invariant under T, there exists a basis of $E(\lambda_i, S)$ with respect to which $T|_{E(\lambda_i,S)}$ has an upper-triangular matrix by Theorem 5.44. Let β_i be such a basis of $E(\lambda_i, S)$, and let β be the union of the β_i. Since $V = E(\lambda_1, S) \oplus \cdots \oplus E(\lambda_m, S)$, β is a basis of V. Moreover, each basis vector in β_i is an eigenvector of S, so $\mathcal{M}(S, \beta)$ is diagonal. Lastly, for each $v \in \beta_i$, $Tv \in E(\lambda_i, S) = \operatorname{span}(\beta_i)$. By choice of β_i, $\mathcal{M}(T|_{E(\lambda_i,S)}, \beta_i)$ is upper-triangular, so $\mathcal{M}(T, \beta)$ is upper-triangular. ■

Exercise 5E8

Suppose $m = 3$ in Example 5.72 and D_x, D_y are the commuting partial differentiation operators on $\mathcal{P}_3(\mathbb{R}^2)$ from that example. Find a basis of $\mathcal{P}_3(\mathbb{R}^2)$ with respect to which D_x and D_y each have an upper-triangular matrix.

Solution. Consider the basis $1, x, y, x^2, xy, y^2, x^3, x^2y, xy^2, y^3$ of $\mathcal{P}_3(\mathbb{R}^2)$. Enumerate the basis vectors as $v_1, \ldots, v_{10}$ in the order given. Then $D_x v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ and $D_y v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ for each $j = 1, \ldots, 10$, so $\mathcal{M}(D_x)$ and $\mathcal{M}(D_y)$ are upper-triangular with respect to this basis. ■

Exercise 5E9

Suppose V is a finite-dimensional nonzero complex vector space. Suppose that $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that S and T commute for all $S, T \in \mathcal{E}$.

(a) Prove that there is a vector in V that is an eigenvector for every element of $\mathcal{E}$.
(b) Prove that there is a basis of V with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

Axler Note. *This exercise extends 5.78 and 5.80, which consider the case in which $\mathcal{E}$ contains only two elements. For this exercise, $\mathcal{E}$ may contain any number of elements, and $\mathcal{E}$ may even be an infinite set.*

Solution.

(a) We proceed by induction on $\dim V$. For the base case $\dim V = 1$, every nonzero vector in V is an eigenvector for every element of $\mathcal{E}$. Now suppose $\dim V > 1$ and the statement holds for all complex vector spaces of dimension less than $\dim V$. If $\mathcal{E} = \varnothing$, then any nonzero vector in V is an eigenvector for every element of $\mathcal{E}$. Since V is complex and finite-dimensional, T has an eigenvalue $\lambda \in \mathbb{C}$. By Exercise 5E3, $E(\lambda, T)$ is invariant under every element of $\mathcal{E}$.

If $E(\lambda, T) = V$, then $T = \lambda I$, so every nonzero vector is an eigenvector of T. If all elements of $\mathcal{E}$ are scalar multiples of the identity, then every nonzero vector is an eigenvector for every element of $\mathcal{E}$. Otherwise, there exists an operator $S \in \mathcal{E}$ that is not a scalar multiple of the identity. Let μ be an eigenvalue of S. Then $E(\mu, S)$ is a nonzero proper subspace of V that is invariant under every element of $\mathcal{E}$ by Exercise 5E3, so $\dim E(\mu, S) < \dim V$. The restrictions $\{R|_{E(\mu,S)} \mid R \in \mathcal{E}\}$ pairwise commute because for any $R_1, R_2 \in \mathcal{E}$,

$$R_1|_{E(\mu,S)}\, R_2|_{E(\mu,S)} = (R_1R_2)|_{E(\mu,S)} = (R_2R_1)|_{E(\mu,S)} = R_2|_{E(\mu,S)}\, R_1|_{E(\mu,S)}.$$

By the inductive hypothesis, there exists a nonzero $v \in E(\mu, S)$ that is an eigenvector for every restriction, hence for every element of $\mathcal{E}$.

(b) We proceed by induction on $\dim V$. For the base case $\dim V = 1$, every operator on V has an upper-triangular matrix with respect to any basis of V. Now suppose $\dim V > 1$ and the statement holds for all complex vector spaces of dimension less than $\dim V$. By part (a), there exists a nonzero $v_1 \in V$ that is an eigenvector for every element of $\mathcal{E}$. Let $U = \operatorname{span}(v_1)$. Since v_1 is an eigenvector for every element of $\mathcal{E}$, U is invariant under every element of $\mathcal{E}$, so the quotient operators $\{T/U \mid T \in \mathcal{E}\}$ on V/U are well-defined. For any $S, T \in \mathcal{E}$,

$$(S/U)(T/U) = (ST)/U = (TS)/U = (T/U)(S/U),$$

so the quotient operators pairwise commute. Since $\dim(V/U) = \dim V - 1$, the inductive hypothesis gives a basis $v_2 + U, \ldots, v_n + U$ of V/U with respect to which every T/U has an upper-triangular matrix. Then $v_1, v_2, \ldots, v_n$ is a basis of V. For any $T \in \mathcal{E}$, $Tv_1 \in \operatorname{span}(v_1)$ since v_1 is an eigenvector of T. For $k \geq 2$, T/U being upper-triangular gives $(T/U)(v_k + U) \in \operatorname{span}(v_2 + U, \ldots, v_k + U)$, so $Tv_k \in \operatorname{span}(v_1, v_2, \ldots, v_k)$. Hence, every element of $\mathcal{E}$ has an upper-triangular matrix with respect to $v_1, \ldots, v_n$. ■

Exercise 5E10

Give an example of two commuting operators S, T on a finite-dimensional real vector space such that $S + T$ has an eigenvalue that does not equal an eigenvalue of S plus an eigenvalue of T and ST has an eigenvalue that does not equal an eigenvalue of S times an eigenvalue of T.

Axler Note. *This exercise shows that 5.81 does not hold on real vector spaces.*

Solution. Consider the operators $S, T \in \mathcal{L}(\mathbb{R}^2)$ defined by

$$S(x, y) = (-y, x) \quad \text{and} \quad T(x, y) = (y, -x).$$

Since both are rotation operators, they commute. Neither S nor T has any real eigenvalues since no nonzero vector in $\mathbb{R}^2$ is a scalar multiple of its image under either operator. However, $S + T$ is the zero operator, which has eigenvalue 0, and $ST = I$, which has eigenvalue 1. ■

6 Inner Product Spaces

6A Inner Products and Norms

Exercise 6A1

Prove or give a counterexample: If $v_1, \ldots, v_m \in V$, then

$$\sum_{j=1}^{m} \sum_{k=1}^{m} \langle v_j, v_k \rangle \geq 0.$$

Solution. By the additivity of inner products in both slots, we have

$$\sum_{j=1}^{m} \sum_{k=1}^{m} \langle v_j, v_k \rangle = \left\langle \sum_{j=1}^{m} v_j, \sum_{k=1}^{m} v_k \right\rangle = \left\| \sum_{j=1}^{m} v_j \right\|^2 \geq 0.$$

■

Exercise 6A2

Suppose $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by

$$\langle u, v \rangle_1 = \langle Su, Sv \rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product on V if and only if S is injective.

Solution. Define $\langle u, v \rangle_1 = \langle Su, Sv \rangle$ for all $u, v \in V$. From the linearity of S and the properties of the regular inner product $\langle \cdot, \cdot \rangle$, we see that $\langle \cdot, \cdot \rangle_1$ satisfies the properties:

- $\langle u, v \rangle_1 = \langle Su, Sv \rangle = \overline{\langle Sv, Su \rangle} = \overline{\langle v, u \rangle_1}$ for all $u, v \in V$.
- $\langle u + w, v \rangle_1 = \langle S(u + w), Sv \rangle = \langle Su + Sw, Sv \rangle = \langle Su, Sv \rangle + \langle Sw, Sv \rangle = \langle u, v \rangle_1 + \langle w, v \rangle_1$ for all $u, v, w \in V$.
- $\langle \alpha u, v \rangle_1 = \langle S(\alpha u), Sv \rangle = \langle \alpha Su, Sv \rangle = \alpha \langle Su, Sv \rangle = \alpha \langle u, v \rangle_1$ for all $u, v \in V$ and $\alpha \in \mathbb{F}$.
- $\langle u, u \rangle_1 = \langle Su, Su \rangle = \|Su\|^2 \geq 0$ for all $u \in V$.

The only property that may not hold is definiteness.

If S is injective, then $\langle u, u \rangle_1 = 0 \implies \|Su\|^2 = 0 \implies Su = 0 \implies u = 0$, so definiteness holds. Conversely, if $\langle \cdot, \cdot \rangle_1$ is an inner product, then definiteness holds, so $Su = 0 \implies \langle u, u \rangle_1 = 0 \implies u = 0$, so S is injective. ■

Exercise 6A3

(a) Show that the function taking an ordered pair $((x_1, x_2), (y_1, y_2))$ of elements of $\mathbb{R}^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on $\mathbb{R}^2$.

(b) Show that the function taking an ordered pair $((x_1, x_2, x_3), (y_1, y_2, y_3))$ of elements of $\mathbb{R}^3$ to $x_1 y_1 + x_3 y_3$ is not an inner product on $\mathbb{R}^3$.

Solution.

(a) It is not linear in the first slot, since

$$\langle (1,0)+(-1,0),(1,0)\rangle = \langle (0,0),(1,0)\rangle = |0\cdot 1| + |0\cdot 0| = 0,$$

but

$$\langle (1,0),(1,0)\rangle + \langle (-1,0),(1,0)\rangle = |1\cdot 1| + |0\cdot 0| + |-1\cdot 1| + |0\cdot 0| = 2.$$

(b) It is not positive definite, since $(0,1,0) \neq (0,0,0)$ but

$$\langle (0,1,0),(0,1,0)\rangle = 0\cdot 0 + 0\cdot 0 = 0.$$

■

Exercise 6A4

Suppose $T \in \mathcal{L}(V)$ is such that $\|Tv\| \leq \|v\|$ for every $v \in V$. Prove that $T - \sqrt{2}I$ is injective.

Solution. Suppose $v \in V$ is such that $(T - \sqrt{2}I)v = 0$. Then $Tv = \sqrt{2}v$, so $\|Tv\| = \sqrt{2}\|v\|$. But $\|Tv\| \leq \|v\|$, so $\sqrt{2}\|v\| \leq \|v\|$, which implies $\|v\| = 0$, so $v = 0$. Thus, $T - \sqrt{2}I$ is injective. ■

Exercise 6A5

Suppose V is a real inner product space.

(a) Show that $\langle u+v, u-v\rangle = \|u\|^2 - \|v\|^2$ for every $u, v \in V$.

(b) Show that if $u, v \in V$ have the same norm, then $u+v$ is orthogonal to $u-v$.

(c) Use (b) to show that the diagonals of a rhombus are perpendicular to each other.

Solution.

(a) We have

$$\begin{aligned}\langle u+v, u-v\rangle &= \langle u, u-v\rangle + \langle v, u-v\rangle \\ &= \langle u,u\rangle - \langle u,v\rangle + \langle v,u\rangle - \langle v,v\rangle \\ &= \|u\|^2 - \|v\|^2.\end{aligned}$$

(b) If $\|u\| = \|v\|$, then $\langle u+v, u-v\rangle = \|u\|^2 - \|v\|^2 = 0$, so $u+v$ is orthogonal to $u-v$.

(c) Let u and v be the vectors corresponding to two adjacent sides of a rhombus. Then $\|u\| = \|v\|$, so $u+v$ is orthogonal to $u-v$, where $u+v$ and $u-v$ represent the diagonals of the rhombus. ■

Exercise 6A6

Suppose $u, v \in V$. Prove that $\langle u, v\rangle = 0 \iff \|u\| \leq \|u+av\|$ for all $a \in \mathbb{F}$.

Solution. Suppose $\langle u,v\rangle = 0$. Then for all $a \in \mathbb{F}$, we have $\langle u, av\rangle = 0$. By the Pythagorean theorem, we have

$$\|u+av\|^2 = \|u\|^2 + \|av\|^2 \geq \|u\|^2,$$

Conversely, suppose that $\|u\| \leq \|u+av\|$ for all $a \in \mathbb{F}$. If $v = 0$, then $\langle u,v\rangle = 0$. Otherwise, consider $v \neq 0$, and let $a = -\frac{\langle u,v\rangle}{\|v\|^2}$. We have $\langle u+av, v\rangle = 0$, so $\langle u+av, av\rangle = 0$. Then by the Pythagorean theorem, we have

$$\|u\|^2 = \|(u+av) - av\|^2 = \|u+av\|^2 + \|av\|^2.$$

Note that $\|u\| \leq \|u + av\|$ implies $\|u\|^2 \leq \|u + av\|^2$. Hence, $\|av\|^2 \leq 0$, which implies $a = 0$ and $\langle u, v\rangle = 0$. ■

Exercise 6A7

Suppose $u, v \in V$. Prove that $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$ if and only if $\|u\| = \|v\|$.

Solution. For all $a, b \in \mathbb{R}$, we have

$$\begin{aligned} \|au + bv\|^2 = \|bu + av\|^2 &\iff \langle au + bv, au + bv\rangle = \langle bu + av, bu + av\rangle \\ &\iff a^2\|u\|^2 + 2ab\,\Re\langle u, v\rangle + b^2\|v\|^2 = b^2\|u\|^2 + 2ab\,\Re\langle u, v\rangle + a^2\|v\|^2 \\ &\iff (a^2 - b^2)(\|u\|^2 - \|v\|^2) = 0. \end{aligned}$$

Since there exist $a, b \in \mathbb{R}$ such that $a^2 - b^2 \neq 0$, we have $\|au + bv\| = \|bu + av\|$ for all $a, b \in \mathbb{R}$ if and only if $\|u\|^2 - \|v\|^2 = 0$, which is equivalent to $\|u\| = \|v\|$. ■

Exercise 6A8

Suppose $a, b, c, x, y \in \mathbb{R}$ and $a^2 + b^2 + c^2 + x^2 + y^2 \leq 1$. Prove that $a + b + c + 4x + 9y \leq 10$.

Solution. Let $u = (a, b, c, x, y)$ and $v = (1, 1, 1, 4, 9)$. Then $\|u\|^2 = a^2 + b^2 + c^2 + x^2 + y^2 \leq 1$ and $\|v\|^2 = 1^2 + 1^2 + 1^2 + 4^2 + 9^2 = 99$. By the Cauchy–Schwarz inequality, we have

$$a + b + c + 4x + 9y = \langle u, v\rangle \leq |\langle u, v\rangle| \leq \|u\|\,\|v\| \leq \sqrt{99} < 10.$$ ■

Exercise 6A9

Suppose $u, v \in V$ and $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 1$. Prove that $u = v$.

Solution. We have

$$\|u - v\|^2 = \langle u - v, u - v\rangle = \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle = 1 - 1 - 1 + 1 = 0,$$

where $\langle v, u\rangle = \overline{\langle u, v\rangle} = \overline{1} = 1$. Therefore, $\|u - v\| = 0$, which implies $u - v = 0$, so $u = v$. ■

Exercise 6A10

Suppose $u, v \in V$ and $\|u\| \leq 1$ and $\|v\| \leq 1$. Prove that

$$\sqrt{1 - \|u\|^2}\,\sqrt{1 - \|v\|^2} \leq 1 - |\langle u, v\rangle|.$$

Solution. Consider the vectors $(\|u\|, \sqrt{1 - \|u\|^2})$ and $(\|v\|, \sqrt{1 - \|v\|^2})$ in $\mathbb{R}^2$. By the Cauchy–Schwarz inequality, we have

$$\|u\|\,\|v\| + \sqrt{1 - \|u\|^2}\,\sqrt{1 - \|v\|^2} \leq \sqrt{\|u\|^2 + 1 - \|u\|^2}\,\sqrt{\|v\|^2 + 1 - \|v\|^2} = 1.$$

By the Cauchy–Schwarz inequality, we also have $|\langle u, v\rangle| \leq \|u\|\,\|v\|$. Therefore,

$$\sqrt{1-\|u\|^2}\,\sqrt{1-\|v\|^2} \leq 1-\|u\|\,\|v\| \leq 1-|\langle u, v\rangle|. \qquad \blacksquare$$

Exercise 6A11

Find vectors $u, v \in \mathbb{R}^2$ such that u is a scalar multiple of $(1,3)$, v is orthogonal to $(1,3)$, and $(1,2) = u + v$.

Solution. Let $u = a(1,3) = (a, 3a)$ for some $a \in \mathbb{R}$. Then $v = (1,2) - u = (1-a, 2-3a)$. Since v is orthogonal to $(1,3)$, we have

$$\langle v, (1,3)\rangle = (1-a) + 3(2-3a) = 7 - 10a = 0,$$

which implies $a = \frac{7}{10}$. Therefore, $u = \left(\frac{7}{10}, \frac{21}{10}\right)$ and $v = \left(\frac{3}{10}, -\frac{1}{10}\right)$. $\blacksquare$

Exercise 6A12

Suppose a, b, c, d are positive numbers.

(a) Prove that $(a+b+c+d)\left(\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}+\dfrac{1}{d}\right) \geq 16$.

(b) For which positive numbers a, b, c, d is the inequality above an equality?

Solution.

(a) Consider the vectors in $\mathbb{R}^4$:

$$\begin{aligned} u &= \left(\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}\right) \implies \|u\|^2 = a+b+c+d, \\ v &= \left(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}}\right) \implies \|v\|^2 = \frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}. \end{aligned}$$

Note that

$$\langle u, v\rangle = \sqrt{a}\cdot\frac{1}{\sqrt{a}} + \sqrt{b}\cdot\frac{1}{\sqrt{b}} + \sqrt{c}\cdot\frac{1}{\sqrt{c}} + \sqrt{d}\cdot\frac{1}{\sqrt{d}} = 4.$$

By the Cauchy–Schwarz inequality, we have $|\langle u, v\rangle| \leq \|u\|\,\|v\|$, so

$$16 = |\langle u, v\rangle|^2 \leq \|u\|^2\,\|v\|^2 = (a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right).$$

(b) By the equality condition of the Cauchy–Schwarz inequality, we have equality if and only if u is a scalar multiple of v. If $u = \alpha v$ for some $\alpha \in \mathbb{R}$, then

$$\begin{cases} \sqrt{a} = \alpha\frac{1}{\sqrt{a}} \implies a = \alpha, \\ \sqrt{b} = \alpha\frac{1}{\sqrt{b}} \implies b = \alpha, \\ \sqrt{c} = \alpha\frac{1}{\sqrt{c}} \implies c = \alpha, \\ \sqrt{d} = \alpha\frac{1}{\sqrt{d}} \implies d = \alpha, \end{cases}$$

so $a = b = c = d$. $\blacksquare$

Exercise 6A13

Show that the square of an average is less than or equal to the average of the squares. More precisely, show that if $a_1, \ldots, a_n \in \mathbb{R}$, then the square of the average of $a_1, \ldots, a_n$ is less than or equal to the average of $a_1^2, \ldots, a_n^2$.

Manual Remark. *This is known as the QM–AM inequality, which is a special case of the power mean inequality.*

Solution. Let $u = (a_1, \ldots, a_n)$ and $v = (1, \ldots, 1) \in \mathbb{R}^n$. Then $\|u\|^2 = a_1^2 + \cdots + a_n^2$ and $\|v\|^2 = n$. By the Cauchy–Schwarz inequality, we have

$$|a_1 + \cdots + a_n|^2 = |\langle u, v\rangle|^2 \leq \|u\|^2 \, \|v\|^2 = n(a_1^2 + \cdots + a_n^2),$$

so

$$\left(\frac{a_1 + \cdots + a_n}{n}\right)^2 = \frac{|a_1 + \cdots + a_n|^2}{n^2} \leq \frac{n(a_1^2 + \cdots + a_n^2)}{n^2} = \frac{a_1^2 + \cdots + a_n^2}{n}. \qquad \blacksquare$$

Exercise 6A14

Suppose $v \in V$ and $v \neq 0$. Prove that $v/\|v\|$ is the unique closest element on the unit sphere of V to v. More precisely, prove that if $u \in V$ and $\|u\| = 1$, then

$$\left\| v - \frac{v}{\|v\|} \right\| \leq \|v - u\|,$$

with equality only if $u = v/\|v\|$.

Solution. Let $u \in V$ with $\|u\| = 1$. Then

$$\begin{aligned}
\|v - u\|^2 &= \|v\|^2 - 2\mathfrak{R}\langle v, u\rangle + \|u\|^2 \\
&\geq \|v\|^2 - 2|\langle v, u\rangle| + 1 \\
&\geq \|v\|^2 - 2\|v\| \, \|u\| + 1 \\
&= \|v\|^2 - 2\|v\| + 1 \\
&= (\|v\| - 1)^2
\end{aligned}$$

Equality holds if and only if $\mathfrak{R}\langle v, u\rangle = |\langle v, u\rangle| = \|v\|$. The second equality implies $v = \alpha u$ for some $\alpha \in \mathbb{F}$. Since $\alpha = \langle v, u\rangle$ and $\mathfrak{R}\langle v, u\rangle = |\langle v, u\rangle|$, we have $\alpha = \|v\| > 0$. Therefore, $u = v/\|v\|$. $\blacksquare$

Exercise 6A15

Suppose u, v are nonzero vectors in $\mathbb{R}^2$. Prove that

$$\langle u, v\rangle = \|u\| \, \|v\| \cos\theta,$$

where θ is the angle between u and v (thinking of u and v as arrows with initial point at the origin).

Axler Hint. *Use the law of cosines on the triangle formed by u, v, and $u - v$.*

Solution. Let θ be the angle between u and v. By the law of cosines, we have

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\,\|v\|\cos\theta.$$

On the other hand, we have

$$\|u - v\|^2 = \langle u - v, u - v\rangle = \langle u, u\rangle - 2\langle u, v\rangle + \langle v, v\rangle = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle.$$

Therefore,

$$\|u\|^2 + \|v\|^2 - 2\|u\|\,\|v\|\cos\theta = \|u\|^2 + \|v\|^2 - 2\langle u, v\rangle,$$

which implies $\langle u, v\rangle = \|u\|\,\|v\|\cos\theta$. ■

Exercise 6A16

The angle between two vectors (thought of as arrows with initial point at the origin) in $\mathbb{R}^2$ or $\mathbb{R}^3$ can be defined geometrically. However, geometry is not as clear in $\mathbb{R}^n$ for $n > 3$. Thus, the angle between two nonzero vectors $x, y \in \mathbb{R}^n$ is defined to be

$$\arccos\frac{\langle x, y\rangle}{\|x\|\,\|y\|},$$

where the motivation for this definition comes from Exercise 6A15. Explain why the Cauchy–Schwarz inequality is needed to show that this definition makes sense.

Solution. The Cauchy–Schwarz inequality states that $|\langle x, y\rangle| \leq \|x\|\,\|y\|$ for all $x, y \in \mathbb{R}^n$. Therefore, if x and y are nonzero vectors in $\mathbb{R}^n$, then

$$-1 \leq -\frac{|\langle x, y\rangle|}{\|x\|\,\|y\|} \leq \frac{\langle x, y\rangle}{\|x\|\,\|y\|} \leq \frac{|\langle x, y\rangle|}{\|x\|\,\|y\|} \leq 1.$$

Since the domain of arccos is $[-1, 1]$, the Cauchy–Schwarz inequality ensures that arccos is defined for the given expression. ■

Exercise 6A17

Prove that

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} k a_k^2\right)\left(\sum_{k=1}^{n} \frac{b_k^2}{k}\right)$$

for all real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

Solution. Define the vectors

$$u = \left(\sqrt{1}a_1, \sqrt{2}a_2, \ldots, \sqrt{n}a_n\right), \quad v = \left(\frac{b_1}{\sqrt{1}}, \frac{b_2}{\sqrt{2}}, \ldots, \frac{b_n}{\sqrt{n}}\right) \in \mathbb{R}^n.$$

Then

$$\|u\|^2 = \sum_{k=1}^{n} k a_k^2, \quad \|v\|^2 = \sum_{k=1}^{n} \frac{b_k^2}{k}, \quad \langle u, v\rangle = \sum_{k=1}^{n} a_k b_k.$$

By the Cauchy–Schwarz inequality, we have

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 = |\langle u, v\rangle|^2 \leq \|u\|^2 \|v\|^2 = \left(\sum_{k=1}^{n} k a_k^2\right)\left(\sum_{k=1}^{n} \frac{b_k^2}{k}\right). \qquad \blacksquare$$

Exercise 6A18

(a) Suppose $f : [1, \infty) \to [0, \infty)$ is continuous. Show that

$$\left(\int_1^\infty f\right)^2 \leq \int_1^\infty x^2 (f(x))^2 \, dx.$$

(b) For which continuous functions $f : [1, \infty) \to [0, \infty)$ is the inequality in (a) an equality with both sides finite?

Solution.

(a) Consider the functions over $[1, \infty)$:

$$u(x) = x f(x), \quad v(x) = \frac{1}{x}.$$

Then

$$\langle u, v\rangle = \int_1^\infty u(x) v(x) \, dx = \int_1^\infty f(x) \, dx, \quad \|u\|^2 = \int_1^\infty (x f(x))^2 \, dx, \quad \|v\|^2 = \int_1^\infty \frac{1}{x^2} \, dx = 1.$$

By the Cauchy–Schwarz inequality, we have

$$\left(\int_1^\infty f\right)^2 = |\langle u, v\rangle|^2 \leq \|u\|^2 \|v\|^2 = \int_1^\infty x^2 (f(x))^2 \, dx.$$

(b) The equality condition of the Cauchy–Schwarz inequality states that we have equality if and only if u is a scalar multiple of v. Note that $f(x) \geq 0$ for all $x \in [1, \infty)$, so the scalar multiple must be nonnegative. Therefore, we have equality if and only if there exists a nonnegative constant c such that $x f(x) = c/x$ for all $x \in [1, \infty)$, which is equivalent to $f(x) = c/x^2$ for all $x \in [1, \infty)$. Moreover, both sides are finite as follows:

$$\int_1^\infty f(x) \, dx = \int_1^\infty \frac{c}{x^2} \, dx = c, \quad \int_1^\infty x^2 (f(x))^2 \, dx = \int_1^\infty x^2 \left(\frac{c}{x^2}\right)^2 dx = c^2. \qquad \blacksquare$$

Exercise 6A19

Suppose $v_1, \ldots, v_n$ is a basis of V and $T \in \mathcal{L}(V)$. Prove that if λ is an eigenvalue of T, then

$$|\lambda|^2 \leq \sum_{j=1}^{n} \sum_{k=1}^{n} |\mathcal{M}(T)_{j,k}|^2,$$

where $\mathcal{M}(T)_{j,k}$ denotes the entry in row j, column k of the matrix of T with respect to the basis $v_1, \ldots, v_n$.

Solution. Let $u \in V$ be an eigenvector of T corresponding to the eigenvalue λ, so $Tu = \lambda u$. Let $A = \mathcal{M}(T)$ be the matrix of T with respect to the basis $v_1, \ldots, v_n$. From $Tu = \lambda u$, the coordinate vector $x \in \mathbb{F}^n$ of u with respect to the basis $v_1, \ldots, v_n$ satisfies $Ax = \lambda x$. Note that $x \neq 0$ since $u \neq 0$.

Applying the Cauchy–Schwarz inequality to the rows of A, we have

$$|\lambda|^2 \|x\|^2 = \|Ax\|^2 = \sum_{j=1}^{n} \left| \sum_{k=1}^{n} A_{j,k} x_k \right|^2 \leq \left(\sum_{j=1}^{n} \sum_{k=1}^{n} |A_{j,k}|^2 \right) \|x\|^2.$$

Dividing both sides by $\|x\|^2$ gives the desired inequality. ■

Exercise 6A20

Prove that if $u, v \in V$, then $\big| \|u\| - \|v\| \big| \leq \|u - v\|$.

Axler Hint. *The inequality above is called the* **reverse triangle inequality**. *For the reverse triangle inequality when $V = \mathbb{C}$, see* Exercise 4-2.

Solution. By the triangle inequality, we have

$$\|u\| = \|(u - v) + v\| \leq \|u - v\| + \|v\|,$$

which implies $\|u\| - \|v\| \leq \|u - v\|$. Similarly, we have

$$\|v\| = \|(v - u) + u\| \leq \|v - u\| + \|u\| = \|u - v\| + \|u\|,$$

which implies $\|v\| - \|u\| \leq \|u - v\|$. Therefore, we have

$$-\|u - v\| \leq \|u\| - \|v\| \leq \|u - v\|,$$

which is equivalent to $\big| \|u\| - \|v\| \big| \leq \|u - v\|$. ■

Exercise 6A21

Suppose $u, v \in V$ are such that

$$\|u\| = 3, \quad \|u + v\| = 4, \quad \|u - v\| = 6.$$

What number does $\|v\|$ equal?

Solution. We have

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \|u\|^2 + 2\Re\langle u, v\rangle + \|v\|^2 = 16, \\ \|u - v\|^2 &= \langle u - v, u - v \rangle = \|u\|^2 - 2\Re\langle u, v\rangle + \|v\|^2 = 36. \end{aligned}$$

Adding the two equations gives

$$2\|u\|^2 + 2\|v\|^2 = 52,$$

so $\|v\|^2 = 26 - 9 = 17$, which implies $\|v\| = \sqrt{17}$. ■

Exercise 6A22

Show that if $u, v \in V$, then
$$\|u+v\|\,\|u-v\| \leq \|u\|^2 + \|v\|^2.$$

Solution. We have

$$\begin{aligned}
\|u+v\|^2\|u-v\|^2 &= \langle u+v, u+v\rangle \langle u-v, u-v\rangle \\
&= (\|u\|^2 + 2\Re\langle u, v\rangle + \|v\|^2)(\|u\|^2 - 2\Re\langle u, v\rangle + \|v\|^2) \\
&= (\|u\|^2 + \|v\|^2)^2 - (2\Re\langle u, v\rangle)^2 \\
&\leq (\|u\|^2 + \|v\|^2)^2,
\end{aligned}$$

where the last inequality follows from the fact that $(2\Re\langle u, v\rangle)^2 \geq 0$. Taking the square root of both sides gives the desired inequality. ■

Exercise 6A23

Suppose $v_1, \ldots, v_m \in V$ are such that $\|v_k\| \leq 1$ for each $k = 1, \ldots, m$. Show that there exist $a_1, \ldots, a_m \in \{1, -1\}$ such that
$$\|a_1v_1 + \cdots + a_mv_m\| \leq \sqrt{m}.$$

Manual Remark. *Note that* $\min\{A, B\}^2 \leq \frac{A^2+B^2}{2}$ *so long as* $A, B \geq 0$. *This follows from the fact that either* $A \leq B$ *or* $B \leq A$ *forces* $\min\{A, B\}^2 \leq A^2$ *and* $\min\{A, B\}^2 \leq B^2$.

Solution. We proceed by induction on m. For the base case $m = 1$, we have $\|v_1\| \leq 1 = \sqrt{1}$, so the statement holds.

Now suppose the statement holds for some $m \geq 1$. Let $v_1, \ldots, v_{m+1} \in V$ be such that $\|v_k\| \leq 1$ for each $k = 1, \ldots, m+1$. By the induction hypothesis, there exist $a_1, \ldots, a_m \in \{1, -1\}$ such that

$$\|a_1v_1 + \cdots + a_mv_m\| \leq \sqrt{m}.$$

Let $u = a_1v_1 + \cdots + a_mv_m$. Then

$$\|u + v_{m+1}\|^2 = \|u\|^2 + 2\Re\langle u, v_{m+1}\rangle + \|v_{m+1}\|^2,$$

and
$$\|u - v_{m+1}\|^2 = \|u\|^2 - 2\Re\langle u, v_{m+1}\rangle + \|v_{m+1}\|^2.$$

Adding these two equations gives

$$\|u + v_{m+1}\|^2 + \|u - v_{m+1}\|^2 = 2\|u\|^2 + 2\|v_{m+1}\|^2.$$

Since $\|v_{m+1}\|^2 \leq 1$, we have
$$2\|u\|^2 + 2\|v_{m+1}\|^2 \leq 2\|u\|^2 + 2.$$

Therefore,

$$\min\{\|u + v_{m+1}\|, \|u - v_{m+1}\|\}^2 \leq \frac{\|u + v_{m+1}\|^2 + \|u - v_{m+1}\|^2}{2} \leq \|u\|^2 + 1 \leq m + 1,$$

where the last inequality follows from the induction hypothesis. Taking the square root of both sides gives

$$\min\{\|u + v_{m+1}\|, \|u - v_{m+1}\|\} \leq \sqrt{m+1}.$$

Therefore, there exists $a_{m+1} \in \{1, -1\}$ such that

$$\|a_1 v_1 + \cdots + a_m v_m + a_{m+1} v_{m+1}\| = \|u + a_{m+1} v_{m+1}\| \leq \sqrt{m+1}. \qquad \blacksquare$$

Exercise 6A24

Prove or give a counterexample: If $\|\cdot\|$ is the norm associated with an inner product on $\mathbb{R}^2$, then there exists $(x, y) \in \mathbb{R}^2$ such that $\|(x, y)\| \neq \max\{|x|, |y|\}$.

Solution. If we define the norm as $\|(x, y)\| = \max\{|x|, |y|\}$ for all $(x, y) \in \mathbb{R}^2$, then take $u = (1, 0)$ and $v = (0, 1)$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 1 + 1 = 2,$$

but

$$2\|u\|^2 + 2\|v\|^2 = 2 + 2 = 4,$$

contradicting the parallelogram equality. $\blacksquare$

Exercise 6A25

Suppose $p > 0$. Prove that there is an inner product on $\mathbb{R}^2$ such that the associated norm is given by

$$\|(x, y)\| = (|x|^p + |y|^p)^{1/p}$$

for all $(x, y) \in \mathbb{R}^2$ if and only if $p = 2$.

Solution. If $p = 2$, then the standard inner product on $\mathbb{R}^2$ gives the associated norm

$$\|(x, y)\| = (|x|^2 + |y|^2)^{1/2}.$$

Now suppose there is an inner product on $\mathbb{R}^2$ such that the associated norm is given by

$$\|(x, y)\| = (|x|^p + |y|^p)^{1/p}$$

for all $(x, y) \in \mathbb{R}^2$. Then the parallelogram equality must hold for this norm. Let $u = (1, 0)$ and $v = (0, 1)$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2^{2/p} + 2^{2/p} = 2 \cdot 2^{2/p} = 2^{1+2/p},$$

and

$$2\|u\|^2 + 2\|v\|^2 = 4.$$

Therefore, we must have

$$2^{1+2/p} = 4,$$

which implies $1 + 2/p = 2$, so $p = 2$. $\blacksquare$

Exercise 6A26

Suppose V is a real inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4}$$

for all $u, v \in V$.

Solution. We have

$$\begin{aligned}
\|u+v\|^2 - \|u-v\|^2 &= \langle u+v, u+v\rangle - \langle u-v, u-v\rangle \\
&= (\|u\|^2 + 2\langle u, v\rangle + \|v\|^2) - (\|u\|^2 - 2\langle u, v\rangle + \|v\|^2) \\
&= 4\langle u, v\rangle.
\end{aligned}$$

Dividing both sides by 4 gives the desired equality. ■

Exercise 6A27

Suppose V is a complex inner product space. Prove that

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i}{4}$$

for all $u, v \in V$.

Solution. We have

$$\begin{aligned}
\|u+v\|^2 - \|u-v\|^2 &= \langle u+v, u+v\rangle - \langle u-v, u-v\rangle \\
&= (\|u\|^2 + 2\mathfrak{R}\langle u, v\rangle + \|v\|^2) - (\|u\|^2 - 2\mathfrak{R}\langle u, v\rangle + \|v\|^2) \\
&= 4\mathfrak{R}\langle u, v\rangle,
\end{aligned}$$

and

$$\begin{aligned}
\|u+iv\|^2 - \|u-iv\|^2 &= \langle u+iv, u+iv\rangle - \langle u-iv, u-iv\rangle \\
&= (\|u\|^2 + 2\mathfrak{R}\langle u, iv\rangle + \|v\|^2) - (\|u\|^2 - 2\mathfrak{R}\langle u, iv\rangle + \|v\|^2) \\
&= 4\mathfrak{R}\langle u, iv\rangle.
\end{aligned}$$

Let $\langle u, v\rangle = a + bi$ for some $a, b \in \mathbb{R}$. Then

$$\langle u, iv\rangle = \bar{i}\langle u, v\rangle = -i(a+bi) = b - ai,$$

so $\mathfrak{R}\langle u, iv\rangle = \mathfrak{I}\langle u, v\rangle$. Therefore, we have

$$\|u+v\|^2 - \|u-v\|^2 + \|u+iv\|^2 i - \|u-iv\|^2 i = 4\mathfrak{R}\langle u, v\rangle + 4\mathfrak{I}\langle u, v\rangle i = 4\langle u, v\rangle.$$

Dividing both sides by 4 gives the desired equality. ■

Exercise 6A28

A norm on a vector space U is a function

$$\|\cdot\| : U \to [0, \infty)$$

such that $\|u\| = 0$ if and only if $u = 0$, $\|\alpha u\| = |\alpha| \, \|u\|$ for all $\alpha \in \mathbb{F}$ and all $u \in U$, and $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in U$. Prove that a norm satisfying the parallelogram equality comes from an inner product (in other words, show that if $\|\cdot\|$ is a norm on U satisfying the parallelogram equality, then there is an inner product $\langle \cdot, \cdot \rangle$ on U such that $\|u\| = \langle u, u \rangle^{1/2}$ for all $u \in U$).
Manual Remark. *This is a highly non-trivial result known as the* Jordan–von Neumann theorem. *The proof is not easy since it requires density of $\mathbb{Q}$ in $\mathbb{R}$ to ensure homogeneity from the rational numbers extends to the real numbers via continuity.*

Solution. Suppose $\|\cdot\|$ is a norm on U satisfying the parallelogram equality. We first work in the case that $\mathbb{F} = \mathbb{R}$. Define

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}$$

for all $u, v \in U$. We now check that $\langle \cdot, \cdot \rangle$ is an inner product on U.

- **Positive-Definiteness**: For any $u \in U$, we have

$$\langle u, u \rangle = \frac{\|u + u\|^2 - \|u - u\|^2}{4} = \frac{\|2u\|^2 - 0}{4} = \frac{4\|u\|^2}{4} = \|u\|^2 \geq 0,$$

since $\|u\| \geq 0$. This establishes that $\|u\| = \langle u, u \rangle^{1/2}$ for all $u \in U$. Moreover, $\langle u, u \rangle = 0$ if and only if $\|u\|^2 = 0$, which is equivalent to $u = 0$.

- **Symmetry**: Note that $\|a\| = \|-a\|$ for any $a \in U$. For any $u, v \in U$, we have

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4} = \frac{\|v + u\|^2 - \|v - u\|^2}{4} = \langle v, u \rangle.$$

- **Additivity in the First Slot**: Let $u, v, w \in U$. Applying the parallelogram equality to the pairs $(u + w, v)$ and $(u - w, v)$ gives

$$\begin{aligned} \|(u + w) + v\|^2 + \|(u + w) - v\|^2 &= 2\|u + w\|^2 + 2\|v\|^2, \\ \|(u - w) + v\|^2 + \|(u - w) - v\|^2 &= 2\|u - w\|^2 + 2\|v\|^2. \end{aligned}$$

Subtracting the second equation from the first gives

$$\|(u + w) + v\|^2 - \|(u - w) + v\|^2 + \|(u + w) - v\|^2 - \|(u - w) - v\|^2 = 2\|u + w\|^2 - 2\|u - w\|^2.$$

Rearranging terms inside the norms gives

$$\|(u + v) + w\|^2 - \|(u + v) - w\|^2 + \|(u - v) + w\|^2 - \|(u - v) - w\|^2 = 2\|u + w\|^2 - 2\|u - w\|^2.$$

Dividing both sides by 4 gives the relation $\langle u + v, w \rangle + \langle u - v, w \rangle = 2\langle u, w \rangle$.

Now apply the parallelogram equality to the pairs $(v + w, u)$ and $(v - w, u)$ to get

$$\begin{aligned} \|(v + w) + u\|^2 + \|(v + w) - u\|^2 &= 2\|v + w\|^2 + 2\|u\|^2, \\ \|(v - w) + u\|^2 + \|(v - w) - u\|^2 &= 2\|v - w\|^2 + 2\|u\|^2. \end{aligned}$$

Subtracting the second equation from the first gives

$$\|(v+w)+u\|^2 - \|(v-w)+u\|^2 + \|(v+w)-u\|^2 - \|(v-w)-u\|^2 = 2\|v+w\|^2 - 2\|v-w\|^2.$$

Rearranging terms inside the norms gives

$$\|(u+v)+w\|^2 - \|(u+v)-w\|^2 + \|(u-v)-w\|^2 - \|(u-v)+w\|^2 = 2\|v+w\|^2 - 2\|v-w\|^2.$$

Dividing both sides by 4 gives the relation $\langle u+v, w\rangle - \langle u-v, w\rangle = 2\langle v, w\rangle$. Adding the two relations gives $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$.

- **Homogeneity in the First Slot**: Let $u, v \in U$ and $\alpha \in \mathbb{R}$. If $\alpha = 0$, then

$$\langle 0, v\rangle = \frac{\|0+v\|^2 - \|0-v\|^2}{4} = \frac{\|v\|^2 - \|v\|^2}{4} = 0 = 0\langle u, v\rangle.$$

We first consider when α is a positive integer. We proceed by induction on α. For the base case $\alpha = 1$, we have $\langle 1 \cdot u, v\rangle = \langle u, v\rangle$. Now suppose the statement holds for some positive integer α. Then

$$\langle (\alpha+1)u, v\rangle = \langle \alpha u + u, v\rangle = \langle \alpha u, v\rangle + \langle u, v\rangle = \alpha\langle u, v\rangle + \langle u, v\rangle = (\alpha+1)\langle u, v\rangle.$$

Therefore, the statement holds for all positive integers α. Next, we consider when α is a negative integer. By direct computation, we have for $\alpha = -1$ that

$$\langle -u, v\rangle = \frac{\|-u+v\|^2 - \|-u-v\|^2}{4} = \frac{\|v-u\|^2 - \|-(u+v)\|^2}{4} = \frac{\|v-u\|^2 - \|u+v\|^2}{4} = -\langle u, v\rangle.$$

We may now consider $\alpha = -\beta$ for some positive integer β. Then

$$\langle \alpha u, v\rangle = \langle -\beta u, v\rangle = \langle -(\beta u), v\rangle = -\langle \beta u, v\rangle = -\beta\langle u, v\rangle = \alpha\langle u, v\rangle.$$

We now consider when α is a rational number. Let $\alpha = p/q$ for some integers p and q with $q > 0$. Then

$$q\langle \alpha u, v\rangle = \langle q(\alpha u), v\rangle = \langle pu, v\rangle = p\langle u, v\rangle,$$

so $\langle \alpha u, v\rangle = \alpha\langle u, v\rangle$.

Finally, we consider when α is an arbitrary real number. By the density of the rational numbers in the real numbers, there exists a sequence $(\alpha_n)_{n=1}^\infty$ of rational numbers such that $\alpha_n \to \alpha$ as $n \to \infty$. By the reverse triangle inequality, we have

$$\Big| \|\alpha_n u + v\| - \|\alpha u + v\| \Big| \le \|(\alpha_n - \alpha)u\| = |\alpha_n - \alpha|\, \|u\|,$$

where $|\alpha_n - \alpha| \to 0$ as $n \to \infty$, so $\|\alpha_n u + v\| \to \|\alpha u + v\|$ as $n \to \infty$. Similarly, we have $\|\alpha_n u - v\| \to \|\alpha u - v\|$ as $n \to \infty$. Then $\langle \alpha_n u, v\rangle \to \langle \alpha u, v\rangle$ as $n \to \infty$. It follows that $\langle \alpha_n u, v\rangle = \alpha_n\langle u, v\rangle \to \alpha\langle u, v\rangle$ as $n \to \infty$, so $\langle \alpha u, v\rangle = \alpha\langle u, v\rangle$.

Now suppose $\mathbb{F} = \mathbb{C}$. Define

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4} + \frac{\|u+iv\|^2 - \|u-iv\|^2}{4} i$$

for all $u, v \in U$. We again check that $\langle \cdot, \cdot \rangle$ is an inner product on U.

- **Positive-Definiteness**: For any $u \in U$, $\Re\langle u, u\rangle$ follows from the real case. Considering $\Im\langle u, u\rangle$, we have that

$$\|u + iu\|^2 = \|(1+i)u\|^2 = |1+i|^2\|u\|^2 = 2\|u\|^2,$$
$$\|u - iu\|^2 = \|(1-i)u\|^2 = |1-i|^2\|u\|^2 = 2\|u\|^2,$$

so $\Im\langle u, u\rangle = 0$. Therefore, $\langle u, u\rangle = \Re\langle u, u\rangle + \Im\langle u, u\rangle i = \Re\langle u, u\rangle \geq 0$. This again establishes that $\|u\| = \langle u, u\rangle^{1/2}$ for all $u \in U$. Moreover, $\langle u, u\rangle = 0$ if and only if $\Re\langle u, u\rangle = 0$, which is equivalent to $u = 0$.
- **Conjugate Symmetry**: For any $u, v \in U$, $\Re\langle u, v\rangle$ follows from the real case. Moreover, note that $\|u+iv\| = \|i(v-iu)\| = |i|\,\|v-iu\| = \|v-iu\|$. Similarly, $\|u-iv\| = \|-i(v+iu)\| = |-i|\,\|v+iu\| = \|v+iu\|$. Thus, for $\Im\langle u, v\rangle$, we have

$$\begin{aligned}\Im\langle u, v\rangle &= \frac{\|u+iv\|^2 - \|u-iv\|^2}{4} \\ &= \frac{\|v-iu\|^2 - \|v+iu\|^2}{4} \\ &= -\Im\langle v, u\rangle.\end{aligned}$$

Therefore, $\langle u, v\rangle = \Re\langle u, v\rangle + \Im\langle u, v\rangle i = \Re\langle v, u\rangle - \Im\langle v, u\rangle i = \overline{\langle v, u\rangle}$.
- **Additivity in the First Slot**: Let $u, v, w \in U$. Then $\Re\langle u+v, w\rangle$ follows from the real case. For $\Im\langle u+v, w\rangle$, the real-case additivity identity applied with w replaced by iw gives

$$\Im\langle u+v, w\rangle = \Im\langle u, w\rangle + \Im\langle v, w\rangle.$$

Therefore,

$$\begin{aligned}\langle u+v, w\rangle &= \Re\langle u+v, w\rangle + \Im\langle u+v, w\rangle i \\ &= \Re\langle u, w\rangle + \Re\langle v, w\rangle + (\Im\langle u, w\rangle + \Im\langle v, w\rangle)i \\ &= (\Re\langle u, w\rangle + \Im\langle u, w\rangle i) + (\Re\langle v, w\rangle + \Im\langle v, w\rangle i) \\ &= \langle u, w\rangle + \langle v, w\rangle.\end{aligned}$$

- **Homogeneity in the First Slot**: Let $u, v \in U$ and $\alpha \in \mathbb{C}$. We first consider $\alpha = i$. Then

$$\begin{aligned}\langle iu, v\rangle &= \frac{\|iu+v\|^2 - \|iu-v\|^2}{4} + \frac{\|iu+iv\|^2 - \|iu-iv\|^2}{4}i \\ &= \frac{\|v-iu\|^2 - \|v+iu\|^2}{4} + \frac{\|i(u+v)\|^2 - \|i(u-v)\|^2}{4}i \\ &= -\Im\langle v, u\rangle + \Re\langle u, v\rangle i = i\langle u, v\rangle.\end{aligned}$$

Now let $\alpha = a + bi$ for some $a, b \in \mathbb{R}$. From the real case, we have

$$\langle au, v\rangle = a\langle u, v\rangle \quad \text{and} \quad \langle ibu, v\rangle = b\langle iu, v\rangle = bi\langle u, v\rangle.$$

Therefore,

$$\langle \alpha u, v\rangle = \langle au + ibu, v\rangle = \langle au, v\rangle + \langle ibu, v\rangle = a\langle u, v\rangle + bi\langle u, v\rangle = \alpha\langle u, v\rangle. \qquad \blacksquare$$

Exercise 6A29

Suppose $V_1, \ldots, V_m$ are inner product spaces. Show that the equation

$$\langle (u_1, \ldots, u_m), (v_1, \ldots, v_m) \rangle = \langle u_1, v_1 \rangle + \cdots + \langle u_m, v_m \rangle$$

defines an inner product on $V_1 \times \cdots \times V_m$.

Axler Note. *In the expression above on the right, for each $k = 1, \ldots, m$, the inner product $\langle u_k, v_k \rangle$ denotes the inner product on V_k. Each of the spaces $V_1, \ldots, V_m$ may have a different inner product, even though the same notation is used here.*

Solution. Let $\langle \cdot, \cdot \rangle_k$ denote the inner product on V_k for each $k = 1, \ldots, m$. To see that $\langle \cdot, \cdot \rangle$ is an inner product on $V_1 \times \cdots \times V_m$, note that the properties of conjugate symmetry, additivity in the first slot, and homogeneity in the first slot follow from the corresponding properties of each $\langle \cdot, \cdot \rangle_k$.

For positive-definiteness, let $(u_1, \ldots, u_m) \in V_1 \times \cdots \times V_m$. Then

$$\langle (u_1, \ldots, u_m), (u_1, \ldots, u_m) \rangle = \langle u_1, u_1 \rangle_1 + \cdots + \langle u_m, u_m \rangle_m \geq 0.$$

Moreover, $\langle (u_1, \ldots, u_m), (u_1, \ldots, u_m) \rangle = 0$ if and only if $\langle u_k, u_k \rangle_k = 0$ for each $k = 1, \ldots, m$, which is equivalent to $u_k = 0$ for each $k = 1, \ldots, m$, so $(u_1, \ldots, u_m) = 0$. ■

Exercise 6A30

Suppose V is a real inner product space. For $u, v, w, x \in V$, define

$$\langle u + iv, w + ix \rangle_{\mathbf{C}} = \langle u, w \rangle + \langle v, x \rangle + (\langle v, w \rangle - \langle u, x \rangle)i.$$

(a) Show that $\langle \cdot, \cdot \rangle_{\mathbf{C}}$ makes $V_{\mathbf{C}}$ into a complex inner product space.

(b) Show that if $u, v \in V$, then

$$\langle u, v \rangle_{\mathbf{C}} = \langle u, v \rangle \quad \text{and} \quad \|u + iv\|_{\mathbf{C}}^2 = \|u\|^2 + \|v\|^2.$$

Axler Hint. *See* Exercise 1B8 *for the definition of the complexification $V_{\mathbf{C}}$.*

Solution.

(a) We first check that $\langle \cdot, \cdot \rangle_{\mathbf{C}}$ is an inner product on $V_{\mathbf{C}}$.

- **Positive-Definiteness**: For any $u + iv \in V_{\mathbf{C}}$, we have

$$\langle u + iv, u + iv \rangle_{\mathbf{C}} = \langle u, u \rangle + \langle v, v \rangle + (\langle v, u \rangle - \langle u, v \rangle)i = \|u\|^2 + \|v\|^2 \geq 0,$$

  since $\|u\|, \|v\| \geq 0$. Moreover, $\langle u + iv, u + iv \rangle_{\mathbf{C}} = 0$ if and only if $\|u\|^2 + \|v\|^2 = 0$, which is equivalent to $u = v = 0$, so $u + iv = 0$.

- **Conjugate Symmetry**: For any $u + iv, w + ix \in V_{\mathbf{C}}$, we have

$$\begin{aligned}\langle w + ix, u + iv \rangle_{\mathbf{C}} &= \langle w, u \rangle + \langle x, v \rangle + (\langle x, u \rangle - \langle w, v \rangle)i \\ &= \overline{\langle u, w \rangle} + \overline{\langle v, x \rangle} + (\overline{\langle u, x \rangle} - \overline{\langle v, w \rangle})i \\ &= \overline{\langle u, w \rangle + \langle v, x \rangle + (\langle v, w \rangle - \langle u, x \rangle)i} = \overline{\langle u + iv, w + ix \rangle_{\mathbf{C}}}.\end{aligned}$$

- **Additivity in the First Slot**: Let $u+iv, w+ix, y+iz \in V_{\mathbb{C}}$. Then

$$\begin{aligned}\langle (u+iv)+(y+iz), w+ix\rangle_{\mathbb{C}} &= \langle (u+y)+i(v+z), w+ix\rangle_{\mathbb{C}} \\ &= \langle u+y, w\rangle + \langle v+z, x\rangle + (\langle v+z, w\rangle - \langle u+y, x\rangle)i\end{aligned}$$

Expanding the inner products gives

$$\begin{aligned}&(\langle u,w\rangle + \langle y,w\rangle) + (\langle v,x\rangle + \langle z,x\rangle) + ((\langle v,w\rangle + \langle z,w\rangle) - (\langle u,x\rangle + \langle y,x\rangle))i \\ &= (\langle u,w\rangle + \langle v,x\rangle + (\langle v,w\rangle - \langle u,x\rangle)i) + (\langle y,w\rangle + \langle z,x\rangle + (\langle z,w\rangle - \langle y,x\rangle)i) \\ &= \langle u+iv, w+ix\rangle_{\mathbb{C}} + \langle y+iz, w+ix\rangle_{\mathbb{C}}.\end{aligned}$$

- **Homogeneity in the First Slot**: Let $u+iv, w+ix \in V_{\mathbb{C}}$ and $\lambda \in \mathbb{R}$. Then

$$\begin{aligned}\langle \lambda(u+iv), w+ix\rangle_{\mathbb{C}} &= \langle (\lambda u) + i(\lambda v), w+ix\rangle_{\mathbb{C}} \\ &= \langle \lambda u, w\rangle + \langle \lambda v, x\rangle + (\langle \lambda v, w\rangle - \langle \lambda u, x\rangle)i \\ &= \lambda\langle u,w\rangle + \lambda\langle v,x\rangle + (\lambda\langle v,w\rangle - \lambda\langle u,x\rangle)i \\ &= \lambda(\langle u,w\rangle + \langle v,x\rangle + (\langle v,w\rangle - \langle u,x\rangle)i) = \lambda\langle u+iv, w+ix\rangle_{\mathbb{C}}.\end{aligned}$$

Moreover, note that $i(u+iv) = -v+iu$. Then

$$\begin{aligned}\langle i(u+iv), w+ix\rangle_{\mathbb{C}} &= \langle -v+iu, w+ix\rangle_{\mathbb{C}} \\ &= \langle -v, w\rangle + \langle u,x\rangle + (\langle u,w\rangle - \langle -v,x\rangle)i \\ &= -\langle v,w\rangle + \langle u,x\rangle + (\langle u,w\rangle + \langle v,x\rangle)i \\ &= i(\langle u,w\rangle + \langle v,x\rangle + (\langle v,w\rangle - \langle u,x\rangle)i) = i\langle u+iv, w+ix\rangle_{\mathbb{C}}.\end{aligned}$$

Hence, homogeneity in the first slot holds for all $\alpha \in \mathbb{C}$ by using additivity in the first slot and the fact that $\alpha = a+bi$ for some $a,b \in \mathbb{R}$.

Hence, $\langle\cdot,\cdot\rangle_{\mathbb{C}}$ is an inner product on $V_{\mathbb{C}}$, so $V_{\mathbb{C}}$ is a complex inner product space.

(b) We have

$$\langle u,v\rangle_{\mathbb{C}} = \langle u+i0, v+i0\rangle_{\mathbb{C}} = \langle u,v\rangle + \langle 0,0\rangle + (\langle 0,v\rangle - \langle u,0\rangle)i = \langle u,v\rangle,$$

and

$$\|u+iv\|_{\mathbb{C}}^2 = \langle u+iv, u+iv\rangle_{\mathbb{C}} = \langle u,u\rangle + \langle v,v\rangle + (\langle v,u\rangle - \langle u,v\rangle)i = \|u\|^2 + \|v\|^2.$$ ■

Exercise 6A31

Suppose $u,v,w \in V$. Prove that

$$\|w - \tfrac{1}{2}(u+v)\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4}.$$

Solution. Set $a = w-u$ and $b = w-v$. Then the left-hand side is

$$\|w - \tfrac{1}{2}(u+v)\|^2 = \|\tfrac{1}{2}(a+b)\|^2 = \frac{1}{4}\|a+b\|^2.$$

Moreover, the right-hand side is

$$\frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4} = \frac{\|a\|^2 + \|b\|^2}{2} - \frac{\|a-b\|^2}{4} = \frac{2\|a\|^2 + 2\|b\|^2 - \|a-b\|^2}{4}.$$

From the parallelogram equality, we have

$$\|a+b\|^2 + \|a-b\|^2 = 2\|a\|^2 + 2\|b\|^2 \implies \|a+b\|^2 = 2\|a\|^2 + 2\|b\|^2 - \|a-b\|^2,$$

so division by 4 gives our desired equality. ∎

Exercise 6A32

Suppose that E is a subset of V with the property that $u, v \in E$ implies $\frac{1}{2}(u+v) \in E$. Let $w \in V$. Show that there is at most one point in E that is closest to w. In other words, show that there is at most one $u \in E$ such that

$$\|w-u\| \leq \|w-x\|$$

for all $x \in E$.

Solution. Suppose $u, v \in E$ are both closest points to w. Then $\|w-u\| = \|w-v\|$. By the property of E, we have $\frac{1}{2}(u+v) \in E$. By Exercise 6A31, we have

$$\|w - \tfrac{1}{2}(u+v)\|^2 = \frac{\|w-u\|^2 + \|w-v\|^2}{2} - \frac{\|u-v\|^2}{4} = \|w-u\|^2 - \frac{\|u-v\|^2}{4}.$$

Since $\frac{1}{2}(u+v) \in E$, we have

$$\|w-u\|^2 \leq \|w - \tfrac{1}{2}(u+v)\|^2 = \|w-u\|^2 - \frac{\|u-v\|^2}{4},$$

so $\|u-v\|^2 = 0$, which implies that $u = v$. ∎

Exercise 6A33

Suppose f, g are differentiable functions from $\mathbb{R}$ to $\mathbb{R}^n$.

(a) Show that $\langle f(t), g(t) \rangle' = \langle f'(t), g(t) \rangle + \langle f(t), g'(t) \rangle$.

(b) Suppose c is a positive number and $\|f(t)\| = c$ for every $t \in \mathbb{R}$. Show that $\langle f'(t), f(t) \rangle = 0$ for every $t \in \mathbb{R}$.

(c) Interpret the result in (b) geometrically in terms of the tangent vector to a curve lying on a sphere in $\mathbb{R}^n$ centered at the origin.

Axler Note. *A function $f : \mathbb{R} \to \mathbb{R}^n$ is called differentiable if there exist differentiable functions $f_1, \ldots, f_n$ from $\mathbb{R}$ to $\mathbb{R}$ such that $f(t) = (f_1(t), \ldots, f_n(t))$ for each $t \in \mathbb{R}$. Furthermore, for each $t \in \mathbb{R}$, the derivative $f'(t) \in \mathbb{R}^n$ is defined by $f'(t) = (f_1'(t), \ldots, f_n'(t))$.*

Solution.

(a) We have

$$\langle f(t), g(t) \rangle' = \left(\sum_{k=1}^{n} f_k(t) g_k(t) \right)' = \sum_{k=1}^{n} \left(f_k'(t) g_k(t) + f_k(t) g_k'(t) \right) = \langle f'(t), g(t) \rangle + \langle f(t), g'(t) \rangle.$$

(b) Suppose $\|f(t)\| = c$ for every $t \in \mathbb{R}$. Then $\langle f(t), f(t) \rangle = c^2$ for every $t \in \mathbb{R}$. Differentiating both sides with respect to t, we get

$$\langle f'(t), f(t) \rangle + \langle f(t), f'(t) \rangle = 2\langle f'(t), f(t) \rangle = 0,$$

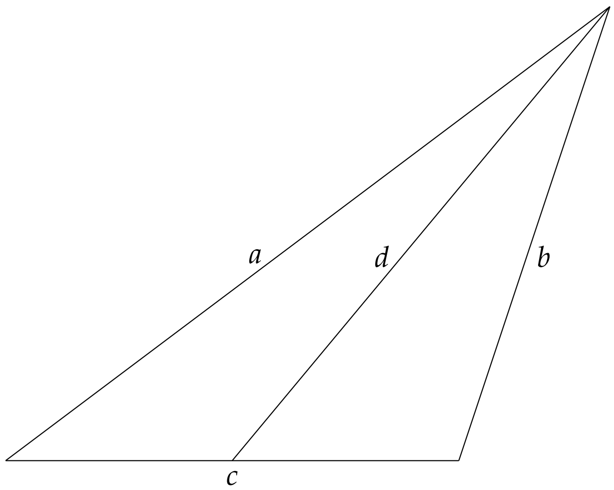
so $\langle f'(t), f(t) \rangle = 0$ for every $t \in \mathbb{R}$.

(c) Geometrically, the result in (b) says that the tangent vector to a curve lying on a sphere in $\mathbb{R}^n$ centered at the origin is orthogonal to the position vector of the point on the curve. ■

Exercise 6A34

Use inner products to prove Apollonius's identity: In a triangle with sides of length a, b, and c, let d be the length of the line segment from the midpoint of the side of length c to the opposite vertex. Then

$$a^2 + b^2 = \tfrac{1}{2}c^2 + 2d^2.$$



Solution. Consider $\triangle ABC$ with $\|C - A\| = a$, $\|C - B\| = b$, and $\|B - A\| = c$. Let M be the midpoint of $\overline{AB}$, so $M = \frac{1}{2}(A + B)$. Then $d = \|C - M\|$. By Exercise 6A31, we have

$$\|C - M\|^2 = \frac{\|C - A\|^2 + \|C - B\|^2}{2} - \frac{\|A - B\|^2}{4} = \frac{a^2 + b^2}{2} - \frac{c^2}{4}.$$

Rearranging terms gives $a^2 + b^2 = \frac{1}{2}c^2 + 2d^2$. ■

Exercise 6A35

Fix a positive integer n. The *Laplacian* Δp of a twice differentiable real-valued function p on $\mathbb{R}^n$ is the function on $\mathbb{R}^n$ defined by

$$\Delta p = \frac{\partial^2 p}{\partial x_1^2} + \cdots + \frac{\partial^2 p}{\partial x_n^2}.$$

The function p is called *harmonic* if $\Delta p = 0$.
A *polynomial* on $\mathbb{R}^n$ is a linear combination (with coefficients in $\mathbb{R}$) of functions of the form $x_1^{m_1} \cdots x_n^{m_n}$, where $m_1, \ldots, m_n$ are nonnegative integers.
Suppose q is a polynomial on $\mathbb{R}^n$. Prove that there exists a harmonic polynomial p on $\mathbb{R}^n$ such that $p(x) = q(x)$ for every $x \in \mathbb{R}^n$ with $\|x\| = 1$.
Axler Note. *The only fact about harmonic functions that you need for this exercise is that if p is a harmonic function on $\mathbb{R}^n$ and $p(x) = 0$ for all $x \in \mathbb{R}^n$ with $\|x\| = 1$, then $p = 0$.*
Axler Hint. *A reasonable guess is that the desired harmonic polynomial p is of the form $q + (1 - \|x\|^2)r$ for some polynomial r. Prove that there is a polynomial r on $\mathbb{R}^n$ such that $q + (1 - \|x\|^2)r$ is harmonic by defining an operator T on a suitable vector space by*

$$Tr = \Delta((1 - \|x\|^2)r)$$

and then showing that T is injective and hence surjective.

Solution. Consider $T \in \mathcal{L}(\mathcal{P}_m(\mathbb{R}^n))$ defined by $Tr = \Delta((1 - \|x\|^2)r)$, where $\deg q = m$. Note that T is well-defined, since multiplication by $1 - \|x\|^2$ increases the degree of a polynomial by 2, and Δ decreases the degree by at most 2. Moreover, linearity of T follows from both linearity of Δ and linearity of multiplication by $1 - \|x\|^2$.

We show that T is injective. Suppose $Tr = 0$ for some $r \in \mathcal{P}_m(\mathbb{R}^n)$. Then $\Delta((1 - \|x\|^2)r) = 0$, so $(1 - \|x\|^2)r$ is harmonic. Since $(1 - \|x\|^2)r(x) = 0$ for all $x \in \mathbb{R}^n$ with $\|x\| = 1$, we have $(1 - \|x\|^2)r = 0$ by the fact about harmonic functions given in the exercise.

Now fix $y \in \mathbb{R}^n$. Evaluating at $x = ty$ for $t \in \mathbb{R}$, we have $(1 - t^2\|y\|^2)r(ty) = 0$ for all $t \in \mathbb{R}$. If $y = 0$, then $r(ty) = r(0) = 0$ for all $t \in \mathbb{R}$. Otherwise, assume $y \neq 0$. Since $1 - t^2\|y\|^2$ is a polynomial in t of degree at most 2, it has at most two roots. Therefore, $r(ty) = 0$ for all $t \in \mathbb{R}$ except for at most two values of t. Since $r(ty)$ is a polynomial in t, it must be identically zero, so $r(ty) = 0$ for all $t \in \mathbb{R}$. Since $y \in \mathbb{R}^n$ was arbitrary, we have $r = 0$, so T is injective. Moreover, finite-dimensionality of $\mathcal{P}_m(\mathbb{R}^n)$ implies that T is surjective. Therefore, there exists $r \in \mathcal{P}_m(\mathbb{R}^n)$ such that $Tr = \Delta((1 - \|x\|^2)r) = -\Delta q$. We then have

$$\Delta(q + (1 - \|x\|^2)r) = \Delta q + \Delta((1 - \|x\|^2)r) = \Delta q - \Delta q = 0,$$

so $p = q + (1 - \|x\|^2)r$ is harmonic. ■

6B Orthonormal Bases

Exercise 6B1

Suppose $e_1, \ldots, e_m$ is a list of vectors in V such that

$$\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$$

for all $a_1, \ldots, a_m \in \mathbb{F}$. Show that $e_1, \ldots, e_m$ is an orthonormal list.
Axler Note. *This exercise provides a converse to 6.24.*

Solution. For each $k \in \{1, \ldots, m\}$, let $a_k = 1$ and $a_j = 0$ for all $j \neq k$. Then

$$\|e_k\|^2 = \|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2 = 1.$$

Next, for distinct $j, k \in \{1, \ldots, m\}$, set $a_j = a_k = 1$ and $a_i = 0$ for all $i \notin \{j, k\}$. Then $\|e_j + e_k\|^2 = 2$ by hypothesis, and expanding gives

$$\|e_j + e_k\|^2 = \|e_j\|^2 + \langle e_j, e_k \rangle + \overline{\langle e_j, e_k \rangle} + \|e_k\|^2 = 2 + 2\Re\langle e_j, e_k \rangle,$$

so $\Re\langle e_j, e_k \rangle = 0$. If $\mathbb{F} = \mathbb{R}$, then $\langle e_j, e_k \rangle = 0$. If $\mathbb{F} = \mathbb{C}$, set $a_j = 1$, $a_k = i$, and all other scalars to 0. Then $\|e_j + ie_k\|^2 = 2$ by hypothesis, and expanding gives

$$\|e_j + ie_k\|^2 = \|e_j\|^2 + i\langle e_k, e_j \rangle + \overline{i\langle e_k, e_j \rangle} + \|e_k\|^2 = 2 - 2\Im\langle e_k, e_j \rangle,$$

so $\Im\langle e_j, e_k \rangle = 0$, and thus $\langle e_j, e_k \rangle = 0$. ■

Exercise 6B2

(a) Suppose $\theta \in \mathbb{R}$. Show that both

$$(\cos\theta, \sin\theta), (-\sin\theta, \cos\theta) \quad \text{and} \quad (\cos\theta, \sin\theta), (\sin\theta, -\cos\theta)$$

are orthonormal bases of $\mathbb{R}^2$.

(b) Show that each orthonormal basis of $\mathbb{R}^2$ is of the form given by one of the two possibilities in (a).

Solution.

(a) Let $v_1 = (\cos\theta, \sin\theta)$ and $v_2 = (-\sin\theta, \cos\theta)$. Then $\|v_1\|^2 = \cos^2\theta + \sin^2\theta = 1$ and $\|v_2\|^2 = \sin^2\theta + \cos^2\theta = 1$. Moreover,

$$\langle v_1, v_2 \rangle = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0.$$

Since v_1, v_2 is an orthonormal list of two vectors in $\mathbb{R}^2$, it is an orthonormal basis of $\mathbb{R}^2$. Similarly, let $w_1 = (\cos\theta, \sin\theta)$ and $w_2 = (\sin\theta, -\cos\theta)$. Then $\|w_1\|^2 = \cos^2\theta + \sin^2\theta = 1$ and $\|w_2\|^2 = \sin^2\theta + \cos^2\theta = 1$. Again,

$$\langle w_1, w_2 \rangle = \cos\theta\sin\theta - \sin\theta\cos\theta = 0.$$

Since w_1, w_2 is an orthonormal list of two vectors in $\mathbb{R}^2$, it is an orthonormal basis of $\mathbb{R}^2$.

(b) Suppose u_1, u_2 is an orthonormal basis of $\mathbb{R}^2$. If $u_1 = (x_1, y_1)$, then $\|u_1\|^2 = x_1^2 + y_1^2 = 1$, so there exists $\theta \in \mathbb{R}$ such that $x_1 = \cos\theta$ and $y_1 = \sin\theta$. If $u_2 = (x_2, y_2)$, then $\|u_2\|^2 = x_2^2 + y_2^2 = 1$ and $\langle u_1, u_2\rangle = x_1x_2 + y_1y_2 = 0$. We now solve the system of equations

$$\begin{cases} x_2^2 + y_2^2 = 1, \\ \cos\theta x_2 + \sin\theta y_2 = 0. \end{cases}$$

If $\sin\theta = 0$, then $\theta = 0$ or $\theta = \pi$, so $\cos\theta = \pm 1$. Then $x_2 = 0$, so $y_2^2 = 1$, and thus $y_2 = \pm 1$. Then $u_2 = (0, \pm 1)$, which is of the form given in (a).

Otherwise, $y_2 = -\frac{\cos\theta}{\sin\theta}x_2$, so

$$x_2^2 + \left(-\frac{\cos\theta}{\sin\theta}x_2\right)^2 = 1 \implies x_2^2 + \frac{\cos^2\theta}{\sin^2\theta}x_2^2 = 1 \implies x_2^2\left(1 + \frac{\cos^2\theta}{\sin^2\theta}\right) = 1.$$

We then have $x_2^2 = \sin^2\theta$, so $x_2 = \pm\sin\theta$. Then $y_2 = -\frac{\cos\theta}{\sin\theta}(\pm\sin\theta) = \mp\cos\theta$. Thus, $u_2 = (\pm\sin\theta, \mp\cos\theta)$, which is of the form given in (a). ■

Exercise 6B3

Suppose $e_1, \dots, e_m$ is an orthonormal list in V and $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2 \iff v \in \operatorname{span}(e_1, \dots, e_m).$$

Solution. If $v \in \operatorname{span}(e_1, \dots, e_m)$, then there exist scalars $a_1, \dots, a_m$ such that $v = a_1e_1 + \cdots + a_me_m$. Then

$$\|v\|^2 = \|a_1e_1 + \cdots + a_me_m\|^2 = |a_1|^2 + \cdots + |a_m|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2.$$

Conversely, suppose $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2$. Let $w = v - (\langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m)$. Note that $\langle w, e_k\rangle = \langle v, e_k\rangle - \langle v, e_k\rangle = 0$ for each $k = 1, \dots, m$, so w is orthogonal to each e_k. Then

$$\|v\|^2 = \left\| w + \sum_{k=1}^{m} \langle v, e_k\rangle e_k \right\|^2 = \|w\|^2 + \sum_{k=1}^{m} |\langle v, e_k\rangle|^2.$$

The assumption of $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2$ gives $\|w\|^2 = 0$, so $w = 0$, and $v \in \operatorname{span}(e_1, \dots, e_m)$. ■

Exercise 6B4

Suppose n is a positive integer. Prove that

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \ldots, \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \ldots, \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g\rangle = \int_{-\pi}^{\pi} fg.$$

Axler Hint. *The following formulas should help.*

$$\begin{aligned}
(\sin x)(\cos y) &= \frac{\sin(x-y)+\sin(x+y)}{2},\\
(\sin x)(\sin y) &= \frac{\cos(x-y)-\cos(x+y)}{2},\\
(\cos x)(\cos y) &= \frac{\cos(x-y)+\cos(x+y)}{2}.
\end{aligned}$$

Solution. Define the vectors

$$f_0 = \frac{1}{\sqrt{2\pi}}, \quad f_k = \frac{\cos kx}{\sqrt{\pi}}, \quad g_k = \frac{\sin kx}{\sqrt{\pi}} \quad \text{for } k = 1, \ldots, n.$$

Then

$$\|f_0\|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} 1 \, \mathrm{d}x = 1.$$

Moreover, note the following for $k = 1, \ldots, n$,

$$\langle f_0, f_k\rangle = \frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{\pi}}\int_{-\pi}^{\pi}\cos kx \, \mathrm{d}x = \frac{1}{\sqrt{2\pi}}\frac{1}{\sqrt{\pi}}\left.\frac{\sin kx}{k}\right|_{-\pi}^{\pi} = 0,$$

and $\langle f_0, g_k\rangle = 0$ because $\sin kx$ is an odd function. Hence, f_0 is orthogonal to each of the other vectors in the list. Using the $\cos kx$ and $\sin kx$ computations in the above integrals, we may simplify the calculations of the norms of f_k and g_k to get

$$\begin{aligned}
\|f_k\|^2 &= \frac{1}{\pi}\int_{-\pi}^{\pi}\cos^2 kx \, \mathrm{d}x = \frac{1}{2\pi}\int_{-\pi}^{\pi}(1+\cos 2kx)\, \mathrm{d}x = 1,\\
\|g_k\|^2 &= \frac{1}{\pi}\int_{-\pi}^{\pi}\sin^2 kx \, \mathrm{d}x = \frac{1}{2\pi}\int_{-\pi}^{\pi}(1-\cos 2kx)\, \mathrm{d}x = 1.
\end{aligned}$$

We now show that the remaining vectors are pairwise orthogonal. For $j, k \in \{1, \ldots, n\}$ with $j \neq k$, we have that $j - k$ and $j + k$ are nonzero integers. Note that $\sin mx = 0$ for all integers m when $x = \pm\pi$.

Then

$$\begin{aligned}
\langle f_j, f_k\rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \cos jx \cos kx \,\mathrm{d}x \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} (\cos(j-k)x + \cos(j+k)x)\,\mathrm{d}x \\
&= \frac{1}{2\pi}\left(\frac{\sin(j-k)x}{j-k} + \frac{\sin(j+k)x}{j+k}\right)\Bigg|_{-\pi}^{\pi} = 0, \\
\langle g_j, g_k\rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \sin jx \sin kx \,\mathrm{d}x \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} (\cos(j-k)x - \cos(j+k)x)\,\mathrm{d}x \\
&= \frac{1}{2\pi}\left(\frac{\sin(j-k)x}{j-k} - \frac{\sin(j+k)x}{j+k}\right)\Bigg|_{-\pi}^{\pi} = 0.
\end{aligned}$$

For $j, k = 1, \ldots, n$, note that $f_j g_k$ is a product of an even function and an odd function, so $\langle f_j, g_k\rangle = 0$. Thus, the list of vectors is orthonormal. ∎

Exercise 6B5

Suppose $f : [-\pi, \pi] \to \mathbb{R}$ is continuous. For each nonnegative integer k, define

$$a_k = \frac{1}{\sqrt{\pi}}\int_{-\pi}^{\pi} f(x)\cos(kx)\,\mathrm{d}x \quad \text{and} \quad b_k = \frac{1}{\sqrt{\pi}}\int_{-\pi}^{\pi} f(x)\sin(kx)\,\mathrm{d}x.$$

Prove that

$$\frac{a_0^2}{2} + \sum_{k=1}^{\infty}(a_k^2 + b_k^2) \le \int_{-\pi}^{\pi} f^2.$$

Axler Note. *The inequality above is actually an equality for all continuous functions $f : [-\pi, \pi] \to \mathbb{R}$. However, proving that this inequality is an equality involves Fourier series techniques beyond the scope of this book.*

Solution. The list of vectors in Exercise 6B4 is an orthonormal list in $C[-\pi, \pi]$. Using the same naming conventions of f_0, f_k, g_k as in Exercise 6B4, we have that

$$\langle f, f_0\rangle = \frac{a_0}{\sqrt{2}}, \quad \langle f, f_k\rangle = a_k, \quad \langle f, g_k\rangle = b_k.$$

By Bessel's inequality, for every $n \ge 1$,

$$\frac{a_0^2}{2} + \sum_{k=1}^{n}(a_k^2 + b_k^2) = |\langle f, f_0\rangle|^2 + \sum_{k=1}^{n}(|\langle f, f_k\rangle|^2 + |\langle f, g_k\rangle|^2) \le \|f\|^2 = \int_{-\pi}^{\pi} f^2.$$

Since the right side is independent of n, taking $n \to \infty$ gives the result. ∎

Exercise 6B6

Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V.

(a) Prove that if $v_1, \ldots, v_n$ are vectors in V such that
$$\|e_k - v_k\| < \frac{1}{\sqrt{n}}$$
for each k, then $v_1, \ldots, v_n$ is a basis of V.

(b) Show that there exist $v_1, \ldots, v_n \in V$ such that
$$\|e_k - v_k\| \leq \frac{1}{\sqrt{n}}$$
for each k, but $v_1, \ldots, v_n$ is not linearly independent.

Axler Note. *This exercise states in* (a) *that an appropriately small perturbation of an orthonormal basis is a basis. Then* (b) *shows that the number* $1/\sqrt{n}$ *on the right side of the inequality in* (a) *cannot be improved upon.*

Solution.

(a) Suppose $v_1, \ldots, v_n$ are linearly dependent. Then there exist scalars $a_1, \ldots, a_n$, not all zero, such that
$$a_1 v_1 + \cdots + a_n v_n = 0.$$
Then
$$\begin{aligned}
\sqrt{\sum_{k=1}^{n} |a_k|^2} &= \left\| \sum_{k=1}^{n} a_k e_k \right\| \\
&= \left\| \sum_{k=1}^{n} a_k (e_k - v_k) \right\| \\
&\leq \sum_{k=1}^{n} |a_k| \|e_k - v_k\| \\
&< \frac{1}{\sqrt{n}} \sum_{k=1}^{n} |a_k| \\
&\leq \frac{1}{\sqrt{n}} \sqrt{n} \sqrt{\sum_{k=1}^{n} |a_k|^2} \\
&= \sqrt{\sum_{k=1}^{n} |a_k|^2}.
\end{aligned}$$
This results in $1 < 1$, which is a contradiction, so $v_1, \ldots, v_n$ must be linearly independent. Since $\dim V = n$, it follows that $v_1, \ldots, v_n$ is a basis of V.

(b) Consider the vector
$$u = \frac{1}{\sqrt{n}}(e_1 + \cdots + e_n), \quad \|u\|^2 = \frac{1}{n}\|e_1 + \cdots + e_n\|^2 = \frac{1}{n}(n) = 1.$$

For each $k = 1, \ldots, n$, define

$$v_k = e_k - \frac{1}{\sqrt{n}}u \implies \|e_k - v_k\| = \|\frac{1}{\sqrt{n}}u\| = \frac{1}{\sqrt{n}}\|u\| = \frac{1}{\sqrt{n}}.$$

Then

$$v_1 + \cdots + v_n = (e_1 - \frac{1}{\sqrt{n}}u) + \cdots + (e_n - \frac{1}{\sqrt{n}}u) = (e_1 + \cdots + e_n) - \frac{n}{\sqrt{n}}u = (e_1 + \cdots + e_n) - \sqrt{n}u = 0,$$

so $v_1, \ldots, v_n$ is linearly dependent. ■

Exercise 6B7

Suppose $T \in \mathcal{L}(\mathbb{R}^3)$ has an upper-triangular matrix with respect to the basis $(1,0,0)$, $(1,1,1)$, $(1,1,2)$. Find an orthonormal basis of $\mathbb{R}^3$ with respect to which T has an upper-triangular matrix.

Solution. Using the Gram–Schmidt procedure, we can produce an orthonormal basis of $\mathbb{R}^3$ from the given basis. Let $v_1 = (1,0,0), v_2 = (1,1,1), v_3 = (1,1,2)$. Set $f_1 = v_1 = (1,0,0)$, so $\|f_1\|^2 = 1^2+0^2+0^2 = 1$. Then

$$f_2 = v_2 - \frac{\langle v_2, f_1\rangle}{\|f_1\|^2}f_1 = (1,1,1) - \frac{1}{1}(1,0,0) = (0,1,1)$$

with $\|f_2\|^2 = 0^2 + 1^2 + 1^2 = 2$. Finally,

$$f_3 = v_3 - \frac{\langle v_3, f_1\rangle}{\|f_1\|^2}f_1 - \frac{\langle v_3, f_2\rangle}{\|f_2\|^2}f_2 = (1,1,2) - (1,0,0) - \frac{3}{2}(0,1,1) = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

with $\|f_3\|^2 = 0^2 + (-\frac{1}{2})^2 + (\frac{1}{2})^2 = \frac{1}{2}$. Then the orthonormal basis of $\mathbb{R}^3$ is

$$e_1 = \frac{f_1}{\|f_1\|} = (1,0,0), \quad e_2 = \frac{f_2}{\|f_2\|} = \frac{1}{\sqrt{2}}(0,1,1), \quad e_3 = \frac{f_3}{\|f_3\|} = \sqrt{2}\left(0, -\frac{1}{2}, \frac{1}{2}\right) = \left(0, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right).$$

By the Gram–Schmidt procedure, we know $\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k)$ for each $k = 1,2,3$. Since $\mathcal{M}(T, (v_1, v_2, v_3))$ is upper-triangular, $\operatorname{span}(v_1, \ldots, v_k)$ is invariant under T for each $k = 1,2,3$. Hence, $\operatorname{span}(e_1, \ldots, e_k)$ is invariant under T for each $k = 1,2,3$, so $\mathcal{M}(T, (e_1, e_2, e_3))$ is upper-triangular. ■

Exercise 6B8

Make $\mathcal{P}_2(\mathbb{R})$ into an inner product space by defining $\langle p, q\rangle = \int_0^1 pq$ for all $p, q \in \mathcal{P}_2(\mathbb{R})$.

(a) Apply the Gram–Schmidt procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

(b) The differentiation operator (the operator that takes p to p') on $\mathcal{P}_2(\mathbb{R})$ has an upper-triangular matrix with respect to the basis $1, x, x^2$, which is not an orthonormal basis. Find the matrix of the differentiation operator on $\mathcal{P}_2(\mathbb{R})$ with respect to the orthonormal basis produced in (a) and verify that this matrix is upper triangular, as expected from the proof of 6.37.

Solution.

(a) Let $v_1 = 1, v_2 = x, v_3 = x^2$. Then $f_1 = v_1 = 1$, so $\|f_1\|^2 = \int_0^1 1^2 = 1$. Next,

$$f_2 = v_2 - \frac{\langle v_2, f_1 \rangle}{\|f_1\|^2} f_1 = x - \int_0^1 x \cdot 1 = x - \frac{1}{2}$$

with $\|f_2\|^2 = \int_0^1 (x - \frac{1}{2})^2 \, dx = \frac{1}{12}$. Finally,

$$f_3 = v_3 - \frac{\langle v_3, f_1 \rangle}{\|f_1\|^2} f_1 - \frac{\langle v_3, f_2 \rangle}{\|f_2\|^2} f_2 = x^2 - \int_0^1 x^2 \cdot 1 \, dx - 12 \left(\int_0^1 x^2 \left(x - \frac{1}{2} \right) dx \right) \left(x - \frac{1}{2} \right) = x^2 - x + \frac{1}{6}$$

with $\|f_3\|^2 = \int_0^1 (x^2 - x + \frac{1}{6})^2 \, dx = \frac{1}{180}$. Then the orthonormal basis of $\mathcal{P}_2(\mathbb{R})$ is

$$e_1 = \frac{f_1}{\|f_1\|} = 1, \quad e_2 = \frac{f_2}{\|f_2\|} = 2\sqrt{3}\left(x - \frac{1}{2} \right), \quad e_3 = \frac{f_3}{\|f_3\|} = 6\sqrt{5}\left(x^2 - x + \frac{1}{6} \right).$$

(b) The differentiation operator D on $\mathcal{P}_2(\mathbb{R})$ has the following matrix with respect to the basis $1, x, x^2$:

$$\mathcal{M}(D, (1, x, x^2)) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$\begin{aligned} De_1 &= D(1) = 0, \\ De_2 &= D\left(2\sqrt{3}\left(x - \frac{1}{2} \right) \right) = 2\sqrt{3} = 2\sqrt{3} e_1, \\ De_3 &= D\left(6\sqrt{5}\left(x^2 - x + \frac{1}{6} \right) \right) = 6\sqrt{5}(2x - 1) = 6\sqrt{5} \cdot \frac{1}{\sqrt{3}} e_2 = 2\sqrt{15} e_2 \end{aligned}$$

Hence, the matrix of D with respect to the orthonormal basis e_1, e_2, e_3 is

$$\mathcal{M}(D, (e_1, e_2, e_3)) = \begin{pmatrix} 0 & 2\sqrt{3} & 0 \\ 0 & 0 & 2\sqrt{15} \\ 0 & 0 & 0 \end{pmatrix},$$

which is upper triangular, as expected. ∎

Exercise 6B9

Suppose $e_1, \ldots, e_m$ is the result of applying the Gram–Schmidt procedure to a linearly independent list $v_1, \ldots, v_m$ in V. Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, \ldots, m$.

Solution. By the Gram–Schmidt procedure, we have

$$e_k = \frac{f_k}{\|f_k\|}, \quad f_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j.$$

From the linear independence of $v_1, \ldots, v_m$, $v_k \notin \operatorname{span}(v_1, \ldots, v_{k-1}) = \operatorname{span}(e_1, \ldots, e_{k-1})$, so $f_k \neq 0$ and $\|f_k\| > 0$. Then by orthogonality of f_k and e_j for $j < k$, we have

$$\langle v_k, e_k \rangle = \frac{1}{\|f_k\|} \langle v_k, f_k \rangle = \frac{1}{\|f_k\|} \left\langle f_k + \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, f_k \right\rangle = \frac{1}{\|f_k\|} \langle f_k, f_k \rangle = \|f_k\| > 0. \quad ∎$$

Exercise 6B10

Suppose $v_1, \ldots, v_m$ is a linearly independent list in V. Explain why the orthonormal list produced by the formulas of the Gram–Schmidt procedure (6.32) is the only orthonormal list $e_1, \ldots, e_m$ in V such that $\langle v_k, e_k \rangle > 0$ and $\operatorname{span}(v_1, \ldots, v_k) = \operatorname{span}(e_1, \ldots, e_k)$ for each $k = 1, \ldots, m$.
Axler Note. *The result in this exercise is used in the proof of 7.58.*

Solution. Suppose $e_1, \ldots, e_m$ is an orthonormal list in V such that $\langle v_k, e_k \rangle > 0$ and $\operatorname{span}(v_1, \ldots, v_k) = \operatorname{span}(e_1, \ldots, e_k)$ for each $k = 1, \ldots, m$. We induce on k to show that e_k agrees with the Gram–Schmidt output.

For $k = 1$, we have $\operatorname{span}(v_1) = \operatorname{span}(e_1)$ and $\|e_1\| = 1$, so $v_1 = \langle v_1, e_1 \rangle e_1$. Taking norms and using $\langle v_1, e_1 \rangle > 0$ gives $\langle v_1, e_1 \rangle = \|v_1\|$, so $e_1 = v_1 / \|v_1\|$, which is the Gram–Schmidt output.

Now suppose the claim holds for $e_1, \ldots, e_k$. Since $v_{k+1} \in \operatorname{span}(e_1, \ldots, e_{k+1})$ and $e_1, \ldots, e_{k+1}$ is orthonormal,

$$v_{k+1} = \sum_{j=1}^{k+1} \langle v_{k+1}, e_j \rangle e_j \implies \langle v_{k+1}, e_{k+1} \rangle e_{k+1} = v_{k+1} - \sum_{j=1}^{k} \langle v_{k+1}, e_j \rangle e_j.$$

By the induction hypothesis, the right-hand side is the Gram–Schmidt vector f_{k+1}. Taking norms and using $\langle v_{k+1}, e_{k+1} \rangle > 0$ gives $\langle v_{k+1}, e_{k+1} \rangle = \|f_{k+1}\|$, so $e_{k+1} = f_{k+1} / \|f_{k+1}\|$, which is the Gram–Schmidt output. ■

Exercise 6B11

Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that $p\left(\frac{1}{2}\right) = \int_0^1 pq$ for every $p \in \mathcal{P}_2(\mathbb{R})$.

Solution. The map $p \mapsto p(\frac{1}{2})$ is a linear functional on $\mathcal{P}_2(\mathbb{R})$. By the Riesz representation theorem, there exists a unique $q \in \mathcal{P}_2(\mathbb{R})$ such that $p(\frac{1}{2}) = \langle p, q \rangle$ for every $p \in \mathcal{P}_2(\mathbb{R})$, given by

$$q = e_1\left(\tfrac{1}{2}\right) e_1 + e_2\left(\tfrac{1}{2}\right) e_2 + e_3\left(\tfrac{1}{2}\right) e_3,$$

where e_1, e_2, e_3 is the orthonormal basis from Exercise 6B8. Evaluating e_1, e_2, e_3 at $\frac{1}{2}$ gives

$$e_1(\tfrac{1}{2}) = 1, \quad e_2(\tfrac{1}{2}) = 2\sqrt{3}\left(\frac{1}{2} - \frac{1}{2}\right) = 0, \quad e_3(\tfrac{1}{2}) = 6\sqrt{5}\left(\frac{1}{4} - \frac{1}{2} + \frac{1}{6}\right) = -\frac{\sqrt{5}}{2}.$$

Hence, the polynomial q is

$$q = e_1 - \frac{\sqrt{5}}{2} e_3 = 1 - 15\left(x^2 - x + \frac{1}{6}\right) = -15x^2 + 15x - \frac{3}{2}.$$ ■

Exercise 6B12

Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that

$$\int_0^1 p(x)\cos(\pi x)\,\mathrm{d}x = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbb{R})$.

Solution. By the Riesz representation theorem, there exists a unique $q \in \mathcal{P}_2(\mathbb{R})$ such that

$$\int_0^1 p(x)\cos(\pi x)\,\mathrm{d}x = \langle p, q\rangle = \int_0^1 pq$$

for every $p \in \mathcal{P}_2(\mathbb{R})$, given by

$$q = \langle e_1, \cos(\pi x)\rangle e_1 + \langle e_2, \cos(\pi x)\rangle e_2 + \langle e_3, \cos(\pi x)\rangle e_3,$$

where e_1, e_2, e_3 is the orthonormal basis from Exercise 6B8. Evaluating the inner products gives

$$\begin{aligned}
\langle e_1, \cos(\pi x)\rangle &= \int_0^1 e_1\cos(\pi x) = \int_0^1 \cos(\pi x) = 0,\\
\langle e_2, \cos(\pi x)\rangle &= \int_0^1 e_2\cos(\pi x) = \int_0^1 2\sqrt{3}\left(x - \frac{1}{2}\right)\cos(\pi x) = \frac{-4\sqrt{3}}{\pi^2},\\
\langle e_3, \cos(\pi x)\rangle &= \int_0^1 e_3\cos(\pi x) = \int_0^1 6\sqrt{5}\left(x^2 - x + \frac{1}{6}\right)\cos(\pi x) = 0.
\end{aligned}$$

Hence,

$$q = \frac{-4\sqrt{3}}{\pi^2}e_2 = \frac{-4\sqrt{3}}{\pi^2}\cdot 2\sqrt{3}\left(x - \frac{1}{2}\right) = \frac{-24}{\pi^2}\left(x - \frac{1}{2}\right).$$

■

Exercise 6B13

Show that a list $v_1, \dots, v_m$ of vectors in V is linearly dependent if and only if the Gram–Schmidt formula in 6.32 produces $f_k = 0$ for some $k \in \{1, \dots, m\}$.

Axler Note. *This exercise gives an alternative to Gaussian elimination techniques for determining whether a list of vectors in an inner product space is linearly dependent.*

Solution. Let f_k, e_k be the vectors obtained from the Gram–Schmidt procedure, with the process continuing as long as $f_k \neq 0$. If $v_1, \dots, v_m$ is linearly dependent, take the smallest k with $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. Then $v_1, \dots, v_{k-1}$ is linearly independent, so $e_1, \dots, e_{k-1}$ is an orthonormal basis of that span. Hence,

$$v_k = \sum_{j=1}^{k-1}\langle v_k, e_j\rangle e_j \implies f_k = v_k - \sum_{j=1}^{k-1}\langle v_k, e_j\rangle e_j = 0.$$

Conversely, if the procedure gives $f_k = 0$ for some minimal k, then

$$v_k = \sum_{j=1}^{k-1}\langle v_k, e_j\rangle e_j \in \operatorname{span}(e_1, \dots, e_{k-1}) = \operatorname{span}(v_1, \dots, v_{k-1}),$$

so $v_1, \dots, v_k$ is a linearly dependent list. ■

Exercise 6B14

Suppose V is a real inner product space and $v_1, \dots, v_m$ is a linearly independent list of vectors in V. Prove that there exist exactly 2^m orthonormal lists $e_1, \dots, e_m$ of vectors in V such that

$$\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(e_1, \dots, e_k)$$

for all $k \in \{1, \dots, m\}$.

Solution. By Exercise 6B10, there is a unique orthonormal list $e_1, \dots, e_m$ such that $\operatorname{span}(v_1, \dots, v_k) = \operatorname{span}(e_1, \dots, e_k)$ and $\langle v_k, e_k \rangle > 0$ for all k. Let $e'_1, \dots, e'_m$ be any orthonormal list satisfying the same span condition. For each k, the vector e'_k lies in $\operatorname{span}(e_1, \dots, e_k)$ and is orthogonal to $\operatorname{span}(e'_1, \dots, e'_{k-1}) = \operatorname{span}(e_1, \dots, e_{k-1})$. Now, write e'_k as

$$e'_k = \sum_{j=1}^{k} c_j e_j$$

for some scalars $c_1, \dots, c_k$. Then $\langle e'_k, e_j \rangle = 0$ for $j < k$ gives $c_j = 0$ for $j < k$, so $e'_k = c_k e_k$. Since $\|e'_k\| = 1$ and V is real, we have $e'_k = \pm e_k$. These m sign choices are independent, giving exactly 2^m such lists. ■

Exercise 6B15

Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are inner products on V such that $\langle u, v \rangle_1 = 0$ if and only if $\langle u, v \rangle_2 = 0$. Prove that there is a positive number c such that $\langle u, v \rangle_1 = c \langle u, v \rangle_2$ for every $u, v \in V$.
Axler Note. *This exercise shows that if two inner products have the same pairs of orthogonal vectors, then each of the inner products is a scalar multiple of the other inner product.*

Solution. If $V = \{0\}$, then the result holds trivially. Otherwise, assume $V \neq \{0\}$, and let $e_1 \in V$ such that $\|e_1\|_2 = 1$. Let $c = \|e_1\|_1^2$. For any $u, v \in V$, $U = \operatorname{span}(u, v, e_1)$ is finite-dimensional. Extend e_1 to an orthonormal basis $e_1, \dots, e_n$ of U with respect to $\langle \cdot, \cdot \rangle_2$. Since $\langle e_j, e_k \rangle_2 = 0$ for $j \neq k$, the same holds for $\langle \cdot, \cdot \rangle_1$.

For each $j > 1$, $\langle e_1 + e_j, e_1 - e_j \rangle_2 = 0$, so $\langle e_1 + e_j, e_1 - e_j \rangle_1 = 0$. Expanding gives

$$0 = \langle e_1 + e_j, e_1 - e_j \rangle_1 = \langle e_1, e_1 \rangle_1 - \langle e_j, e_j \rangle_1 = c - \langle e_j, e_j \rangle_1,$$

so $\langle e_j, e_j \rangle_1 = c > 0$ for all j. Write $u = a_1 e_1 + \cdots + a_n e_n$ and $v = b_1 e_1 + \cdots + b_n e_n$ for some scalars $a_1, \dots, a_n, b_1, \dots, b_n$. Then

$$\langle u, v \rangle_1 = \sum_{j=1}^{n} a_j \overline{b_j} \langle e_j, e_j \rangle_1 = c \sum_{j=1}^{n} a_j \overline{b_j} = c \langle u, v \rangle_2.$$ ■

Exercise 6B16

Suppose V is finite-dimensional. Suppose $\langle\cdot,\cdot\rangle_1$, $\langle\cdot,\cdot\rangle_2$ are inner products on V with corresponding norms $\|\cdot\|_1$ and $\|\cdot\|_2$. Prove that there exists a positive number c such that $\|v\|_1 \le c\|v\|_2$ for every $v \in V$.

Solution. If $V = \{0\}$, then the result holds trivially. Otherwise, assume $V \neq \{0\}$, and let $e_1, \ldots, e_n$ be an orthonormal basis of V with respect to $\langle\cdot,\cdot\rangle_2$. For any $v \in V$, write $v = a_1 e_1 + \cdots + a_n e_n$ for some scalars $a_1, \ldots, a_n$. Then

$$\|v\|_1 \le \sum_{j=1}^{n} |a_j| \|e_j\|_1 \le \left(\sum_{j=1}^{n} |a_j|^2 \right)^{1/2} \left(\sum_{j=1}^{n} \|e_j\|_1^2 \right)^{1/2} = \|v\|_2 \left(\sum_{j=1}^{n} \|e_j\|_1^2 \right)^{1/2}.$$

Taking $c = \left(\sum_{j=1}^{n} \|e_j\|_1^2\right)^{1/2} > 0$ gives the desired inequality. ■

Exercise 6B17

Suppose $\mathbb{F} = \mathbb{C}$ and V is finite-dimensional. Prove that if T is an operator on V such that 1 is the only eigenvalue of T and $\|Tv\| \le \|v\|$ for all $v \in V$, then T is the identity operator.

Solution. By Schur's theorem, there exists an orthonormal basis $e_1, \ldots, e_n$ of V such that $\mathcal{M}(T)$ is upper-triangular. Since 1 is the only eigenvalue of T, the diagonal entries are all 1, so

$$Te_k = e_k + \sum_{i=1}^{k-1} a_{i,k} e_i$$

for each k. By orthonormality,

$$\|Te_k\|^2 = \left\| e_k + \sum_{i=1}^{k-1} a_{i,k} e_i \right\|^2 = 1 + \sum_{i=1}^{k-1} |a_{i,k}|^2.$$

Since $\|Te_k\| \le \|e_k\| = 1$ by hypothesis, we have $\sum_{i=1}^{k-1} |a_{i,k}|^2 = 0$, forcing each $a_{i,k} = 0$. Thus, $Te_k = e_k$ for all k, so $T = I$. ■

Exercise 6B18

Suppose $u_1, \ldots, u_m$ is a linearly independent list in V. Show that there exists $v \in V$ such that $\langle u_k, v\rangle = 1$ for all $k \in \{1, \ldots, m\}$.

Solution. Let $U = \operatorname{span}(u_1, \ldots, u_m)$. Since $u_1, \ldots, u_m$ is a basis of U, there exists a unique linear functional $f : U \to \mathbb{F}$ satisfying $f(u_k) = 1$ for all $k \in \{1, \ldots, m\}$. Because U is a finite-dimensional inner product space (with the inner product inherited from V), the Riesz representation theorem provides a unique $v \in U$ such that $f(u) = \langle u, v\rangle$ for all $u \in U$. In particular, $\langle u_k, v\rangle = f(u_k) = 1$ for all k. ■

Exercise 6B19

Suppose $v_1, \ldots, v_n$ is a basis of V. Prove that there exists a basis $u_1, \ldots, u_n$ of V such that

$$\langle v_j, u_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k. \end{cases}$$

Solution. For each $k \in \{1, \ldots, n\}$, since $v_1, \ldots, v_n$ is a basis of V, there is a unique linear functional $f_k : V \to \mathbb{F}$ satisfying $f_k(v_j) = \delta_{jk}$ for all $j \in \{1, \ldots, n\}$. By the Riesz representation theorem, there exists a unique $u_k \in V$ such that $f_k(v) = \langle v, u_k \rangle$ for all $v \in V$. In particular, $\langle v_j, u_k \rangle = f_k(v_j) = \delta_{jk}$. The list $u_1, \ldots, u_n$ is linearly independent because if $c_1 u_1 + \cdots + c_n u_n = 0$, then for each j,

$$0 = \left\langle v_j, \sum_{k=1}^{n} c_k u_k \right\rangle = \overline{c_j},$$

which implies $c_j = 0$ for all j. Hence, $u_1, \ldots, u_n$ is a basis of V with the desired properties. ■

Exercise 6B20

Suppose $\mathbb{F} = \mathbb{C}$, V is finite-dimensional, and $\mathcal{E} \subseteq \mathcal{L}(V)$ is such that $ST = TS$ for all $S, T \in \mathcal{E}$. Prove that there is an orthonormal basis of V with respect to which every element of $\mathcal{E}$ has an upper-triangular matrix.

Axler Note. *This exercise strengthens Exercise 5E9(b) (in the context of inner product spaces) by asserting that the basis in that exercise can be chosen to be orthonormal.*

Solution. By Exercise 5E9(b), there exists a basis $v_1, \ldots, v_n$ of V such that every element of $\mathcal{E}$ has an upper-triangular matrix with respect to this basis. Equivalently, every $T \in \mathcal{E}$ satisfies $T(\operatorname{span}(v_1, \ldots, v_k)) \subseteq \operatorname{span}(v_1, \ldots, v_k)$ for each $k \in \{1, \ldots, n\}$.

Apply the Gram–Schmidt procedure to $v_1, \ldots, v_n$ to obtain an orthonormal basis $e_1, \ldots, e_n$ of V. By the key property of Gram–Schmidt,

$$\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(v_1, \ldots, v_k)$$

for each k. Hence, every $T \in \mathcal{E}$ still satisfies $T(\operatorname{span}(e_1, \ldots, e_k)) \subseteq \operatorname{span}(e_1, \ldots, e_k)$ for each k, which means every element of $\mathcal{E}$ has an upper-triangular matrix with respect to the orthonormal basis $e_1, \ldots, e_n$. ■

Exercise 6B21

Suppose $\mathbb{F} = \mathbb{C}$, V is finite-dimensional, $T \in \mathcal{L}(V)$, and all eigenvalues of T have absolute value less than 1. Let $\varepsilon > 0$. Prove that there exists a positive integer m such that $\|T^m v\| \leq \varepsilon \|v\|$ for every $v \in V$.

Manual Remark. *This exercise is quite difficult with a long solution. A high level overview is this: Schur's theorem gives an orthonormal basis of V with respect to which $\mathcal{M}(T)$ is upper-triangular. This reduces the problem to showing each basis vector is contracted by T^m for sufficiently large m, which is done by induction on the index of the basis vector. The final step is to use the triangle inequality to show that T^m contracts every vector in V.*

Solution. If $V = \{0\}$, then the result holds trivially. Otherwise, assume $V \neq \{0\}$, and let $\dim V = n \geq 1$. By Schur's theorem, there exists an orthonormal basis $e_1, \ldots, e_n$ of V such that $\mathcal{M}(T)$ is upper-triangular whose diagonal entries are the eigenvalues of T. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of T, so $|\lambda_j| < 1$ for each j. Write $Te_j = \lambda_j e_j + v_j$ for each j, where $v_j \in \operatorname{span}(e_1, \ldots, e_{j-1})$ and $v_1 = 0$. Let

$$\lambda = \max_j |\lambda_j| < 1,$$

and choose α such that $\lambda < \alpha < 1$.

We prove by induction on j that there exists some $C_j > 0$ such that $\|T^k e_j\| \leq C_j \alpha^k$ for all $k \geq 0$. For the base case $j = 1$, we have $Te_1 = \lambda_1 e_1$, which gives

$$\|T^k e_1\| = |\lambda_1|^k \leq \lambda^k < \alpha^k,$$

so we can take $C_1 = 1$.

Assume the inductive hypothesis holds for all indices less than j. Consider

$$v_j = \sum_{i=1}^{j-1} a_i e_i.$$

By the inductive hypothesis, $\|T^k e_i\| \leq C_i \alpha^k$ for all $i < j$, so

$$\|T^k v_j\| \leq \sum_{i=1}^{j-1} |a_i| \, \|T^k e_i\| \leq B\alpha^k, \quad B = \sum_{i=1}^{j-1} |a_i| C_i.$$

Repeatedly applying T where $Te_j = \lambda_j e_j + v_j$ gives

$$T^k e_j = \lambda_j^k e_j + \sum_{i=0}^{k-1} \lambda_j^{k-1-i} T^i v_j.$$

Taking norms and using the triangle inequality gives

$$\|T^k e_j\| \leq \lambda^k + B \sum_{i=0}^{k-1} \lambda^{k-1-i} \alpha^i = \lambda^k + B \frac{\alpha^k - \lambda^k}{\alpha - \lambda} < \left(1 + \frac{B}{\alpha - \lambda}\right) \alpha^k,$$

so we can take $C_j = 1 + \frac{B}{\alpha - \lambda}$ to complete the inductive step.

Let $C = \max\{C_1, \ldots, C_n\}$. For any $v \in V$, write $v = a_1 e_1 + \cdots + a_n e_n$ for some scalars $a_1, \ldots, a_n$. Then $|a_j| \leq \|v\|$ by orthonormality, so

$$\|T^k v\| \leq \sum_{j=1}^{n} |a_j| \, \|T^k e_j\| \leq nC\alpha^k \|v\|.$$

Choosing such m that $nC\alpha^m \leq \varepsilon$ gives $\|T^m v\| \leq \varepsilon \|v\|$ for every $v \in V$. ■

Exercise 6B22

Suppose $C[-1,1]$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$\langle f, g\rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1,1]$. Let φ be the linear functional on $C[-1,1]$ defined by $\varphi(f) = f(0)$. Show that there does not exist $g \in C[-1,1]$ such that

$$\varphi(f) = \langle f, g\rangle$$

for every $f \in C[-1,1]$.

Axler Note. *This exercise shows that the Riesz representation theorem (6.42) does not hold on infinite-dimensional vector spaces without additional hypotheses on V and φ.*

Solution. Suppose for contradiction that there exists $g \in C[-1,1]$ such that $f(0) = \int_{-1}^{1} fg$ for every $f \in C[-1,1]$. Take $f(x) = x^2 g(x)$, which is continuous. Then

$$0 = f(0) = \int_{-1}^{1} x^2 (g(x))^2 \, dx.$$

The integrand is continuous and nonnegative, so $x^2(g(x))^2 = 0$ for all $x \in [-1,1]$. Hence, $g(x) = 0$ for all $x \in [-1,1] \setminus \{0\}$, and continuity forces $g(0) = 0$, so $g \equiv 0$. But then $f(0) = 0$ for every $f \in C[-1,1]$, which is false (take $f = 1$), a contradiction. Hence, no such g exists. ■

Exercise 6B23

For all $u, v \in V$, define $d(u,v) = \|u - v\|$.

(a) Show that d is a metric on V.
(b) Show that if V is finite-dimensional, then d is a complete metric on V (meaning that every Cauchy sequence converges).
(c) Show that every finite-dimensional subspace of V is a closed subset of V (with respect to the metric d).

Axler Note. *This exercise requires familiarity with metric spaces.*

Solution.

(a) We verify the metric axioms for d:

- **Non-negativity and Identity**: For all $u, v \in V$, $\|u - v\| \geq 0$ and $\|u - v\| = 0$ if and only if $u = v$.
- **Symmetry**: For all $u, v \in V$, we have

$$d(u,v) = \|u - v\| = \|-(v-u)\| = \|v - u\| = d(v,u).$$

- **Triangle Inequality**: For all $u, v, w \in V$, we have

$$d(u,w) = \|u - w\| = \|(u-v) + (v-w)\| \leq \|u - v\| + \|v - w\| = d(u,v) + d(v,w).$$

Therefore, (V, d) is a metric space.

(b) If V is finite-dimensional, let $e_1, \ldots, e_n$ be an orthonormal basis of V. A Cauchy sequence $(v_k)_{k\geq 1}$ in V can be written as $v_k = a_{k,1}e_1 + \cdots + a_{k,n}e_n$ for scalars $a_{k,j}$. By the Cauchy–Schwarz inequality, for each j,

$$|a_{k,j} - a_{m,j}| = |\langle v_k - v_m, e_j\rangle| \leq \|v_k - v_m\|,$$

so each coordinate sequence $(a_{k,j})_{k\geq 1}$ is Cauchy in $\mathbb{F}$. Since $\mathbb{F}$ is complete, each $(a_{k,j})$ converges to some $a_j \in \mathbb{F}$. Define $v = a_1e_1 + \cdots + a_ne_n \in V$. Since $e_1, \ldots, e_n$ is orthonormal,

$$\|v_k - v\|^2 = \sum_{j=1}^{n} |a_{k,j} - a_j|^2 \to 0,$$

so $v_k \to v$ in V, and (V, d) is complete.

(c) Let U be a finite-dimensional subspace of V, and let $(u_k)_{k\geq 1}$ be a sequence in U converging to some $v \in V$. Since (u_k) converges, it is Cauchy. Restricting the inner product on V to U, we see that (u_k) is also a Cauchy sequence in U. By part (b), U is complete, so (u_k) converges to some $u \in U$. Since (u_k) converges to v in V and to u in U, we have $v = u \in U$. Therefore, U is closed in V. ■

6C Orthogonal Complements and Minimization Problems

Exercise 6C1

Suppose $v_1, \ldots, v_m \in V$. Prove that

$$\{v_1, \ldots, v_m\}^\perp = (\operatorname{span}(v_1, \ldots, v_m))^\perp.$$

Solution. Suppose $v \in \{v_1, \ldots, v_m\}^\perp$. Then $\langle v, v_k \rangle = 0$ for each $k = 1, \ldots, m$. For any $u \in \operatorname{span}(v_1, \ldots, v_m)$, we can write $u = a_1 v_1 + \cdots + a_m v_m$ for some scalars $a_1, \ldots, a_m$. Then

$$\langle v, u \rangle = \sum_{k=1}^{m} \overline{a_k} \langle v, v_k \rangle = 0,$$

so $v \in (\operatorname{span}(v_1, \ldots, v_m))^\perp$.

Conversely, suppose $v \in (\operatorname{span}(v_1, \ldots, v_m))^\perp$. Since $\{v_1, \ldots, v_m\} \subseteq \operatorname{span}(v_1, \ldots, v_m)$, Theorem 6.48(e) shows that $\operatorname{span}(v_1, \ldots, v_m)^\perp \subseteq \{v_1, \ldots, v_m\}^\perp$, so $v \in \{v_1, \ldots, v_m\}^\perp$. ■

Exercise 6C2

Suppose U is a subspace of V with basis $u_1, \ldots, u_m$ and

$$u_1, \ldots, u_m, v_1, \ldots, v_n$$

is a basis of V. Prove that if the Gram–Schmidt procedure is applied to the basis of V above, producing a list $e_1, \ldots, e_m, f_1, \ldots, f_n$, then $e_1, \ldots, e_m$ is an orthonormal basis of U and $f_1, \ldots, f_n$ is an orthonormal basis of $U^\perp$.

Solution. Let $e_1, \ldots, e_m, f_1, \ldots, f_n$ be the result of applying the Gram–Schmidt procedure to the basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V. By the properties of the Gram–Schmidt procedure, $\operatorname{span}(e_1, \ldots, e_k) = \operatorname{span}(u_1, \ldots, u_k)$ for each $k \leq m$. In particular, $\operatorname{span}(e_1, \ldots, e_m) = \operatorname{span}(u_1, \ldots, u_m) = U$, so $e_1, \ldots, e_m$ is an orthonormal basis of U.

Since each f_j is constructed to be orthogonal to all previous vectors in the Gram–Schmidt process, it follows that each f_j is orthogonal to every e_k for $k = 1, \ldots, m$. Hence, $f_1, \ldots, f_n \in U^\perp$. Additionally, $f_1, \ldots, f_n$ is an orthonormal list in $U^\perp$ because the Gram–Schmidt procedure produces orthonormal vectors. By Theorem 6.49, $V = U \oplus U^\perp$, so $\dim U + \dim U^\perp = \dim V$. Since $\dim U = m$ and $\dim V = m + n$, we have $\dim U^\perp = n$. Therefore, $f_1, \ldots, f_n$ is an orthonormal basis of $U^\perp$. ■

Exercise 6C3

Suppose U is the subspace of $\mathbb{R}^4$ defined by

$$U = \operatorname{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find an orthonormal basis of U and an orthonormal basis of $U^\perp$.

Solution. Let $v_1 = (1, 2, 3, -4)$ and $v_2 = (-5, 4, 3, 2)$. Applying the Gram–Schmidt procedure to v_1 and

v_2, we first set $f_1 = v_1$ with $\|f_1\|^2 = 30$. Then

$$f_2 = v_2 - \frac{\langle v_2, f_1\rangle}{\|f_1\|^2} f_1 = (-5,4,3,2) - \frac{4}{30}(1,2,3,-4) = \frac{1}{15}(-77,56,39,38),$$

where

$$\|f_2\|^2 = \frac{1}{225}((-77)^2 + 56^2 + 39^2 + 38^2) = \frac{12030}{225}.$$

Then an orthonormal basis of U is given by

$$e_1 = \frac{1}{\sqrt{30}}(1,2,3,-4), \quad e_2 = \frac{1}{\sqrt{12030}}(-77,56,39,38).$$

An orthonormal basis for $U^\perp$ consists of vectors orthogonal to both v_1 and v_2. To find such vectors, consider the system of equations

$$\begin{cases} x + 2y + 3z - 4w = 0 \\ -5x + 4y + 3z + 2w = 0 \end{cases}$$

The first equation yields $x = -2y - 3z + 4w$, and substituting into the second equation gives

$$-5(-2y - 3z + 4w) + 4y + 3z + 2w = 0 \implies 14y + 18z - 18w = 0 \implies 7y = -9z + 9w.$$

Choosing $z = w = 1$ gives $y = 0$, and then $x = -2(0) - 3(1) + 4(1) = 1$. Choosing $z = 0$ and $w = 7$ gives $y = 9$ and then $x = -2(9) - 3(0) + 4(7) = -18 + 28 = 10$. Hence, we have two vectors $u_1 = (1,0,1,1)$ and $u_2 = (10,9,0,7)$ in $U^\perp$. Applying the Gram–Schmidt procedure to u_1 and u_2, we first set $g_1 = u_1$ with $\|g_1\|^2 = 3$. Then

$$g_2 = u_2 - \frac{\langle u_2, g_1\rangle}{\|g_1\|^2} g_1 = (10,9,0,7) - \frac{17}{3}(1,0,1,1) = \frac{1}{3}(13,27,-17,4),$$

where

$$\|g_2\|^2 = \frac{1}{9}(13^2 + 27^2 + (-17)^2 + 4^2) = \frac{1203}{9}.$$

Then an orthonormal basis of $U^\perp$ is given by

$$e_1' = \frac{1}{\sqrt{3}}(1,0,1,1), \quad e_2' = \frac{1}{\sqrt{1203}}(13,27,-17,4). \qquad \blacksquare$$

Exercise 6C4

Suppose $e_1, \dots, e_n$ is a list of vectors in V with $\|e_k\| = 1$ for each $k = 1, \dots, n$ and

$$\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2$$

for all $v \in V$. Prove that $e_1, \dots, e_n$ is an orthonormal basis of V.
Axler Note. *This exercise provides a converse to 6.30(b).*

Solution. Setting $v = e_j$, we have

$$1 = \|e_j\|^2 = |\langle e_j, e_1\rangle|^2 + \cdots + |\langle e_j, e_n\rangle|^2 = |\langle e_j, e_j\rangle|^2 + \sum_{k \neq j} |\langle e_j, e_k\rangle|^2.$$

Since $\|e_j\| = 1$, we have $|\langle e_j, e_j\rangle|^2 = 1$, so

$$1 = 1 + \sum_{k \neq j} |\langle e_j, e_k\rangle|^2 \implies \sum_{k \neq j} |\langle e_j, e_k\rangle|^2 = 0,$$

which forces $|\langle e_j, e_k\rangle|^2 = 0$ for each $k \neq j$, so $\langle e_j, e_k\rangle = 0$ for each $j \neq k$. Hence, $e_1, \ldots, e_n$ is an orthonormal list. To show that $e_1, \ldots, e_n$ is a basis of V, let $v \in V$ be arbitrary. Consider

$$u = v - \sum_{k=1}^{n} \langle v, e_k\rangle e_k.$$

Then for each j:

$$\langle u, e_j\rangle = \langle v, e_j\rangle - \sum_{k=1}^{n} \langle v, e_k\rangle\langle e_k, e_j\rangle = \langle v, e_j\rangle - \langle v, e_j\rangle = 0$$

by orthonormality. Thus, u is orthogonal to each e_j, so by the given condition,

$$\|u\|^2 = |\langle u, e_1\rangle|^2 + \cdots + |\langle u, e_n\rangle|^2 = 0,$$

which implies $u = 0$. Therefore, $v \in \operatorname{span}(e_1, \ldots, e_n)$. Since $e_1, \ldots, e_n$ is an orthonormal list and $\operatorname{span}(e_1, \ldots, e_n) = V$, it follows that $e_1, \ldots, e_n$ is an orthonormal basis of V. ■

Exercise 6C5

Suppose that V is finite-dimensional and U is a subspace of V. Show that $P_{U^\perp} = I - P_U$, where I is the identity operator on V.

Solution. For every $v \in V$, $v = P_U v + P_{U^\perp} v$ by the definition of orthogonal projection. Then

$$P_{U^\perp} v = v - P_U v = Iv - P_U v = (I - P_U)v.$$

Hence, $P_{U^\perp} = I - P_U$. ■

Exercise 6C6

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$T = TP_{(\operatorname{null} T)^\perp} = P_{\operatorname{range} T} T.$$

Solution. We first show that $T = TP_{(\operatorname{null} T)^\perp}$. For any $v \in V$, write $v = u + w$ for some $u \in \operatorname{null} T$ and $w \in (\operatorname{null} T)^\perp$. Then

$$TP_{(\operatorname{null} T)^\perp} v = TP_{(\operatorname{null} T)^\perp}(u + w) = TP_{(\operatorname{null} T)^\perp} u + TP_{(\operatorname{null} T)^\perp} w = T0 + Tw = Tw.$$

Since $u \in \operatorname{null} T$, we have $Tu = 0$, so $Tv = T(u + w) = Tu + Tw = 0 + Tw = Tw$. Hence, $T = TP_{(\operatorname{null} T)^\perp}$.

Next, we show that $T = P_{\operatorname{range} T} T$. For any $v \in V$, we have $Tv \in \operatorname{range} T$, so $P_{\operatorname{range} T} Tv = Tv$. Therefore, $T = P_{\operatorname{range} T} T$. ■

Exercise 6C7

Suppose that X and Y are finite-dimensional subspaces of V. Prove that $P_X P_Y = 0$ if and only if $\langle x, y\rangle = 0$ for all $x \in X$ and all $y \in Y$.

Solution. $\langle x, y\rangle = 0$ for all $x \in X, y \in Y \iff Y \subseteq X^\perp \iff \operatorname{range} P_Y \subseteq \operatorname{null} P_X \iff P_X P_Y = 0$. ∎

Exercise 6C8

Suppose U is a finite-dimensional subspace of V and $v \in V$. Define a linear functional $\varphi\colon U \to \mathbb{F}$ by

$$\varphi(u) = \langle u, v\rangle$$

for all $u \in U$. By the Riesz representation theorem (6.42) as applied to the inner product space U, there exists a unique vector $w \in U$ such that

$$\varphi(u) = \langle u, w\rangle$$

for all $u \in U$. Show that $w = P_U v$.

Solution. By the Riesz representation theorem, we have $w \in U$. For any $u \in U$, we have

$$\langle u, v - w\rangle = \langle u, v\rangle - \langle u, w\rangle = \varphi(u) - \varphi(u) = 0.$$

Hence, $v - w \in U^\perp$, so $w = P_U v$. ∎

Exercise 6C9

Suppose V is finite-dimensional. Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and every vector in $\operatorname{null} P$ is orthogonal to every vector in $\operatorname{range} P$. Prove that there exists a subspace U of V such that $P = P_U$.

Solution. Let $U = \operatorname{range} P$. By Exercise 3B27, $P^2 = P$ implies $V = U \oplus \operatorname{null} P$. Since every $v \in \operatorname{null} P$ is orthogonal to every $u \in U$, we have $\operatorname{null} P \subseteq U^\perp$. From the fundamental theorem of linear maps,

$$\dim \operatorname{null} P = \dim V - \dim U = \dim U^\perp.$$

Hence, $\operatorname{null} P = U^\perp$, so $V = U \oplus U^\perp$. Then for any $x \in V$, we can write $x = u + w$ for some $u \in U$ and $w \in U^\perp$. Since $u \in \operatorname{range} P$ and $P^2 = P$, we have $Pu = u$. Then $Px = P(u + w) = Pu + Pw = u + 0 = u = P_U x$, so $P = P_U$. ∎

Exercise 6C10

Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$ is such that $P^2 = P$ and

$$\|Pv\| \le \|v\|$$

for every $v \in V$. Prove that there exists a subspace U of V such that $P = P_U$.

Solution. Let $U = \operatorname{range} P$. From Exercise 3B27, $P^2 = P$ implies $V = U \oplus \operatorname{null} P$. Let $w \in \operatorname{null} P$ and $u \in U$. For every $\alpha \in \mathbb{F}$,

$$\|u\| = \|P(u + \alpha w)\| \leq \|u + \alpha w\|.$$

By Exercise 6A6, $\langle u, w \rangle = 0$. Therefore, $\operatorname{null} P \subseteq U^\perp$. From the fundamental theorem of linear maps,

$$\dim \operatorname{null} P = \dim V - \dim U = \dim U^\perp.$$

Hence, $\operatorname{null} P = U^\perp$, so $V = U \oplus U^\perp$. For any $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp = \operatorname{null} P$. Then $Pv = Pu + Pw = Pu = u = P_U v$, where $Pu = u$ since $u \in \operatorname{range} P$ and $P^2 = P$. Hence, $P = P_U$. ■

Exercise 6C11

Suppose $T \in \mathcal{L}(V)$ and U is a finite-dimensional subspace of V. Prove that

$$U \text{ is invariant under } T \iff P_U T P_U = T P_U.$$

Solution. Suppose U is invariant under T. For any $v \in V$, we have $P_U v \in U$, so $T P_U v \in U$ by invariance. Since P_U acts as the identity on U, it follows that $P_U T P_U v = T P_U v$.

Conversely, suppose $P_U T P_U = T P_U$. For any $u \in U$, we have $P_U u = u$, so

$$Tu = T P_U u = P_U T P_U u \in \operatorname{range} P_U = U,$$

so U is invariant under T. ■

Exercise 6C12

Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and U is a subspace of V. Prove that

$$U \text{ and } U^\perp \text{ are both invariant under } T \iff P_U T = T P_U.$$

Solution. Suppose U and $U^\perp$ are both invariant under T. For any $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Then

$$P_U T v = P_U(Tu + Tw) = P_U Tu + P_U Tw = Tu + 0 = Tu,$$

since $Tu \in U$ and $Tw \in U^\perp$. On the other hand,

$$T P_U v = T P_U(u + w) = Tu,$$

so $P_U T v = T P_U v$ for all $v \in V$, and hence $P_U T = T P_U$.

Conversely, suppose $P_U T = T P_U$. For any $u \in U$, we have $P_U(Tu) = T(P_U u) = Tu$, so $Tu \in U$, and U is invariant under T. For any $w \in U^\perp$, we have $P_U(Tw) = T(P_U w) = T(0) = 0$, so $Tw \in U^\perp$, and $U^\perp$ is invariant under T. ■

Exercise 6C13

Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. For each $v \in V$, let φ_v denote the linear functional on V defined by

$$\varphi_v(u) = \langle u, v \rangle$$

for all $u \in V$.

(a) Show that $v \mapsto \varphi_v$ is an injective linear map from V to V'.

(b) Use (a) and a dimension-counting argument to show that $v \mapsto \varphi_v$ is an isomorphism from V onto V'.

Axler Note. *The purpose of this exercise is to give an alternative proof of the Riesz representation theorem (6.42 and 6.58) when $\mathbb{F} = \mathbb{R}$. Thus, you should not use the Riesz representation theorem as a tool in your solution.*

Solution.

(a) Define the map $\Phi : V \to V'$ by $\Phi(v) = \varphi_v$ for all $v \in V$. To show that Φ is linear, let $v_1, v_2 \in V$ and $a, b \in \mathbb{R}$. Then for any $u \in V$,

$$\begin{aligned}
\Phi(av_1 + bv_2)(u) &= \varphi_{av_1+bv_2}(u) \\
&= \langle u, av_1 + bv_2 \rangle \\
&= a\langle u, v_1 \rangle + b\langle u, v_2 \rangle \\
&= a\varphi_{v_1}(u) + b\varphi_{v_2}(u) \\
&= a\Phi(v_1)(u) + b\Phi(v_2)(u).
\end{aligned}$$

Since this holds for all $u \in V$, we have $\Phi(av_1 + bv_2) = a\Phi(v_1) + b\Phi(v_2)$, so Φ is linear. Injectivity follows from the fact that if $\Phi(v) = 0$, then $\varphi_v(u) = 0$ for all $u \in V$, which implies $\langle u, v \rangle = 0$ for all $u \in V$. In particular, taking $u = v$ gives $\langle v, v \rangle = 0$, so $v = 0$.

(b) By part (a), Φ is an injective linear map from V to V'. Since V is finite-dimensional, we have $\dim V = \dim V'$. Hence, Φ is also surjective, and therefore an isomorphism from V onto V'. ■

Exercise 6C14

Suppose that $e_1, \ldots, e_n$ is an orthonormal basis of V. Explain why the dual basis (see 3.112) of $e_1, \ldots, e_n$ is $e_1, \ldots, e_n$ under the identification of V' with V provided by the Riesz representation theorem (6.58).

Solution. Let $\varphi_1, \ldots, \varphi_n$ be the dual basis of $e_1, \ldots, e_n$. By the Riesz representation theorem, for each j there exists a unique $v_j \in V$ such that $\varphi_j(u) = \langle u, v_j \rangle$ for all $u \in V$. Define $\psi_j \in V'$ by $\psi_j(u) = \langle u, e_j \rangle$. Then for any k,

$$\psi_j(e_k) = \langle e_k, e_j \rangle = \delta_{jk} = \varphi_j(e_k).$$

Since ψ_j and φ_j agree on a basis of V, they are equal. By uniqueness in the Riesz representation theorem, $v_j = e_j$. Hence, the dual basis $\varphi_1, \ldots, \varphi_n$ corresponds to $e_1, \ldots, e_n$ under the identification of V' with V provided by the Riesz representation theorem. ■

Exercise 6C15

In $\mathbb{R}^4$, let

$$U = \operatorname{span}((1,1,0,0), (1,1,1,2)).$$

Find $u \in U$ such that $\|u - (1,2,3,4)\|$ is as small as possible.

Solution. Applying the Gram–Schmidt procedure to the basis $v_1 = (1,1,0,0)$ and $v_2 = (1,1,1,2)$ of U, we first set $f_1 = v_1$ with $\|f_1\|^2 = 2$. Then

$$f_2 = v_2 - \frac{\langle v_2, f_1 \rangle}{\|f_1\|^2} f_1 = (1,1,1,2) - \frac{2}{2}(1,1,0,0) = (0,0,1,2),$$

with $\|f_2\|^2 = 5$. Projecting $(1,2,3,4)$ onto U gives

$$P_U(1,2,3,4) = \frac{\langle (1,2,3,4), f_1 \rangle}{\|f_1\|^2} f_1 + \frac{\langle (1,2,3,4), f_2 \rangle}{\|f_2\|^2} f_2 = \frac{3}{2}(1,1,0,0) + \frac{11}{5}(0,0,1,2) = \left(\frac{3}{2}, \frac{3}{2}, \frac{11}{5}, \frac{22}{5}\right). \quad \blacksquare$$

Exercise 6C16

Suppose $C[-1,1]$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$\langle f, g \rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1,1]$. Let U be the subspace of $C[-1,1]$ defined by

$$U = \{f \in C[-1,1] \mid f(0) = 0\}.$$

(a) Show that $U^\perp = \{0\}$.
(b) Show that 6.49 and 6.52 do not hold without the finite-dimensional hypothesis.

Appendix. *See* Appendix 6C16 *for an analysis-focused solution to part (a).*

Solution.

(a) Let $g \in U^\perp$, and let $f(x) = x^2 g(x)$. Then f is continuous on $[-1,1]$ and $f(0) = 0$, so $f \in U$. Hence,

$$0 = \langle f, g \rangle = \int_{-1}^{1} x^2 g(x)^2 \, dx.$$

Since $x^2 g(x)^2 \geq 0$ for all $x \in [-1,1]$, it follows that $x^2 g(x)^2 = 0$ for all $x \in [-1,1]$. Therefore, $g(x) = 0$ for all $x \in [-1,1] \setminus \{0\}$, and by continuity, $g(0) = 0$ as well. Hence, $g = 0$, so $U^\perp = \{0\}$.

(b) 6.49 asserts that $V = U \oplus U^\perp$, but $U^\perp = \{0\}$, so $U \oplus U^\perp = U \neq C[-1,1]$ since the constant function 1 is not in U. 6.52 asserts that $(U^\perp)^\perp = U$, but $(U^\perp)^\perp = \{0\}^\perp = C[-1,1] \neq U$. $\blacksquare$

Exercise 6C17

Find $p \in \mathcal{P}_3(\mathbb{R})$ such that $p(0) = 0$, $p'(0) = 0$, and $\int_0^1 |2 + 3x - p(x)|^2 \, dx$ is as small as possible.

Solution. The conditions $p(0) = 0$ and $p'(0) = 0$ imply that $p(x) = ax^2 + bx^3$ for some scalars $a, b \in \mathbb{R}$. Let $f(x) = 2 + 3x$ and $g(x) = ax^2 + bx^3$. To minimize the integral, we project f onto $\operatorname{span}(x^2, x^3)$ in $C[0,1]$ to get the resulting system of equations:

$$0 = \langle f - g, x^2 \rangle = \int_0^1 (2 + 3x - ax^2 - bx^3)x^2 \, dx = \frac{2}{3} + \frac{3}{4} - \frac{a}{5} - \frac{b}{6},$$
$$0 = \langle f - g, x^3 \rangle = \int_0^1 (2 + 3x - ax^2 - bx^3)x^3 \, dx = \frac{1}{2} + \frac{3}{5} - \frac{a}{6} - \frac{b}{7}.$$

Solving this system yields $a = 24$ and $b = -\frac{203}{10}$, so the polynomial $p(x) = 24x^2 - \frac{203}{10}x^3$ minimizes the integral. ■

Exercise 6C18

Find $p \in \mathcal{P}_5(\mathbb{R})$ that makes $\int_{-\pi}^{\pi} |\sin x - p(x)|^2 \, dx$ as small as possible.

Axler Note. *The polynomial 6.65 is an excellent approximation to the answer to this exercise, but here you are asked to find the exact solution, which involves powers of π. A computer that can perform symbolic integration should help.*

Solution. Similar to Exercise 6C17, we project $\sin x$ onto $\operatorname{span}(1, x, x^2, x^3, x^4, x^5)$ in $C[-\pi, \pi]$. Note that $\sin x$ is orthogonal to 1, x^2, and x^4 on $[-\pi, \pi]$ because it is an odd function. Hence, we only need to consider the projection onto $\operatorname{span}(x, x^3, x^5)$. Using the Gram–Schmidt procedure, we can find an orthonormal basis of $\operatorname{span}(x, x^3, x^5)$ and then compute the projection of $\sin x$ onto this subspace. Calculating the orthogonal basis, we begin with $f_1 = x$ with $\|f_1\|^2 = \frac{2\pi^3}{3}$. Then

$$f_2 = x^3 - \frac{\langle x^3, f_1 \rangle}{\|f_1\|^2} f_1 = x^3 - \frac{3\pi^2}{5}x, \quad \|f_2\|^2 = \frac{8\pi^7}{175}.$$

Finally,

$$f_3 = x^5 - \frac{\langle x^5, f_1 \rangle}{\|f_1\|^2} f_1 - \frac{\langle x^5, f_2 \rangle}{\|f_2\|^2} f_2 = x^5 - \frac{10\pi^2}{9}x^3 + \frac{5\pi^4}{21}x, \quad \|f_3\|^2 = \frac{128\pi^{11}}{43659}.$$

Projecting $\sin x$ onto $U = \operatorname{span}(x, x^3, x^5)$ gives

$$P_U \sin x = \frac{\langle \sin x, f_1 \rangle}{\|f_1\|^2} f_1 + \frac{\langle \sin x, f_2 \rangle}{\|f_2\|^2} f_2 + \frac{\langle \sin x, f_3 \rangle}{\|f_3\|^2} f_3,$$

where the inner products are computed as follows:

$$\langle \sin x, f_1 \rangle = \int_{-\pi}^{\pi} x \sin x \, dx = 2\pi,$$
$$\langle \sin x, f_2 \rangle = \int_{-\pi}^{\pi} \left(x^3 - \frac{3\pi^2}{5}x \right) \sin x \, dx = \frac{4\pi}{5}(\pi^2 - 15),$$
$$\langle \sin x, f_3 \rangle = \int_{-\pi}^{\pi} \left(x^5 - \frac{10\pi^2}{9}x^3 + \frac{5\pi^4}{21}x \right) \sin x \, dx = \frac{2}{63}(8\pi^5 - 840\pi^3 + 7560\pi).$$

The coefficients of f_1, f_2, and f_3 in the projection are then

$$\frac{\langle \sin x, f_1 \rangle}{\|f_1\|^2} = \frac{3}{\pi^2}, \quad \frac{\langle \sin x, f_2 \rangle}{\|f_2\|^2} = \frac{35(\pi^2 - 15)}{2\pi^6}, \quad \frac{\langle \sin x, f_3 \rangle}{\|f_3\|^2} = \frac{693(\pi^4 - 105\pi^2 + 945)}{8\pi^{10}}.$$

Hence, the polynomial $p(x) = P_U \sin x$ that minimizes the integral is $p(x) = \alpha x + \beta x^3 + \gamma x^5$, where

$$\begin{aligned}\alpha &= \frac{105}{8\pi^6}(\pi^4 - 153\pi^2 + 1485),\\ \beta &= -\frac{315}{4\pi^8}(\pi^4 - 125\pi^2 + 1155),\\ \gamma &= \frac{693}{8\pi^{10}}(\pi^4 - 105\pi^2 + 945).\end{aligned}$$

■

Exercise 6C19

Suppose V is finite-dimensional and $P \in \mathcal{L}(V)$ is an orthogonal projection of V onto some subspace of V. Prove that $P^\dagger = P$.

Solution. For $u, v \in V$, write $u = u_U + u_{U^\perp}$ and $v = v_U + v_{U^\perp}$ with $u_U, v_U \in U = \operatorname{range} P$ and $u_{U^\perp}, v_{U^\perp} \in U^\perp = \operatorname{null} P$. Then $Pu = u_U$ and $Pv = v_U$, so

$$\langle Pu, v\rangle = \langle u_U, v_U + v_{U^\perp}\rangle = \langle u_U, v_U\rangle$$

since $u_U \in U$ and $v_{U^\perp} \in U^\perp$. On the other hand,

$$\langle u, Pv\rangle = \langle u_U + u_{U^\perp}, v_U\rangle = \langle u_U, v_U\rangle$$

since $u_{U^\perp} \in U^\perp$ and $v_U \in U$. Hence, $\langle Pu, v\rangle = \langle u, Pv\rangle$ for all $u, v \in V$, so $P^\dagger = P$. ■

Exercise 6C20

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Show that

$$\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp \quad \text{and} \quad \operatorname{range} T^\dagger = (\operatorname{null} T)^\perp.$$

Solution. Let P be the orthogonal projection of W onto range T and let $\tilde{T} : (\operatorname{null} T)^\perp \to \operatorname{range} T$ denote the restriction of T to $(\operatorname{null} T)^\perp$, so that $T^\dagger = \tilde{T}^{-1}P$. Since $\tilde{T}^{-1} : \operatorname{range} T \to (\operatorname{null} T)^\perp$ is a bijection, and P maps W onto range T,

$$\operatorname{null} T^\dagger = \operatorname{null}(\tilde{T}^{-1}P) = \operatorname{null} P = (\operatorname{range} T)^\perp,$$

where the middle equality uses injectivity of $\tilde{T}^{-1}$, and

$$\operatorname{range} T^\dagger = \tilde{T}^{-1}(P(W)) = \tilde{T}^{-1}(\operatorname{range} T) = (\operatorname{null} T)^\perp,$$

where the second equality uses surjectivity of P onto range T. ■

Exercise 6C21

Suppose $T \in \mathcal{L}(\mathbb{F}^3, \mathbb{F}^2)$ is defined by

$$T(a,b,c) = (a+b+c,\ 2b+3c).$$

(a) For $(x,y) \in \mathbb{F}^2$, find a formula for $T^\dagger(x,y)$.
(b) Verify that the equation $TT^\dagger = P_{\operatorname{range} T}$ from 6.69(b) holds with the formula for $T^\dagger$ obtained in (a).
(c) Verify that the equation $T^\dagger T = P_{(\operatorname{null} T)^\perp}$ from 6.69(c) holds with the formula for $T^\dagger$ obtained in (a).

Solution.

(a) From $a+b+c = 2b+3c = 0$, we see that $\operatorname{null} T = \operatorname{span}(1,-3,2)$, so

$$(\operatorname{null} T)^\perp = \{(a,b,c) \mid a - 3b + 2c = 0\}.$$

Moreover, $\operatorname{range} T = \mathbb{F}^2$, so $P_{\operatorname{range} T} = I$. Let $T^\dagger(x,y) = (a,b,c)$. Then $T(a,b,c) = (x,y)$ and $(a,b,c) \in (\operatorname{null} T)^\perp$. Hence, we have the system of equations

$$\begin{cases} x = a+b+c, \\ y = 2b+3c, \\ 0 = a-3b+2c. \end{cases}$$

This yields the solution

$$T^\dagger(x,y) = \left(\frac{13x-5y}{14}, \frac{3x+y}{14}, \frac{-2x+4y}{14}\right).$$

(b) For $(x,y) \in \mathbb{F}^2$, we have

$$TT^\dagger(x,y) = T\left(\frac{13x-5y}{14}, \frac{3x+y}{14}, \frac{-2x+4y}{14}\right) = (x,y),$$

as required.

(c) We have $P_{(\operatorname{null} T)^\perp}$ being the projection onto the plane $a-3b+2c=0$. Hence,

$$P_{(\operatorname{null} T)^\perp}(a,b,c) = (a,b,c) - \frac{a-3b+2c}{14}(1,-3,2) = \left(\frac{13a+3b-2c}{14}, \frac{3a+5b+6c}{14}, \frac{-2a+6b+10c}{14}\right).$$

For $(a,b,c) \in \mathbb{F}^3$, we have

$$T^\dagger T(a,b,c) = T^\dagger(a+b+c, 2b+3c) = \left(\frac{13a+3b-2c}{14}, \frac{3a+5b+6c}{14}, \frac{-2a+6b+10c}{14}\right),$$

as required. ■

Exercise 6C22

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$TT^\dagger T = T \quad \text{and} \quad T^\dagger T T^\dagger = T^\dagger.$$

Axler Note. *Both formulas above clearly hold if T is invertible because in that case we can replace $T^\dagger$ with T^{-1}.*

Solution. Let $V = \operatorname{null} T \oplus (\operatorname{null} T)^\perp$ so that any $v \in V$ can be written as $v = u + w$ with $u \in \operatorname{null} T$ and $w \in (\operatorname{null} T)^\perp$. For the first equality, we have $Tv = Tw \in \operatorname{range} T$. Then $T^\dagger Tw = w$ since $T^\dagger$ restricted to $\operatorname{range} T$ is the inverse of T restricted to $(\operatorname{null} T)^\perp$. Hence,

$$TT^\dagger Tv = TT^\dagger Tw = Tw = Tv$$

for all $v \in V$, so $TT^\dagger T = T$.

For the second equality, let $w \in W$ and write $w = x + y$ with $x \in \operatorname{range} T$ and $y \in (\operatorname{range} T)^\perp$. Then $T^\dagger w = T^\dagger x \in (\operatorname{null} T)^\perp$, so

$$T^\dagger T T^\dagger w = T^\dagger T(T^\dagger x) = T^\dagger x = T^\dagger w.$$

Hence, $T^\dagger T T^\dagger = T^\dagger$. ∎

Exercise 6C23

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$(T^\dagger)^\dagger = T.$$

Axler Note. *The equation above is analogous to the equation $(T^{-1})^{-1} = T$ that holds if T is invertible.*

Solution. By Exercise 6C20, $\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp$ and $\operatorname{range} T^\dagger = (\operatorname{null} T)^\perp$. Applying 6.68 to $T^\dagger$ in place of T and substituting via Exercise 6C20, for $v \in V$,

$$(T^\dagger)^\dagger v = (T^\dagger|_{(\operatorname{null} T^\dagger)^\perp})^{-1} P_{\operatorname{range} T^\dagger}\, v = (T^\dagger|_{\operatorname{range} T})^{-1} P_{(\operatorname{null} T)^\perp}\, v.$$

By 6.67, $T|_{(\operatorname{null} T)^\perp}$ is an isomorphism onto $\operatorname{range} T$ whose inverse is $T^\dagger|_{\operatorname{range} T}$, so $(T^\dagger|_{\operatorname{range} T})^{-1} = T|_{(\operatorname{null} T)^\perp}$. Hence, $v - P_{(\operatorname{null} T)^\perp} v \in \operatorname{null} T$,

$$(T^\dagger)^\dagger v = T(P_{(\operatorname{null} T)^\perp} v) = Tv.$$ ∎

7 Operators on Inner Product Spaces

7A Self-Adjoint and Normal Operators

Exercise 7A1

Suppose n is a positive integer. Define $T \in \mathcal{L}(\mathbb{F}^n)$ by

$$T(z_1, \ldots, z_n) = (0, z_1, \ldots, z_{n-1}).$$

Find a formula for $T^*(z_1, \ldots, z_n)$.

Exercise 7A2

Suppose $T \in \mathcal{L}(V, W)$. Prove that

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0.$$

Exercise 7A3

Suppose $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. Prove that

$$\lambda \text{ is an eigenvalue of } T \iff \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

Exercise 7A4

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V. Prove that

$$U \text{ is invariant under } T \iff U^\perp \text{ is invariant under } T^*.$$

Exercise 7A5

Suppose $T \in \mathcal{L}(V, W)$. Suppose $e_1, \ldots, e_n$ is an orthonormal basis of V and $f_1, \ldots, f_m$ is an orthonormal basis of W. Prove that

$$\|Te_1\|^2 + \cdots + \|Te_n\|^2 = \|T^* f_1\|^2 + \cdots + \|T^* f_m\|^2.$$

Axler Note. *The numbers $\|Te_1\|^2, \ldots, \|Te_n\|^2$ in the equation above depend on the orthonormal basis $e_1, \ldots, e_n$, but the right side of the equation does not depend on $e_1, \ldots, e_n$. Thus, the equation above shows that the sum on the left side does not depend on which orthonormal basis $e_1, \ldots, e_n$ is used.*

Exercise 7A6

Suppose $T \in \mathcal{L}(V, W)$. Prove that

(a) T is injective $\iff$ T^* is surjective;

(b) T is surjective $\iff$ T^* is injective.

Exercise 7A7

Prove that if $T \in \mathcal{L}(V, W)$, then

(a) $\dim \operatorname{null} T^* = \dim \operatorname{null} T + \dim W - \dim V$;

(b) $\dim \operatorname{range} T^* = \dim \operatorname{range} T$.

Exercise 7A8

Suppose A is an m-by-n matrix with entries in $\mathbb{F}$. Use (b) in Exercise 7 to prove that the row rank of A equals the column rank of A.

Axler Note. *This exercise asks for yet another alternative proof of a result that was previously proved in 3.57 and 3.133.*

Exercise 7A9

Prove that the product of two self-adjoint operators on V is self-adjoint if and only if the two operators commute.

Exercise 7A10

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is self-adjoint if and only if

$$\langle Tv, v\rangle = \langle T^*v, v\rangle$$

for all $v \in V$.

Exercise 7A11

Define an operator $S : \mathbb{F}^2 \to \mathbb{F}^2$ by $S(w, z) = (-z, w)$.

(a) Find a formula for S^*.

(b) Show that S is normal but not self-adjoint.

(c) Find all eigenvalues of S.

Axler Note. *If $\mathbb{F} = \mathbb{R}$, then S is the operator on $\mathbb{R}^2$ of counterclockwise rotation by $90°$.*

Exercise 7A12

An operator $B \in \mathcal{L}(V)$ is called **skew** if
$$B^* = -B.$$
Suppose that $T \in \mathcal{L}(V)$. Prove that T is normal if and only if there exist commuting operators A and B such that A is self-adjoint, B is a skew operator, and $T = A + B$.

Exercise 7A13

Suppose $\mathbb{F} = \mathbb{R}$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}T = T^*$ for all $T \in \mathcal{L}(V)$.

(a) Find all eigenvalues of $\mathcal{A}$.
(b) Find the minimal polynomial of $\mathcal{A}$.

Exercise 7A14

Define an inner product on $\mathcal{P}_2(\mathbb{R})$ by $\langle p, q \rangle = \int_0^1 pq$. Define an operator $T \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ by
$$T(ax^2 + bx + c) = bx.$$

(a) Show that with this inner product, the operator T is not self-adjoint.
(b) The matrix of T with respect to the basis $1, x, x^2$ is
$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$
This matrix equals its conjugate transpose, even though T is not self-adjoint. Explain why this is not a contradiction.

Exercise 7A15

Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that

(a) T is self-adjoint $\iff$ T^{-1} is self-adjoint;
(b) T is normal $\iff$ T^{-1} is normal.

Exercise 7A16

Suppose $\mathbb{F} = \mathbb{R}$.

(a) Show that the set of self-adjoint operators on V is a subspace of $\mathcal{L}(V)$.
(b) What is the dimension of the subspace of $\mathcal{L}(V)$ in (a) [in terms of $\dim V$]?

Exercise 7A17

Suppose $\mathbb{F} = \mathbb{C}$. Show that the set of self-adjoint operators on V is not a subspace of $\mathcal{L}(V)$.

Exercise 7A18

Suppose $\dim V \geq 2$. Show that the set of normal operators on V is not a subspace of $\mathcal{L}(V)$.

Exercise 7A19

Suppose $T \in \mathcal{L}(V)$ and $\|T^*v\| \leq \|Tv\|$ for every $v \in V$. Prove that T is normal.
Axler Note. *This exercise fails on infinite-dimensional inner product spaces, leading to what are called* hyponormal operators, *which have a well-developed theory.*

Exercise 7A20

Suppose $P \in \mathcal{L}(V)$ is such that $P^2 = P$. Prove that the following are equivalent.

(a) P is self-adjoint.
(b) P is normal.
(c) There is a subspace U of V such that $P = P_U$.

Exercise 7A21

Suppose $D : \mathcal{P}_8(\mathbb{R}) \to \mathcal{P}_8(\mathbb{R})$ is the differentiation operator defined by $Dp = p'$. Prove that there does not exist an inner product on $\mathcal{P}_8(\mathbb{R})$ that makes D a normal operator.

Exercise 7A22

Give an example of an operator $T \in \mathcal{L}(\mathbb{R}^3)$ such that T is normal but not self-adjoint.

Exercise 7A23

Suppose T is a normal operator on V. Suppose also that $v, w \in V$ satisfy the equations

$$\|v\| = \|w\| = 2, \quad Tv = 3v, \quad Tw = 4w.$$

Show that $\|T(v + w)\| = 10$.

Exercise 7A24

Suppose $T \in \mathcal{L}(V)$ and

$$a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1} + z^m$$

is the minimal polynomial of T. Prove that the minimal polynomial of T^* is

$$\overline{a_0} + \overline{a_1} z + \overline{a_2} z^2 + \cdots + \overline{a_{m-1}} z^{m-1} + z^m.$$

Axler Note. *This exercise shows that the minimal polynomial of T^* equals the minimal polynomial of T if $\mathbf{F} = \mathbf{R}$.*

Exercise 7A25

Suppose $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if T^* is diagonalizable.

Exercise 7A26

Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by $Tv = \langle v, u \rangle x$ for every $v \in V$.

(a) Prove that if V is a real vector space, then T is self-adjoint if and only if the list u, x is linearly dependent.

(b) Prove that T is normal if and only if the list u, x is linearly dependent.

Exercise 7A27

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that

$$\operatorname{null} T^k = \operatorname{null} T \quad \text{and} \quad \operatorname{range} T^k = \operatorname{range} T$$

for every positive integer k.

Exercise 7A28

Suppose $T \in \mathcal{L}(V)$ is normal. Prove that if $\lambda \in \mathbf{F}$, then the minimal polynomial of T is not a polynomial multiple of $(x - \lambda)^2$.

Exercise 7A29

Prove or give a counterexample: If $T \in \mathcal{L}(V)$ and there is an orthonormal basis $e_1, \ldots, e_n$ of V such that $\|Te_k\| = \|T^* e_k\|$ for each $k = 1, \ldots, n$, then T is normal.

Exercise 7A30

Suppose that $T \in \mathcal{L}(\mathbb{F}^3)$ is normal and $T(1,1,1) = (2,2,2)$. Suppose $(z_1, z_2, z_3) \in \operatorname{null} T$. Prove that $z_1 + z_2 + z_3 = 0$.

Exercise 7A31

Fix a positive integer n. In the inner product space of continuous real-valued functions on $[-\pi, \pi]$ with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg$, let

$$V = \operatorname{span}(1, \cos x, \cos 2x, \ldots, \cos nx, \sin x, \sin 2x, \ldots, \sin nx).$$

(a) Define $D \in \mathcal{L}(V)$ by $Df = f'$. Show that $D^* = -D$. Conclude that D is normal but not self-adjoint.
(b) Define $T \in \mathcal{L}(V)$ by $Tf = f''$. Show that T is self-adjoint.

Exercise 7A32

Suppose $T : V \to W$ is a linear map. Show that under the standard identification of V with V' (see 6.58) and the corresponding identification of W with W', the adjoint map $T^* : W \to V$ corresponds to the dual map $T' : W' \to V'$. More precisely, show that

$$T'(\varphi_w) = \varphi_{T^*w}$$

for all $w \in W$, where φ_w and φ_{T^*w} are defined as in 6.58.

A Appendix: Alternate Proofs of Exercises

A1 Vector Spaces

Exercise 1C13

Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Coolempire93. Conversely, suppose that $U_1 \cup U_2 \cup U_3$ is a subspace of V but none of the subspaces contains the other two. Let $W_1 = U_2 \cup U_3$ and $W_2 = U_1 \cup U_3$. By the assumption, we cannot have $W_1 \subseteq U_1$ or $W_2 \subseteq U_2$, so there exist $x \in U_1 \setminus W_1$ and $y \in U_2 \setminus W_2$. Now, since $U_1 \cup U_2 \cup U_3$ is a vector space, we have $x + \alpha y \in U_1 \cup U_2 \cup U_3$ for any nonzero $\alpha \in \mathbb{F}$. But if $x + \alpha y \in U_1$, then $(x + \alpha y) - x = \alpha y \in U_1$, and as α is nonzero, this means that $y \in U_1$, a contradiction. Similarly, if $x + \alpha y \in U_2$, then $(x + \alpha y) - \alpha y = x \in U_2$, a contradiction. Therefore, we must have $x + \alpha y \in U_3$ for all nonzero α to satisfy the closure of the vector space.

As such, $x + y \in U_3$ and $x - y \in U_3$; that is, $(x + y) + (x - y) = 2x \in U_3$ and $(x + y) - (x - y) = 2y \in U_3$, contradicting the original definitions of x and $y^{\dagger}$. Therefore, one of the subspaces must contain the other two.

† *Final remark.* This is why the field must have more than two elements. Since part of the field axioms is that $0 \neq 1$, if the field only has two elements, it must be that $1 = -1$ (consider addition mod 2 as the addition operation), so $2x \in U_3$ would not mean that $x \in U_3$, as $2x = 0x = 0$. ■

A2 Finite-Dimensional Vector Spaces

Exercise 2C8

Suppose $v_1, \ldots, v_m$ is linearly independent in V and $w \in V$. Prove that

$$\dim \operatorname{span}(v_1 + w, \ldots, v_m + w) \geq m - 1.$$

Civil Service Pigeon. Let $S = \operatorname{span}(v_1 + w, \ldots, v_m + w)$. Since each $v_i = (v_i + w) - w$, we have

$$S + \operatorname{span}(w) = \operatorname{span}(v_1, \ldots, v_m, w).$$

Since $v_1, \ldots, v_m$ are linearly independent, we have

$$m \leq \dim(S + \operatorname{span}(w)) \leq \dim S + \dim \operatorname{span}(w) \leq \dim S + 1 \implies \dim S \geq m - 1. \qquad \blacksquare$$

A3 Linear Maps

Exercise 3B29

Suppose $p \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $5q'' + 3q' = p$.

Coolempire93. Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$ and suppose there exists a polynomial $q(x) = b_0 + b_1x + \cdots + b_mx^m$ satisfying $5q'' + 3q' = p$. If $\deg q = m \geq 2$, we can say q' has degree $m-1$ and q'' has degree $m-2$, so $5q'' + 3q'$ has degree $m-1$. To satisfy the given equation, then, we must have $m - 1 = n = \deg p$, and we may write $q(x) = b_0 + b_1x + \cdots + b_{n+1}x^{n+1}$. Before we deal with the general case, we will quickly consider when $n = 0$ (which would force $m < 2$):

If $p(x) = a_0$, then let $q(x) = b_0 + b_1x$. Because $5q'' + 3q' = p$, we have $5q'' + 3q' = 5(0) + 3b_1 = a_0$, so we set $b_1 = \frac{a_0}{3}$ and we have a polynomial q such that $5q'' + 3q' = p$ for any value of b_0. As we will see below, the constant term of q never depends on p.

For the remainder of this proof, we take $m = n + 1$ and suppose $n \geq 1$. We have

$$\begin{aligned} q'(x) &= b_1 + 2b_2x + \cdots + (n+1)b_{n+1}x^n \\ q''(x) &= 2b_2 + 6b_3x + \cdots + n(n+1)b_{n+1}x^{n-1} \end{aligned}$$

Equating coefficients of $5q'' + 3q'$ with those of p, we obtain the system

$$\begin{cases} a_0 = 3b_1 + 10b_2 \\ a_1 = 6b_2 + 30b_3 \\ \vdots \\ a_{n-1} = 3nb_n + 5n(n+1)b_{n+1} \\ a_n = 3(n+1)b_{n+1} \end{cases}.$$

That is, $a_i = 3(i+1)b_{i+1} + 5(i+1)(i+2)b_{i+2}$ for $0 \leq i < n$ and $a_n = 3(n+1)b_{n+1}$. Finally, we solve for the coefficients of q by back-substitution. Noting that

$$b_j = \frac{a_{j-1} - 5j(j+1)b_{j+1}}{3j} = \frac{a_{j-1}}{3j} - \frac{5}{3}(j+1)b_{j+1},$$

we have

$$\begin{cases} b_{n+1} = \frac{a_n}{3(n+1)} \\ b_n = \frac{a_{n-1}}{3n} - \frac{5}{3}(n+1)b_{n+1} = \frac{a_{n-1}}{3n} - \frac{5}{9}a_n \\ b_{n-1} = \frac{a_{n-2}}{3(n-1)} - \frac{5}{3}nb_n = \frac{a_{n-2}}{3(n-1)} - \frac{5}{9}a_{n-1} + \frac{25}{27}na_n \\ \vdots \\ b_1 = \frac{a_0 - 10b_2}{3} \end{cases}$$

which has a valid solution† since no division by 0 occurs. Finally, we note that b_0 is a free variable, so there is an infinite family of polynomials $q \in \mathcal{P}(\mathbb{R})$ with the given coefficients such that $5q'' + 3q' = p$. ■

† Solving this recurrence is outside the scope of this course, but, specifically, the coefficients of q are

$$\begin{aligned} b_j &= \frac{a_{j-1}}{3j} - \frac{5}{9}a_j + \frac{25}{27}(j+1)a_{j+1} - \frac{125}{81}(j+1)(j+2)a_{j+2} + \cdots + (-1)^{n-j+1}\frac{5^{n-j+1}}{3^{n-j+2}}(j+1)(j+2)\cdots na_n \\ &= \frac{1}{3}\left(\frac{a_{j-1}}{j} + \sum_{i=j}^{n}\left(-\frac{5}{3}\right)^{i-j+1}\frac{i!}{j!}a_i\right). \end{aligned}$$

Exercise 3D19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has the same matrix with respect to every basis of V if and only if T is a scalar multiple of the identity operator.

The manual solution provides a constructive proof utilizing sums of the form $v_1 + v_k$ to eliminate the k-th column, along with $v_2 + v_1$ to eliminate the 1st column. Here, we present different constructions. Each proof assumes the declaration of the matrix to be A and that $\dim V \geq 2$.

This method shows that we can swap two *arbitrary* basis vectors to force the same value of all off-diagonal entries as well as the same value of all diagonal entries. Then, we can negate one basis vector to force all off-diagonal entries to be 0 via the linear independence of the basis vectors.

Coolempire93. Let i, j be arbitrary with $1 \leq i, j \leq n$ and $i \neq j$. Consider the basis $w_1, \ldots, w_n$ of V defined by $w_i = v_j$, $w_j = v_i$ and $w_k = v_k$ otherwise. Then

$$\begin{aligned} T(w_j - v_i) &= Tw_j - Tv_i \\ T(0) &= A_{1,j}w_1 + A_{2,j}w_2 + \cdots + A_{n,j}w_n - (A_{1,i}v_1 + A_{2,i}v_2 + \cdots + A_{n,i}v_n) \\ 0 &= (A_{j,j} - A_{i,i})v_i + (A_{i,j} - A_{j,i})v_j + \sum_{\substack{k \neq i \\ k \neq j}} (A_{k,j} - A_{k,i})v_k. \end{aligned}$$

By linear independence of $v_1, \ldots, v_n$, we have $A_{i,i} = A_{j,j}$, $A_{i,j} = A_{j,i}$ and $A_{k,j} = A_{k,i}$ for all $k \neq i, j$. Because i and j were arbitrary, we have $A_{1,1} = A_{2,2} = \cdots = A_{n,n} = \lambda$, as well as A is symmetric, and all off-diagonal entries match across rows, so we can say $A_{i,j} = \gamma$ for all $i \neq j$. To finish, consider the basis $u_1, \ldots, u_n$ defined by $u_1 = -v_1$ and $u_k = v_k$ otherwise. Then

$$\begin{aligned} T(v_1 + u_1) &= Tv_1 + Tu_1 \\ T(0) &= \lambda v_1 + \gamma(v_2 + \cdots + v_n) + (\lambda u_1 + \gamma(u_2 + \cdots + u_n)) \\ 0 &= 2\gamma(v_2 + \cdots + v_n). \end{aligned}$$

Again by linear independence of $v_1, \ldots, v_n$, we have $2\gamma = 0$ and thus $\gamma = 0$. Therefore, $A = \lambda I$ and T is a scalar multiple of the identity. ■

This method shows that we can negate one *arbitrary* basis vector to force all off-diagonal entries to be 0 via the linear independence of the basis vectors. Then, we can swap v_1 with any other basis vector to force all diagonal entries to be the same.

Coolempire93. Let j be arbitrary and consider the basis $w_1, \ldots, w_n$ of V defined by $w_j = -v_j$ and $w_i = v_i$ otherwise. Then

$$\begin{aligned} T(w_j + v_j) &= Tw_j + Tv_j \\ T(0) &= A_{1,j}w_1 + A_{2,j}w_2 + \cdots + A_{n,j}w_n + A_{1,j}v_1 + A_{2,j}v_2 + \cdots + A_{n,j}v_n \\ 0 &= 2\sum_{k \neq j} A_{k,j}v_k. \end{aligned}$$

By linear independence of $v_1, \ldots, v_n$, we have $A_{k,j} = 0$ for all $k \neq j$. Since j was arbitrary, we have that all off-diagonal entries of A are 0.

We turn to one more basis to prove that all diagonal entries of A must be the same. Once again, let j be arbitrary, and this time consider the basis $u_1, \dots, u_n$ defined by $u_1 = v_j$, $u_j = v_1$ and $u_i = v_i$ otherwise. Then

$$\begin{aligned} T(v_1 - u_j) &= Tv_1 - Tu_j \\ T(0) &= A_{1,1}v_1 - A_{j,j}u_j \\ 0 &= (A_{1,1} - A_{j,j})v_1. \end{aligned}$$

As v_1 belongs to a basis for V, it is nonzero, and thus as j was arbitrary, $A_{1,1} = A_{2,2} = \cdots = A_{n,n} = \lambda$ for some scalar λ, and we have $A = \lambda I$, i.e., T is a multiple of the identity operator. ■

Exercise 3D20

Suppose $q \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{R})$ such that

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$$

for all $x \in \mathbb{R}$.

Coolempire93. We go by the same technique as in *Appendix 3B29*.

Let $q(x) = a_0 + a_1x + \cdots + a_nx^n$ and suppose that there exists a polynomial $p(x) = b_0 + b_1x + \cdots + b_mx^m$ satisfying the constraint $q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$. If q is constant, then let $p = q = a_0$. We have $(x^2 + x)p''(x) + 2xp'(x) + p(3) = (x^2 + x)0 + 2x(0) + a_0 = a_0 = q(x)$. If q is linear, say $q(x) = a_0 + a_1x$, then let $p(x) = b_0 + b_1x$. We want

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3) = (x^2 + x)0 + 2x(b_1) + b_0 + 3b_1 = 2b_1x + 3b_1 + b_0,$$

so letting $b_1 = \frac{a_1}{2}$ and $b_0 = a_0 - \frac{3a_1}{2}$ yields a polynomial p satisfying the given equation. Otherwise, consider when $n \geq 2$. If p has degree $m \geq 2$, then p' has degree $m - 1$ and p'' has degree $m - 2$, so $(x^2 + x)p'' + 2xp'$ has degree m. To satisfy the constraint, we must have $m = n \geq 2$, so we may write $p(x) = b_0 + b_1x + \cdots + b_nx^n$. Evaluating, we obtain

$$\begin{aligned} p'(x) &= b_1 + 2b_2x + \cdots + nb_nx^{n-1} \implies 2xp'(x) = 2b_1x + 4b_2x^2 + \cdots + 2nb_nx^n \\ p''(x) &= 2b_2 + 6b_3x + \cdots + n(n-1)b_nx^{n-2} \implies \begin{cases} x^2p''(x) = 2b_2x^2 + 6b_3x^3 + \cdots + n(n-1)b_nx^n \\ xp''(x) = 2b_2x + 6b_3x^2 + \cdots + n(n-1)b_nx^{n-1} \end{cases}. \end{aligned}$$

Equating coefficients of $(x^2 + x)p'' + 2xp' + p(3)$ with those of q, we have that

$$\begin{cases} a_0 = p(3) = b_0 + 3b_1 + 9b_2 + \cdots + 3^nb_n \\ a_i = i(i+1)(b_i + b_{i+1}) \qquad \text{for } 1 \leq i \leq n-1 \\ a_n = n(n-1)b_n + 2nb_n = n(n+1)b_n \end{cases},$$

with the formula for a_i coming from the fact that the coefficient of x^i in $(x^2 + x)p''(x) + 2xp'(x)$ is

$$(i(i-1)b_i + i(i+1)b_{i+1}) + 2ib_i = (i^2 - i + 2i)b_i + i(i+1)b_{i+1} = i(i+1)(b_i + b_{i+1}).$$

As before, we solve for the coefficients of p by back-substitution. Noting that for $1 \leq i \leq n$, we have

$$b_i = \frac{a_i}{i(i+1)} - b_{i+1},$$

we obtain the system

$$\begin{cases} b_n = \frac{a_n}{n(n+1)} \\ b_{n-1} = \frac{a_{n-1}}{n(n-1)} - b_n = \frac{a_{n-1}}{n(n-1)} - \frac{a_n}{n(n+1)} \\ b_{n-2} = \frac{a_{n-2}}{(n-1)(n-2)} - b_{n-1} = \frac{a_{n-2}}{(n-1)(n-2)} - \frac{a_{n-1}}{n(n-1)} + \frac{a_n}{n(n+1)} \\ \vdots \\ b_1 = \frac{a_1}{2} - b_2 \\ b_0 = a_0 - 3b_1 - 9b_2 - \cdots - 3^n b_n \end{cases}$$

which has a valid solution† since no division by 0 occurs. ■

† As before, solving this recurrence is outside the scope of this course, but, specifically, the coefficients of p are

$$b_i = \frac{a_i}{i(i+1)} - \frac{a_{i+1}}{(i+1)(i+2)} + \cdots + (-1)^{n-i}\frac{a_n}{n(n+1)} = \sum_{j=i}^{n}(-1)^{j-i}\frac{a_j}{j(j+1)}$$

for $1 \leq i \leq n$ and $b_0 = a_0 - \sum_{j=1}^{n} 3^j b_j$.

Exercise 3E11

Suppose $U = \{(x_1, x_2, \ldots) \in \mathbb{F}^\infty \mid x_k \neq 0 \text{ for only finitely many } k\}$.

(a) Show that U is a subspace of $\mathbb{F}^\infty$.

(b) Prove that $\mathbb{F}^\infty/U$ is infinite-dimensional.

Proof.

(a) Since $(0, 0, \ldots) \in U$, U is nonempty. Suppose x, $y \in U$ and $c \in \mathbb{F}$. Let S_x and S_y be the sets of indices where x and y have nonzero terms, respectively. By definition, S_x and S_y are finite. $cx + y$ can only have nonzero terms at indices in $S_x \cup S_y$. Since the union of two finite sets is finite, $cx + y \in U$, which makes U a subspace of $\mathbb{F}^\infty$.

(b) Partition $\mathbb{Z}^+$ into a countably infinite family of disjoint infinite subsets $A_1, A_2, \ldots$. For each n, define $v_n \in \mathbb{F}^\infty$ such that its j-th term is 1 if $j \in A_n$ and 0 otherwise.

Suppose that $\sum_{i=1}^{m} c_i(v_i + U) = 0 + U$ for some positive integer m and scalars $c_1, \ldots, c_m \in \mathbb{F}$. Then, $w = \sum_{i=1}^{m} c_i v_i \in U$. By construction, for $i \in \{1, \ldots, m\}$, the j-th term of w is c_i for all $j \in A_i$. Since $w \in U$, it only has a finite number of nonzero terms. But each A_i is infinite, so this forces $c_i = 0$ for all $i \in \{1, \ldots, m\}$. Hence, $v_1 + U, v_2 + U, \ldots$ are linearly independent in $\mathbb{F}^\infty/U$.

Because $\mathbb{F}^\infty/U$ contains an infinite sequence of linearly independent vectors, it is infinite-dimensional.

■

Exercise 3F11

Suppose $v_1, \ldots, v_n$ is a basis of V and $\varphi_1, \ldots, \varphi_n$ is the corresponding dual basis of V'. Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n.$$

Civil Service Pigeon. Because $\varphi_1, \ldots, \varphi_n$ is the dual basis, $\varphi_i(v_j) = \delta_{i,j}$. For each basis vector v_j,

$$\left(\sum_{i=1}^{n} \psi(v_i)\varphi_i\right)(v_j) = \sum_{i=1}^{n} \psi(v_i)\varphi_i(v_j) = \psi(v_j).$$

Since the two linear functionals agree on a basis of V, they are equal on all of V. ∎

A4 Polynomials

Exercise 4A9

Prove that every polynomial of odd degree with real coefficients has a real zero.

blanketism. Let $p(x)$ be a polynomial of odd degree with real coefficients. As $x \to \infty$, $p(x) \to \infty$ or $p(x) \to -\infty$ depending on the leading coefficient. Similarly, as $x \to -\infty$, $p(x) \to -\infty$ or $p(x) \to \infty$. By the intermediate value theorem, since $p(x)$ is continuous and takes on both positive and negative values, there must exist some $x \in \mathbb{R}$ such that $p(x) = 0$. Therefore, p has a real zero. ■

Exercise 4A11

Suppose $p \in \mathcal{P}(\mathbb{C})$. Define $q : \mathbb{C} \to \mathbb{C}$ by

$$q(z) = p(z)\overline{p(\bar{z})}.$$

Prove that q is a polynomial with real coefficients.

blanketism. If $p = 0$, then $q = 0$ is a polynomial with real coefficients. Otherwise, let $n = \deg p$. Then we may write

$$p(z) = \alpha(z - \lambda_1)(z - \lambda_2)\cdots(z - \lambda_n),$$

where $\alpha \in \mathbb{C}$ and $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{C}$ are the roots of p. Then

$$q(z) = p(z)\overline{p(\bar{z})} = \alpha(z - \lambda_1)(z - \lambda_2)\cdots(z - \lambda_n) \cdot \bar{\alpha}(z - \overline{\lambda_1})(z - \overline{\lambda_2})\cdots(z - \overline{\lambda_n}).$$

Note that the product contains the term $\alpha\bar{\alpha} = |\alpha|^2 \in \mathbb{R}$, as well as the pairs

$$(z - \lambda_i)(z - \overline{\lambda_i}) = z^2 - 2\Re(\lambda_i)z + |\lambda_i|^2.$$

Each such pair contributes a quadratic polynomial with real coefficients. Since the product of polynomials with real coefficients also has real coefficients, it follows that $q(z)$ is a polynomial with real coefficients. ■

A5 Eigenvalues and Eigenvectors

Exercise 5A27

Suppose that V is finite-dimensional and $k \in \{1, \ldots, \dim V - 1\}$. Suppose $T \in \mathcal{L}(V)$ is such that every subspace of V of dimension k is invariant under T. Prove that T is a scalar multiple of the identity operator.

Civil Service Pigeon. Let $v \in V$ be nonzero. For any $u \notin \operatorname{span}(v)$, we can extend v, u to a basis and construct a k-dimensional subspace that contains v but not u since $k \in \{1, \ldots, \dim V - 1\}$. So, the intersection of all k-dimensional subspaces containing v is precisely $\operatorname{span}(v)$.

The intersection of invariant subspaces is invariant, so $\operatorname{span}(v)$ is invariant. This means that every nonzero vector is an eigenvector of T, meaning that T is a scalar multiple of the identity by Exercise 5A26. ∎

Exercise 5A42

Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \ldots, x_n) = (x_1, 2x_2, 3x_3, \ldots, nx_n)$.

(a) Find all eigenvalues and eigenvectors of T.
(b) Find all subspaces of $\mathbb{F}^n$ that are invariant under T.

Civil Service Pigeon. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{F}^n$.

(a) For each $k \in \{1, \ldots, n\}$, we have $Te_k = ke_k$. Hence, the eigenvalues of T are $1, 2, \ldots, n$ and the eigenvectors corresponding to k are the nonzero scalar multiples of e_k. Note that this list of eigenvalues (and eigenvectors) is exhaustive because $\dim(\mathbb{F}^n) = n$.

(b) Let $U \subseteq \mathbb{F}^n$ be an invariant subspace under T. Take any $u = \sum_{i=1}^{n} \alpha_i e_i \in U$ and suppose $\alpha_k \neq 0$ for some k. By Lagrange interpolation, there exists a polynomial p such that $p(i) = \delta_{i,k}$ for all $i \in \{1, \ldots, n\}$. Since U is invariant under T, it is also invariant under $p(T)$, and so

$$p(T)u = \sum_{i=1}^{n} \alpha_i p(T) e_i = \sum_{i=1}^{n} \alpha_i p(i) e_i = \alpha_k e_k \in U \implies e_k \in U.$$

Repeating this argument for every index with a nonzero coefficient in some element of U shows that U contains each of the basis vectors $e_1, \ldots, e_n$. Hence, $U = \operatorname{span}\{e_i : i \in S\}$ for some $S \subseteq \{1, \ldots, n\}$.

Conversely, any such span is invariant because $Te_i = ie_i$ belongs to that span. There are 2^n such subspaces, and they exhaust all invariant subspaces of T. ∎

Exercise 5A43

Suppose that V is finite-dimensional, $\dim V > 1$, and $T \in \mathcal{L}(V)$. Prove that $\{p(T) \mid p \in \mathcal{P}(\mathbb{F})\} \neq \mathcal{L}(V)$.

blanketism. Let $v_1, \ldots, v_n$ be a basis for V, with $n = \dim V$. Consider the operators $T_1, T_2 \in \mathcal{L}(V)$ defined by

$$T_1 v_1 = v_2, \quad T_1 v_i = 0 \text{ for } i = 2, \ldots, n,$$

and

$$T_2 v_1 = v_1, \quad T_2 v_i = 0 \text{ for } i = 2, \ldots, n.$$

Then $T_2 T_1 v_1 = T_2(T_1 v_1) = T_2 v_2 = 0$, and $T_1 T_2 v_1 = T_1(T_2 v_1) = T_1 v_1 = v_2$. Hence, $T_2 T_1 \neq T_1 T_2$, so T_1 and T_2 do not commute. Let $S = \{p(T) \mid p \in \mathcal{P}(\mathbb{F})\}$. Let $p, q \in \mathcal{P}$ be defined as

$$p(z) = a_0 + a_1 z + \cdots + a_m z^m, \quad q(z) = b_0 + b_1 z + \cdots + b_k z^k.$$

Since $p(T)q(T) = q(T)p(T)$, it follows that all operators in S commute. However, if $T_1 = r(T)$ and $T_2 = s(T)$ for some polynomials $r, s \in \mathcal{P}$, then T_1 and T_2 would commute, which is a contradiction. Hence, T_1 and T_2 cannot both be in S, so $S \neq \mathcal{L}(V)$. ■

Exercise 5B29

Show that every operator on a finite-dimensional vector space of dimension at least two has an invariant subspace of dimension two.

Axler Note. Exercise 5C6 *will give an improvement of this result when* $\mathbb{F} = \mathbb{C}$.

Civil Service Pigeon. If $\mathbb{F} = \mathbb{C}$, the result is immediate: every operator on a finite-dimensional complex vector space has an upper-triangular matrix with respect to some basis $v_1, \ldots, v_n$. Then $\operatorname{span}(v_1, v_2)$ is a 2-dimensional invariant subspace. Hence, we restrict to $\mathbb{F} = \mathbb{R}$ from now on.

Assume $\dim V \geq 2$ and let $T \in \mathcal{L}(V)$ have minimal polynomial p. Over $\mathbb{R}$, p factors into linear and irreducible quadratic factors.

Case 1: p has an irreducible quadratic factor. Write $p(z) = (z^2 + az + b)m(z)$. Pick $v \in V$ such that $x = m(T)v \neq 0$; then $(T^2 + aT + bI)x = 0$. Set $U = \operatorname{span}(x, Tx)$. Clearly, T maps U to itself since $T(Tx) = -bx - aTx \in U$. If x and Tx were linearly dependent, x would be an eigenvector and its eigenvalue would be a real root of $z^2 + az + b$, contradicting irreducibility. Hence $\dim U = 2$.

Case 2: p splits completely into linear factors. Then T is upper-triangularizable over $\mathbb{R}$. Let v_1, $\ldots$, v_n be a basis with respect to which the matrix of T is upper-triangular. Then, $Tv_1 \in \operatorname{span}(v_1)$ and $Tv_2 \in \operatorname{span}(v_1, v_2)$. Therefore $\operatorname{span}(v_1, v_2)$ is a 2-dimensional invariant subspace.

These two cases cover all possibilities over $\mathbb{R}$, and so combining with trivial complex case finishes the proof. ■

Exercise 5C3

Suppose $T \in \mathcal{L}(V)$ is invertible and $v_1, \ldots, v_n$ is a basis of V with respect to which the matrix of T is upper triangular, with $\lambda_1, \ldots, \lambda_n$ on the diagonal. Show that the matrix of T^{-1} is also upper triangular with respect to the basis $v_1, \ldots, v_n$, with

$$\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}$$

on the diagonal.

blanketism. Let A be the matrix of T with respect to the basis $v_1, \ldots, v_n$. Since T is invertible, A is also invertible. Let $A = (a_{ij})$ be the upper-triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal, and let $B = (b_{ij})$ be the matrix of T^{-1} with respect to the same basis. Denote b_j as the j-th column of B. Then we have

$$Ab_j = e_j,$$

where e_j is the j-th column of the identity matrix. In the nth row, we have $\lambda_n b_{nj} = \delta_{nj}$, hence $b_{nj} = 0$ for $j < n$ and $b_{nn} = \lambda_n^{-1}$. Now, consider the $(n-1)$-th row. We have

$$a_{n-1,n-1}b_{n-1,j} + a_{n-1,n}b_{n,j} = \delta_{n-1,j}.$$

For $j < n-1$, we have $b_{n,j} = 0$ and $\delta_{n-1,j} = 0$, so $b_{n-1,j} = 0$. For $j = n-1$, we have $b_{n,n-1} = 0$ and $\delta_{n-1,n-1} = 1$, so $b_{n-1,n-1} = a_{n-1,n-1}^{-1} = \lambda_{n-1}^{-1}$. Assume inductively that for some $k \in \{1, \ldots, n-1\}$, we have $b_{ij} = 0$ for all $i > j$ and $b_{ii} = \lambda_i^{-1}$ for all $i \geq k+1$. Now, consider the k-th row. We have

$$a_{kk}b_{kj} + a_{k,k+1}b_{k+1,j} + \cdots + a_{kn}b_{nj} = \delta_{kj}.$$

For $j < k$, we have $b_{ij} = 0$ for all $i > j \geq k$, so the left-hand side reduces to $a_{kk}b_{kj} = 0$, hence $b_{kj} = 0$. For $j = k$, we have $b_{ij} = 0$ for all $i > k$, so the left-hand side reduces to $a_{kk}b_{kk} = 1$, hence $b_{kk} = a_{kk}^{-1} = \lambda_k^{-1}$. By induction, we conclude that B is upper-triangular with $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$ on the diagonal. ■

A6 Inner Product Spaces

Exercise 6C16

Suppose $C[-1,1]$ is the vector space of continuous real-valued functions on the interval $[-1,1]$ with inner product given by

$$\langle f, g\rangle = \int_{-1}^{1} fg$$

for all $f, g \in C[-1,1]$. Let U be the subspace of $C[-1,1]$ defined by

$$U = \{f \in C[-1,1] \mid f(0) = 0\}.$$

(a) Show that $U^{\perp} = \{0\}$.
(b) Show that 6.49 and 6.52 do not hold without the finite-dimensional hypothesis.

blanketism. A more analysis-based solution to part (a) is as follows.

Let $g \in U^{\perp}$, so $\langle f, g\rangle = 0$ for all $f \in U$. For any $\varepsilon > 0$, define $f_\varepsilon \in C[-1,1]$ by $f_\varepsilon(x) = g(x)$ for $|x| \geq \varepsilon$, $f_\varepsilon(0) = 0$, with f_ε linear on $[-\varepsilon, 0]$ and on $[0, \varepsilon]$ so that f_ε is continuous at $x = 0$ and $x = \pm\varepsilon$. Then $f_\varepsilon \in U$, so $\langle f_\varepsilon, g\rangle = 0$. Let $M = \max_{[-1,1]} |g|$. Since $f_\varepsilon = g$ outside $[-\varepsilon, \varepsilon]$,

$$|\langle g, g\rangle| = |\langle f_\varepsilon - g, g\rangle| = \left|\int_{-\varepsilon}^{\varepsilon} (f_\varepsilon - g)g\right| \leq \int_{-\varepsilon}^{\varepsilon} |f_\varepsilon - g|\,|g| \leq 2M \cdot M \cdot 2\varepsilon = 4\varepsilon M^2.$$

Since $\varepsilon > 0$ is arbitrary, $\langle g, g\rangle = 0$, so $g = 0$. Hence, $U^{\perp} = \{0\}$. ■

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